NPLEX Part I Competencies

Built on the NABNE Part I blueprint and written to USMLE Step 1 depth. Each blueprint unit lists what you should be able to do, tagged by where it comes from.

200
questions in 50 cases of 4

236
study units on the blueprint

1,687
specific competencies

213
USMLE extensions added

Where each competency comes from

NPLEX Required by the NPLEX blueprint

NPLEX + USMLE On both outlines

USMLE extension Step 1 topic added inside the closest NPLEX unit

Weighting by body system

SYSTEM	EXAM %	QUESTIONS	CASES	UNITS	COMPETENCIES	USMLE EXTENSIONS	WEEK
Cardio Cardiovascular System	12%	24	6	28	195	12	1
GI Gastrointestinal System	12%	24	6	24	162	17	2
MSK Musculoskeletal System	10%	20	5	20	140	19	3
Neuro Neurological System	10%	20	5	29	221	35	4
Repro Reproductive System	10%	20	5	23	192	21	5
Urology Urinary System	10%	20	5	19	119	15	6

SYSTEM	EXAM %	QUESTIONS	CASES	UNITS	COMPETENCIES	USMLE EXTENSIONS	WEEK
Endo Endocrine System	8%	16	4	19	125	14	7
Immuno Immunological System	8%	16	4	20	129	5	8
Pulm Pulmonary System and Upper Respiratory Tract	8%	16	4	21	147	21	9
Heme Hematopoietic System	6%	12	3	13	107	24	10
Derm Integumentary System	6%	12	3	20	150	30	11
Total	100%	200	50	236	1,687	213	

Structure/Function 60% (Anatomy, Physiology, Biochemistry & Genetics). Disease/Dysfunction 40% (Microbiology & Immunology, Pathology). You must pass both.

Cardio Cardiovascular System

12% · 24 questions · Week 1

HOW THE BODY WORKS

CV-1 Cardiovascular embryology

Describe embryological development of the cardiovascular system, including valves, chambers, and blood vessels.

CV-1.1 Describe how paired endocardial tubes from splanchnic mesoderm fuse into the primitive heart tube by the end of week 3 and loop to the right in week 4, and relate defective ciliary dynein (Kartagener syndrome) to dextrocardia. NPLEX · Anatomy · DOK 2

CV-1.2 Trace atrial septation through the septum primum, ostium primum, ostium secundum, septum secundum, and foramen ovale, and predict the defect when each step fails (patent foramen ovale, ostium secundum ASD, ostium primum ASD). NPLEX + USMLE · Anatomy · DOK 3

CV-1.3 Explain how fused endocardial cushions divide the atrioventricular canal and help form the membranous septum and the mitral and tricuspid valves, and relate endocardial cushion defects to Down syndrome. NPLEX + USMLE · Anatomy · DOK 2

CV-1.4 Describe how neural crest cells build the spiral aorticopulmonary septum, and predict transposition of the great vessels from failed spiraling, persistent truncus arteriosus from failed septation, and tetralogy of Fallot from anterosuperior infundibular septum displacement. NPLEX + USMLE · Anatomy · DOK 3

CV-1.5 Trace fetal circulation through the umbilical vein, ductus venosus, foramen ovale, ductus arteriosus, and umbilical arteries, and explain why the best oxygenated blood reaches the brain. NPLEX · Anatomy · DOK 2

CV-1.6 Explain closure of the foramen ovale and ductus arteriosus at birth (falling pulmonary resistance, rising left atrial pressure, rising O2, falling prostaglandin E2) and name the adult remnants of fetal vessels. NPLEX + USMLE · Physiology · DOK 2

CV-1.7 Relate the aortic arch derivatives (third arch to common and proximal internal carotid, fourth to aortic arch and right subclavian, sixth to proximal pulmonary arteries and ductus arteriosus) to the different paths of the right and left recurrent laryngeal nerves. NPLEX · Anatomy · DOK 2

CV-2 Heart and vessel histology

Describe microscopic anatomy of the heart and blood vessels.

CV-2.1 Describe the endocardium, myocardium, epicardium, and fibrous cardiac skeleton, and explain how the skeleton electrically insulates the atria from the ventricles except at the AV bundle. NPLEX · Anatomy · DOK 2

CV-2.2 Identify the features of cardiac muscle cells (branched, one or two central nuclei, intercalated discs with fascia adherens, desmosomes, and gap junctions, diads at the Z line) and contrast them with skeletal muscle. NPLEX · Anatomy · DOK 2

CV-2.3 Distinguish Purkinje fibers (large, pale, glycogen rich, few myofibrils, subendocardial) from working myocytes, and identify the atrial granules that store atrial natriuretic peptide. NPLEX · Anatomy · DOK 1

CV-2.4 Compare the tunica intima, media, and adventitia of elastic arteries, muscular arteries, arterioles, continuous, fenestrated, and sinusoidal capillaries, and veins, and relate each wall structure to its function. NPLEX · Anatomy · DOK 2

CV-2.5 Identify endothelial products (nitric oxide, prostacyclin, endothelin-1, and von Willebrand factor and P-selectin stored in Weibel-Palade bodies) and predict how endothelial dysfunction shifts vessel tone and clotting. NPLEX · Physiology · DOK 2

CV-3 Heart, great vessels, and pericardium

Describe location, structure, and boundaries of the heart, major vessels, and pericardium.

CV-3.1 Locate the heart in the middle mediastinum with its apex at the left fifth intercostal space, midclavicular line, and identify the chambers that form the right border, left border, and anterior surface on a chest radiograph. NPLEX · Anatomy · DOK 1

CV-3.2 Relate chamber position to clinical findings: the right ventricle lies most anterior behind the sternum, and the left atrium lies most posterior against the esophagus, so left atrial enlargement can cause dysphagia. NPLEX · Anatomy · DOK 2

CV-3.3 Describe the fibrous and serous (parietal and visceral) pericardium and the transverse and oblique sinuses, and explain why pericardial pain refers to the shoulder through the phrenic nerve (C3 to C5). NPLEX + USMLE · Anatomy · DOK 2

CV-3.4 Identify the great vessels (SVC, IVC, pulmonary trunk and arteries, four pulmonary veins, aortic arch branches from right to left) and the ligamentum arteriosum with the left recurrent laryngeal nerve hooking under the arch. NPLEX | Anatomy | DOK 1

CV-3.5 Identify the internal features of each chamber (crista terminalis, pectinate muscles, fossa ovalis, coronary sinus opening, trabeculae carneae, papillary muscles, moderator band, conus arteriosus) and compare left and right ventricular wall thickness. NPLEX | Anatomy | DOK 1

CV-3.6 Predict injuries from blunt and penetrating chest trauma: aortic tear at the isthmus near the ligamentum arteriosum after sudden deceleration, myocardial contusion, ventricular puncture, and hemopericardium causing tamponade. USMLE extension | Pathology | DOK 2

CV-4 Heart valves and the cardiac cycle

Describe location and explain heart valve function in relation to the cardiac cycle.

CV-4.1 Identify the cusps of each heart valve, the anchoring of atrioventricular cusps by chordae tendineae to papillary muscles, and the chest wall site where each valve is best heard. NPLEX | Anatomy | DOK 1

CV-4.2 Relate valve opening and closing to pressure differences across each phase of the Wiggers diagram (isovolumetric contraction, ejection, isovolumetric relaxation, rapid and reduced filling, atrial systole) and to the timing of S1 and S2. NPLEX | Physiology | DOK 3

CV-4.3 Interpret a left ventricular pressure-volume loop, identifying mitral and aortic valve opening and closing, and predict how the loop changes in aortic stenosis, aortic regurgitation, and mitral regurgitation. NPLEX + USMLE | Physiology | DOK 3

CV-4.4 Explain the jugular venous a, c, and v waves and x and y descents, and predict cannon a waves in AV dissociation and large v waves in tricuspid regurgitation. NPLEX + USMLE | Physiology | DOK 3

CV-4.5 Explain physiologic, wide, fixed, and paradoxical splitting of S2 and give a cause of each (right bundle branch block or pulmonic stenosis, atrial septal defect, left bundle branch block or aortic stenosis). NPLEX + USMLE | Physiology | DOK 3

CV-4.6 Distinguish S3 (rapid filling of a volume-overloaded ventricle) from S4 (atrial contraction into a stiff ventricle), and identify functional flow murmurs in pregnancy, anemia, fever, and childhood. NPLEX + USMLE | Physiology | DOK 2

CV-4.7 Classify murmurs by timing and location (systolic ejection, holosystolic, early diastolic, mid-diastolic, continuous), match each to its lesion, and predict changes with inspiration, squatting, handgrip, and Valsalva. NPLEX + USMLE | Physiology | DOK 3

CV-5 Cardiac muscle contraction

Explain physiological basis of contraction in cardiac muscle.

CV-5.1 Explain the phases of the ventricular myocyte action potential (fast Na+ upstroke, transient K+ outflow, L-type Ca2+ plateau, K+ repolarization, inward rectifier resting potential near -90 mV) and why the long refractory period prevents tetany. NPLEX | Physiology | DOK 2

CV-5.2 Trace excitation-contraction coupling from T-tubule L-type Ca2+ channels through calcium-induced calcium release by ryanodine receptors (RyR2) to Ca2+ binding of troponin C and cross-bridge cycling. NPLEX | Physiology | DOK 2

CV-5.3 Explain relaxation by SERCA (regulated by phospholamban), the Na+/Ca2+ exchanger, and the sarcolemmal Ca2+-ATPase, and predict why ATP loss in ischemia impairs both contraction and relaxation. NPLEX | Physiology | DOK 3

CV-5.4 Explain how beta-1 adrenergic stimulation acts through cAMP and protein kinase A to phosphorylate L-type Ca2+ channels, phospholamban, and troponin I, raising contractility and speeding relaxation. NPLEX | Physiology | DOK 2

CV-5.5 Explain the cardiac length-tension relationship at the sarcomere level (filament overlap and troponin C Ca2+ sensitivity) and contrast how force is graded in cardiac versus skeletal muscle. NPLEX | Physiology | DOK 2

CV-6 Cardiac cycle and its regulation

Explain functions and regulatory mechanisms of the cardiac cycle.

CV-6.1 Calculate cardiac output, stroke volume, ejection fraction, and mean arterial pressure from given values, and use the Fick principle to calculate cardiac output from O2 consumption and arteriovenous O2 content. NPLEX | Physiology | DOK 2

CV-6.2 Predict how changes in preload, afterload, and contractility alter stroke volume, end-systolic volume, and the position of the Frank-Starling curve. NPLEX + USMLE · Physiology · DOK 3

CV-6.3 Interpret combined cardiac function and venous return curves and predict the new operating point after hemorrhage, volume infusion, venoconstriction, increased contractility, or a change in total peripheral resistance. NPLEX · Physiology · DOK 3

CV-6.4 Apply the law of Laplace to explain why concentric hypertrophy normalizes wall stress and a dilated ventricle needs more O2, and list the determinants of myocardial O2 demand (heart rate, contractility, wall stress). NPLEX + USMLE · Physiology · DOK 3

CV-6.5 Trace the baroreceptor reflex (carotid sinus through CN IX, aortic arch through CN X, nucleus tractus solitarius) and predict heart rate and vascular resistance changes after standing, hemorrhage, and carotid sinus massage. NPLEX + USMLE · Physiology · DOK 3

CV-6.6 Explain hormonal control of blood volume and pressure by the renin-angiotensin-aldosterone system, ADH, and the natriuretic peptides ANP and BNP. NPLEX · Physiology · DOK 2

CV-6.7 Explain the mechanisms of orthostatic hypotension (volume depletion, autonomic failure such as diabetic neuropathy, prolonged bed rest) and define it as a fall of 20 mmHg systolic or 10 mmHg diastolic within 3 minutes of standing. USMLE extension · Physiology · DOK 2

CV-7 Conduction system and the ECG

Describe location, function, autonomic regulation, and electrical measurement of the conduction system.

CV-7.1 Locate the SA node, AV node, bundle of His, bundle branches, and Purkinje fibers, trace an impulse through them, and explain why AV nodal delay allows the atria to finish filling the ventricles. NPLEX + USMLE · Anatomy · DOK 2

CV-7.2 Explain the pacemaker action potential (funny current through HCN channels, T-type and L-type Ca2+ currents, K+ repolarization) and how norepinephrine and acetylcholine change the slope of phase 4. NPLEX · Physiology · DOK 2

CV-7.3 Relate the blood supply of the SA and AV nodes (from the RCA in most hearts) to sinus bradycardia and AV block during an inferior myocardial infarction. NPLEX + USMLE · Anatomy · DOK 3

CV-7.4 Interpret a normal ECG: identify the P wave, PR interval, QRS complex, ST segment, T wave, and QT interval, calculate heart rate, and estimate the mean QRS axis from leads I and aVF. NPLEX + USMLE · Physiology · DOK 2

CV-7.5 Distinguish first degree, Mobitz I, Mobitz II, and third degree AV block and right and left bundle branch block on ECG, and predict which AV blocks arise in the AV node and which arise below it. NPLEX + USMLE · Physiology · DOK 3

CV-7.6 Identify premature beats, atrial fibrillation, atrial flutter, multifocal atrial tachycardia, paroxysmal supraventricular tachycardia, ventricular tachycardia, torsades de pointes, and ventricular fibrillation on ECG, and relate each to its mechanism and risk of cardiac arrest. NPLEX + USMLE · Physiology · DOK 3

CV-7.7 Explain how an accessory pathway in Wolff-Parkinson-White syndrome produces a short PR interval and delta wave, and describe sick sinus syndrome as SA node dysfunction causing sinus bradycardia, sinus pauses, or alternating bradycardia and tachycardia. USMLE extension · Physiology · DOK 3

CV-8 Coronary circulation

Describe location and branching patterns of coronary arteries and circulatory pathways of blood supply.

CV-8.1 Trace the origins and courses of the right coronary artery, left main, left anterior descending, and circumflex arteries, and identify the myocardial regions each supplies. NPLEX + USMLE · Anatomy · DOK 2

CV-8.2 Define coronary dominance by the origin of the posterior descending artery and predict why the posteromedial papillary muscle, with its single supply, is the one that ruptures after an inferior infarction. NPLEX + USMLE · Anatomy · DOK 3

CV-8.3 Trace cardiac venous drainage through the great, middle, and small cardiac veins to the coronary sinus, and the anterior cardiac and smallest cardiac veins that bypass it. NPLEX · Anatomy · DOK 1

CV-8.4 Explain why left ventricular perfusion occurs mainly in diastole, why the myocardium extracts about 70% of arterial O2 at rest, and why tachycardia and high wall stress make the subendocardium ischemic first. NPLEX + USMLE · Physiology · DOK 3

CV-8.5 Explain local metabolic control of coronary flow by adenosine and other metabolites and why increased myocardial O2 demand must be met by increased flow rather than greater extraction. NPLEX · Physiology · DOK 2

CV-8.6 Explain how an anomalous left coronary artery arising from the pulmonary artery causes myocardial ischemia in infancy once pulmonary artery pressure falls after birth. USMLE extension Pathology DOK 3

CV-9 Blood distribution to body regions

Describe anatomical patterns of blood distribution to somatic and visceral areas.

CV-9.1 Identify the branches of the aortic arch, descending thoracic aorta, and abdominal aorta with their vertebral levels (celiac trunk T12, superior mesenteric L1, renal about L1, inferior mesenteric L3, bifurcation L4). NPLEX Anatomy DOK 1

CV-9.2 Trace the arterial supply of the brain through the internal carotid and vertebrobasilar systems and identify the vessels that form the circle of Willis. NPLEX Anatomy DOK 1

CV-9.3 Trace the main arteries of the upper limb (subclavian, axillary, brachial, radial, ulnar) and lower limb (external iliac, femoral, popliteal, anterior and posterior tibial, fibular, dorsalis pedis) and locate the palpable pulse points. NPLEX Anatomy DOK 2

CV-9.4 Trace venous return through the SVC and IVC, the left renal vein crossing anterior to the aorta, and the hepatic portal vein, and identify the portosystemic anastomoses that dilate in portal hypertension. NPLEX Anatomy DOK 2

CV-9.5 Identify watershed zones (splenic flexure, rectosigmoid junction, border zones between cerebral artery territories) and explain why they infarct first during systemic hypotension. NPLEX Pathology DOK 3

CV-9.6 Compare the share of resting cardiac output received by the gut and liver, kidneys, skeletal muscle, brain, and heart, and relate each to its O2 extraction. NPLEX Physiology DOK 2

CV-10 Lymphatic vessels, tissues, and organs

Describe location, structure, circulatory pathways, and functions of lymphatic vessels, tissues, and organs.

CV-10.1 Describe the structure of lymphatic capillaries (blind ends, overlapping endothelial flaps, anchoring filaments) and explain how they return about 2 to 3 L of filtered fluid and protein to the blood each day. NPLEX Anatomy DOK 2

CV-10.2 Trace lymph from collecting vessels through the cisterna chyli and thoracic duct to the left venous angle, and identify the regions drained by the right lymphatic duct. NPLEX Anatomy DOK 1

CV-10.3 Explain the forces that move lymph forward: valves, rhythmic contraction of lymphangions, the skeletal muscle pump, breathing, and rising interstitial fluid pressure. NPLEX Physiology DOK 2

CV-10.4 Describe the structure and functions of lymph nodes, spleen (red and white pulp), thymus (cortex and medulla), and mucosa-associated lymphoid tissue, including where B and T cells mature and are selected. NPLEX Anatomy DOK 2

CV-10.5 Explain the causes and features of lymphedema (node dissection, radiation, tumor, Wuchereria bancrofti filariasis), including swelling that becomes firm and nonpitting as fibrosis develops, and chylothorax from thoracic duct injury. NPLEX + USMLE Pathology DOK 2

CV-11 Hemodynamics and blood flow regulation

Describe forces involved in circulation of blood and lymph, and regulation of blood flow.

CV-11.1 Apply the relationship flow equals pressure gradient divided by resistance, and Poiseuille's law, to predict how changes in vessel radius, length, and blood viscosity alter resistance and flow. NPLEX Physiology DOK 3

CV-11.2 Calculate total resistance for vessels arranged in series and in parallel, and explain why arterioles set total peripheral resistance and why parallel organ circuits can adjust flow independently. NPLEX Physiology DOK 2

CV-11.3 Relate blood velocity to total cross-sectional area, and predict turbulence, bruits, and flow murmurs from Reynolds number factors such as anemia, high velocity, and stenosis. NPLEX Physiology DOK 3

CV-11.4 Explain how arterial compliance and stroke volume set pulse pressure, and predict isolated systolic hypertension with a wide pulse pressure from arterial stiffening with age. NPLEX + USMLE Physiology DOK 3

CV-11.5 Apply the Starling equation to predict edema from raised capillary hydrostatic pressure, low plasma oncotic pressure, increased capillary permeability, or lymphatic obstruction. NPLEX Physiology DOK 3

CV-11.6 Distinguish metabolic, myogenic, endothelial, neural, and hormonal control of arteriolar tone, and identify the main control of flow in the coronary, cerebral, skeletal muscle, skin, and pulmonary beds. NPLEX Physiology DOK 3

CV-12 Exercise and the cardiovascular system

Explain acute and adaptive effects of exercise on the cardiovascular system.

CV-12.1 Explain how central command and muscle mechanoreceptor and metaboreceptor reflexes start the exercise response, with early vagal withdrawal followed by rising sympathetic drive. NPLEX Physiology DOK 2

CV-12.2 Predict changes in heart rate, stroke volume, cardiac output, arteriovenous O2 difference, and O2 uptake during graded exercise using the Fick principle. NPLEX Physiology DOK 3

CV-12.3 Explain the redistribution of cardiac output during exercise: metabolic dilation in active muscle, sympathetic constriction in gut and kidney, rising skin flow for heat loss, and stable brain flow. NPLEX Physiology DOK 2

CV-12.4 Compare blood pressure and total peripheral resistance responses in dynamic (aerobic) exercise and static (isometric) exercise, and relate each to volume or pressure load on the heart. NPLEX Physiology DOK 2

CV-12.5 Distinguish the trained endurance athlete's heart (eccentric remodeling, larger stroke volume, resting bradycardia from vagal tone, higher maximal O2 uptake) from pathologic hypertrophy such as hypertrophic cardiomyopathy. NPLEX Physiology DOK 3

CV-13 Cardiovascular biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to cardiovascular function and pathology.

CV-13.1 Trace the exogenous and endogenous lipoprotein pathways (chylomicrons with apo B-48, VLDL, IDL, LDL with apo B-100, HDL with apo A-I) and the roles of lipoprotein lipase, apo C-II, apo E, LCAT, and CETP. NPLEX + USMLE Biochemistry & Genetics DOK 2

CV-13.2 Classify familial dyslipidemias by defect and lipid pattern: lipoprotein lipase or apo C-II deficiency (type I, chylomicrons, pancreatitis), LDL receptor defects (type II), apo E2 (type III, remnants), and hepatic VLDL overproduction (type IV). NPLEX + USMLE Biochemistry & Genetics DOK 3

CV-13.3 Explain how HMG-CoA reductase controls cholesterol synthesis, how SREBP raises LDL receptor and HMG-CoA reductase expression when cell sterol is low, and how LDL enters cells by receptor-mediated endocytosis. NPLEX Biochemistry & Genetics DOK 2

CV-13.4 Explain the fuels of the heart (mostly fatty acid beta-oxidation through the carnitine shuttle at rest, plus glucose, lactate, and ketones) and the switch to anaerobic glycolysis with lactate build-up during ischemia. NPLEX Biochemistry & Genetics DOK 2

CV-13.5 Explain the role of thiamine pyrophosphate in pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase, and relate thiamine deficiency to wet beriberi (high-output heart failure, vasodilation, edema) in alcohol use disorder and malnutrition. NPLEX + USMLE Biochemistry & Genetics DOK 3

CV-13.6 Explain how deficiency of B6, B12, or folate raises homocysteine and vascular risk, and how vitamin C deficiency (scurvy) and copper deficiency (lysyl oxidase, Menkes disease) weaken vessel walls. NPLEX Biochemistry & Genetics DOK 2

CV-13.7 Predict ECG changes of hypokalemia (U waves, flat T waves), hyperkalemia (peaked T waves, wide QRS), hypocalcemia (long QT), and hypercalcemia (short QT), and relate hypomagnesemia to torsades de pointes. NPLEX + USMLE Biochemistry & Genetics DOK 3

CV-14 Cardiovascular genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying cardiovascular disease.

CV-14.1 Explain the genetics of familial hypercholesterolemia: autosomal dominant LDLR, APOB, or gain-of-function PCSK9 variants, the gene-dosage difference between heterozygotes and homozygotes, and loss of feedback on liver cholesterol synthesis. NPLEX + USMLE Biochemistry & Genetics DOK 3

CV-14.2 Relate hypertrophic cardiomyopathy to autosomal dominant sarcomere gene mutations (MYH7, MYBPC3), myocyte disarray, and the variable expressivity and incomplete penetrance seen among relatives. NPLEX + USMLE Biochemistry & Genetics DOK 2

CV-14.3 Relate Marfan syndrome (FBN1 on chromosome 15, increased TGF-beta signaling), Loeys-Dietz syndrome (TGFBR1 or TGFBR2), and vascular Ehlers-Danlos syndrome (COL3A1) to aortic dilation, dissection, mitral valve prolapse, and arterial rupture. NPLEX + USMLE Biochemistry & Genetics DOK 3

CV-14.4 Explain congenital long QT syndrome (loss of KCNQ1 or KCNH2, gain of SCN5A), compare Jervell and Lange-Nielsen and Romano-Ward inheritance, and relate Brugada syndrome (SCN5A loss) to sudden cardiac death. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

CV-14.5 Associate genetic syndromes with heart defects: Down syndrome (endocardial cushion defects), Turner syndrome (bicuspid aortic valve, coarctation), 22q11.2 deletion (tetralogy, truncus arteriosus), Williams syndrome (supravalvular aortic stenosis), and VACTERL association. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

CV-14.6 Relate Duchenne and Becker muscular dystrophy (X-linked dystrophin), Friedreich ataxia (frataxin GAA repeat), and mitochondrial DNA disorders with heteroplasmy (Kearns-Sayre syndrome) to cardiomyopathy or conduction block. NPLEX | Biochemistry & Genetics | DOK 2

CV-14.7 Explain why adult cardiomyocytes, which have largely left the cell cycle, respond to pressure or volume load by hypertrophy with reactivated fetal genes (ANP, BNP, beta-myosin heavy chain), and why infarcts heal by scar. NPLEX | Biochemistry & Genetics | DOK 2

CV-15 Heart and lung interactions

Explain relationship between cardiovascular and pulmonary systems.

CV-15.1 Compare pressures and resistance in the pulmonary and systemic circuits, and explain how recruitment and distension lower pulmonary vascular resistance as cardiac output rises. NPLEX | Physiology | DOK 2

CV-15.2 Contrast hypoxic pulmonary vasoconstriction with hypoxic vasodilation in systemic arterioles, and predict pulmonary hypertension from chronic global hypoxia in COPD or at high altitude. NPLEX | Physiology | DOK 3

CV-15.3 Explain how breathing changes venous return and ventricular filling, including normal S2 splitting, pulsus paradoxus, and the drop in cardiac output with positive pressure ventilation. NPLEX | Physiology | DOK 3

CV-15.4 Predict pulmonary edema when pulmonary capillary pressure exceeds plasma oncotic pressure in left heart failure or mitral stenosis, and right ventricular failure (cor pulmonale) from chronic lung disease or pulmonary embolism. NPLEX + USMLE | Physiology | DOK 3

CV-15.5 Explain the gravity-dependent distribution of blood flow and ventilation from apex to base, the resulting V/Q ratios, and the conditions that create zone 1 (no flow). NPLEX | Physiology | DOK 2

CV-15.6 Explain why hypoxemia from a right-to-left shunt barely improves with 100% O2, and identify nonrespiratory roles of the pulmonary circulation (ACE conversion of angiotensin I, bradykinin breakdown, filtering of small clots). NPLEX | Physiology | DOK 3

CV-16 Immune responses in cardiovascular disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

CV-16.1 Explain atherosclerosis as chronic inflammation: endothelial VCAM-1, monocyte and T-cell recruitment, oxidized LDL uptake by scavenger receptors into foam cells, IL-1 beta release, and metalloproteinases that thin the fibrous cap. NPLEX + USMLE | Microbiology & Immunology | DOK 3

CV-16.2 Explain acute rheumatic fever as molecular mimicry between group A streptococcal M protein and cardiac antigens, and identify Aschoff bodies, Anitschkow cells, pancarditis, ASO titers, and the Jones major criteria. NPLEX + USMLE | Microbiology & Immunology | DOK 3

CV-16.3 Classify vasculitides by immune mechanism: immune complex (polyarteritis nodosa with hepatitis B, IgA vasculitis, cryoglobulinemia), ANCA-associated (GPA, microscopic polyangiitis, EGPA), and granulomatous T-cell (giant cell and Takayasu arteritis). NPLEX + USMLE | Microbiology & Immunology | DOK 3

CV-16.4 Explain immune-mediated pericardial, valvular, and conduction disease: Dressler syndrome after infarction, Libman-Sacks endocarditis in lupus, and congenital heart block from maternal anti-Ro (SSA) antibodies. NPLEX | Microbiology & Immunology | DOK 2

CV-16.5 Distinguish hyperacute, acute cellular, acute antibody-mediated, and chronic heart transplant rejection (cardiac allograft vasculopathy) by immune mechanism and timing. NPLEX | Microbiology & Immunology | DOK 2

CV-16.6 Describe how viral myocarditis moves from direct myocyte injury to a cytotoxic T-cell response and dilated cardiomyopathy, and identify eosinophilic hypersensitivity myocarditis and rare myocarditis after mRNA COVID-19 vaccination. NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-16.7 Explain how lost T-cell surveillance in AIDS or transplant immunosuppression lets HHV-8 driven Kaposi sarcoma grow and why lesions often regress when immunity returns, and relate atrial myxoma IL-6 release to fever and weight loss. NPLEX + USMLE | Microbiology & Immunology | DOK 3

CV-17 Cardiovascular pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

CV-17.1 Compare endocarditis bacteria by morphology and virulence factors: Staphylococcus aureus (gram-positive cocci in clusters, coagulase, protein A, clumping factor), viridans streptococci (cocci in chains, alpha-hemolytic, optochin resistant, dextran), Enterococcus faecalis, and Staphylococcus epidermidis (biofilm on prosthetic material). NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-17.2 Identify causes of culture-negative endocarditis (HACEK organisms, Coxiella burnetii, Bartonella) and relate Streptococcus gallolyticus endocarditis to colon cancer. NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-17.3 Describe the Trypanosoma cruzi life cycle (triatomine bug feces, nondividing blood trypomastigotes, intracellular amastigotes in heart and smooth muscle) and its transmission by transfusion, transplant, pregnancy, and food. NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-17.4 Describe Borrelia burgdorferi (spirochete, Ixodes tick, white-footed mouse reservoir, OspA to OspC switching, VlsE variation) and Rickettsia rickettsii (Dermacentor tick, obligate intracellular, endothelial infection, actin-based spread). NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-17.5 Compare flaviviruses (yellow fever, dengue; Aedes mosquitoes; positive-sense RNA) with filoviruses (Ebola, Marburg; negative-sense RNA; fruit bat reservoir; body fluid spread), and explain antibody-dependent enhancement in dengue. NPLEX + USMLE | Microbiology & Immunology | DOK 3

CV-17.6 Explain how coxsackie B virus (nonenveloped positive-sense RNA picornavirus, fecal-oral spread) enters cardiomyocytes through the coxsackie-adenovirus receptor and cleaves dystrophin with protease 2A, and name other myocarditis viruses (adenovirus, parvovirus B19, HHV-6). NPLEX + USMLE | Microbiology & Immunology | DOK 2

CV-17.7 Identify other microbial causes of heart and vessel disease: diphtheria toxin myocarditis, tuberculous constrictive pericarditis, and tertiary syphilis aortitis from obliterative endarteritis of the vasa vasorum. NPLEX | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

CV-18-A Hypertensive heart diseases

1 Pulmonary hypertension **2** Systemic hypertension

CV-18-A.1 Define stage 1 (130 to 139 or 80 to 89 mmHg) and stage 2 (140/90 mmHg or higher) hypertension, and explain primary hypertension as polygenic with risk factors such as age, obesity, sodium intake, and alcohol. NPLEX + USMLE | Pathology | DOK 2

CV-18-A.2 Identify secondary causes of hypertension (renal artery stenosis from atherosclerosis or fibromuscular dysplasia, primary aldosteronism, pheochromocytoma, Cushing syndrome, coarctation, sleep apnea) and the key clue for each. NPLEX + USMLE | Pathology | DOK 3

CV-18-A.3 Describe hypertensive damage: hyaline arteriolosclerosis, concentric left ventricular hypertrophy with diastolic dysfunction and atrial enlargement, nephrosclerosis, retinopathy, and Charcot-Bouchard microaneurysms causing intracerebral hemorrhage. NPLEX + USMLE | Pathology | DOK 3

CV-18-A.4 Distinguish hypertensive emergency (severe elevation, often 180/120 mmHg or higher, with acute organ damage) from chronic hypertension, and identify hyperplastic onion-skin arteriolosclerosis with fibrinoid necrosis. NPLEX + USMLE | Pathology | DOK 3

CV-18-A.5 Define pulmonary hypertension (mean pulmonary artery pressure over 20 mmHg at rest) and classify its causes, including left heart disease (most common), lung disease and hypoxia, chronic thromboembolism, and heritable BMPR2 pulmonary arterial hypertension. NPLEX + USMLE | Pathology | DOK 2

CV-18-A.6 Explain pulmonary arterial hypertension pathogenesis (low NO and prostacyclin, high endothelin-1, smooth muscle proliferation, plexiform lesions) and predict a loud P2, right ventricular heave, right axis deviation, and cor pulmonale. NPLEX + USMLE | Pathology | DOK 3

CV-18-B Congestive heart failure

1 Left-sided **2** Right-sided

CV-18-B.1 Explain the causes of left-sided heart failure and relate pulmonary venous congestion to dyspnea, orthopnea, paroxysmal nocturnal dyspnea, basal crackles, and hemosiderin-laden macrophages (heart failure cells) in the lung. NPLEX + USMLE | Pathology | DOK 3

CV-18-B.2 Explain the causes of right-sided heart failure (most often left-sided failure; also cor pulmonale and right ventricular infarction) and relate systemic venous congestion to raised JVP, hepatojugular reflux, pitting edema, and nutmeg liver. NPLEX + USMLE | Pathology | DOK 3

CV-18-B.3 Distinguish heart failure with reduced ejection fraction (40% or lower, systolic) from preserved ejection fraction (50% or higher, diastolic), and interpret an S3, raised BNP, and chest radiograph signs of congestion. NPLEX + USMLE | Pathology | DOK 3

CV-18-B.4 Explain how sympathetic activation, the renin-angiotensin-aldosterone system, and ADH first support output but then drive fluid retention, remodeling, and progression, while natriuretic peptides partly oppose them. NPLEX + USMLE | Physiology | DOK 3

CV-18-B.5 Predict complications of heart failure: acute pulmonary edema, cardiogenic shock, atrial fibrillation and ventricular arrhythmias, mural thrombi with embolism, cardiorenal syndrome, and cardiac cachexia. NPLEX + USMLE | Pathology | DOK 3

CV-18-B.6 Explain high-output heart failure in severe anemia, thyrotoxicosis, arteriovenous fistula, wet beriberi, and Paget disease of bone, and contrast its low systemic vascular resistance with low-output failure. USMLE extension | Physiology | DOK 3

CV-18-B.7 Predict sudden cardiogenic pulmonary edema and hypotension from acute mitral regurgitation after rupture of chordae tendineae or a papillary muscle. USMLE extension | Pathology | DOK 3

CV-18-C Ischemic heart disease

1 Angina pectoris **2** Chronic ischemic heart disease **3** Myocardial infarction (MI)

CV-18-C.1 Identify coronary atherosclerosis as the cause of most ischemic heart disease, list its risk factors, and name other causes (vasospasm, emboli, vasculitis, increased demand from tachycardia, hypertrophy, or aortic stenosis). NPLEX + USMLE | Pathology | DOK 2

CV-18-C.2 Distinguish stable angina (fixed stenosis of about 70% or more, exertional, ST depression), unstable angina (plaque disruption with partial thrombus, rest pain, no troponin rise), and Prinzmetal angina (vasospasm, transient ST elevation). NPLEX + USMLE | Pathology | DOK 3

CV-18-C.3 Trace the pathogenesis of myocardial infarction from plaque rupture and occlusive thrombus to irreversible injury after about 20 to 40 minutes, and distinguish transmural (often STEMI) from subendocardial (often NSTEMI) infarction. NPLEX + USMLE | Pathology | DOK 3

CV-18-C.4 Predict the occluded coronary artery from ECG lead changes (LAD in V1 to V4; circumflex in I, aVL, V5, V6; RCA in II, III, aVF) and interpret the timing of troponin and CK-MB rise and fall. NPLEX + USMLE | Pathology | DOK 3

CV-18-C.5 Relate the microscopic timeline of infarction (wavy fibers and coagulative necrosis in day 1, neutrophils at 1 to 3 days, macrophages at 3 to 7 days, granulation tissue by 1 to 2 weeks, scar by 2 months) to complications. NPLEX + USMLE | Pathology | DOK 3

CV-18-C.6 Predict complications of infarction by timing: early arrhythmia and cardiogenic shock, fibrinous pericarditis, papillary muscle, septal, or free wall rupture (most often 3 to 7 days), ventricular aneurysm with mural thrombus, and Dressler syndrome. NPLEX + USMLE | Pathology | DOK 3

CV-18-C.7 Describe chronic ischemic heart disease (ischemic cardiomyopathy): healed infarct scars, hypertrophy of surviving myocardium, left ventricular dilation, progressive heart failure, arrhythmias, and sudden cardiac death. NPLEX + USMLE | Pathology | DOK 2

CV-18-D Valvular heart diseases

1 Aortic stenosis/insufficiency **2** Mitral stenosis/insufficiency **3** Endocarditis **4** Mitral valve prolapse (MVP) **5** Rheumatic heart disease **6** Carcinoid heart disease

CV-18-D.1 Explain the causes of aortic stenosis (age-related calcification, bicuspid valve, rheumatic disease) and its findings: crescendo-decrescendo murmur radiating to the carotids, pulsus parvus et tardus, concentric hypertrophy, syncope, angina, and dyspnea. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.2 Explain the causes of aortic regurgitation (aortic root dilation from hypertension, Marfan, or syphilis; bicuspid valve; endocarditis; rheumatic disease) and its findings: early decrescendo diastolic murmur, wide pulse pressure, bounding pulses, eccentric hypertrophy. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.3 Explain mitral stenosis (almost always rheumatic) and its findings: opening snap with diastolic rumble, left atrial enlargement, atrial fibrillation with mural thrombus, pulmonary hypertension, and dysphagia or hoarseness from atrial compression. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.4 Explain the causes of mitral regurgitation (prolapse, papillary muscle ischemia or rupture, left ventricular dilation, rheumatic disease, endocarditis) and its holosystolic apical murmur radiating to the axilla, S3, and left atrial volume overload. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.5 Distinguish acute (Staphylococcus aureus, normal valves) from subacute (viridans streptococci, damaged valves) infective endocarditis, identify the valves most involved, and interpret Janeway lesions, Osler nodes, Roth spots, and splinter hemorrhages. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.6 Distinguish nonbacterial thrombotic (marantic) endocarditis in cancer and hypercoagulable states and Libman-Sacks endocarditis in lupus (small sterile vegetations on either or both leaflet surfaces) from infective endocarditis. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.7 Explain mitral valve prolapse (myxomatous degeneration, links to Marfan, Ehlers-Danlos, and ADPKD), its mid-systolic click and late systolic murmur that move earlier with standing or Valsalva, and its complications. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.8 Explain how repeated rheumatic fever produces chronic rheumatic heart disease, with commissural fusion, thickened and shortened chordae, fish-mouth mitral stenosis, and combined mitral and aortic involvement. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.9 Explain carcinoid heart disease: serotonin from liver metastases of a neuroendocrine tumor reaches the right heart, causing tricuspid regurgitation and pulmonic stenosis, while lung monoamine oxidase spares the left heart; urine 5-HIAA is high. NPLEX + USMLE | Pathology | DOK 3

CV-18-D.10 Identify complications of prosthetic heart valves: valve thrombosis and embolism, mechanical hemolysis with schistocytes, structural failure of bioprosthetic valves, and endocarditis, often from Staphylococcus epidermidis. USMLE extension | Pathology | DOK 2

CV-18-E Primary myocardial diseases

1 Cardiomyopathies (dilated, restrictive, hypertrophic) **2** Myocarditis

CV-18-E.1 Explain dilated cardiomyopathy causes (familial TTN variants, alcohol, prior myocarditis, doxorubicin, peripartum, Chagas, thiamine deficiency, hemochromatosis) and findings: four-chamber dilation, systolic dysfunction, S3, and functional mitral and tricuspid regurgitation. NPLEX + USMLE | Pathology | DOK 3

CV-18-E.2 Explain hypertrophic cardiomyopathy: asymmetric septal hypertrophy, myocyte disarray, outflow obstruction from systolic anterior motion of the mitral valve, a murmur louder with Valsalva and softer with squatting, and sudden death in young athletes. NPLEX + USMLE | Pathology | DOK 3

CV-18-E.3 Explain restrictive cardiomyopathy causes (amyloidosis, sarcoidosis, hemochromatosis, radiation, endomyocardial fibrosis, Loeffler endocarditis) and its diastolic filling failure with right-sided congestion, and identify the low-voltage ECG despite a thick ventricular wall in amyloidosis. NPLEX + USMLE | Pathology | DOK 3

CV-18-E.4 Compare dilated, hypertrophic, and restrictive cardiomyopathy by chamber size, wall thickness, ejection fraction, and type of dysfunction (systolic versus diastolic). NPLEX + USMLE | Pathology | DOK 3

CV-18-E.5 Explain the causes of myocarditis (coxsackie B and other viruses, Chagas, Lyme, diphtheria, drug hypersensitivity, lupus, giant cell myocarditis) and its lymphocytic infiltrate with myocyte necrosis, raised troponin, arrhythmia, and progression to dilated cardiomyopathy. NPLEX + USMLE | Pathology | DOK 3

CV-18-E.6 Explain stress (takotsubo) cardiomyopathy, a catecholamine-linked transient apical ballooning with near-normal coronary arteries, usually in postmenopausal women after acute stress, and tachycardia-induced cardiomyopathy that improves once the rate is controlled. USMLE extension | Pathology | DOK 2

CV-18-F Pericardial disease

1 Metastatic disease **2** Pericardial effusions **3** Pericarditis (primary and secondary)

CV-18-F.1 Explain acute (primary) pericarditis, usually viral such as coxsackie B, and its sharp pleuritic chest pain relieved by sitting forward, friction rub, and diffuse ST elevation with PR depression. NPLEX + USMLE | Pathology | DOK 3

CV-18-F.2 Identify causes of secondary pericarditis (early post-infarction, Dressler syndrome, uremia, lupus, rheumatoid arthritis, radiation, tuberculosis, bacteria, cancer, cardiac surgery) and match them to serous, fibrinous, purulent, hemorrhagic, or caseous exudate. NPLEX + USMLE | Pathology | DOK 2

CV-18-F.3 Explain constrictive pericarditis (tuberculosis, prior viral pericarditis, surgery, radiation) and its Kussmaul sign, pericardial knock, and calcification, and distinguish it from restrictive cardiomyopathy. NPLEX + USMLE | Pathology | DOK 3

CV-18-F.4 Explain the causes of serous and hemorrhagic pericardial effusion and its findings (muffled heart sounds, low-voltage QRS, electrical alternans, enlarged globular heart on radiograph), and why the rate of fluid build-up matters more than volume. NPLEX + USMLE · Pathology · DOK 3

CV-18-F.5 Explain cardiac tamponade, including traumatic or post-rupture hemopericardium, and predict Beck triad (hypotension, distended neck veins, distant heart sounds), pulsus paradoxus, and equal diastolic pressures in all chambers. NPLEX + USMLE · Pathology · DOK 3

CV-18-F.6 Identify metastatic disease as far more common than primary cardiac tumors, name the usual sources (lung, breast, melanoma, lymphoma, leukemia), and relate pericardial spread to bloody effusion with malignant cells and tamponade. NPLEX + USMLE · Pathology · DOK 2

CV-18-G Congenital heart conditions

1 Bicuspid aortic valve **2** Patent ductus arteriosus **3** Septal defects (interventricular, atrial) **4** Tetralogy of Fallot

CV-18-G.1 Explain left-to-right shunt physiology and Eisenmenger syndrome: pulmonary vascular remodeling raises pressure until the shunt reverses, causing late cyanosis, clubbing, polycythemia, and right ventricular hypertrophy. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.2 Explain ventricular septal defect (most often membranous) and predict its harsh holosystolic murmur at the left lower sternal border, the spontaneous closure of many small defects, and heart failure with large defects. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.3 Distinguish ostium secundum ASD (most common), ostium primum ASD (Down syndrome), and patent foramen ovale, and predict a fixed split S2, pulmonic flow murmur, and paradoxical embolism. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.4 Explain patent ductus arteriosus (sixth arch derivative; risk with prematurity and congenital rubella) and its continuous machine-like murmur, bounding pulses, and differential cyanosis of the toes once Eisenmenger syndrome develops. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.5 Explain tetralogy of Fallot (pulmonary infundibular stenosis, right ventricular hypertrophy, overriding aorta, VSD) from anterosuperior infundibular septum displacement, its link to 22q11.2 deletion, boot-shaped heart, and relief of tet spells by squatting. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.6 Explain bicuspid aortic valve (about 1 to 2% of people, linked to Turner syndrome and coarctation) and predict early calcific aortic stenosis, aortic regurgitation, endocarditis, and ascending aortic dilation or dissection. NPLEX + USMLE · Pathology · DOK 3

CV-18-G.7 Explain coarctation of the aorta (linked to Turner syndrome and bicuspid valve) and predict arm hypertension, weak delayed femoral pulses, and rib notching from enlarged intercostal collateral arteries. USMLE extension · Pathology · DOK 3

CV-18-G.8 Explain transposition of the great vessels (failed spiraling of the aorticopulmonary septum, linked to maternal diabetes) and why survival depends on mixing through a PDA, ASD, or VSD. USMLE extension · Pathology · DOK 3

CV-18-H Hemodynamic conditions

1 Embolism **2** Hemorrhage **3** Infarction **4** Edema **5** Shock **6** Thrombosis

CV-18-H.1 Apply Virchow triad to explain thrombus formation, distinguish pale arterial thrombi from red venous thrombi, identify lines of Zahn, and predict thrombus fates (propagation, embolization, dissolution, organization and recanalization). NPLEX · Pathology · DOK 3

CV-18-H.2 Identify inherited hypercoagulable states (factor V Leiden, prothrombin G20210A, antithrombin, protein C or S deficiency) and acquired ones (antiphospholipid syndrome, cancer, pregnancy, estrogen, immobility, surgery). NPLEX + USMLE · Pathology · DOK 2

CV-18-H.3 Classify emboli: venous thromboemboli to the lung, systemic emboli from mural thrombi to brain, legs, gut, kidney, and spleen, fat embolism 1 to 3 days after long-bone fracture, amniotic fluid embolism, and gas embolism. NPLEX + USMLE · Pathology · DOK 3

CV-18-H.4 Distinguish white infarcts (solid organs with end-arterial supply) from red infarcts (venous occlusion, loose tissue, dual supply, reperfusion), coagulative from liquefactive necrosis, and the factors that decide infarct size. NPLEX · Pathology · DOK 3

CV-18-H.5 Define petechiae, purpura, ecchymosis, and hematoma, and predict the effect of hemorrhage by volume and site, including shock after acute loss of more than about 20% of blood volume. NPLEX Pathology DOK 2

CV-18-H.6 Distinguish transudate from exudate edema, relate each to Starling forces or inflammation, and contrast pitting edema with nonpitting edema of lymphedema. NPLEX Pathology DOK 3

CV-18-H.7 Classify shock as hypovolemic, cardiogenic, obstructive, or distributive (septic, anaphylactic, neurogenic) and predict cardiac output, systemic vascular resistance, and pulmonary capillary wedge pressure in each. NPLEX + USMLE Pathology DOK 3

CV-18-H.8 Explain the stages of shock (compensated, progressive with lactic acidosis, irreversible), septic shock mediators (endotoxin, TNF, IL-1, nitric oxide), and resulting multiorgan dysfunction (acute tubular necrosis, ARDS, DIC). NPLEX + USMLE Pathology DOK 3

CV-18-I Vascular conditions

1 Aneurysm **2** Aortic dissection **3** Arteriosclerosis and atherosclerosis **4** Familial hypercholesterolemia **5** Giant cell arteritis (temporal arteritis) **6** Peripheral arterial disease (PAD) **7** Pulmonary embolism (PE) **8** Raynaud phenomenon (primary and secondary) **9** Thromboangiitis obliterans **10** Thrombosis, deep vein (DVT) **11** Varicose veins **12** Vasculitis

CV-18-I.1 Distinguish the three patterns of arteriosclerosis: atherosclerosis, arteriolosclerosis (hyaline and hyperplastic), and Monckeberg medial calcific sclerosis, which does not narrow the lumen. NPLEX Pathology DOK 2

CV-18-I.2 Trace atherosclerosis from endothelial injury to fatty streak to fibrous plaque, identify the arteries most affected (abdominal aorta, coronary, popliteal, carotid), and predict complications (rupture, thrombosis, embolism, aneurysm). NPLEX + USMLE Pathology DOK 3

CV-18-I.3 Explain familial hypercholesterolemia findings: LDL often above 190 mg/dL in heterozygotes, coronary disease in childhood or the teens in homozygotes, tendon xanthomas, xanthelasma, and early corneal arcus. NPLEX + USMLE Pathology DOK 3

CV-18-I.4 Distinguish abdominal aortic aneurysm (atherosclerosis, older male smokers, infrarenal, rupture risk rising with diameter) from thoracic aneurysm (cystic medial degeneration, tertiary syphilis), berry aneurysms, and mycotic aneurysms. NPLEX + USMLE Pathology DOK 3

CV-18-I.5 Explain aortic dissection (intimal tear into the media; hypertension, Marfan, bicuspid valve, Turner, cocaine), its tearing back pain, unequal arm pressures, widened mediastinum, Stanford A versus B, and complications such as tamponade and aortic regurgitation. NPLEX + USMLE Pathology DOK 3

CV-18-I.6 Explain peripheral arterial disease (claudication, rest pain, ulcers, ankle-brachial index 0.90 or lower, Leriche syndrome) and atheroembolic disease with blue toes and cholesterol clefts after arterial catheterization. NPLEX + USMLE Pathology DOK 3

CV-18-I.7 Explain pulmonary embolism from deep leg vein thrombi and predict sudden dyspnea, pleuritic pain, tachycardia, hypoxemia with raised A-a gradient, acute right ventricular strain, and why most emboli do not cause lung infarction. NPLEX + USMLE Pathology DOK 3

CV-18-I.8 Explain deep vein thrombosis risk factors through Virchow triad and its unilateral warm, swollen, painful leg, and distinguish it from superficial thrombophlebitis, including migratory thrombophlebitis with cancer (Trousseau sign). NPLEX + USMLE Pathology DOK 3

CV-18-I.9 Explain varicose veins and chronic venous insufficiency (incompetent valves, risk with age, pregnancy, obesity, prolonged standing) and predict edema, stasis dermatitis, and ulcers near the medial malleolus. NPLEX + USMLE Pathology DOK 2

CV-18-I.10 Distinguish primary Raynaud phenomenon (benign vasospasm in young women) from secondary Raynaud with scleroderma, lupus, or Buerger disease, and explain the white, blue, red color sequence. NPLEX + USMLE Pathology DOK 3

CV-18-I.11 Explain thromboangiitis obliterans (heavy smokers, often young men; segmental thrombosing inflammation of medium and small arteries extending to veins and nerves) and predict digital ulcers and gangrene unless tobacco use stops. NPLEX + USMLE Pathology DOK 3

CV-18-I.12 Explain giant cell arteritis (age over 50, granulomatous inflammation of carotid branches, headache, jaw claudication, high ESR, skip lesions on biopsy), its risk of blindness, and its link to polymyalgia rheumatica. NPLEX + USMLE Pathology DOK 3

CV-18-I.13 Explain Takayasu arteritis (granulomatous large-vessel vasculitis of the aortic arch and branches, usually young Asian women) and predict weak upper limb pulses and unequal arm blood pressures. NPLEX + USMLE Pathology DOK 3

CV-18-I.14 Explain polyarteritis nodosa (hepatitis B link, transmural necrotizing inflammation of medium arteries with lesions of different ages, renal and mesenteric microaneurysms, lung sparing, ANCA negative). NPLEX + USMLE Pathology DOK 3

CV-18-I.15 Identify Kawasaki disease in a child under 5 (fever, conjunctivitis, strawberry tongue, rash, swollen hands and feet, cervical lymphadenopathy) and predict coronary artery aneurysm as its key complication. NPLEX + USMLE Pathology DOK 3

CV-18-I.16 Distinguish ANCA-associated small-vessel vasculitis: granulomatosis with polyangiitis (PR3–ANCA, sinus, lung, and kidney), microscopic polyangiitis (MPO-ANCA, no granulomas), and eosinophilic granulomatosis with polyangiitis (asthma, eosinophilia, MPO-ANCA). NPLEX + USMLE Pathology DOK 3

CV-18-I.17 Identify IgA vasculitis (palpable purpura on buttocks and legs, arthralgia, abdominal pain, IgA nephropathy, often after an upper respiratory infection in children) and Behcet syndrome (oral and genital ulcers, uveitis). NPLEX + USMLE Pathology DOK 2

CV-18-J Vascular neoplasms

1 Hemangiomas **2** Kaposi sarcoma

CV-18-J.1 Distinguish capillary hemangioma, including the infantile (strawberry) type that appears in the first weeks of life, grows fast, regresses, and is GLUT1 positive, from cavernous hemangioma, which does not regress and is the most common benign liver tumor. NPLEX + USMLE Pathology DOK 2

CV-18-J.2 Distinguish pyogenic granuloma (including the pregnancy gingival form), cherry hemangioma, and bacillary angiomatosis (Bartonella in immunocompromised hosts, neutrophilic infiltrate) from hemangiomas and Kaposi sarcoma. NPLEX Pathology DOK 3

CV-18-J.3 Explain Kaposi sarcoma pathogenesis: HHV-8 infection, spindle cells forming slit-like vascular spaces, viral cyclin D homolog and LANA effects on p53 and Rb, and growth favored by immune suppression; lesions progress from patch to plaque to nodule. NPLEX + USMLE Pathology DOK 3

CV-18-J.4 Compare the classic, endemic African, transplant-associated, and AIDS-associated forms of Kaposi sarcoma by population, sites, and course. NPLEX + USMLE Pathology DOK 2

CV-18-J.5 Identify atrial myxoma as the most common primary cardiac tumor in adults (pedunculated, usually left atrial, ball-valve obstruction, emboli, constitutional symptoms) and cardiac rhabdomyoma with tuberous sclerosis in children. USMLE extension Pathology DOK 3

CV-18-J.6 Identify angiosarcoma risk factors (vinyl chloride, arsenic, Thorotrast for liver tumors; chronic lymphedema after mastectomy) and its aggressive course. USMLE extension Pathology DOK 2

CV-18-K Infectious vascular diseases

1 Bacterial endocarditis **2** Chagas disease **3** Lyme disease **4** Rocky Mountain spotted fever **5** Viral hemorrhagic fever (yellow fever, Dengue fever, filoviruses) **6** Viral myocarditis

CV-18-K.1 Explain how bacteremia seeds damaged endothelium or fibrin-platelet deposits to form vegetations, and predict the likely organism by setting: native valve, IV drug use, prosthetic valve, dental work, or GI and GU procedures. NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.2 Predict complications of bacterial endocarditis: valve destruction and ring abscess, septic emboli (systemic in left-sided, lung in right-sided disease), immune complex glomerulonephritis, and mycotic aneurysm. NPLEX + USMLE Pathology DOK 3

CV-18-K.3 Distinguish acute Chagas disease (often mild; Romana sign) from chronic Chagas years later (dilated cardiomyopathy with apical aneurysm and conduction block, megaesophagus, megacolon) in people from Latin America. NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.4 Distinguish the stages of Lyme disease (erythema migrans; then facial palsy, meningitis, and carditis with AV block; then large joint arthritis and late neurologic disease) and relate risk to Northeast and upper Midwest exposure. NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.5 Explain Rocky Mountain spotted fever as rickettsial vasculitis and predict fever, headache, a rash spreading from wrists and ankles to the trunk with palms and soles involved, thrombocytopenia, and possible shock and multiorgan failure. NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.6 Compare yellow fever (midzonal liver necrosis, Councilman bodies, jaundice, black vomit) with dengue (fever, severe muscle and joint pain, thrombocytopenia, plasma leak in severe dengue). NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.7 Explain the shared mechanism of viral hemorrhagic fevers (endothelial damage, cytokine storm, consumptive coagulopathy) and the high fatality and body fluid spread of Ebola and Marburg filoviruses. NPLEX + USMLE Microbiology & Immunology DOK 3

CV-18-K.8 Predict viral myocarditis in a young adult after a respiratory or GI illness (chest pain, arrhythmia, raised troponin, heart failure) and identify lymphocytic infiltrate with myocyte necrosis on biopsy. NPLEX + USMLE Pathology DOK 3

HOW THE BODY WORKS

GI-1 Gastrointestinal embryology

Describe embryological development of the gastrointestinal tract and glands.

GI-1.1 Identify the foregut, midgut, and hindgut derivatives and their arteries (celiac trunk, superior mesenteric, inferior mesenteric), and explain why visceral pain from each is referred to the epigastric, periumbilical, or suprapubic region. NPLEX · Anatomy · DOK 2

GI-1.2 Describe physiologic herniation of the midgut in week 6, its 270 degree counterclockwise rotation around the superior mesenteric artery, and its return by week 10, and predict the results of malrotation and of failed return (omphalocele). NPLEX + USMLE · Anatomy · DOK 3

GI-1.3 Explain how the tracheoesophageal septum divides the foregut into the esophagus and the respiratory diverticulum, and relate incomplete division to esophageal atresia and tracheoesophageal fistula. NPLEX + USMLE · Anatomy · DOK 2

GI-1.4 Describe development of the liver, gallbladder, and bile ducts from the hepatic diverticulum and of the pancreas from dorsal and ventral buds, and relate abnormal ventral bud migration to annular pancreas and failed duct fusion to pancreas divisum. NPLEX + USMLE · Anatomy · DOK 2

GI-1.5 Explain how neural crest cells migrate to form the myenteric and submucosal plexuses, and relate failed migration (RET mutations) to the aganglionic distal segment of Hirschsprung disease. NPLEX + USMLE · Anatomy · DOK 2

GI-1.6 Describe recanalization of the duodenal lumen and relate its failure to duodenal atresia, with bilious vomiting, the double bubble sign, and an association with Down syndrome. NPLEX · Anatomy · DOK 2

GI-2 Gastrointestinal histology

Describe microscopic anatomy of the gastrointestinal tract and glands.

GI-2.1 Identify the four layers of the gastrointestinal wall (mucosa, submucosa, muscularis externa, serosa or adventitia) and locate the submucosal (Meissner) and myenteric (Auerbach) plexuses within them. NPLEX · Anatomy · DOK 1

GI-2.2 Compare the epithelium of the esophagus (nonkeratinized stratified squamous), stomach and intestine (simple columnar), and anal canal (columnar above the pectinate line, stratified squamous below), and relate the gastroesophageal junction to Barrett metaplasia. NPLEX + USMLE · Anatomy · DOK 2

GI-2.3 Identify gastric parietal cells (HCl, intrinsic factor), chief cells (pepsinogen), mucous neck cells, and antral G cells (gastrin), and predict which secretions fall when autoimmune gastritis destroys the body and fundus mucosa. NPLEX + USMLE · Anatomy · DOK 3

GI-2.4 Distinguish the duodenum (Brunner glands in the submucosa), jejunum (tall plicae circulares), ileum (Peyer patches), and colon (crypts rich in goblet cells, no villi) on histologic sections. NPLEX · Anatomy · DOK 2

GI-2.5 Describe the hepatic lobule and acinus, including the portal triad, sinusoids, space of Disse, Kupffer cells, and stellate (Ito) cells, and explain why zone 3 is most vulnerable to ischemia and to acetaminophen injury. NPLEX + USMLE · Anatomy · DOK 3

GI-2.6 Distinguish pancreatic acini with zymogen granules from islets of Langerhans, and compare serous and mucous salivary acini, identifying the parotid as a mainly serous gland. NPLEX · Anatomy · DOK 1

GI-3 Gastrointestinal organs and glands

Describe location, structure, and boundaries of organs and glands of the gastrointestinal system.

GI-3.1 Identify the parts of the stomach (cardia, fundus, body, antrum, pylorus) and the four parts of the duodenum, and locate the major duodenal papilla in the second part. NPLEX · Anatomy · DOK 1

GI-3.2 Classify abdominal organs as intraperitoneal or retroperitoneal, naming the retroperitoneal gut structures (duodenum parts 2 to 4, pancreas except the tail, ascending and descending colon, and part of the rectum). NPLEX · Anatomy · DOK 2

GI-3.3 Describe the liver lobes, bare area, and porta hepatis, and trace bile from the hepatic ducts to the common bile duct, which joins the pancreatic duct at the ampulla of Vater under the sphincter of Oddi. NPLEX · Anatomy · DOK 2

GI-3.4 Distinguish the colon from the small intestine by taeniae coli, haustra, and omental appendices, and locate the base of the appendix at McBurney point where the taeniae converge. NPLEX Anatomy DOK 2

GI-3.5 Locate the parotid, submandibular, and sublingual glands and their ducts, including the parotid duct opening opposite the upper second molar, and explain why parotid disease can involve the facial nerve passing through the gland. NPLEX Anatomy DOK 2

GI-4 The gut in the body cavities

Describe gastrointestinal system in relation to oral, mediastinal, and abdominopelvic cavities.

GI-4.1 Trace the esophagus from the pharynx through the superior and posterior mediastinum to the stomach, naming its three narrowings: the cricopharyngeus, the crossing of the aortic arch and left main bronchus, and the diaphragmatic hiatus at T10. NPLEX Anatomy DOK 2

GI-4.2 Relate the innervation of the tongue (CN V3 and CN VII to the anterior two thirds, CN IX to the posterior third, CN XII motor except palatoglossus by CN X) to the deficit produced by a given nerve lesion. NPLEX Anatomy DOK 3

GI-4.3 Describe the greater and lesser sacs, the greater and lesser omentum, and the omental (epiploic) foramen, and identify the hepatoduodenal ligament contents (portal vein, proper hepatic artery, common bile duct). NPLEX Anatomy DOK 2

GI-4.4 Identify the structures carried by the falciform (ligamentum teres), gastrohepatic (gastric arteries), gastrosplenic (short gastric vessels), and splenorenal (splenic artery and vein, pancreatic tail) ligaments. NPLEX Anatomy DOK 1

GI-4.5 Describe the boundaries of the inguinal canal and Hesselbach triangle, and distinguish indirect (lateral to the inferior epigastric vessels, through the deep ring), direct (medial, through the triangle), and femoral (below the inguinal ligament) hernias. NPLEX + USMLE Anatomy DOK 3

GI-4.6 Locate the dependent peritoneal spaces where fluid, pus, or blood collects (hepatorenal pouch of Morison when supine, rectouterine or rectovesical pouch when upright, subphrenic spaces) and relate them to abscess formation. NPLEX + USMLE Anatomy DOK 2

GI-5 Gastrointestinal blood supply

Describe location, structure, and circulatory pathways of blood supply of the gastrointestinal system.

GI-5.1 Trace the branches of the celiac trunk (left gastric, splenic, common hepatic) to the stomach, liver, spleen, pancreas, and duodenum, and identify the gastroduodenal artery as the vessel eroded by a posterior duodenal ulcer. NPLEX + USMLE Anatomy DOK 2

GI-5.2 Describe the territories of the superior and inferior mesenteric arteries and the marginal artery, and predict ischemic colitis at the watershed zones (splenic flexure, rectosigmoid junction) after hypotension. NPLEX + USMLE Anatomy DOK 3

GI-5.3 Trace portal blood from the superior mesenteric and splenic veins, which join behind the pancreatic neck, through the liver sinusoids and central veins to the hepatic veins and inferior vena cava. NPLEX Anatomy DOK 1

GI-5.4 Identify the portosystemic anastomoses (left gastric to esophageal veins draining to azygos, paraumbilical to abdominal wall veins, superior to middle and inferior rectal veins) and predict varices and caput medusae in portal hypertension. NPLEX + USMLE Anatomy DOK 3

GI-5.5 Compare the anal canal above and below the pectinate line by epithelium, arterial supply, venous drainage (portal versus systemic), lymph drainage (internal iliac versus superficial inguinal), and pain sensation (visceral versus somatic). NPLEX + USMLE Anatomy DOK 2

GI-5.6 Explain the dual blood supply of the liver (most flow from the portal vein, the rest from the hepatic artery) and why hepatic infarction is uncommon while the bile ducts depend on hepatic arterial flow. NPLEX Physiology DOK 2

GI-6 Motility, digestion, and absorption

Explain mechanisms, functions, regulation, and factors affecting mastication, deglutition, digestion, absorption, peristalsis, and defecation.

GI-6.1 Describe mastication by the muscles of mastication (CN V3), the voluntary oral and reflex pharyngeal and esophageal phases of swallowing coordinated by the medulla, and vagal nitric oxide and VIP relaxation of the lower esophageal sphincter and proximal stomach. NPLEX Physiology DOK 2

GI-6.2 Explain how slow waves from the interstitial cells of Cajal set contraction frequency, how the myenteric plexus coordinates peristalsis (contraction above and relaxation below a bolus), and how motilin drives the fasting migrating motor complex. NPLEX Physiology DOK 2

GI-6.3 Explain the cephalic, gastric, and intestinal phases of gastric acid secretion, including parietal cell stimulation by acetylcholine, gastrin, and histamine, inhibition by somatostatin, and the H+/K+ ATPase as the final step. NPLEX Physiology DOK 2

GI-6.4 Relate gastrin, cholecystokinin, secretin, GIP, and motilin to their source cells (G, I, S, K, M) and stimuli, and predict the hormonal response to a meal that delivers acid and fat to the duodenum. NPLEX Physiology DOK 3

GI-6.5 Explain carbohydrate, protein, and fat digestion by salivary and pancreatic amylase, brush border disaccharidases, enteropeptidase activation of trypsinogen, and pancreatic lipase acting with bile salts. NPLEX Physiology DOK 2

GI-6.6 Describe the defecation reflex (rectal distension, reflex relaxation of the internal anal sphincter, voluntary control of the external sphincter by the pudendal nerve) and the gastrocolic reflex, and predict incontinence from loss of each element. NPLEX + USMLE Physiology DOK 3

GI-6.7 Explain how vomiting is triggered through the chemoreceptor trigger zone in the area postrema and the medullary vomiting center, and predict the hypochloremic hypokalemic metabolic alkalosis of prolonged vomiting. NPLEX + USMLE Physiology DOK 3

GI-7 Products of digestion

Explain composition, function, transport, and regulation of products of digestion.

GI-7.1 Explain monosaccharide absorption (SGLT1 sodium cotransport of glucose and galactose, GLUT5 uptake of fructose, GLUT2 basolateral exit) and why glucose plus sodium in oral rehydration solution increases water absorption. NPLEX Physiology DOK 2

GI-7.2 Explain absorption of free amino acids by sodium-linked transporters and of dipeptides and tripeptides by H+ cotransport (PepT1), and relate the neutral amino acid transport defect of Hartnup disease to pellagra-like findings. NPLEX Biochemistry & Genetics DOK 2

GI-7.3 Trace fat absorption from micelles to re-esterification in enterocytes, packaging with apoB-48 into chylomicrons, and entry into lacteals, and predict steatorrhea, acanthocytes, and vitamin E deficiency in abetalipoproteinemia. NPLEX Physiology DOK 3

GI-7.4 Identify where iron (duodenum), folate (jejunum), and vitamin B12 and bile acids (terminal ileum) are absorbed, and predict the deficiencies that follow duodenal disease or ileal resection. NPLEX + USMLE Physiology DOK 3

GI-7.5 Explain the enterohepatic circulation of bile acids with active reabsorption in the terminal ileum, and predict steatorrhea and cholesterol gallstone risk when ileal disease or resection interrupts it. NPLEX Physiology DOK 3

GI-7.6 Compare secretory diarrhea (cholera toxin raising cAMP and opening CFTR chloride channels), osmotic diarrhea (unabsorbed lactose), and inflammatory diarrhea, and use the stool osmotic gap and response to fasting to distinguish them. NPLEX + USMLE Physiology DOK 3

GI-8 Digestive biochemistry and dietary requirements

Explain biochemistry of digestive processes, including endogenous production of chemical energy and chemical composition and dietary requirements of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors.

GI-8.1 Identify the essential amino acids and essential fatty acids (linoleic and alpha-linolenic acid) required in the diet, and compare kwashiorkor (protein deficiency, edema from hypoalbuminemia, fatty liver) with marasmus (calorie deficiency, wasting without edema). NPLEX + USMLE Biochemistry & Genetics DOK 2

GI-8.2 Explain how fat malabsorption (pancreatic insufficiency, cholestasis, celiac disease) causes deficiency of vitamins A, D, E, and K, and predict night blindness, osteomalacia, hemolytic anemia with neuropathy, and a prolonged PT. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-8.3 Trace vitamin B12 absorption (release by acid and pepsin, haptocorrin binding, pancreatic protease cleavage, intrinsic factor binding, ileal uptake) and distinguish B12 from folate deficiency by methylmalonic acid and homocysteine levels. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-8.4 Relate deficiencies of thiamine (Wernicke encephalopathy, beriberi), niacin (pellagra), pyridoxine (sideroblastic anemia, seizures), and vitamin C (scurvy from impaired collagen hydroxylation) to the cofactor role of each vitamin. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

GI-8.5 Explain endogenous energy production in the fed and fasting states, including hepatic glycogenolysis, gluconeogenesis from alanine, lactate, and glycerol, and hepatic ketogenesis, and why the liver cannot use the ketones it makes. NPLEX | Biochemistry & Genetics | DOK 2

GI-8.6 Relate deficiencies of minerals absorbed in the gut to their findings: iron (microcytic anemia), zinc (poor wound healing, dysgeusia, acrodermatitis), and copper (anemia, neutropenia, myelopathy), including copper deficiency from excess zinc or gastric bypass. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

GI-8.7 Predict the findings of vitamin A toxicity (intracranial hypertension, liver injury, teratogenicity) and vitamin D toxicity (hypercalcemia, hypercalciuria, kidney stones). USMLE extension | Biochemistry & Genetics | DOK 2

GI-9 Liver detoxification, bilirubin, and gland functions

Explain non-digestive functions of salivary glands, liver, and gallbladder, including bilirubin metabolism and detoxification pathways.

GI-9.1 Trace bilirubin from heme breakdown (heme oxygenase, biliverdin reductase) through albumin transport, hepatic conjugation by UGT1A1, canalicular excretion by MRP2, and conversion by gut bacteria to urobilinogen and stercobilin. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

GI-9.2 Interpret conjugated and unconjugated serum bilirubin, urine bilirubin, and urine urobilinogen to distinguish hemolytic, hepatocellular, and obstructive jaundice. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

GI-9.3 Explain phase I (cytochrome P450) and phase II (glucuronidation, sulfation, glutathione conjugation) biotransformation, and use acetaminophen overdose to show how glutathione depletion lets NAPQI cause zone 3 necrosis. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

GI-9.4 Explain carbon tetrachloride hepatotoxicity as cytochrome P450 conversion to the trichloromethyl free radical, which causes lipid peroxidation, fatty change, and centrilobular necrosis. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

GI-9.5 Explain the synthetic and metabolic functions of the liver (albumin, clotting factors, urea cycle disposal of ammonia), and predict hypoalbuminemia, a prolonged PT, and hyperammonemia in liver failure. NPLEX + USMLE | Physiology | DOK 3

GI-9.6 Describe the protective functions of saliva (bicarbonate buffering, IgA, lysozyme, lactoferrin, enamel remineralization) and predict dental caries when salivary flow falls. NPLEX + USMLE | Physiology | DOK 2

GI-9.7 Explain how the gallbladder concentrates bile by absorbing sodium and water and contracts under cholecystokinin, and relate stasis and concentration to gallstone formation. NPLEX | Physiology | DOK 2

GI-10 Gastrointestinal genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying gastrointestinal disease.

GI-10.1 Explain hereditary hemochromatosis (autosomal recessive HFE C282Y mutation, inappropriately low hepcidin, excess iron absorption) and predict cirrhosis, diabetes, skin bronzing, cardiomyopathy, arthropathy, and hepatocellular carcinoma risk. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-10.2 Explain Wilson disease (autosomal recessive ATP7B mutation, failed biliary copper excretion and ceruloplasmin loading) and relate it to low serum ceruloplasmin, Kayser-Fleischer rings, basal ganglia degeneration, and hepatitis or cirrhosis. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-10.3 Explain how the PiZZ alpha-1 antitrypsin variant misfolds and is retained in the hepatocyte endoplasmic reticulum as PAS-positive, diastase-resistant globules causing cirrhosis, while low serum levels cause panacinar emphysema. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-10.4 Compare the adenoma to carcinoma sequence (APC loss, KRAS activation, TP53 loss) with the mismatch repair pathway (MLH1, MSH2, microsatellite instability) of Lynch syndrome in colorectal cancer. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-10.5 Distinguish familial adenomatous polyposis (autosomal dominant APC, hundreds of adenomas), Gardner and Turcot variants, MUTYH-associated polyposis (autosomal recessive), and Peutz-Jeghers syndrome (STK11, hamartomas, mucocutaneous pigmentation). NPLEX + USMLE Biochemistry & Genetics DOK 2

GI-10.6 Relate CFTR mutations in cystic fibrosis to meconium ileus, exocrine pancreatic insufficiency with steatorrhea and fat-soluble vitamin deficiency, and focal biliary cirrhosis. NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-11 Immune responses in gastrointestinal disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

GI-11.1 Describe mucosal immunity in the gut, including antigen sampling by M cells over Peyer patches, secretory IgA transported by the polymeric Ig receptor, and oral tolerance, and relate selective IgA deficiency to recurrent mucosal infections. NPLEX Microbiology & Immunology DOK 2

GI-11.2 Explain celiac disease immunopathogenesis: tissue transglutaminase deamidates gliadin, HLA-DQ2 or DQ8 presents it to CD4 T cells, and IgA anti-tissue transglutaminase and anti-endomysial antibodies appear. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-11.3 Explain how NOD2 variants, impaired handling of commensal bacteria, and Th1 and Th17 responses produce transmural granulomatous inflammation in Crohn disease. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-11.4 Match autoimmune gastrointestinal diseases to their autoantibodies: autoimmune gastritis (parietal cell and intrinsic factor), autoimmune hepatitis (ANA, anti-smooth muscle, anti-LKM1), and primary biliary cholangitis (antimitochondrial). NPLEX + USMLE Microbiology & Immunology DOK 2

GI-11.5 Explain how chronic Helicobacter pylori antigen stimulation drives B cell proliferation in gastric MALT lymphoma, and why early lesions can regress when the organism is eradicated. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-11.6 Explain that hepatocyte injury in hepatitis B is caused mainly by cytotoxic CD8 T cells rather than the virus itself, and relate HBsAg immune complexes to polyarteritis nodosa and membranous nephropathy. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-11.7 Explain why hepatitis B vaccination with recombinant HBsAg produces anti-HBs without anti-HBc, so serology separates vaccine immunity from resolved infection. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12 Gastrointestinal pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

GI-12.1 Distinguish toxin-mediated causes of diarrhea and vomiting: preformed Staphylococcus aureus and Bacillus cereus toxins (onset within hours), cholera toxin and the ETEC heat-labile toxin (both raise cAMP), and Clostridioides difficile toxins A and B. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.2 Compare invasive enteric bacteria by key features: Shigella (nonmotile, very low infectious dose), Salmonella (motile, H2S producing), and Campylobacter jejuni (comma-shaped, grows at 42 degrees C (107.6 degrees F), poultry). NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.3 Describe Yersinia enterocolitica (mesenteric adenitis mimicking appendicitis), EHEC O157:H7 (Shiga toxin, bloody diarrhea, hemolytic uremic syndrome), and Listeria monocytogenes (deli meats and soft cheese, actin-based cell-to-cell spread). NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.4 Compare the viral causes of enteritis: rotavirus (segmented dsRNA, villus damage in infants), norovirus (ssRNA calicivirus, outbreaks), and the enteroviruses coxsackievirus and echovirus (ssRNA picornaviruses, fecal-oral). NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.5 Compare hepatitis viruses: HAV (picornavirus, fecal-oral), HBV (partially dsDNA hepadnavirus with reverse transcriptase), HCV (flavivirus, high antigenic variation), HDV (defective, requires HBsAg), and HEV (fecal-oral, severe in pregnancy). NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.6 Describe Helicobacter pylori as a curved, flagellated, microaerophilic gram-negative rod whose urease makes ammonia to buffer acid, and identify the urea breath test, stool antigen, and biopsy as ways to detect it. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-12.7 Compare intestinal parasites: Giardia (cysts in water, fatty diarrhea), Entamoeba histolytica (flask-shaped ulcers, liver abscess), Cryptosporidium, Cyclospora, and Cystoisospora (acid-fast oocysts), and Strongyloides (skin-penetrating larvae, autoinfection, hyperinfection with corticosteroids). NPLEX + USMLE Microbiology & Immunology DOK 2

DISEASES AND CONDITIONS

GI-13-A Salivary gland disease

1 Parotitis

GI-13-A.1 Distinguish mumps parotitis (paramyxovirus, often bilateral, raised serum amylase, with orchitis, pancreatitis, or aseptic meningitis) from acute bacterial parotitis (Staphylococcus aureus, dehydrated or postoperative patients, pus at the parotid duct). NPLEX + USMLE Pathology DOK 3

GI-13-A.2 Explain why salivary stones form most often in the submandibular gland (more viscous saliva, upward duct course), and predict painful swelling with meals and secondary sialadenitis. USMLE extension Pathology DOK 2

GI-13-B Pancreatic disease

1 Pancreatitis

GI-13-B.1 Explain acute pancreatitis as premature intrapancreatic activation of trypsin and other enzymes, and identify its causes, led by gallstones and alcohol, plus hypertriglyceridemia, hypercalcemia, ERCP, mumps, trauma, and drugs such as thiazides. NPLEX + USMLE Pathology DOK 2

GI-13-B.2 Interpret the findings of acute pancreatitis: epigastric pain radiating to the back, lipase above three times normal, fat necrosis with calcium soap formation and hypocalcemia, and Cullen and Grey Turner signs of retroperitoneal hemorrhage. NPLEX + USMLE Pathology DOK 3

GI-13-B.3 Predict complications of acute pancreatitis, including pseudocyst (lined by granulation tissue, not epithelium), necrosis with infection or abscess, ARDS, DIC, and hypovolemic shock. NPLEX + USMLE Pathology DOK 3

GI-13-B.4 Describe chronic pancreatitis (alcohol, cystic fibrosis, ductal obstruction, hereditary PRSS1 and SPINK1 mutations) with fibrosis and calcifications, and predict exocrine insufficiency (steatorrhea, fat-soluble vitamin deficiency) and diabetes. NPLEX + USMLE Pathology DOK 3

GI-13-C Hepatic diseases and disorders

1 Cholestasis **2** Cirrhosis **3** Gilbert syndrome **4** Hepatitis (non-infectious) **5** Portal hypertension

GI-13-C.1 Distinguish intrahepatic causes of cholestasis (drugs, pregnancy, total parenteral nutrition, primary biliary cholangitis) from extrahepatic obstruction, and interpret raised conjugated bilirubin, alkaline phosphatase, and GGT, pruritus, pale stools, and xanthomas. NPLEX + USMLE Pathology DOK 3

GI-13-C.2 Compare primary biliary cholangitis (middle-aged women, antimitochondrial antibody, granulomatous destruction of small intrahepatic ducts) with primary sclerosing cholangitis (ulcerative colitis, onion-skin fibrosis, beaded ducts, cholangiocarcinoma risk). NPLEX + USMLE Pathology DOK 3

GI-13-C.3 Explain how activated hepatic stellate cells lay down collagen in the space of Disse to produce bridging fibrosis and regenerative nodules in cirrhosis, and identify its main causes (alcohol, chronic HBV and HCV, fatty liver disease, metabolic disease). NPLEX + USMLE Pathology DOK 2

GI-13-C.4 Predict the findings of cirrhosis from hepatocellular failure (jaundice, coagulopathy, hypoalbuminemia, hyperestrogenism with spider angiomas, palmar erythema, gynecomastia, and hepatic encephalopathy with asterixis) and from portal hypertension. NPLEX + USMLE Pathology DOK 3

GI-13-C.5 Explain hepatorenal syndrome (splanchnic vasodilation and renal vasoconstriction with histologically normal kidneys) and hepatopulmonary syndrome (intrapulmonary vascular dilation causing hypoxemia) as complications of cirrhosis. NPLEX + USMLE Pathology DOK 3

GI-13-C.6 Compare Gilbert syndrome (mildly reduced UGT1A1 activity, benign unconjugated hyperbilirubinemia during fasting, stress, or illness) with Crigler-Najjar syndrome type I (absent UGT1A1, kernicterus in infancy) and type II (partial deficiency, milder course). NPLEX + USMLE Biochemistry & Genetics DOK 3

GI-13-C.7 Describe the progression of alcohol-related liver disease from steatosis to alcoholic hepatitis (hepatocyte ballooning, Mallory-Denk bodies, neutrophils) to cirrhosis, and interpret an AST to ALT ratio greater than 2. NPLEX + USMLE Pathology DOK 3

GI-13-C.8 Explain metabolic dysfunction-associated steatotic liver disease (MASLD, formerly nonalcoholic fatty liver disease) as a result of insulin resistance and obesity, and distinguish simple steatosis from steatohepatitis (MASH) that can progress to cirrhosis and hepatocellular carcinoma. NPLEX + USMLE Pathology DOK 2

GI-13-C.9 Distinguish autoimmune hepatitis (young women, hypergammaglobulinemia, interface hepatitis with plasma cells) from drug and toxin hepatitis, such as acetaminophen zone 3 necrosis and isoniazid hepatitis. NPLEX + USMLE Pathology DOK 3

GI-13-C.10 Classify causes of portal hypertension as prehepatic (portal vein thrombosis), intrahepatic (cirrhosis, schistosomiasis), or posthepatic (Budd-Chiari syndrome, right heart failure), and predict ascites, splenomegaly, varices, and caput medusae. NPLEX + USMLE Pathology DOK 3

GI-13-C.11 Distinguish Dubin-Johnson syndrome (MRP2 defect, conjugated hyperbilirubinemia, black pigmented liver) from Rotor syndrome (similar but milder, liver not pigmented) as benign inherited causes of conjugated hyperbilirubinemia. USMLE extension Biochemistry & Genetics DOK 2

GI-13-D Gallbladder diseases

1 Cholecystitis **2** Cholelithiasis

GI-13-D.1 Explain acute calculous cholecystitis from persistent cystic duct obstruction with chemical then bacterial inflammation, interpret right upper quadrant pain, fever, Murphy sign, and leukocytosis, and contrast acalculous cholecystitis in critically ill patients. NPLEX + USMLE Pathology DOK 3

GI-13-D.2 Identify chronic cholecystitis with a calcified porcelain gallbladder as a risk for gallbladder adenocarcinoma. NPLEX + USMLE Pathology DOK 1

GI-13-D.3 Compare cholesterol stones (bile supersaturated with cholesterol, gallbladder stasis; risks include female sex, obesity, pregnancy, estrogen, rapid weight loss) with black pigment stones (hemolysis) and brown pigment stones (biliary infection). NPLEX + USMLE Pathology DOK 2

GI-13-D.4 Predict the consequence of a gallstone by its location: biliary colic, choledocholithiasis with obstructive jaundice, ascending cholangitis (fever, jaundice, right upper quadrant pain), gallstone pancreatitis, and gallstone ileus through a cholecystoenteric fistula. NPLEX + USMLE Pathology DOK 3

GI-13-D.5 Explain Mirizzi syndrome as a stone impacted in the cystic duct or gallbladder neck compressing the common hepatic duct, producing obstructive jaundice. USMLE extension Pathology DOK 2

GI-13-E Deficiency and malabsorption conditions

1 Achlorhydria **2** Gluten-sensitive enteropathy (celiac disease) **3** Enzyme deficiencies **4** Lactase deficiency

GI-13-E.1 Explain the causes of achlorhydria (autoimmune atrophic gastritis, chronic H. pylori pangastritis, gastrectomy) and predict hypergastrinemia with ECL cell hyperplasia and carcinoid risk, impaired iron and B12 absorption, and bacterial overgrowth. NPLEX + USMLE Pathology DOK 3

GI-13-E.2 Describe celiac disease findings: villous atrophy, crypt hyperplasia, and intraepithelial lymphocytes mainly in the distal duodenum and proximal jejunum, iron deficiency anemia, dermatitis herpetiformis, and risk of enteropathy-associated T cell lymphoma. NPLEX + USMLE Pathology DOK 3

GI-13-E.3 Distinguish malabsorption from pancreatic enzyme deficiency (steatorrhea, low fecal elastase, normal D-xylose absorption) from mucosal disease (abnormal D-xylose absorption), and relate enteropeptidase deficiency to failed protease activation. NPLEX + USMLE Pathology DOK 3

GI-13-E.4 Explain how primary (adult-type) or secondary (after viral enteritis or celiac disease) lactase deficiency causes osmotic diarrhea, bloating, and flatulence after dairy, with a positive hydrogen breath test and normal villi in the primary form. NPLEX + USMLE Pathology DOK 3

GI-13-E.5 Predict the consequences of short bowel syndrome after extensive resection, including diarrhea, steatorrhea, B12 and bile acid loss after ileal resection, and calcium oxalate kidney stones from enteric hyperoxaluria. USMLE extension | Pathology | DOK 3

GI-13-E.6 Explain how bacterial overgrowth in a blind loop after gastric surgery deconjugates bile acids and consumes vitamin B12, causing steatorrhea and B12 deficiency with normal or raised folate. USMLE extension | Pathology | DOK 3

GI-13-F Obstructive gastrointestinal diseases

1 Achalasia **2** Adynamic ileus **3** Hernia **4** Intussusception/volvulus **5** Megacolon

GI-13-F.1 Explain achalasia as loss of inhibitory (nitric oxide and VIP) neurons of the myenteric plexus, producing incomplete LES relaxation and aperistalsis, with dysphagia to solids and liquids, a bird-beak barium swallow, and Chagas disease as a secondary cause. NPLEX + USMLE | Pathology | DOK 3

GI-13-F.2 Distinguish adynamic (paralytic) ileus (after abdominal surgery, hypokalemia, peritonitis, opioids; absent bowel sounds, diffuse gas) from mechanical obstruction (adhesions, hernias, tumors, impaction; high-pitched bowel sounds, dilated loops proximal to a transition point). NPLEX + USMLE | Pathology | DOK 3

GI-13-F.3 Distinguish sliding from paraesophageal hiatal hernia, and explain how an incarcerated abdominal wall hernia becomes strangulated as venous congestion leads to arterial compromise and bowel infarction. NPLEX + USMLE | Pathology | DOK 3

GI-13-F.4 Compare intussusception (telescoping, usually ileocolic in young children, a lead point such as hypertrophied Peyer patches or Meckel diverticulum, currant jelly stool) with volvulus (twisting on the mesentery, sigmoid in older adults, midgut with malrotation in infants). NPLEX + USMLE | Pathology | DOK 3

GI-13-F.5 Distinguish causes of megacolon: Hirschsprung disease (absent ganglion cells in the rectum, delayed passage of meconium, empty rectal vault), Chagas disease (acquired loss of the plexus), and toxic megacolon complicating ulcerative colitis or C. difficile colitis. NPLEX + USMLE | Pathology | DOK 3

GI-13-F.6 Explain Zenker diverticulum as a false pharyngoesophageal diverticulum above the cricopharyngeus from cricopharyngeal dysfunction, causing dysphagia, regurgitation of undigested food, and halitosis. USMLE extension | Pathology | DOK 2

GI-13-F.7 Describe gastroparesis (delayed gastric emptying from diabetic autonomic neuropathy, with early satiety and vomiting) and irritable bowel syndrome (recurrent pain linked to defecation and altered stool form without structural disease). USMLE extension | Physiology | DOK 2

GI-13-G Inflammatory gastrointestinal diseases

1 Appendicitis **2** Barrett esophagus **3** Diverticular disease **4** Enteritis **5** Esophageal/gastric/duodenal ulcers **6** Esophagitis (non-infectious) **7** Gastritis **8** Gastroesophageal reflux disease (GERD) **9** Inflammatory bowel disease (Crohn disease, ulcerative colitis)

GI-13-G.1 Explain appendicitis as luminal obstruction (fecalith, lymphoid hyperplasia) leading to distension, ischemia, and bacterial invasion, and trace pain from the periumbilical region (visceral) to McBurney point (parietal peritoneum) before perforation. NPLEX + USMLE | Pathology | DOK 3

GI-13-G.2 Describe Barrett esophagus as intestinal metaplasia with goblet cells replacing distal squamous epithelium after chronic reflux, and relate it to the dysplasia to adenocarcinoma sequence. NPLEX + USMLE | Pathology | DOK 2

GI-13-G.3 Explain how a low-fiber diet and high intraluminal pressure create false diverticula in the sigmoid colon where vasa recta penetrate the wall, and distinguish painless bleeding in diverticulosis from diverticulitis with left lower quadrant pain, fever, abscess, or fistula. NPLEX + USMLE | Pathology | DOK 3

GI-13-G.4 Identify noninfectious causes of enteritis and colitis, including radiation, NSAIDs, and microscopic colitis (collagenous with a thick subepithelial collagen band, lymphocytic with excess intraepithelial lymphocytes) causing chronic watery diarrhea with normal-looking mucosa. NPLEX + USMLE | Pathology | DOK 2

GI-13-G.5 Compare gastric and duodenal ulcers by mechanism (H. pylori, NSAID loss of prostaglandin protection, Zollinger-Ellison), pain timing with meals, and malignancy risk, and predict hemorrhage or perforation with free air under the diaphragm. NPLEX + USMLE | Pathology | DOK 3

GI-13-G.6 Explain acute erosive gastritis and stress ulcers, including Curling ulcer after severe burns (hypovolemia and mucosal ischemia) and Cushing ulcer after brain injury (vagal stimulation raising acid), and NSAID and alcohol injury. NPLEX + USMLE | Pathology | DOK 2

GI-13-G.7 Distinguish reflux esophagitis (basal cell hyperplasia, elongated papillae) from eosinophilic esophagitis (atopy, 15 or more eosinophils per high-power field, rings and furrows, food impaction) and pill esophagitis (doxycycline, NSAIDs, potassium chloride). NPLEX + USMLE Pathology DOK 3

GI-13-G.8 Compare chronic H. pylori gastritis (antrum, intestinal metaplasia, risk of adenocarcinoma and MALT lymphoma) with autoimmune gastritis (body and fundus, parietal cell loss, achlorhydria, pernicious anemia). NPLEX + USMLE Pathology DOK 3

GI-13-G.9 Explain GERD from transient LES relaxation, low LES tone, and hiatal hernia (risks include obesity and pregnancy), and predict heartburn, chronic cough, hoarseness, and complications of stricture and Barrett esophagus. NPLEX + USMLE Pathology DOK 3

GI-13-G.10 Compare Crohn disease (mouth to anus, often terminal ileum, skip lesions, transmural inflammation, noncaseating granulomas, cobblestone mucosa) with ulcerative colitis (rectum extending proximally, continuous mucosal inflammation, crypt abscesses, pseudopolyps). NPLEX + USMLE Pathology DOK 3

GI-13-G.11 Predict complications of inflammatory bowel disease: fistulas, strictures, B12 and bile acid malabsorption, and oxalate stones in Crohn disease; toxic megacolon and primary sclerosing cholangitis in ulcerative colitis; colorectal cancer, erythema nodosum, uveitis, and arthritis in both. NPLEX + USMLE Pathology DOK 3

GI-13-G.12 Distinguish Mallory-Weiss syndrome (mucosal laceration at the gastroesophageal junction with hematemesis after retching) from Boerhaave syndrome (transmural esophageal rupture with chest pain, pneumomediastinum, and subcutaneous emphysema). USMLE extension Pathology DOK 3

GI-13-G.13 Distinguish anal fissure (posterior midline tear, painful bright red bleeding) from anorectal abscess and fistula (Crohn disease association), proctitis, and rectal prolapse. USMLE extension Pathology DOK 2

GI-13-H Congenital gastrointestinal disease

1 Esophageal atresia **2** Esophageal webs and rings **3** Meckel diverticulum **4** Pyloric stenosis

GI-13-H.1 Describe esophageal atresia with a distal tracheoesophageal fistula (the most common type), predicting polyhydramnios, drooling, choking and cyanosis with feeds, failure to pass a nasogastric tube, and air in the stomach, plus VACTERL associations. NPLEX + USMLE Pathology DOK 3

GI-13-H.2 Distinguish esophageal webs (upper esophagus; Plummer-Vinson syndrome with iron deficiency anemia, dysphagia, and squamous cell carcinoma risk) from Schatzki rings (lower esophagus at the squamocolumnar junction, intermittent solid food dysphagia). NPLEX + USMLE Pathology DOK 3

GI-13-H.3 Describe Meckel diverticulum as a persistent vitelline duct forming a true diverticulum of the ileum, apply the rule of twos, and predict painless rectal bleeding from ectopic gastric mucosa or obstruction from intussusception. NPLEX + USMLE Pathology DOK 3

GI-13-H.4 Explain infantile hypertrophic pyloric stenosis, predicting nonbilious projectile vomiting at 2 to 6 weeks, a palpable olive-shaped mass, and hypochloremic hypokalemic metabolic alkalosis, with first-born males and macrolide exposure as risks. NPLEX + USMLE Pathology DOK 3

GI-13-H.5 Explain biliary atresia as progressive obliteration of the extrahepatic bile ducts, presenting in the first weeks of life with persistent conjugated jaundice, acholic stools, dark urine, and progression to cirrhosis. USMLE extension Pathology DOK 3

GI-13-H.6 Distinguish gastroschisis (defect usually right of the umbilicus, bowel not covered by peritoneum) from omphalocele (midline, covered by peritoneum and amnion, associated with trisomies and Beckwith-Wiedemann syndrome), and explain malrotation with Ladd bands. USMLE extension Anatomy DOK 3

GI-13-H.7 Explain cleft lip as failed fusion of the maxillary and medial nasal processes and cleft palate as failed fusion of the palatine shelves, and predict feeding difficulty. USMLE extension Anatomy DOK 2

GI-13-I Conditions of the abdominal cavity

1 Ascites **2** Peritonitis/adhesions

GI-13-I.1 Explain ascites in cirrhosis from portal hypertension, hypoalbuminemia, and splanchnic vasodilation with renin-angiotensin-aldosterone activation and sodium retention, and interpret a serum-ascites albumin gradient of 1.1 g/dL or more as portal hypertension. NPLEX + USMLE Pathology DOK 3

GI-13-I.2 Distinguish spontaneous bacterial peritonitis (cirrhotic ascites, usually E. coli or Klebsiella, ascitic neutrophils of 250/mm3 or more) from secondary peritonitis (perforated viscus, polymicrobial, rigid abdomen). NPLEX + USMLE Pathology DOK 3

GI-13-I.3 Explain how fibrous adhesions form after abdominal surgery or peritonitis and predict small bowel obstruction as their main complication. NPLEX + USMLE Pathology DOK 2

GI-13-J Gastrointestinal vascular diseases

1 Esophageal varices **2** Hemorrhoids **3** Infarction **4** Vascular ectasias of the colon

GI-13-J.1 Explain esophageal varices as dilated submucosal veins fed by the left gastric vein draining to the azygos system in portal hypertension, and predict painless massive hematemesis as a major cause of death in cirrhosis. NPLEX + USMLE Pathology DOK 3

GI-13-J.2 Distinguish internal hemorrhoids (above the pectinate line, painless bright red bleeding, prolapse) from external hemorrhoids (below the line, painful when thrombosed), and identify straining, constipation, pregnancy, and low-fiber diet as risks. NPLEX + USMLE Pathology DOK 3

GI-13-J.3 Compare acute mesenteric ischemia (often an embolus from atrial fibrillation to the superior mesenteric artery, pain out of proportion to the exam, raised lactate) with chronic mesenteric ischemia (postprandial pain, food fear, weight loss) and ischemic colitis. NPLEX + USMLE Pathology DOK 3

GI-13-J.4 Describe angiodysplasia (dilated, tortuous submucosal vessels mostly in the cecum and right colon of older adults) as a cause of painless lower GI bleeding, and relate it to aortic stenosis (Heyde syndrome) and end-stage kidney disease. NPLEX + USMLE Pathology DOK 2

GI-13-J.5 Explain necrotizing enterocolitis in premature infants as mucosal ischemia with bacterial invasion, predicting bloody stools, abdominal distension, and pneumatosis intestinalis on radiograph. USMLE extension Pathology DOK 3

GI-13-K Gastrointestinal neoplasms

1 Esophageal **2** Gastric **3** Intestinal (gastrinoma) **4** Liver **5** Oral (leukoplakia) **6** Pancreas **7** Colorectal

GI-13-K.1 Compare esophageal squamous cell carcinoma (upper and middle third; alcohol, tobacco, achalasia, caustic injury, Plummer-Vinson) with adenocarcinoma (distal third; GERD, Barrett esophagus, obesity), and predict progressive dysphagia to solids then liquids with weight loss. NPLEX + USMLE Pathology DOK 3

GI-13-K.2 Distinguish intestinal-type gastric adenocarcinoma (H. pylori, nitrosamines, atrophic gastritis, gland formation) from diffuse type (signet ring cells, linitis plastica, CDH1 loss), and identify metastatic signs (Virchow node, Sister Mary Joseph nodule, Krukenberg tumor). NPLEX + USMLE Pathology DOK 3

GI-13-K.3 Identify gastric MALT lymphoma as an extranodal marginal zone B cell lymphoma driven by chronic H. pylori infection, and gastrointestinal stromal tumor as a KIT (CD117) mutant tumor of interstitial cells of Cajal, most often gastric. NPLEX + USMLE Pathology DOK 2

GI-13-K.4 Explain Zollinger-Ellison syndrome from a gastrin-secreting tumor of the duodenum or pancreas, predicting multiple or distal ulcers, diarrhea, steatorrhea from lipase inactivation by acid, a paradoxical gastrin rise after secretin, and MEN1 association. NPLEX + USMLE Pathology DOK 3

GI-13-K.5 Explain why carcinoid tumors of the small intestine cause carcinoid syndrome (flushing, diarrhea, wheezing, right-sided valve disease) mainly after liver metastasis, and identify raised urinary 5-HIAA as its marker. NPLEX + USMLE Pathology DOK 3

GI-13-K.6 Explain hepatocellular carcinoma risk factors (chronic HBV even without cirrhosis, HCV, alcohol, hemochromatosis, aflatoxin B1 with TP53 mutation) and findings (raised alpha-fetoprotein, hepatomegaly, hematogenous spread). NPLEX + USMLE Pathology DOK 3

GI-13-K.7 Identify metastases as the most common malignancy of the liver and peritoneum, and distinguish benign liver lesions: cavernous hemangioma (most common) and hepatocellular adenoma (oral contraceptive or anabolic steroid use, rupture risk). NPLEX + USMLE Pathology DOK 2

GI-13-K.8 Distinguish oral leukoplakia (white plaque that does not scrape off, may harbor dysplasia) from candidiasis (scrapes off) and hairy leukoplakia (EBV, lateral tongue, immunosuppression), and identify tobacco and alcohol as oral squamous cell carcinoma risks and HPV-16 as an oropharyngeal carcinoma risk. NPLEX + USMLE Pathology DOK 3

GI-13-K.9 Explain pancreatic ductal adenocarcinoma (KRAS mutation, smoking, chronic pancreatitis), predicting painless jaundice with a palpable gallbladder (Courvoisier sign) in head tumors, migratory thrombophlebitis, weight loss, and raised CA 19-9. NPLEX + USMLE Pathology DOK 3

GI-13-K.10 Compare right-sided colon cancer (occult bleeding, iron deficiency anemia) with left-sided cancer (napkin-ring narrowing, change in bowel habit, obstruction), and identify risks (age, IBD, polyposis and Lynch syndromes) and CEA as a recurrence marker. NPLEX + USMLE Pathology DOK 3

GI-13-K.11 Classify colonic polyps as hyperplastic, adenomatous (tubular, tubulovillous, villous, with malignant risk rising with size and villous component), sessile serrated, or hamartomatous (juvenile, Peutz-Jeghers). NPLEX + USMLE Pathology DOK 2

GI-13-K.12 Identify pleomorphic adenoma as the most common salivary tumor (usually parotid, mixed epithelial and stromal tissue, recurs if incompletely excised), Warthin tumor (smokers, lymphoid stroma), and mucoepidermoid carcinoma as the most common salivary malignancy. USMLE extension Pathology DOK 2

GI-13-K.13 Describe cholangiocarcinoma (primary sclerosing cholangitis, liver flukes such as Clonorchis sinensis) and gallbladder adenocarcinoma (gallstones), predicting painless obstructive jaundice, and identify anal squamous cell carcinoma below the pectinate line as HPV-related. USMLE extension Pathology DOK 3

GI-13-L Infectious gastrointestinal diseases

1 Enterocolitis **2** Esophagitis **3** Gastroenteritis **4** Gingivitis/periodontitis **5** Oral thrush **6** Stomatitis **7** Viral hepatitis

GI-13-L.1 Explain pseudomembranous colitis: antibiotic disruption of colonic flora (clindamycin, fluoroquinolones, cephalosporins), damage by C. difficile toxins A and B, yellow-white pseudomembranes, watery diarrhea, and toxic megacolon risk, diagnosed by stool toxin or toxin gene testing. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.2 Distinguish causes of bloody enterocolitis (Shigella, Campylobacter, Salmonella, EHEC, Entamoeba histolytica) and predict their complications: hemolytic uremic syndrome after EHEC O157:H7, Guillain-Barre syndrome after Campylobacter, and reactive arthritis. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.3 Distinguish infectious esophagitis in immunocompromised patients: Candida (white pseudomembranes, yeast and pseudohyphae), HSV-1 (punched-out ulcers, intranuclear inclusions), and CMV (linear ulcers, inclusions in endothelial and stromal cells). NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.4 Match acute watery gastroenteritis to its setting: rotavirus (unvaccinated infants), norovirus (cruise ship and institutional outbreaks), ETEC (traveler's diarrhea), Vibrio cholerae (contaminated water, rice-water stool), and Bacillus cereus (reheated rice). NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.5 Explain how plaque biofilm causes reversible gingivitis that can progress to periodontitis with loss of periodontal ligament and alveolar bone (Porphyromonas gingivalis), and relate acid from Streptococcus mutans to dental caries and tooth abscess. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-13-L.6 Describe oral thrush as Candida albicans overgrowth (budding yeast and pseudohyphae, white plaques that scrape off) and predict it in infants, inhaled corticosteroid or antibiotic use, diabetes, and HIV with low CD4 counts. NPLEX + USMLE Microbiology & Immunology DOK 2

GI-13-L.7 Distinguish herpetic gingivostomatitis (primary HSV-1, fever with gum and lip vesicles), herpangina (coxsackie A, posterior pharyngeal vesicles), hand-foot-and-mouth disease (coxsackie A16), and aphthous ulcers (Crohn disease, Behcet disease). NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.8 Interpret hepatitis B serology to identify acute infection (HBsAg, IgM anti-HBc), the window period (IgM anti-HBc alone), chronic infection (HBsAg over 6 months, IgG anti-HBc), resolved infection, and vaccination (anti-HBs alone). NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.9 Predict outcomes of viral hepatitis: chronic HBV in most infected neonates but few adults, chronic HCV in most adults, cirrhosis and hepatocellular carcinoma with HBV or HCV, and HCV-related mixed cryoglobulinemia. NPLEX + USMLE Microbiology & Immunology DOK 3

GI-13-L.10 Distinguish pyogenic liver abscess (biliary or portal spread, often E. coli or Klebsiella) from amebic liver abscess (Entamoeba histolytica, anchovy paste fluid, travel history). USMLE extension Microbiology & Immunology DOK 3

HOW THE BODY WORKS

MS-1 Musculoskeletal embryology

Describe embryological development of the musculoskeletal system, including muscle, bone, and joints.

MS-1.1 Trace the fate of somites from paraxial mesoderm into sclerotome (vertebrae, ribs), myotome (skeletal muscle), and dermatome (dermis), and relate epaxial muscles to dorsal rami and hypaxial muscles to ventral rami. NPLEX · Anatomy · DOK 2

MS-1.2 Explain resegmentation, in which each vertebral body forms from the caudal half of one sclerotome and the cranial half of the next, and identify the notochord remnant as the nucleus pulposus. NPLEX · Anatomy · DOK 2

MS-1.3 Relate limb patterning signals to limb defects: FGF from the apical ectodermal ridge (proximal to distal outgrowth), sonic hedgehog from the zone of polarizing activity (anterior to posterior patterning; ectopic expression causes polydactyly), HOX genes (segment identity), and thalidomide (phocomelia). NPLEX + USMLE · Anatomy · DOK 3

MS-1.4 Explain why the upper limb rotates laterally and the lower limb rotates medially during development, and use this to predict the position of flexor and extensor compartments and the spiral dermatome pattern of the leg. NPLEX · Anatomy · DOK 2

MS-1.5 Compare intramembranous ossification (flat bones of the skull vault, most of the clavicle) with endochondral ossification (hyaline cartilage model, primary center in the diaphysis, secondary centers in the epiphyses), and describe how joints form from interzone mesenchyme by cavitation. NPLEX · Anatomy · DOK 2

MS-1.6 Match pharyngeal arch muscles to their nerves: first arch (muscles of mastication, CN V3), second arch (muscles of facial expression, CN VII), third arch (stylopharyngeus, CN IX), fourth and sixth arches (pharynx and larynx, CN X). NPLEX · Anatomy · DOK 1

MS-2 Muscle, bone, and joint histology

Describe microscopic anatomy of the musculoskeletal system, including skeletal, cardiac, and smooth muscle; compact and spongy bone; and fibrous, cartilaginous, and synovial joints.

MS-2.1 Identify the A band, I band, H zone, M line, and Z line of the sarcomere, and predict which bands shorten during contraction (I band and H zone) and which stays constant (A band). NPLEX · Anatomy · DOK 2

MS-2.2 Distinguish skeletal muscle (multinucleate, peripheral nuclei, triads), cardiac muscle (branched, central nuclei, intercalated discs with gap junctions and desmosomes, dyads), and smooth muscle (fusiform, central nucleus, dense bodies, no striations) on histology. NPLEX · Anatomy · DOK 2

MS-2.3 Describe compact bone (osteons with central Haversian canals, Volkmann canals, concentric lamellae, osteocytes in lacunae linked by canaliculi) and spongy bone (trabeculae surrounding marrow), and distinguish woven from lamellar bone. NPLEX · Anatomy · DOK 1

MS-2.4 Compare osteoblasts (mesenchymal origin, secrete type I collagen osteoid and alkaline phosphatase), osteocytes (mechanosensing, FGF23 secretion), and osteoclasts (monocyte and macrophage lineage, multinucleated, ruffled border, acid and cathepsin K in Howship lacunae). NPLEX · Anatomy · DOK 2

MS-2.5 Describe the epiphyseal growth plate zones (reserve, proliferative, hypertrophic, calcification, ossification) and predict how defects in chondrocyte proliferation (achondroplasia) or mineralization (rickets) alter the plate. NPLEX · Anatomy · DOK 3

MS-2.6 Distinguish fibrous joints (sutures, syndesmoses, gomphoses), cartilaginous joints (hyaline synchondroses, fibrocartilage symphyses such as the pubic symphysis and intervertebral disc), and synovial joints (avascular articular hyaline cartilage, synoviocytes producing hyaluronan and lubricin). NPLEX · Anatomy · DOK 2

MS-3 Axial and appendicular skeleton

Describe location and structure, and explain function of vertebrae, skull bones, vertebral column, pectoral girdle, upper extremity, pelvic girdle, and lower extremity.

MS-3.1 Identify regional vertebral features: cervical transverse foramina and bifid spines, the atlas (no body) and axis (dens), thoracic costal facets, and large lumbar bodies, and explain how each relates to regional function. NPLEX · Anatomy · DOK 1

MS-3.2 Distinguish the normal cervical and lumbar lordosis and thoracic kyphosis from scoliosis (lateral curvature with vertebral rotation) and excessive kyphosis, and predict how severe thoracic scoliosis restricts lung expansion. NPLEX + USMLE | Anatomy | DOK 2

MS-3.3 Locate the major skull foramina and their contents (superior orbital fissure, foramen rotundum, foramen ovale, internal acoustic meatus, jugular foramen, foramen magnum), and explain why a pterion fracture can tear the middle meningeal artery. NPLEX | Anatomy | DOK 2

MS-3.4 Describe the clavicle, scapula, humerus, radius, ulna, and carpal bones, and predict the nerve at risk with fractures of the humeral surgical neck (axillary), radial groove (radial), and medial epicondyle (ulnar). NPLEX | Anatomy | DOK 3

MS-3.5 Describe the hip bone (ilium, ischium, pubis fused at the acetabulum), femur, tibia, fibula, and foot arches, relate a collapsed medial longitudinal arch to flat foot (pes planus), and predict common fibular nerve injury with a fibular neck fracture. NPLEX + USMLE | Anatomy | DOK 3

MS-4 Joints: structure, innervation, and function

Describe location, structure, and innervation of joints, and explain functions of different joint types.

MS-4.1 Classify joints by function (synarthrosis, amphiarthrosis, diarthrosis) and synovial joints by type (hinge, pivot, ball and socket, saddle, condyloid, plane), giving an example of each such as the atlantoaxial pivot and first carpometacarpal saddle. NPLEX | Anatomy | DOK 1

MS-4.2 Apply Hilton law (nerves to muscles acting on a joint also supply the joint) to explain articular proprioception and nociception, and why hip disease can present as knee pain through the obturator and femoral nerves. NPLEX | Anatomy | DOK 2

MS-4.3 Explain why the glenohumeral joint depends on the rotator cuff for stability, and predict axillary nerve injury and loss of deltoid abduction after anterior shoulder dislocation. NPLEX | Anatomy | DOK 3

MS-4.4 Relate knee ligaments and menisci to clinical tests: anterior drawer and Lachman (ACL), posterior drawer (PCL), valgus stress (MCL), varus stress (LCL), and explain why the medial meniscus, attached to the MCL, is injured more often. NPLEX | Anatomy | DOK 3

MS-4.5 Describe the temporomandibular joint (synovial joint with an articular disc, hinge and gliding movement, innervated by auriculotemporal and masseteric branches of CN V3) and relate disc displacement to clicking and jaw pain. NPLEX + USMLE | Anatomy | DOK 2

MS-5 Muscles: origin, insertion, action, innervation

Describe origin, insertion, main action, and innervation of muscles in body regions: head and neck; upper and lower extremities; back, thorax, abdomen, and pelvis.

MS-5.1 Describe the rotator cuff muscles: supraspinatus (initial abduction, suprascapular nerve), infraspinatus (lateral rotation, suprascapular nerve), teres minor (lateral rotation, axillary nerve), subscapularis (medial rotation, subscapular nerves), and identify supraspinatus as most often injured. NPLEX + USMLE | Anatomy | DOK 2

MS-5.2 Predict upper limb deficits from nerve injury: musculocutaneous (weak elbow flexion), radial (wrist drop), median (loss of thenar opposition, ape hand), ulnar (claw hand, weak finger abduction), and long thoracic (winged scapula from serratus anterior paralysis). NPLEX | Anatomy | DOK 3

MS-5.3 Predict lower limb deficits from nerve injury: femoral (weak knee extension), obturator (weak adduction), superior gluteal (Trendelenburg sign with pelvic drop on the opposite side), common fibular (foot drop, weak eversion), and tibial (weak plantarflexion and inversion). NPLEX | Anatomy | DOK 3

MS-5.4 Describe the action and innervation of head and neck muscles: muscles of mastication (CN V3), muscles of facial expression (CN VII), tongue muscles (CN XII, except palatoglossus by CN X), and sternocleidomastoid and trapezius (CN XI). NPLEX | Anatomy | DOK 2

MS-5.5 Describe trunk muscles: erector spinae (extension, dorsal rami), diaphragm (inspiration, phrenic nerve C3 to C5), anterior abdominal wall layers (thoracoabdominal and subcostal nerves), and levator ani (pelvic floor support). NPLEX | Anatomy | DOK 2

MS-6 Musculoskeletal connective tissue

Describe embryology and structure, and explain function of connective tissues of the musculoskeletal system.

MS-6.1 Trace musculoskeletal connective tissue origins: limb bones, cartilage, and tendons from lateral plate (somatic) mesoderm, axial skeleton from sclerotome, and craniofacial bone and cartilage largely from neural crest mesenchyme. NPLEX | Anatomy | DOK 2

MS-6.2 Match collagen types to tissues: type I (bone, tendon, ligament, fibrocartilage), type II (hyaline cartilage, nucleus pulposus), type III (reticular fibers, vessels, granulation tissue), and type IV (basement membrane). NPLEX · Biochemistry & Genetics · DOK 1

MS-6.3 Sequence collagen synthesis (Gly-X-Y chains, proline and lysine hydroxylation requiring vitamin C, triple helix formation, propeptide cleavage, lysyl oxidase cross-linking requiring copper) and predict the defect at each step in scurvy, osteogenesis imperfecta, and Menkes disease. NPLEX · Biochemistry & Genetics · DOK 3

MS-6.4 Explain how articular cartilage resists compression (aggrecan with chondroitin and keratan sulfate on hyaluronan binding water) and why avascular cartilage and poorly vascular tendons heal slowly. NPLEX · Anatomy · DOK 2

MS-6.5 Describe elastic fibers (elastin cores on fibrillin-1 microfibrils) in elastic ligaments such as the ligamentum flavum, and explain how fibrillin-1 normally sequesters TGF-beta. NPLEX · Biochemistry & Genetics · DOK 2

MS-7 Muscle contraction

Explain mechanisms and factors affecting contraction of skeletal, smooth, and cardiac muscle.

MS-7.1 Trace skeletal muscle excitation-contraction coupling from T-tubule depolarization through the dihydropyridine receptor and its mechanical link to ryanodine receptor 1, calcium release from the sarcoplasmic reticulum, calcium binding to troponin C, and calcium reuptake by SERCA. NPLEX · Physiology · DOK 2

MS-7.2 Sequence the cross-bridge cycle (ATP binding releases myosin from actin, hydrolysis cocks the head, phosphate release drives the power stroke) and explain rigor mortis as the loss of ATP. NPLEX · Physiology · DOK 2

MS-7.3 Explain how skeletal muscle force is graded by motor unit recruitment (size principle) and summation to tetanus, and interpret length-tension and force-velocity relationships. NPLEX · Physiology · DOK 3

MS-7.4 Explain calcium-induced calcium release through L-type channels and ryanodine receptor 2 in cardiac muscle, why its long refractory period prevents tetany, and how sympathetic stimulation increases contractility via phospholamban phosphorylation. NPLEX · Physiology · DOK 3

MS-7.5 Explain smooth muscle contraction through calcium-calmodulin activation of myosin light chain kinase, relaxation by myosin light chain phosphatase, the latch state, and nitric oxide induced relaxation through cGMP. NPLEX · Physiology · DOK 2

MS-7.6 Compare type I fibers (slow, oxidative, red, rich in mitochondria and myoglobin, fatigue resistant) with type II fibers (fast, glycolytic, fatigable), and predict adaptations to endurance training (greater oxidative capacity) and resistance training (type II fiber hypertrophy). NPLEX · Physiology · DOK 2

MS-8 Musculoskeletal biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to musculoskeletal function and pathology.

MS-8.1 Trace vitamin D activation from 7-dehydrocholesterol in UVB-exposed skin through hepatic 25-hydroxylation and renal 1-alpha-hydroxylation, and explain why serum 25-hydroxyvitamin D reflects vitamin D status. NPLEX + USMLE · Biochemistry & Genetics · DOK 2

MS-8.2 Explain how PTH (bone resorption via osteoblast RANKL, distal tubule calcium reabsorption, phosphaturia, 1-alpha-hydroxylase activation), calcitriol, calcitonin, and osteocyte FGF23 regulate serum calcium and phosphate, and how osteoprotegerin limits osteoclast activation. NPLEX · Physiology · DOK 3

MS-8.3 Compare muscle fuel use across exercise duration (phosphocreatine via creatine kinase, anaerobic glycolysis to lactate, oxidative metabolism of glucose and fatty acids) and explain the Cori cycle and creatinine as a marker of muscle mass. NPLEX · Biochemistry & Genetics · DOK 2

MS-8.4 Explain McArdle disease (autosomal recessive muscle glycogen phosphorylase deficiency) and predict exercise intolerance, cramps, myoglobinuria, failure of blood lactate to rise with exercise, and the second wind phenomenon. NPLEX + USMLE · Biochemistry & Genetics · DOK 3

MS-8.5 Relate carnitine shuttle defects such as carnitine palmitoyltransferase II deficiency to muscle weakness and exertional rhabdomyolysis during prolonged exercise or fasting, when muscle depends on fatty acid oxidation. NPLEX · Biochemistry & Genetics · DOK 3

MS-8.6 Explain mineral and cofactor roles in bone and muscle: magnesium for PTH secretion, copper for lysyl oxidase cross-linking, vitamin K for gamma-carboxylation of osteocalcin, and calcium hydroxyapatite as the bone mineral. NPLEX · Biochemistry & Genetics · DOK 2

MS-9 Musculoskeletal genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic defects underlying musculoskeletal disease.

MS-9.1 Relate transcription factors to skeletal development: RUNX2 for osteoblast differentiation (haploinsufficiency causes cleidocranial dysplasia with absent clavicles and delayed suture closure), SOX9 for chondrocyte differentiation, and HOX genes for segment identity. NPLEX | Biochemistry & Genetics | DOK 2

MS-9.2 Explain how FGFR3 gain-of-function mutations inhibit chondrocyte proliferation in achondroplasia, and why most cases are new autosomal dominant mutations associated with advanced paternal age. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

MS-9.3 Explain the dominant-negative effect of glycine substitutions in COL1A1 or COL1A2 in severe osteogenesis imperfecta, compared with null alleles that halve normal collagen output and cause the milder type I form. NPLEX | Biochemistry & Genetics | DOK 3

MS-9.4 Apply the reading frame rule to dystrophin: frameshift deletions abolish dystrophin (Duchenne), while in-frame deletions produce a truncated, partly functional protein (Becker), and explain why the very large DMD gene has a high new mutation rate. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

MS-9.5 Explain myotonic dystrophy type 1 as an autosomal dominant CTG repeat expansion in DMPK showing anticipation, and predict myotonia, distal weakness, cataracts, frontal balding, testicular atrophy, and cardiac conduction defects. NPLEX | Biochemistry & Genetics | DOK 3

MS-9.6 Explain maternal inheritance and heteroplasmy in mitochondrial myopathies, and identify ragged red fibers on Gomori trichrome stain in disorders such as MERRF and MELAS. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

MS-9.7 Relate cell cycle and oncogene defects to bone and soft tissue sarcomas: RB1 loss (osteosarcoma after hereditary retinoblastoma), TP53 loss in Li-Fraumeni syndrome, and the EWSR1–FLI1 fusion transcription factor from t(11;22) in Ewing sarcoma. NPLEX | Biochemistry & Genetics | DOK 3

MS-10 Muscle and nerve interactions

Explain relationship between musculoskeletal and neurological systems.

MS-10.1 Trace neuromuscular transmission (calcium entry, SNARE-mediated acetylcholine release, nicotinic receptor end plate potential, acetylcholinesterase) and localize the defect in myasthenia gravis, Lambert-Eaton syndrome, and botulism. NPLEX | Physiology | DOK 3

MS-10.2 Distinguish myasthenia gravis (anti-acetylcholine receptor antibodies, weakness that worsens with use, ptosis and diplopia) from Lambert-Eaton syndrome (anti-presynaptic calcium channel antibodies, strength improving with repeated effort, small cell lung cancer). NPLEX + USMLE | Physiology | DOK 3

MS-10.3 Compare muscle spindles (Ia afferents, stretch reflex, gamma motor neuron tuning) with Golgi tendon organs (Ib afferents, tension sensing), and match reflexes to roots: biceps C5 to C6, triceps C7, patellar L4, Achilles S1. NPLEX | Physiology | DOK 2

MS-10.4 Distinguish neurogenic weakness (fasciculations, grouped atrophy, fiber type grouping, as in spinal muscular atrophy from SMN1 deletion) from myopathic weakness (proximal weakness, elevated CK), and identify type II fiber atrophy from disuse or corticosteroids. NPLEX + USMLE | Pathology | DOK 3

MS-10.5 Explain why hypocalcemia increases nerve and muscle excitability (lower sodium channel threshold), producing tetany with Chvostek and Trousseau signs, and why hypokalemia and hyperkalemia both cause muscle weakness. NPLEX | Physiology | DOK 3

MS-11 Immune responses in musculoskeletal disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

MS-11.1 Explain rheumatoid arthritis pathogenesis: HLA-DR4 associated autoimmunity to citrullinated peptides, CD4 T cell and cytokine (TNF, IL-1, IL-6) driven synovitis, and pannus with RANKL-driven osteoclasts eroding cartilage and bone. NPLEX + USMLE | Microbiology & Immunology | DOK 3

MS-11.2 Distinguish rheumatoid arthritis (symmetric MCP and PIP synovitis sparing the DIP, morning stiffness over 1 hour, ulnar deviation, swan neck deformity, nodules, anti-CCP and rheumatoid factor) from osteoarthritis, and identify Felty syndrome (RA, splenomegaly, neutropenia). NPLEX + USMLE | Pathology | DOK 3

MS-11.3 Match autoantibodies to rheumatic diseases: anti-CCP (rheumatoid arthritis), anti-dsDNA and anti-Smith (SLE), anti-Scl-70 (diffuse systemic sclerosis), anticentromere (limited systemic sclerosis), anti-U1 RNP (mixed connective tissue disease), anti-Jo-1 (myositis), anti-SSA and anti-SSB (Sjogren syndrome). NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-11.4 Describe musculoskeletal features of SLE (nonerosive arthritis), systemic sclerosis (sclerodactyly, calcinosis, flexion contractures, Raynaud phenomenon), and mixed connective tissue disease (overlapping SLE, scleroderma, and myositis features with anti-U1 RNP). NPLEX + USMLE · Pathology · DOK 2

MS-11.5 Classify hypersensitivity in musculoskeletal disease: type II in rheumatic fever polyarthritis (molecular mimicry) and Lambert-Eaton syndrome (antibodies against small cell lung cancer cross-reacting with presynaptic calcium channels), type III in SLE arthritis, and type IV in polymyositis. NPLEX · Microbiology & Immunology · DOK 3

MS-11.6 Explain innate immune activation in joints: urate crystals activating the NLRP3 inflammasome to release IL-1 beta in gout, and neutrophil proteases destroying articular cartilage within days in septic arthritis. NPLEX · Microbiology & Immunology · DOK 3

MS-11.7 Describe postinfectious and postvaccine arthritis: HLA-B27 associated reactive arthritis after enteric or chlamydial infection, symmetric small joint arthritis after parvovirus B19 or rubella infection, and transient arthralgia in adult women after rubella vaccination. NPLEX · Microbiology & Immunology · DOK 2

MS-12 Musculoskeletal pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

MS-12.1 Describe Staphylococcus aureus (gram-positive cocci in clusters, catalase and coagulase positive) and explain how protein A, collagen and fibronectin adhesins, and biofilm on hardware make it the most common cause of osteomyelitis and septic arthritis. NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.2 Describe Neisseria gonorrhoeae (gram-negative diplococci; pili with antigenic variation, IgA protease) and the disseminated gonococcal infection pattern of tenosynovitis, pustular dermatitis, and migratory polyarthralgia or purulent monoarthritis in sexually active young adults. NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.3 Match osteomyelitis organisms to settings: Salmonella (sickle cell disease), Pseudomonas aeruginosa (injection drug use, puncture through a shoe sole), Pasteurella multocida (cat and dog bites), Staphylococcus epidermidis (prosthetic material), and Mycobacterium tuberculosis (vertebrae). NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.4 Describe Clostridium perfringens (gram-positive, anaerobic, spore-forming rod) and explain how alpha toxin (phospholipase C, lecithinase) destroys cell membranes, producing myonecrosis, gas, and hemolysis in contaminated deep wounds. NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.5 Describe Streptococcus pyogenes (beta-hemolytic gram-positive cocci in chains) and explain how M protein resists phagocytosis while hyaluronidase, streptokinase, and superantigen pyrogenic exotoxins promote spread along fascia and toxic shock. NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.6 Describe Borrelia burgdorferi (spirochete) and its transmission cycle, in which Ixodes larvae acquire it from white-footed mice and nymphs transmit it after about 36 hours or more of attachment, and the late Lyme arthritis of large joints, especially the knee. NPLEX + USMLE · Microbiology & Immunology · DOK 2

MS-12.7 Explain how Mycobacterium tuberculosis spreads hematogenously to vertebral bodies (Pott disease), forming caseating granulomas that collapse adjacent vertebrae and disc space, producing gibbus deformity and psoas abscess. NPLEX + USMLE · Microbiology & Immunology · DOK 3

DISEASES AND CONDITIONS

MS-13-A Musculoskeletal nutritional deficiencies

1 Osteomalacia **2** Rickets **3** Scurvy

MS-13-A.1 Explain how vitamin D deficiency or phosphate loss causes defective mineralization of osteoid, producing rickets before growth plate closure and osteomalacia after it. NPLEX + USMLE · Pathology · DOK 2

MS-13-A.2 Identify risk factors for vitamin D deficiency: limited sun exposure, darker skin, unsupplemented exclusive breastfeeding, fat malabsorption, liver or kidney disease, and anticonvulsant use. NPLEX + USMLE · Pathology · DOK 1

MS-13-A.3 Describe osteomalacia findings: diffuse bone pain, proximal muscle weakness, Looser zones (pseudofractures), and labs showing low or normal calcium, low phosphate, high alkaline phosphatase, high PTH, and low 25-hydroxyvitamin D. NPLEX + USMLE Pathology DOK 3

MS-13-A.4 Describe rickets findings: rachitic rosary, craniotabes, frontal bossing, bowed legs (genu varum), widened and cupped metaphyses on radiograph, and distinguish X-linked hypophosphatemic rickets caused by FGF23 excess. NPLEX + USMLE Pathology DOK 3

MS-13-A.5 Explain how vitamin C deficiency impairs prolyl and lysyl hydroxylation, producing unstable collagen triple helices, fragile vessels, and poor wound healing. NPLEX + USMLE Pathology DOK 2

MS-13-A.6 Describe scurvy findings and risks: perifollicular hemorrhage, corkscrew hairs, swollen bleeding gums, poor wound healing, hemarthrosis, and painful subperiosteal hemorrhage in children, in people with diets lacking fruit and vegetables or with alcohol use disorder. NPLEX + USMLE Pathology DOK 3

MS-13-B Inflammatory musculoskeletal diseases

1 Ankylosing spondylitis **2** Bursitis **3** Fibromyalgia **4** Gout **5** Myositis (dermatomyositis, polymyositis) **6** Polymyalgia rheumatica (PMR) **7** Reactive arthritis (Reiter syndrome) **8** Tendonopathy

MS-13-B.1 Explain ankylosing spondylitis as an HLA-B27 associated seronegative spondyloarthropathy with enthesitis and sacroiliitis, and predict inflammatory back pain in young men that improves with exercise, reduced chest expansion, and bamboo spine from syndesmophytes. NPLEX + USMLE Pathology DOK 3

MS-13-B.2 Identify extra-articular complications of ankylosing spondylitis: anterior uveitis, aortitis with aortic regurgitation, restrictive lung disease, and vertebral fractures in a fused spine. NPLEX + USMLE Pathology DOK 2

MS-13-B.3 Describe bursitis by site and cause, including prepatellar bursitis from kneeling, olecranon bursitis, subacromial and trochanteric bursitis from overuse, and septic bursitis from Staphylococcus aureus. NPLEX + USMLE Pathology DOK 2

MS-13-B.4 Explain fibromyalgia as a disorder of central pain sensitization, and identify widespread pain for at least 3 months with fatigue and unrefreshing sleep, most often in women, with normal inflammatory markers and CK. NPLEX + USMLE Pathology DOK 2

MS-13-B.5 Explain hyperuricemia in gout from urate underexcretion (most cases, thiazides, kidney disease) or overproduction (Lesch-Nyhan HGPRT deficiency, tumor lysis, glucose-6-phosphatase deficiency), with xanthine oxidase converting hypoxanthine and xanthine to uric acid. NPLEX + USMLE Biochemistry & Genetics DOK 3

MS-13-B.6 Identify gout findings: acute monoarthritis of the first MTP joint (podagra) triggered by alcohol or purine-rich meals, needle-shaped negatively birefringent urate crystals, tophi, punched-out erosions, and uric acid kidney stones. NPLEX + USMLE Pathology DOK 3

MS-13-B.7 Distinguish dermatomyositis (complement-mediated capillary injury, perifascicular atrophy, heliotrope rash, Gottron papules, association with internal cancer) from polymyositis (endomysial CD8 T cells), both with proximal weakness, high CK, and possible anti-Jo-1 interstitial lung disease. NPLEX + USMLE Pathology DOK 3

MS-13-B.8 Interpret muscle biopsy and laboratory findings to separate inflammatory myositis (high CK, inflammatory infiltrates, myofiber necrosis) from polymyalgia rheumatica (normal CK and strength) and steroid myopathy. NPLEX + USMLE Pathology DOK 3

MS-13-B.9 Describe polymyalgia rheumatica in adults over 50: pain and stiffness of the shoulder and hip girdles, elevated ESR, normal CK, and its association with giant cell (temporal) arteritis and risk of vision loss. NPLEX + USMLE Pathology DOK 2

MS-13-B.10 Explain reactive arthritis as an HLA-B27 associated sterile arthritis following Chlamydia trachomatis or enteric infection (Shigella, Salmonella, Yersinia, Campylobacter), presenting with arthritis, conjunctivitis, and urethritis, plus enthesitis and keratoderma blennorrhagicum. NPLEX + USMLE Pathology DOK 3

MS-13-B.11 Explain tendinopathy as overuse-induced collagen disorganization with limited inflammation (tendinosis), and identify lateral epicondylitis (extensor carpi radialis brevis origin), medial epicondylitis, and supraspinatus impingement with a painful arc. NPLEX + USMLE Pathology DOK 3

MS-13-B.12 Describe de Quervain tenosynovitis (abductor pollicis longus and extensor pollicis brevis, positive Finkelstein test), plantar fasciitis, and Achilles tendinopathy with rupture risk increased by fluoroquinolones. NPLEX + USMLE Pathology DOK 2

MS-13-B.13 Distinguish psoriatic arthritis (DIP involvement, dactylitis, nail pitting, pencil-in-cup deformity, axial disease linked to HLA-B27) from rheumatoid arthritis. USMLE extension Pathology DOK 3

MS-13-B.14 Describe juvenile idiopathic arthritis (arthritis before age 16 lasting at least 6 weeks), including the oligoarticular ANA-positive form with silent anterior uveitis and the systemic form with daily spiking fever and salmon-colored rash. USMLE extension Pathology DOK 2

MS-13-B.15 Distinguish pseudogout (calcium pyrophosphate deposition, rhomboid weakly positively birefringent crystals, chondrocalcinosis, knee and wrist, linked to hyperparathyroidism and hemochromatosis) from gout. USMLE extension Pathology DOK 3

MS-13-B.16 Describe adhesive capsulitis (capsular inflammation and fibrosis with loss of both active and passive shoulder motion, associated with diabetes) and ganglion cysts (myxoid degeneration near a joint capsule or tendon sheath, often the dorsal wrist, lacking a true lining). USMLE extension Pathology DOK 2

MS-13-C Metabolic musculoskeletal diseases

1 Osteopetrosis

MS-13-C.1 Explain osteopetrosis as failure of osteoclast bone resorption (for example carbonic anhydrase II deficiency with renal tubular acidosis, or CLCN7 mutations), producing dense but brittle bone. NPLEX Pathology DOK 2

MS-13-C.2 Predict osteopetrosis complications: fractures, pancytopenia from marrow obliteration with extramedullary hematopoiesis and hepatosplenomegaly, and cranial nerve palsies (vision and hearing loss) from narrowed foramina, more severe in the infantile autosomal recessive form. NPLEX Pathology DOK 3

MS-13-C.3 Interpret serum calcium, phosphate, alkaline phosphatase, and PTH to distinguish osteoporosis (all normal), osteomalacia, Paget disease (isolated high alkaline phosphatase), primary hyperparathyroidism, and renal osteodystrophy. USMLE extension Pathology DOK 3

MS-13-C.4 Explain renal osteodystrophy: phosphate retention and low calcitriol in chronic kidney disease cause secondary hyperparathyroidism, with osteitis fibrosa cystica (subperiosteal resorption, brown tumors) and defective mineralization. USMLE extension Pathology DOK 3

MS-13-D Congenital musculoskeletal diseases

1 Marfan syndrome **2** Muscular dystrophy **3** Osteogenesis imperfecta

MS-13-D.1 Explain Marfan syndrome as an autosomal dominant FBN1 (fibrillin-1) defect with excess TGF-beta signaling, and identify tall stature, arachnodactyly, arm span exceeding height, pectus deformity, joint hypermobility, and upward lens dislocation. NPLEX + USMLE Biochemistry & Genetics DOK 3

MS-13-D.2 Predict Marfan cardiovascular complications (aortic root dilation, aortic dissection, mitral valve prolapse) and distinguish Marfan from homocystinuria (downward lens dislocation, intellectual disability, thrombosis). NPLEX + USMLE Pathology DOK 3

MS-13-D.3 Describe Duchenne muscular dystrophy (X-linked, absent dystrophin): onset before age 5, Gowers sign, calf pseudohypertrophy from fibrofatty replacement, markedly elevated CK, dilated cardiomyopathy, and respiratory failure. NPLEX + USMLE Pathology DOK 3

MS-13-D.4 Compare Becker muscular dystrophy (later onset, milder course, truncated dystrophin) with Duchenne, and use dystrophin immunostaining and genetic testing to separate them. NPLEX + USMLE Pathology DOK 3

MS-13-D.5 Explain osteogenesis imperfecta as defective type I collagen (COL1A1 or COL1A2), and identify multiple fractures with minimal trauma, blue sclerae, conductive hearing loss, dentinogenesis imperfecta, and joint laxity. NPLEX + USMLE Pathology DOK 3

MS-13-D.6 Distinguish osteogenesis imperfecta from nonaccidental trauma in a child with multiple fractures using sclerae, dentition, family history, and fracture pattern. NPLEX + USMLE Pathology DOK 3

MS-13-D.7 Describe Ehlers-Danlos syndrome as a group of collagen disorders (classic type V collagen, vascular type III collagen) with skin hyperextensibility, joint hypermobility and dislocation, easy bruising, and arterial or bowel rupture in the vascular type. NPLEX + USMLE Biochemistry & Genetics DOK 3

MS-13-D.8 Identify achondroplasia findings: rhizomelic limb shortening with a normal-length trunk, macrocephaly with frontal bossing, and normal intelligence. USMLE extension Pathology DOK 1

MS-13-E Degenerative musculoskeletal diseases

1 Osteoarthritis (OA)/degenerative joint disease (DJD) **2** Osteoporosis **3** Paget disease **4** Avascular necrosis

MS-13-E.1 Explain osteoarthritis pathogenesis (chondrocyte injury, proteoglycan loss, cartilage fibrillation, eburnation, subchondral sclerosis and cysts, osteophytes) and risk factors (age, obesity, prior joint injury, mechanical load). NPLEX + USMLE · Pathology · DOK 3

MS-13-E.2 Identify osteoarthritis findings: pain worse with use, morning stiffness under 30 minutes, Heberden (DIP) and Bouchard (PIP) nodes, asymmetric weight-bearing joint involvement, and radiographic joint space narrowing with osteophytes. NPLEX + USMLE · Pathology · DOK 3

MS-13-E.3 Explain osteoporosis as reduced bone mass with normal mineralization, driven by low peak bone mass, estrogen loss increasing RANKL-mediated resorption, aging, and secondary causes (glucocorticoids, hyperthyroidism, immobilization, alcohol, smoking). NPLEX + USMLE · Pathology · DOK 3

MS-13-E.4 Interpret osteoporosis findings: vertebral compression, hip, and distal radius fractures, height loss and kyphosis, normal serum calcium, phosphate, and alkaline phosphatase, and DEXA T-scores of minus 2.5 or lower (osteopenia between minus 1 and minus 2.5). NPLEX + USMLE · Pathology · DOK 3

MS-13-E.5 Explain Paget disease of bone as osteoclast overactivity followed by disorganized osteoblastic repair (lytic, mixed, and sclerotic phases) producing thick but weak mosaic lamellar bone, with isolated elevation of alkaline phosphatase. NPLEX + USMLE · Pathology · DOK 3

MS-13-E.6 Predict Paget disease complications in older adults: bone pain, enlarging skull, hearing loss, fractures, high-output heart failure from arteriovenous shunting, and osteosarcoma. NPLEX + USMLE · Pathology · DOK 3

MS-13-E.7 Explain avascular necrosis as bone infarction from interrupted blood supply, and identify causes (glucocorticoids, alcohol, sickle cell disease, SLE, decompression sickness, femoral neck and scaphoid fractures) and the femoral head as the most common site. NPLEX + USMLE · Pathology · DOK 3

MS-13-E.8 Distinguish pediatric hip disorders: developmental dysplasia of the hip (newborn, Ortolani and Barlow tests), Legg-Calve-Perthes disease (idiopathic femoral head osteonecrosis in young children), and slipped capital femoral epiphysis (obese adolescent, epiphysis slips off the femoral neck). USMLE extension · Pathology · DOK 3

MS-13-E.9 Relate disc degeneration to posterolateral nucleus pulposus herniation compressing the lower root (L4 to L5 herniation affects L5), and distinguish spondylosis (degenerative change) from spondylolisthesis (forward vertebral slip). USMLE extension · Anatomy · DOK 3

MS-13-E.10 Distinguish adolescent knee disorders: Osgood-Schlatter disease (traction apophysitis of the tibial tubercle), chondromalacia patellae (softened patellar cartilage with anterior knee pain), and osteochondritis dissecans (separation of a subchondral bone and cartilage fragment). USMLE extension · Pathology · DOK 2

MS-13-E.11 Describe Dupuytren contracture as myofibroblast proliferation in the palmar fascia causing flexion contracture of the ring and little fingers. USMLE extension · Pathology · DOK 1

MS-13-F Musculoskeletal neoplasms

1 Chondrosarcoma **2** Ewing sarcoma **3** Osteoid osteoma **4** Osteosarcoma **5** Rhabdomyosarcoma

MS-13-F.1 Describe chondrosarcoma as a malignant cartilage-forming tumor of older adults in the pelvis, proximal femur, and shoulder girdle, which can arise from enchondroma or osteochondroma and often carries IDH1 or IDH2 mutations. NPLEX + USMLE · Pathology · DOK 2

MS-13-F.2 Describe Ewing sarcoma in children and adolescents: diaphyseal or pelvic tumor of small round blue cells, CD99 positive, EWSR1–FLI1 from t(11;22), onion-skin periosteal reaction, and fever that can mimic osteomyelitis. NPLEX + USMLE · Pathology · DOK 3

MS-13-F.3 Describe osteoid osteoma (under 2 cm cortical nidus with a sclerotic rim in young males, night pain relieved by NSAIDs because the nidus produces prostaglandin E2) and distinguish osteoblastoma (larger, vertebral). NPLEX + USMLE · Pathology · DOK 2

MS-13-F.4 Describe osteosarcoma as the most common primary malignant bone tumor apart from myeloma: malignant osteoid, metaphysis around the knee, peaks in adolescents and in older adults with Paget disease or prior radiation. NPLEX + USMLE · Pathology · DOK 3

MS-13-F.5 Interpret osteosarcoma imaging (Codman triangle, sunburst pattern), relate risk to RB1 and TP53 loss, and predict hematogenous metastasis to lung. NPLEX + USMLE · Pathology · DOK 3

MS-13-F.6 Describe rhabdomyosarcoma as the most common soft tissue sarcoma of childhood, arising in the head and neck or genitourinary tract (sarcoma botryoides), with desmin and myogenin positive rhabdomyoblasts and PAX3 or PAX7 to FOXO1 fusions in the alveolar type. NPLEX + USMLE · Pathology · DOK 3

MS-13-F.7 Distinguish benign cartilage tumors: osteochondroma (EXT1 or EXT2, metaphyseal bony stalk with a cartilage cap) and enchondroma (intramedullary hyaline cartilage in hand bones, multiple in Ollier disease and Maffucci syndrome), with risk of chondrosarcoma. USMLE extension · Pathology · DOK 2

MS-13-F.8 Describe giant cell tumor of bone (epiphyseal soap-bubble lesion in young adults, RANKL-expressing stromal cells with osteoclast-like giant cells) and tenosynovial giant cell tumor (synovial or tendon sheath tumor overexpressing CSF1). USMLE extension · Pathology · DOK 2

MS-13-F.9 Explain why metastases are the most common malignant bone tumors, distinguish osteoblastic (prostate), osteolytic (kidney, thyroid, lung), and mixed (breast) lesions, and predict pathologic fracture and hypercalcemia as complications. USMLE extension · Pathology · DOK 3

MS-13-F.10 Compare soft tissue sarcomas by lineage marker, rhabdomyosarcoma (desmin, myogenin) and leiomyosarcoma (smooth muscle actin, desmin; uterus and retroperitoneum in adults), and explain their hematogenous spread to lung. USMLE extension · Pathology · DOK 2

MS-13-F.11 Describe hypertrophic pulmonary osteoarthropathy (digital clubbing, periosteal new bone formation, and arthritis) as a paraneoplastic syndrome of lung cancer. USMLE extension · Pathology · DOK 2

MS-13-G Infectious musculoskeletal diseases

1 Septic (infectious) arthritis **2** Necrotizing fasciitis **3** Osteomyelitis **4** Wet and gas gangrene

MS-13-G.1 Explain septic arthritis pathogenesis (hematogenous seeding or direct inoculation, mostly Staphylococcus aureus, Neisseria gonorrhoeae in sexually active adults) and risk factors (rheumatoid arthritis, prosthetic joints, injection drug use, diabetes). NPLEX + USMLE · Microbiology & Immunology · DOK 3

MS-13-G.2 Interpret synovial fluid analysis: noninflammatory (under 2,000 WBC/mm3), inflammatory, and septic (usually over 50,000 WBC/mm3 with neutrophils, positive Gram stain or culture), plus crystal identification, to evaluate an acute hot swollen joint. NPLEX + USMLE · Pathology · DOK 3

MS-13-G.3 Explain necrotizing fasciitis as rapidly spreading infection along fascial planes, type I polymicrobial (diabetes, perineal Fournier gangrene) or type II Streptococcus pyogenes, with pain out of proportion to exam, crepitus, bullae, and systemic toxicity. NPLEX + USMLE · Microbiology & Immunology · DOK 3

MS-13-G.4 Explain why hematogenous osteomyelitis in children localizes to the metaphysis of long bones, and contrast contiguous spread from diabetic foot ulcers and vertebral osteomyelitis with discitis in adults. NPLEX + USMLE · Pathology · DOK 3

MS-13-G.5 Identify osteomyelitis findings and complications: sequestrum, involucrum, Brodie abscess, elevated ESR and CRP, MRI as the most sensitive imaging, chronic draining sinus, and squamous cell carcinoma in a chronic sinus tract. NPLEX + USMLE · Pathology · DOK 3

MS-13-G.6 Distinguish dry gangrene (coagulative necrosis after arterial occlusion, sharply demarcated) from wet gangrene (superimposed bacterial infection with liquefactive necrosis), and explain the higher risk in diabetes and peripheral artery disease. NPLEX + USMLE · Pathology · DOK 3

MS-13-G.7 Describe gas gangrene (clostridial myonecrosis) after deep contaminated wounds: rapid spread, severe pain, crepitus from gas, bronze skin, hemolysis, and shock from Clostridium perfringens alpha toxin. NPLEX + USMLE · Pathology · DOK 3

MS-13-G.8 Describe infective myositis: pyomyositis (Staphylococcus aureus abscess in large muscles) and trichinellosis (Trichinella spiralis larvae encysting in skeletal muscle after undercooked pork or wild game, with myalgia, periorbital edema, and eosinophilia). USMLE extension · Microbiology & Immunology · DOK 2

MS-13-H Trauma

1 Injury to the musculoskeletal system

MS-13-H.1 Sequence fracture healing (hematoma and inflammation, soft callus, hard callus, remodeling) and predict delayed union from poor blood supply, infection, instability, smoking, diabetes, and malnutrition. NPLEX + USMLE · Pathology · DOK 3

MS-13-H.2 Predict complications of specific fractures: scaphoid and femoral neck (avascular necrosis from retrograde or retinacular blood supply), supracondylar humerus (brachial artery and median nerve), and long bone fractures (fat embolism with hypoxemia, confusion, and petechiae). NPLEX + USMLE · Pathology · DOK 3

MS-13-H.3 Distinguish sprain (ligament), strain (muscle or tendon), contusion (soft tissue hemorrhage), and dislocation (loss of joint congruity), and explain why ligaments heal slowly. NPLEX + USMLE · Pathology · DOK 2

MS-13-H.4 Explain compartment syndrome (rising pressure in a closed fascial compartment after fracture or crush, causing ischemia) and identify pain out of proportion, pain with passive stretch, preserved pulses early, and Volkmann ischemic contracture as a late result. NPLEX + USMLE | Pathology | DOK 3

MS-13-H.5 Explain rhabdomyolysis from crush injury, extreme exertion, or statins, and predict very high CK, myoglobinuria (dipstick positive for blood without red cells), hyperkalemia, and acute kidney injury. USMLE extension | Pathology | DOK 3

MS-13-H.6 Describe myositis ossificans as heterotopic bone in muscle after trauma, and distinguish its zonal pattern (mature bone at the periphery) from osteosarcoma. USMLE extension | Pathology | DOK 2

HOW THE BODY WORKS

NE-1 Neural tube embryology

Describe embryological development of the neural tube and derivatives.

NE-1.1 Describe neurulation, including notochord induction of the neural plate, elevation and fusion of the neural folds, and closure of the anterior and posterior neuropores during the fourth week. NPLEX + USMLE · Anatomy · DOK 2

NE-1.2 Relate the primary brain vesicles (prosencephalon, mesencephalon, rhombencephalon) and secondary vesicles (telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon) to the adult structures and ventricles each forms. NPLEX · Anatomy · DOK 1

NE-1.3 Distinguish neuroectoderm derivatives (CNS neurons, astrocytes, oligodendrocytes, ependyma, retina) from neural crest derivatives (sensory and autonomic ganglia, Schwann cells, adrenal medulla, melanocytes) and mesoderm-derived microglia. NPLEX · Anatomy · DOK 2

NE-1.4 Distinguish spina bifida occulta, meningocele, myelomeningocele, and anencephaly, and relate open neural tube defects to elevated maternal serum and amniotic fluid alpha-fetoprotein and to periconceptional folate deficiency. NPLEX + USMLE · Pathology · DOK 3

NE-1.5 Explain holoprosencephaly as failed separation of the prosencephalon into two hemispheres, linked to sonic hedgehog pathway defects and trisomy 13, and relate it to midline facial defects such as cyclopia and cleft lip. NPLEX + USMLE · Pathology · DOK 3

NE-1.6 Compare Chiari I malformation (cerebellar tonsils below the foramen magnum, associated with syringomyelia), Chiari II malformation (vermis and medulla herniation with lumbosacral myelomeningocele and hydrocephalus), and Dandy-Walker malformation (vermis agenesis with cystic fourth ventricle). NPLEX + USMLE · Anatomy · DOK 3

NE-2 Neuron and nerve histology

Describe microscopic anatomy of motor and sensory neurons and nerves.

NE-2.1 Identify the parts of a multipolar motor neuron (soma with Nissl substance, dendrites, axon hillock, axon, synaptic terminals) and compare it with pseudounipolar dorsal root ganglion neurons and bipolar retinal and olfactory neurons. NPLEX · Anatomy · DOK 1

NE-2.2 Describe the connective tissue layers of a peripheral nerve (endoneurium around each axon, perineurium around each fascicle, epineurium around the whole nerve) and explain the perineurium's role as a blood-nerve barrier. NPLEX · Anatomy · DOK 2

NE-2.3 Compare myelination by Schwann cells (one internode of one axon in the PNS) and oligodendrocytes (internodes of many axons in the CNS), and relate Schwann cell injury to Guillain-Barré syndrome and oligodendrocyte injury to multiple sclerosis and PML. NPLEX + USMLE · Anatomy · DOK 2

NE-2.4 Identify the functions of astrocytes (blood-brain barrier support, potassium buffering, glial scar), microglia (resident CNS macrophages that form multinucleated giant cells in HIV infection), and ependymal cells lining the ventricles. NPLEX · Anatomy · DOK 1

NE-2.5 Describe Wallerian degeneration distal to an axon transection and central chromatolysis in the cell body, and explain why peripheral axons can regenerate along Schwann cell tubes while CNS axons generally do not. NPLEX + USMLE · Pathology · DOK 2

NE-2.6 Identify red neurons (shrunken cell body, eosinophilic cytoplasm, pyknotic nucleus) as the histologic marker of irreversible neuronal injury, appearing about 12 to 24 hours after hypoxia or ischemia. NPLEX + USMLE · Pathology · DOK 1

NE-3 Brain and spinal cord structures

Describe location and structure, and explain function of neural structures in cranial cavity and vertebral canal.

NE-3.1 Locate the primary motor, primary somatosensory, primary visual, and primary auditory cortices and the Broca and Wernicke areas, and predict the deficit produced by a lesion of each. NPLEX + USMLE · Anatomy · DOK 3

NE-3.2 Describe the direct and indirect basal ganglia pathways, including dopamine acting on D1 and D2 receptors, and predict how loss of substantia nigra pars compacta neurons (Parkinson disease) or caudate neurons (Huntington disease) changes movement. NPLEX + USMLE · Physiology · DOK 3

NE-3.3 Explain cerebellar function in coordination and distinguish a vermis lesion (truncal ataxia, wide-based gait) from a hemisphere lesion (ipsilateral limb ataxia, intention tremor, dysdiadochokinesia). NPLEX Anatomy DOK 3

NE-3.4 Identify the thalamic relay nuclei (VPL for body sensation, VPM for face sensation and taste, LGN for vision, MGN for hearing, VL for motor) and relate thalamic infarction to contralateral sensory loss and thalamic pain syndrome. NPLEX + USMLE Anatomy DOK 2

NE-3.5 Identify the brainstem level of the cranial nerve nuclei (midbrain III and IV; pons V to VIII; medulla IX, X, XII) and the spinal cord's gray horns, white columns, enlargements, and conus medullaris ending near L1 to L2. NPLEX Anatomy DOK 1

NE-3.6 Explain how damage to the ascending reticular activating system of the upper brainstem or to both cerebral hemispheres causes coma, and distinguish coma, delirium, and brain death (irreversible loss of all brain and brainstem function). NPLEX + USMLE Physiology DOK 3

NE-4 Cerebrospinal fluid compartments and meninges

Describe location and structure, and explain function of cerebrospinal fluid compartments and meninges.

NE-4.1 Describe the meninges (dura, arachnoid, pia) and the epidural, subdural, and subarachnoid spaces, and identify which space an arterial epidural, venous subdural, and aneurysmal subarachnoid hemorrhage occupies. NPLEX + USMLE Anatomy DOK 2

NE-4.2 Identify the falx cerebri and tentorium cerebelli and predict the structures compressed in subfalcine (anterior cerebral artery), uncal (CN III, posterior cerebral artery, cerebral peduncle), and tonsillar (medullary respiratory centers) herniation. NPLEX + USMLE Anatomy DOK 3

NE-4.3 Describe the subarachnoid cisterns, including the lumbar cistern below the conus medullaris where lumbar puncture is done at L3 to L4 or L4 to L5, and the arachnoid granulations that return CSF to the dural venous sinuses. NPLEX Anatomy DOK 2

NE-4.4 Explain the blood-brain barrier (endothelial tight junctions, basement membrane, astrocyte foot processes) and the blood-CSF barrier (choroid plexus epithelium), and identify circumventricular organs that lack a barrier, such as the area postrema and posterior pituitary. NPLEX Physiology DOK 2

NE-4.5 Interpret CSF opening pressure, cell count and type, protein, and glucose to distinguish bacterial, viral, fungal, and tuberculous meningitis, and recognize albuminocytologic dissociation in Guillain-Barré syndrome. NPLEX + USMLE Pathology DOK 3

NE-5 Sensory receptors, pathways, and reflexes

Describe location and explain functions of sensory receptors and associated anatomical pathways for somatic and visceral sensory perception and reflexes.

NE-5.1 Classify Meissner corpuscles, Pacinian corpuscles, Merkel discs, Ruffini endings, free nerve endings, muscle spindles, and Golgi tendon organs by stimulus, adaptation rate, and afferent fiber type. NPLEX Physiology DOK 2

NE-5.2 Trace the dorsal column medial lemniscus and spinothalamic pathways from receptor to cortex, naming where each decussates, and predict the sensory loss from a lesion at a given level. NPLEX + USMLE Anatomy DOK 3

NE-5.3 Predict the findings of Brown-Séquard hemisection (ipsilateral weakness and loss of vibration and position sense, contralateral loss of pain and temperature) and syringomyelia (bilateral cape-like loss of pain and temperature with preserved touch). NPLEX + USMLE Pathology DOK 3

NE-5.4 Describe the monosynaptic stretch reflex (spindle, Ia afferent, alpha motor neuron), the Golgi tendon organ Ib reflex, and the withdrawal and crossed extensor reflexes, and identify the roots of the biceps (C5), triceps (C7), patellar (L4), and Achilles (S1) reflexes. NPLEX Physiology DOK 2

NE-5.5 Explain referred pain as convergence of visceral and somatic afferents on the same spinal segments, using cardiac pain felt in the left arm and diaphragmatic irritation felt in the shoulder (C3 to C5) as examples. NPLEX Physiology DOK 2

NE-5.6 Explain pain modulation by gate control and by descending inhibition from the periaqueductal gray using endogenous opioids, and identify postherpetic neuralgia, phantom limb pain, and thalamic pain as neuropathic pain from nervous system injury. NPLEX + USMLE Physiology DOK 2

NE-5.7 Distinguish migraine (unilateral pulsating headache with nausea and photophobia, often with aura, involving trigeminovascular activation and CGRP release), cluster headache (brief severe periorbital pain with ipsilateral lacrimation and rhinorrhea), and tension headache (bilateral band-like pressure). USMLE extension Physiology DOK 3

NE-6 Special senses

Describe location, structure, pathways, and explain functions of special senses (visual, auditory, gustatory, olfactory, vestibular) and associated glands.

NE-6.1 Trace the visual pathway from retina through optic nerve, chiasm, optic tract, lateral geniculate nucleus, and optic radiations (Meyer loop) to the calcarine cortex, and predict the visual field defect from a lesion at each site. NPLEX + USMLE Anatomy DOK 3

NE-6.2 Describe the pupillary light reflex (CN II afferent, Edinger-Westphal nucleus and CN III efferent) and accommodation, and interpret anisocoria, a relative afferent pupillary defect, and an Argyll Robertson pupil. NPLEX + USMLE Physiology DOK 3

NE-6.3 Explain phototransduction in rods and cones (light isomerizing retinal in rhodopsin, transducin activation, cGMP breakdown by phosphodiesterase, closure of cGMP-gated sodium channels, hyperpolarization) and relate vitamin A deficiency to night blindness. NPLEX + USMLE Biochemistry & Genetics DOK 2

NE-6.4 Describe aqueous humor production by the ciliary epithelium and drainage through the trabecular meshwork and canal of Schlemm, and describe tear film production by the lacrimal gland (CN VII parasympathetic fibers) and meibomian glands. NPLEX + USMLE Anatomy DOK 2

NE-6.5 Trace sound transmission through the tympanic membrane, ossicles, oval window, and cochlea, explain hair cell transduction in potassium-rich endolymph and tonotopy (base high frequency, apex low), and distinguish conductive from sensorineural loss with Weber and Rinne tests. NPLEX + USMLE Physiology DOK 3

NE-6.6 Describe the semicircular canals (angular acceleration), utricle and saccule (linear acceleration and gravity), and vestibulo-ocular reflex, and relate vestibular imbalance to vertigo and nystagmus and sensory conflict to motion sickness. NPLEX + USMLE Physiology DOK 2

NE-6.7 Trace taste (CN VII anterior two thirds, CN IX posterior third, to the nucleus solitarius and VPM) and smell (olfactory neurons through the cribriform plate to the olfactory bulb, no thalamic relay), and predict anosmia after cribriform fracture. NPLEX Anatomy DOK 2

NE-7 Somatic and visceral motor and sensory components

Describe location, pathways, functions of somatic motor and sensory components and visceral motor and sensory components.

NE-7.1 Trace the lateral corticospinal tract from the precentral gyrus through the posterior limb of the internal capsule, cerebral peduncle, and pyramidal decussation to the anterior horn alpha motor neuron. NPLEX Anatomy DOK 2

NE-7.2 Distinguish upper from lower motor neuron lesions by tone, reflexes, Babinski sign, fasciculations, and atrophy, and apply this to ALS (mixed signs) and poliomyelitis and spinal muscular atrophy (lower motor neuron signs). NPLEX + USMLE Pathology DOK 3

NE-7.3 Compare the one-neuron somatic motor pathway (acetylcholine on nicotinic receptors of skeletal muscle) with the two-neuron visceral motor pathway (preganglionic and postganglionic neurons) to smooth muscle, cardiac muscle, and glands. NPLEX Anatomy DOK 2

NE-7.4 Predict the findings of subacute combined degeneration (dorsal columns and lateral corticospinal tracts), tabes dorsalis (dorsal columns and dorsal roots), anterior spinal artery infarction (dorsal columns spared), and complete cord transection. NPLEX + USMLE Pathology DOK 3

NE-7.5 Describe visceral sensory pathways, including pain afferents that travel with sympathetic nerves and reflex afferents from carotid sinus baroreceptors (CN IX) and aortic arch baroreceptors (CN X) to the nucleus solitarius. NPLEX Anatomy DOK 2

NE-8 Cranial and spinal nerves

Describe location and explain function of cranial and spinal nerves.

NE-8.1 Identify each cranial nerve's skull foramen, components, and functions, and predict the deficit from injury, such as a down-and-out eye with ptosis and mydriasis (CN III), failed abduction (CN VI), or tongue deviation toward the lesion (CN XII). NPLEX + USMLE Anatomy DOK 3

NE-8.2 Distinguish a central (upper motor neuron) facial lesion that weakens only the contralateral lower face from a CN VII lesion that weakens the entire ipsilateral face, including the forehead. NPLEX + USMLE | Anatomy | DOK 3

NE-8.3 Describe the brachial plexus (roots, trunks, divisions, cords, branches) and predict deficits from Erb palsy (C5 to C6), Klumpke palsy (C8 to T1), and injuries to the axillary, radial, median, ulnar, and long thoracic nerves. NPLEX + USMLE | Anatomy | DOK 3

NE-8.4 Predict deficits from injuries to lower limb nerves: common fibular at the fibular neck (foot drop), tibial, femoral, obturator, and superior gluteal (Trendelenburg sign). NPLEX + USMLE | Anatomy | DOK 3

NE-8.5 Use dermatome landmarks (C6 thumb, T4 nipple, T10 umbilicus, L4 knee and medial leg, L5 dorsum of foot, S1 lateral foot) to localize a spinal nerve root lesion. NPLEX | Anatomy | DOK 2

NE-8.6 Explain internuclear ophthalmoplegia from a medial longitudinal fasciculus lesion (impaired adduction on lateral gaze with nystagmus of the abducting eye, convergence preserved) and its association with multiple sclerosis. NPLEX + USMLE | Anatomy | DOK 3

NE-9 Autonomic nervous system

Describe location and pathways, and explain functions of autonomic nervous system.

NE-9.1 Compare sympathetic (thoracolumbar, T1 to L2) and parasympathetic (craniosacral, CN III, VII, IX, X and S2 to S4) outflow, including where preganglionic and postganglionic cell bodies and ganglia lie. NPLEX | Anatomy | DOK 2

NE-9.2 Identify the transmitter and receptor at each autonomic synapse (acetylcholine at nicotinic ganglia, norepinephrine at adrenergic targets, acetylcholine at muscarinic targets and sweat glands) and the G protein linked to alpha1, alpha2, beta, M1, M2, and M3 receptors. NPLEX | Physiology | DOK 2

NE-9.3 Trace the oculosympathetic pathway (hypothalamus, ciliospinal center at C8 to T2, superior cervical ganglion, carotid plexus) and predict ptosis, miosis, and anhidrosis in Horner syndrome from a Pancoast tumor or carotid dissection. NPLEX + USMLE | Anatomy | DOK 3

NE-9.4 Identify the parasympathetic ganglia of the head and their targets: ciliary (CN III, pupillary sphincter and ciliary muscle), pterygopalatine (CN VII, lacrimal gland), submandibular (CN VII, submandibular and sublingual glands), and otic (CN IX, parotid). NPLEX | Anatomy | DOK 1

NE-9.5 Explain autonomic dysreflexia after spinal cord injury at or above T6, linking a stimulus below the lesion (such as a full bladder) to unopposed sympathetic discharge with severe hypertension and reflex bradycardia. NPLEX + USMLE | Physiology | DOK 3

NE-10 CNS blood supply and CSF flow

Describe pathways of blood supply and origin and flow of cerebrospinal fluid for central nervous system.

NE-10.1 Trace the circle of Willis from the internal carotid and vertebrobasilar systems and identify the territories supplied by the anterior, middle, and posterior cerebral arteries and the lenticulostriate arteries. NPLEX + USMLE | Anatomy | DOK 2

NE-10.2 Predict the stroke syndrome from occlusion of the ACA (contralateral leg), MCA (contralateral face and arm, aphasia if dominant, neglect if nondominant), PCA (homonymous hemianopia with macular sparing), and lenticulostriate arteries (pure motor lacunar stroke). NPLEX + USMLE | Pathology | DOK 3

NE-10.3 Predict brainstem syndromes from occlusion of the PICA (lateral medullary syndrome with dysphagia, hoarseness, and crossed pain loss), anterior spinal artery (medial medullary syndrome), AICA (lateral pontine syndrome), and basilar artery (locked-in syndrome). NPLEX + USMLE | Pathology | DOK 3

NE-10.4 Trace CSF from the choroid plexus through the interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, and foramina of Luschka and Magendie to the subarachnoid space, and predict which ventricles dilate with obstruction at each site. NPLEX + USMLE | Anatomy | DOK 3

NE-10.5 Describe venous drainage through the dural venous sinuses to the internal jugular veins, and relate tearing of bridging veins to subdural hematoma and dural sinus thrombosis to raised intracranial pressure. NPLEX + USMLE | Anatomy | DOK 2

NE-10.6 Explain autoregulation of cerebral blood flow and its sensitivity to $PaCO_2$ (hypercapnia dilates, hypocapnia constricts), and use cerebral perfusion pressure (MAP minus ICP) to explain the Cushing reflex of hypertension, bradycardia, and irregular breathing. NPLEX + USMLE | Physiology | DOK 3

NE-11 Association cortex

Describe pathways and explain functions and patterns of activity for association cortex.

NE-11.1 Describe prefrontal association cortex functions (executive function, planning, judgment, personality) and predict disinhibition, apathy, and poor judgment after frontal lobe damage, as in frontotemporal dementia. NPLEX + USMLE | Physiology | DOK 3

NE-11.2 Explain the role of parietal association cortex in spatial attention, and predict hemispatial neglect after a nondominant (usually right) parietal lesion and Gerstmann syndrome (agraphia, acalculia, finger agnosia, left-right confusion) after a dominant angular gyrus lesion. NPLEX | Physiology | DOK 3

NE-11.3 Distinguish Broca aphasia (nonfluent, comprehension intact), Wernicke aphasia (fluent, comprehension impaired), conduction aphasia (repetition impaired, arcuate fasciculus), and global aphasia by lesion site. NPLEX | Anatomy | DOK 3

NE-11.4 Compare the dorsal (occipitoparietal, spatial location and motion) and ventral (occipitotemporal, object and face recognition) visual association streams, and predict prosopagnosia after fusiform gyrus damage. NPLEX | Physiology | DOK 2

NE-11.5 Relate EEG patterns to arousal and sleep stages (beta awake, alpha relaxed with eyes closed, theta in N1, sleep spindles and K complexes in N2, delta in N3, low-voltage beta-like activity in REM). NPLEX + USMLE | Physiology | DOK 2

NE-12 Hypothalamic and limbic function

Explain mechanisms, factors affecting, function, and control of hypothalamic and limbic pathways.

NE-12.1 Identify hypothalamic nuclei and functions: supraoptic and paraventricular (ADH, oxytocin), arcuate (dopamine, GHRH), lateral (hunger), ventromedial (satiety), anterior (heat loss), posterior (heat conservation), and suprachiasmatic (circadian rhythm). NPLEX | Anatomy | DOK 2

NE-12.2 Explain hypothalamic thermoregulation, including how pyrogenic cytokines (IL-1, IL-6, TNF) raise the set point through prostaglandin E2 in fever, and predict hyperphagia after ventromedial lesions and anorexia after lateral lesions. NPLEX | Physiology | DOK 3

NE-12.3 Describe the limbic system (hippocampus, amygdala, cingulate gyrus, fornix, mammillary bodies) and relate hippocampal damage to anterograde amnesia, mammillary body damage to Wernicke-Korsakoff syndrome, and bilateral amygdala damage to Klüver-Bucy syndrome. NPLEX + USMLE | Anatomy | DOK 3

NE-12.4 Explain how the suprachiasmatic nucleus, driven by retinal light input, sets circadian rhythm through pineal melatonin, and relate loss of lateral hypothalamic orexin (hypocretin) neurons to narcolepsy with cataplexy. NPLEX + USMLE | Physiology | DOK 2

NE-12.5 Distinguish REM sleep behavior disorder (loss of REM atonia, often preceding Parkinson disease or Lewy body dementia), N3 parasomnias of childhood (sleepwalking, sleep terrors), and restless legs syndrome (urge to move at rest, linked to iron deficiency). USMLE extension | Physiology | DOK 3

NE-13 Synaptic transmission and action potentials

Explain mechanisms, factors affecting, function, and control of synaptic transmission, graded potentials, action potential, and axon conduction.

NE-13.1 Explain the resting membrane potential using the Na+/K+ ATPase, potassium leak channels, and the Nernst equation, and predict how changes in extracellular potassium alter neuronal and muscle excitability. NPLEX | Physiology | DOK 3

NE-13.2 Describe the phases of the action potential (voltage-gated sodium channel activation and inactivation, delayed potassium efflux, afterhyperpolarization), threshold, and the absolute and relative refractory periods. NPLEX | Physiology | DOK 2

NE-13.3 Compare graded potentials (EPSPs and IPSPs) with action potentials and explain temporal and spatial summation at the axon hillock. NPLEX | Physiology | DOK 2

NE-13.4 Explain how myelin speeds conduction by saltatory conduction between nodes of Ranvier, how axon diameter affects velocity, and why demyelination in multiple sclerosis and Guillain-Barré syndrome slows or blocks conduction. NPLEX + USMLE | Physiology | DOK 3

NE-13.5 Describe neuromuscular transmission (calcium entry, SNARE-mediated vesicle release, acetylcholine on nicotinic receptors, acetylcholinesterase) and predict how botulinum toxin, myasthenia gravis, and Lambert-Eaton syndrome each disrupt it. NPLEX + USMLE | Physiology | DOK 3

NE-13.6 Classify seizures as focal (with or without impaired awareness) or generalized (tonic-clonic, absence with 3 Hz spike-and-wave), explain them as excess synchronous neuronal firing from an excitation and inhibition imbalance, and describe febrile seizures in young children. USMLE extension | Physiology | DOK 3

NE-13.7 Explain hypokalemic periodic paralysis as an inherited skeletal muscle ion channelopathy causing episodes of flaccid weakness with low serum potassium, often triggered by carbohydrate-rich meals or rest after exercise. USMLE extension | Physiology | DOK 2

NE-14 Neurological biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to neurological function and pathology.

NE-14.1 Explain why the brain depends on glucose (entering through GLUT1 at the blood-brain barrier and GLUT3 on neurons) and shifts to ketone bodies in prolonged fasting, and predict neuroglycopenic symptoms of hypoglycemia. NPLEX | Biochemistry & Genetics | DOK 2

NE-14.2 Explain thiamine pyrophosphate as a cofactor for pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, and transketolase, and relate deficiency to Wernicke encephalopathy, dry beriberi, and worsening when glucose is given before thiamine. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-14.3 Explain vitamin B12 as a cofactor for methionine synthase and methylmalonyl-CoA mutase, and use elevated methylmalonic acid to distinguish B12 deficiency, which damages myelin, from folate deficiency, which spares the nervous system. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-14.4 Relate neurologic findings to deficiencies of vitamin B6 (seizures and neuropathy from impaired GABA and transmitter synthesis), niacin (pellagra with dementia), and vitamin E (spinocerebellar ataxia and dorsal column degeneration). NPLEX + USMLE | Biochemistry & Genetics | DOK 2

NE-14.5 Explain how hyperammonemia from liver failure or urea cycle defects injures the brain, linking astrocyte glutamine synthetase activity to astrocyte swelling and Alzheimer type II astrocytes. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-14.6 Describe the lipid composition of myelin (cholesterol, galactocerebroside, sulfatide, sphingomyelin) and relate defects in lysosomal sphingolipid breakdown to neurodegeneration in the sphingolipidoses and leukodystrophies. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

NE-14.7 Explain how lead inhibits ALA dehydratase and ferrochelatase, and relate lead poisoning to encephalopathy in children and to wrist and foot drop from peripheral motor neuropathy in adults. NPLEX | Biochemistry & Genetics | DOK 3

NE-15 Neurotransmitter synthesis and breakdown

Explain biochemistry of neurotransmitter synthesis, function, and degradation.

NE-15.1 Trace catecholamine synthesis from tyrosine through L-DOPA (tyrosine hydroxylase, rate limiting), dopamine, norepinephrine (dopamine beta-hydroxylase, vitamin C dependent), and epinephrine (PNMT, induced by cortisol). NPLEX | Biochemistry & Genetics | DOK 2

NE-15.2 Describe catecholamine breakdown by MAO and COMT to homovanillic acid (from dopamine) and vanillylmandelic acid and metanephrines (from norepinephrine and epinephrine), and relate elevated urinary HVA and VMA to neuroblastoma. NPLEX | Biochemistry & Genetics | DOK 2

NE-15.3 Trace serotonin synthesis from tryptophan by tryptophan hydroxylase in the raphe nuclei, its conversion to melatonin in the pineal gland, and its breakdown by MAO to 5-hydroxyindoleacetic acid. NPLEX | Biochemistry & Genetics | DOK 2

NE-15.4 Explain GABA synthesis from glutamate by pyridoxal phosphate-dependent glutamate decarboxylase, contrast glutamate (excitatory, NMDA and AMPA receptors) with GABA and glycine (inhibitory), and relate glutamate excitotoxicity to ischemic neuronal death. NPLEX | Biochemistry & Genetics | DOK 3

NE-15.5 Describe acetylcholine synthesis by choline acetyltransferase from acetyl-CoA and choline and its breakdown by acetylcholinesterase, and relate loss of cholinergic neurons in the nucleus basalis of Meynert to Alzheimer disease. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

NE-15.6 Identify the main source nuclei of dopamine (substantia nigra, ventral tegmental area), norepinephrine (locus coeruleus), and serotonin (raphe), and relate decreased dopamine to Parkinson disease and decreased GABA and acetylcholine to Huntington disease. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

NE-16 Neurogenetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying neurological disease.

NE-16.1 Explain trinucleotide repeat expansion in Huntington disease (CAG, HTT), fragile X syndrome (CGG, FMR1 hypermethylation), myotonic dystrophy (CTG, DMPK), and Friedreich ataxia (GAA, frataxin), including each inheritance pattern and anticipation in the first three. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-16.2 Explain the chromosomal mechanisms of Down syndrome (maternal meiotic nondisjunction, Robertsonian translocation, mosaicism) and relate an extra copy of the APP gene on chromosome 21 to early-onset Alzheimer disease. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-16.3 Explain how loss of tumor suppressor genes causes autosomal dominant neurocutaneous syndromes: NF1 (neurofibromin, chromosome 17), NF2 (merlin, chromosome 22), tuberous sclerosis (TSC1 hamartin, TSC2 tuberin, mTOR activation), and von Hippel-Lindau (VHL, chromosome 3). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-16.4 Describe Sturge-Weber syndrome as a sporadic somatic GNAQ mutation causing a facial port-wine stain in the CN V1 distribution, ipsilateral leptomeningeal angiomas, seizures, and glaucoma. USMLE extension | Biochemistry & Genetics | DOK 2

NE-16.5 Explain spinal muscular atrophy as autosomal recessive loss of SMN1 causing degeneration of anterior horn motor neurons, with a floppy infant, tongue fasciculations, and areflexia in the severe infantile form. USMLE extension | Biochemistry & Genetics | DOK 3

NE-16.6 Relate driver gene defects to nervous system tumors: two-hit RB1 tumor suppressor loss in retinoblastoma, IDH mutation with 1p/19q codeletion in oligodendroglioma, and MYCN oncogene amplification predicting aggressive neuroblastoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

NE-17 Nervous and endocrine system interactions

Explain relationship of neurological system to endocrine system.

NE-17.1 Describe the hypophyseal portal system and the hypothalamic hormones (TRH, CRH, GnRH, GHRH, somatostatin, dopamine) that control the anterior pituitary, and explain why dopamine is the main inhibitor of prolactin. NPLEX | Physiology | DOK 2

NE-17.2 Explain ADH and oxytocin synthesis in the supraoptic and paraventricular nuclei with axonal transport to the posterior pituitary, and predict central diabetes insipidus or SIADH after head trauma, neurosurgery, or a CNS tumor. NPLEX + USMLE | Physiology | DOK 3

NE-17.3 Predict the effects of a pituitary adenoma or craniopharyngioma compressing the optic chiasm (bitemporal hemianopia) and pituitary stalk (hyperprolactinemia from loss of dopamine inhibition). NPLEX | Pathology | DOK 3

NE-17.4 Explain the neuroendocrine stress response linking the hypothalamus to the sympathetic nervous system and adrenal medulla (catecholamines) and to the HPA axis (CRH, ACTH, cortisol). NPLEX | Physiology | DOK 2

NE-17.5 Relate thyroid hormone to nervous system function, including impaired brain development in congenital hypothyroidism, delayed reflex relaxation in hypothyroidism, and tremor and hyperreflexia in hyperthyroidism. NPLEX | Physiology | DOK 2

NE-18 Immune responses in neurological disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

NE-18.1 Describe how the blood-brain barrier and resident microglia shape CNS innate immunity, and explain how cytokine release (IL-1, TNF) in bacterial meningitis increases barrier permeability and causes cerebral edema. NPLEX + USMLE | Microbiology & Immunology | DOK 2

NE-18.2 Explain multiple sclerosis as T cell-mediated (Th1 and Th17) autoimmune injury to CNS myelin, with HLA-DR2 association, CSF oligoclonal IgG bands, and periventricular white matter plaques. NPLEX + USMLE | Microbiology & Immunology | DOK 3

NE-18.3 Explain molecular mimicry in Guillain-Barré syndrome after Campylobacter jejuni infection, with antibodies against peripheral nerve gangliosides, and relate anti-GQ1b antibodies to the Miller Fisher variant (ophthalmoplegia, ataxia, areflexia). NPLEX + USMLE | Microbiology & Immunology | DOK 3

NE-18.4 Classify myasthenia gravis (antibodies to postsynaptic acetylcholine receptors, associated with thymoma and thymic hyperplasia) and Lambert-Eaton syndrome (antibodies to presynaptic calcium channels) as type II hypersensitivity, and MS and ADEM as T cell-mediated. NPLEX + USMLE | Microbiology & Immunology | DOK 3

NE-18.5 Explain neurologic paraneoplastic syndromes as immune responses to tumor antigens shared with neurons: anti-Hu (small cell lung cancer, sensory neuronopathy, limbic encephalitis), anti-Yo (cerebellar degeneration), and anti-NMDA receptor encephalitis (ovarian teratoma). NPLEX + USMLE Microbiology & Immunology DOK 3

NE-18.6 Explain how conjugate polysaccharide vaccines against H. influenzae type b, S. pneumoniae, and N. meningitidis produce T cell-dependent responses, and compare inactivated (Salk) with live oral (Sabin) polio vaccines for mucosal IgA immunity. NPLEX Microbiology & Immunology DOK 2

NE-19 Neurological pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

NE-19.1 Compare the morphology and capsule-based virulence of Streptococcus pneumoniae (gram-positive lancet-shaped diplococci), Neisseria meningitidis (gram-negative diplococci with lipooligosaccharide endotoxin), and Haemophilus influenzae type b (gram-negative coccobacillus with polyribosylribitol phosphate capsule). NPLEX + USMLE Microbiology & Immunology DOK 2

NE-19.2 Describe the neonatal meningitis pathogens Streptococcus agalactiae (group B, maternal vaginal colonization), Escherichia coli with the K1 capsule, and Listeria monocytogenes (gram-positive rod, tumbling motility, actin-based cell spread, unpasteurized dairy and deli meats). NPLEX + USMLE Microbiology & Immunology DOK 2

NE-19.3 Contrast botulinum toxin (cleaves SNARE proteins, blocking acetylcholine release and causing flaccid paralysis) with tetanospasmin (retrograde transport to spinal inhibitory interneurons, blocking GABA and glycine release and causing spastic paralysis). NPLEX + USMLE Microbiology & Immunology DOK 3

NE-19.4 Describe DNA viruses of the nervous system: herpesviruses (enveloped dsDNA, latency in sensory ganglia, spread along axons) and JC polyomavirus (nonenveloped dsDNA, latent in the kidney, reactivated in immunosuppression). NPLEX + USMLE Microbiology & Immunology DOK 2

NE-19.5 Describe RNA viruses of the nervous system: enteroviruses (nonenveloped positive-sense picornaviruses, fecal-oral), West Nile virus (flavivirus spread by Culex mosquitoes from bird reservoirs), and rabies virus (bullet-shaped negative-sense rhabdovirus, retrograde axonal transport). NPLEX + USMLE Microbiology & Immunology DOK 2

NE-19.6 Identify Cryptococcus neoformans (encapsulated yeast in pigeon droppings, seen with India ink, detected by capsular antigen), Mucor and Rhizopus (broad nonseptate hyphae favored by ketoacidosis), and Coccidioides (spherules, southwestern US soil). NPLEX + USMLE Microbiology & Immunology DOK 2

NE-19.7 Compare Treponema pallidum (spirochete detected with darkfield, VDRL, and FTA-ABS), Borrelia burgdorferi (spirochete spread by Ixodes ticks), and Mycobacterium leprae (acid-fast, prefers cooler body sites, infects Schwann cells, armadillo reservoir in the US). NPLEX + USMLE Microbiology & Immunology DOK 2

DISEASES AND CONDITIONS

NE-20-A Neurological vascular disease

1 Cerebral infarction (thrombosis, embolism, common obstructions)

2 Cerebral vascular accident (CVA) (ischemic/hemorrhagic) **3** Ischemic-hypoxic encephalopathy

4 Intracranial hemorrhage (cerebral, subarachnoid, epidural/subdural) **5** Vascular lesions of spinal cord

NE-20-A.1 Distinguish thrombotic cerebral infarction (atherosclerosis at the carotid bifurcation, MCA, or basilar artery) from embolic infarction (cardiac sources such as atrial fibrillation, often with hemorrhagic conversion) and lacunar infarcts from hypertensive arteriolosclerosis. NPLEX + USMLE Pathology DOK 3

NE-20-A.2 Describe the time course of cerebral infarction: red neurons by 12 to 24 hours, neutrophils in the first days, macrophages clearing debris over weeks, and reactive gliosis around a fluid-filled cyst from liquefactive necrosis. NPLEX + USMLE Pathology DOK 2

NE-20-A.3 Distinguish ischemic stroke, hemorrhagic stroke, and transient ischemic attack by mechanism, list shared risk factors (hypertension, atrial fibrillation, diabetes, smoking, dyslipidemia), and explain why noncontrast CT is used first to exclude hemorrhage. NPLEX + USMLE Pathology DOK 3

NE-20-A.4 Explain global hypoxic-ischemic encephalopathy, identifying the most vulnerable neurons (hippocampal CA1 pyramidal cells, cerebellar Purkinje cells, neocortical pyramidal cells with laminar necrosis) and watershed infarcts between arterial territories after severe hypotension. NPLEX + USMLE Pathology DOK 3

NE-20-A.5 Compare epidural hematoma (middle meningeal artery tear with temporal bone fracture, lucid interval, biconvex and limited by sutures) with subdural hematoma (bridging vein tear in elderly, alcohol use, or shaken infants, crescent-shaped and crossing sutures). NPLEX + USMLE | Pathology | DOK 3

NE-20-A.6 Explain subarachnoid hemorrhage from a ruptured saccular aneurysm, most often near the anterior communicating artery, with sudden worst headache and xanthochromic CSF, risk factors ADPKD and Ehlers-Danlos syndrome, and complications vasospasm and hydrocephalus. NPLEX + USMLE | Pathology | DOK 3

NE-20-A.7 Explain hypertensive intraparenchymal hemorrhage from Charcot-Bouchard microaneurysms in the basal ganglia, thalamus, pons, and cerebellum, and contrast it with lobar hemorrhage from cerebral amyloid angiopathy in older adults. NPLEX + USMLE | Pathology | DOK 3

NE-20-A.8 Describe the spinal cord blood supply (one anterior and two posterior spinal arteries, artery of Adamkiewicz) and predict anterior spinal artery syndrome after aortic surgery or dissection: paralysis and loss of pain and temperature with vibration and position sense spared. NPLEX + USMLE | Pathology | DOK 3

NE-20-A.9 Describe arteriovenous malformations (tangles of arteries draining directly into veins without a capillary bed) and other vascular malformations as causes of intracerebral hemorrhage and seizures in young adults. USMLE extension | Pathology | DOK 2

NE-20-A.10 Explain carotid and vertebral artery dissection as a cause of stroke in young adults after neck trauma, and subclavian steal syndrome (proximal subclavian stenosis reversing vertebral flow, with arm claudication and unequal arm blood pressures). USMLE extension | Pathology | DOK 3

NE-20-A.11 Explain cerebral venous sinus thrombosis in hypercoagulable states such as pregnancy and estrogen use, and hypertensive encephalopathy and posterior reversible encephalopathy syndrome as vasogenic edema, mainly of posterior white matter, during severe hypertension. USMLE extension | Pathology | DOK 3

NE-20-B Degenerative and demyelination diseases

1 Alzheimer disease **2** Amyotrophic lateral sclerosis (ALS)

3 Extrapyramidal diseases (spinocerebellar degeneration, Huntington chorea, Parkinsonism)

4 Guillain-Barré syndrome **5** Multiple sclerosis (MS) **6** Acute disseminated encephalomyelitis

NE-20-B.1 Explain Alzheimer disease pathogenesis: amyloid precursor protein cleavage by beta and gamma secretase into amyloid beta plaques, hyperphosphorylated tau tangles, and risk from age, APOE e4, Down syndrome, and APP, PSEN1, or PSEN2 mutations. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.2 Identify Alzheimer disease findings: cortical atrophy with hydrocephalus ex vacuo, hippocampal involvement, neuritic plaques, neurofibrillary tangles, amyloid angiopathy, and a course beginning with short-term memory loss. NPLEX + USMLE | Pathology | DOK 2

NE-20-B.3 Explain ALS as degeneration of upper and lower motor neurons (some familial cases with SOD1 or C9orf72 mutations) causing mixed UMN and LMN signs with sensation, bladder control, and eye movements spared, and death from respiratory failure. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.4 Explain Parkinson disease as loss of substantia nigra dopaminergic neurons with alpha-synuclein Lewy bodies, causing resting tremor, rigidity, bradykinesia, and postural instability, and identify secondary parkinsonism from MPTP and dopamine-blocking drugs. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.5 Explain Huntington disease as an autosomal dominant CAG expansion in HTT on chromosome 4 with caudate atrophy and GABAergic neuron loss, causing chorea, dementia, and depression in midlife, with anticipation through paternal transmission. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.6 Explain Friedreich ataxia (autosomal recessive GAA expansion in frataxin, mitochondrial iron accumulation) as spinocerebellar degeneration of dorsal columns, spinocerebellar and corticospinal tracts, with hypertrophic cardiomyopathy, diabetes, kyphoscoliosis, and pes cavus. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.7 Explain Guillain-Barré syndrome as acute inflammatory demyelinating polyradiculoneuropathy after infection, causing ascending symmetric flaccid weakness with areflexia, and complications of respiratory failure and autonomic instability. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.8 Describe multiple sclerosis risk factors (women aged 20 to 40, higher latitude), relapsing-remitting course, and findings including optic neuritis, internuclear ophthalmoplegia, sensory loss, heat-worsened symptoms, and periventricular MRI plaques. NPLEX + USMLE | Pathology | DOK 3

NE-20-B.9 Distinguish acute disseminated encephalomyelitis (monophasic, mostly in children, after viral infection or rarely vaccination, perivenular demyelination) from relapsing multiple sclerosis. NPLEX | Pathology | DOK 3

NE-20-B.10 Distinguish dementia with Lewy bodies (cortical Lewy bodies, visual hallucinations, fluctuating cognition, parkinsonism), frontotemporal dementia (early personality change, tau Pick bodies or TDP-43 inclusions), and progressive supranuclear palsy (tau, vertical gaze palsy, early falls). USMLE extension Pathology DOK 3

NE-20-B.11 Distinguish vascular dementia (stepwise decline after multiple infarcts or diffuse small vessel disease) and mild cognitive impairment (measurable decline with daily function preserved) from Alzheimer disease. USMLE extension Pathology DOK 3

NE-20-B.12 Describe transverse myelitis as acute inflammation across a spinal cord segment, often after infection or with MS, causing bilateral weakness, a sensory level, and bladder dysfunction. USMLE extension Pathology DOK 2

NE-20-B.13 Distinguish essential tremor (often familial, postural and action tremor that may improve with alcohol) from parkinsonian resting tremor and cerebellar intention tremor, and describe cervical dystonia (torticollis) as sustained involuntary neck muscle contraction. USMLE extension Pathology DOK 3

NE-20-C Diseases of increased intracranial fluid

1 Cerebral edema **2** Hydrocephalus

NE-20-C.1 Compare vasogenic edema (blood-brain barrier disruption from tumor, abscess, or trauma), cytotoxic edema (cell swelling from ischemia or hypoxia), and interstitial edema (hydrocephalus), and predict papilledema, Cushing triad, and herniation as complications. NPLEX + USMLE Pathology DOK 3

NE-20-C.2 Distinguish noncommunicating hydrocephalus (aqueductal stenosis, Chiari II, Dandy-Walker, tumor) from communicating hydrocephalus (arachnoid granulation scarring after meningitis or subarachnoid hemorrhage) and from hydrocephalus ex vacuo, and explain head enlargement in infants. NPLEX + USMLE Pathology DOK 3

NE-20-C.3 Describe normal pressure hydrocephalus in older adults as enlarged ventricles without consistently raised opening pressure, causing urinary incontinence, magnetic gait, and dementia that may improve after CSF drainage. NPLEX + USMLE Pathology DOK 3

NE-20-C.4 Describe idiopathic intracranial hypertension in obese women of childbearing age (headache, papilledema, vision loss, CN VI palsy, normal imaging, raised opening pressure) and its associations with vitamin A excess and tetracyclines. USMLE extension Pathology DOK 3

NE-20-D Metabolic and nutritional neurological diseases

1 Hepatic encephalopathy **2** Peripheral neuropathy **3** Vitamin B12 deficiency **4** Wernicke-Korsakoff syndrome

NE-20-D.1 Explain hepatic encephalopathy from ammonia and other gut-derived toxins bypassing the liver in cirrhosis or portosystemic shunts, its precipitants (GI bleeding, infection, high protein intake, constipation, hypokalemia), and findings of confusion and asterixis. NPLEX + USMLE Pathology DOK 3

NE-20-D.2 Classify peripheral neuropathy by pattern (distal symmetric, mononeuropathy, mononeuritis multiplex) and by axonal or demyelinating injury, and list major causes: diabetes, alcohol, B12 and thiamine deficiency, vasculitis, leprosy, HIV, and neurotoxic chemotherapy. NPLEX + USMLE Pathology DOK 3

NE-20-D.3 Explain diabetic neuropathy (sorbitol accumulation, advanced glycation end products, microvascular injury) and its findings: stocking-glove sensory loss, autonomic dysfunction (gastroparesis, orthostatic hypotension), neuropathic foot ulcers, and pupil-sparing CN III palsy. NPLEX + USMLE Pathology DOK 3

NE-20-D.4 Identify causes of B12 deficiency (pernicious anemia, strict vegan diet, gastrectomy, ileal disease, Diphyllobothrium latum) and its neurologic findings: subacute combined degeneration with lost vibration and position sense, positive Romberg sign, spasticity, neuropathy, and dementia. NPLEX + USMLE Pathology DOK 3

NE-20-D.5 Explain Wernicke encephalopathy (confusion, ophthalmoplegia, ataxia) and its progression to Korsakoff syndrome (amnesia with confabulation), with mammillary body and medial thalamic damage, in alcohol use disorder, malnutrition, or hyperemesis. NPLEX + USMLE Pathology DOK 3

NE-20-D.6 Describe Charcot-Marie-Tooth disease as a group of inherited motor and sensory neuropathies (most often autosomal dominant PMP22 duplication) causing distal weakness, foot drop, pes cavus, and hammer toes. USMLE extension Pathology DOK 2

NE-20-D.7 Explain osmotic demyelination syndrome after overly rapid correction of chronic hyponatremia, with demyelination of the central pons causing quadriparesis, dysarthria, dysphagia, and possible locked-in syndrome. USMLE extension Pathology DOK 3

NE-20-E Congenital and genetic neurological diseases

1 Down syndrome **2** Leukodystrophies **3** Phenylketonuria (PKU) **4** Storage diseases **5** Wilson disease

NE-20-E.1 Describe the neurologic and associated features of Down syndrome: intellectual disability, hypotonia, early-onset Alzheimer disease, atlantoaxial instability, and associated duodenal atresia, atrioventricular septal defects, and increased leukemia risk. NPLEX + USMLE Pathology DOK 3

NE-20-E.2 Compare the leukodystrophies: X-linked adrenoleukodystrophy (ABCD1, very long chain fatty acid buildup from impaired peroxisomal oxidation, adrenal insufficiency), metachromatic leukodystrophy (arylsulfatase A, sulfatides), and Krabbe disease (galactocerebrosidase, globoid cells). NPLEX + USMLE Biochemistry & Genetics DOK 3

NE-20-E.3 Explain phenylketonuria as autosomal recessive phenylalanine hydroxylase or BH4 deficiency, making tyrosine essential, and relate phenylalanine buildup to intellectual disability, seizures, musty odor, fair skin, and microcephaly in infants of untreated mothers. NPLEX + USMLE Biochemistry & Genetics DOK 3

NE-20-E.4 Distinguish Tay-Sachs disease (hexosaminidase A, GM2 ganglioside, cherry-red macula, no hepatosplenomegaly) from Niemann-Pick disease (sphingomyelinase, cherry-red macula with hepatosplenomegaly and foam cells). NPLEX + USMLE Biochemistry & Genetics DOK 3

NE-20-E.5 Describe the neurologic features of Fabry disease (X-linked alpha-galactosidase A deficiency, painful neuropathy, angiokeratomas, early stroke), Gaucher disease (glucocerebrosidase, neuronopathic types 2 and 3), and Hurler (alpha-L-iduronidase, corneal clouding) versus Hunter syndrome (X-linked, no corneal clouding). NPLEX + USMLE Biochemistry & Genetics DOK 3

NE-20-E.6 Explain Wilson disease as autosomal recessive ATP7B mutation impairing biliary copper excretion, with low ceruloplasmin, copper deposits in the basal ganglia causing tremor, dystonia, and psychiatric change, Kayser-Fleischer rings, and liver disease. NPLEX + USMLE Pathology DOK 3

NE-20-E.7 Describe fragile X syndrome (FMR1 CGG expansion with hypermethylation and silencing) as the most common inherited cause of intellectual disability, with autism features, long face, large ears, and macroorchidism. USMLE extension Biochemistry & Genetics DOK 2

NE-20-E.8 Explain maple syrup urine disease (branched-chain alpha-ketoacid dehydrogenase deficiency, sweet urine odor, neonatal encephalopathy) and ornithine transcarbamylase deficiency (X-linked hyperammonemia with orotic aciduria) as inborn errors that injure the brain. USMLE extension Biochemistry & Genetics DOK 3

NE-20-E.9 Explain acute intermittent porphyria (autosomal dominant porphobilinogen deaminase deficiency) with attacks of abdominal pain, neuropathy, and psychiatric symptoms, elevated urine porphobilinogen, and triggers such as alcohol, fasting, and cytochrome P450 inducers. USMLE extension Biochemistry & Genetics DOK 3

NE-20-E.10 Identify causes of congenital microcephaly, including congenital Zika, cytomegalovirus, rubella, and toxoplasma infection, fetal alcohol syndrome, and maternal phenylketonuria. USMLE extension Pathology DOK 2

NE-20-F Neurological neoplasms

1 Medulloblastoma **2** Meningiomas **3** Neuroblastoma **4** Neuroglial tumors (astrocytomas, oligodendrogliomas) **5** Neuronal tumors **6** Tumors of peripheral nerves (schwannoma, neurofibromatosis)

NE-20-F.1 Describe medulloblastoma as the most common malignant brain tumor of childhood: a small round blue cell tumor of the cerebellum with Homer Wright rosettes that can block the fourth ventricle and seed the CSF as drop metastases. NPLEX + USMLE Pathology DOK 3

NE-20-F.2 Describe meningioma as a dural-based extra-axial tumor of arachnoid cap cells, common in adult women, with whorls and psammoma bodies, associations with NF2 and prior radiation, and seizures or focal deficits from compression. NPLEX + USMLE Pathology DOK 2

NE-20-F.3 Describe neuroblastoma as a neural crest tumor of young children, usually in the adrenal medulla, presenting as an abdominal mass with elevated urinary HVA and VMA, Homer Wright rosettes, opsoclonus-myoclonus, and worse prognosis with MYCN amplification. NPLEX Pathology DOK 3

NE-20-F.4 Describe glioblastoma as the most common primary malignant brain tumor in adults: an IDH-wildtype astrocytic tumor that can cross the corpus callosum, with pseudopalisading necrosis, microvascular proliferation, GFAP staining, and poor survival. NPLEX + USMLE Pathology DOK 3

NE-20-F.5 Distinguish pilocytic astrocytoma (children, cerebellum, cystic with mural nodule, Rosenthal fibers, BRAF alteration, good prognosis) from IDH-mutant diffuse astrocytoma and oligodendroglioma (frontal lobe, fried-egg cells, chicken-wire capillaries, calcification, seizures). NPLEX + USMLE | Pathology | DOK 3

NE-20-F.6 Describe neuronal and mixed neuronal-glial tumors such as ganglioglioma, a slow-growing temporal lobe tumor of children and young adults that presents with seizures, and central neurocytoma within the lateral ventricles. NPLEX | Pathology | DOK 2

NE-20-F.7 Describe schwannoma (S-100 positive Schwann cells, Antoni A areas with Verocay bodies and Antoni B areas) and predict hearing loss, tinnitus, and imbalance from a vestibular schwannoma at the cerebellopontine angle, bilateral in NF2. NPLEX + USMLE | Pathology | DOK 3

NE-20-F.8 Describe neurofibromatosis type 1: café-au-lait macules, cutaneous neurofibromas, plexiform neurofibromas with risk of malignant peripheral nerve sheath tumor, Lisch nodules, optic pathway glioma, and pheochromocytoma. NPLEX + USMLE | Pathology | DOK 2

NE-20-F.9 Describe brain metastases as the most common intracranial tumors in adults (from lung, breast, melanoma, and other primaries, often at the gray-white junction) and primary CNS lymphoma (diffuse large B cell, EBV-related in AIDS). USMLE extension | Pathology | DOK 2

NE-20-F.10 Describe retinoblastoma (leukocoria in young children, bilateral in hereditary RB1 cases, later osteosarcoma risk) and uveal melanoma as the most common primary intraocular malignancy in adults. USMLE extension | Pathology | DOK 2

NE-20-F.11 Describe the CNS lesions of tuberous sclerosis (cortical tubers, subependymal nodules and giant cell astrocytoma, seizures, with ash-leaf spots and cardiac rhabdomyoma) and von Hippel-Lindau disease (cerebellar and retinal hemangioblastomas, renal cell carcinoma). USMLE extension | Pathology | DOK 2

NE-20-G Infections of the CNS and PNS

1 Arboviruses **2** Botulism **3** Brain abscess **4** Encephalitis **5** Fungal infections **6** Herpes viruses **7** Leprosy **8** Meningitis **9** Neurosyphilis **10** Poliomyelitis **11** Prion disease **12** Progressive multifocal leukoencephalopathy (PML) **13** Subacute sclerosing panencephalitis (SSPE) **14** Tetanus **15** Varicella zoster virus (VZV)

NE-20-G.1 Describe arboviral encephalitis (West Nile, St. Louis, and eastern and western equine encephalitis viruses) transmitted by mosquitoes from bird reservoirs, with humans as dead-end hosts, and West Nile acute flaccid paralysis in older adults. NPLEX + USMLE | Microbiology & Immunology | DOK 2

NE-20-G.2 Distinguish foodborne botulism (preformed toxin), infant botulism (ingested spores, honey, floppy baby with constipation), and wound botulism, all causing descending symmetric flaccid paralysis starting with cranial nerves, without fever or sensory loss. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.3 Explain brain abscess spread from contiguous infection (sinusitis, otitis, mastoiditis, dental) or hematogenous seeding (endocarditis, lung infection, right-to-left cardiac shunts), common organisms (streptococci, staphylococci, anaerobes), and a ring-enhancing lesion with fever and focal deficits. NPLEX | Pathology | DOK 3

NE-20-G.4 Describe the histology of viral encephalitis (perivascular lymphocytic cuffs, microglial nodules, neuronophagia), and rabies acquired from bat, raccoon, or dog bites, with Negri bodies, agitation, and hydrophobia. NPLEX + USMLE | Pathology | DOK 2

NE-20-G.5 Describe parasitic CNS infection: Toxoplasma gondii (ring-enhancing lesions in AIDS), Taenia solium neurocysticercosis (ingested eggs, brain cysts, seizures), and Naegleria fowleri (freshwater exposure, cribriform plate entry, rapidly fatal meningoencephalitis). NPLEX + USMLE | Microbiology & Immunology | DOK 2

NE-20-G.6 Describe cryptococcal meningitis in AIDS (raised opening pressure, positive capsular antigen) and rhinocerebral mucormycosis in diabetic ketoacidosis (necrotic nasal or palatal eschar spreading to the orbit and brain). NPLEX + USMLE | Pathology | DOK 3

NE-20-G.7 Distinguish CNS herpesvirus diseases: HSV-1 hemorrhagic temporal lobe encephalitis, HSV-2 meningitis and neonatal encephalitis, CMV encephalitis in advanced AIDS and congenital periventricular calcifications, and EBV-associated primary CNS lymphoma. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.8 Compare tuberculoid leprosy (strong Th1 response, few bacilli, few anesthetic hypopigmented plaques, thickened nerves) with lepromatous leprosy (Th2 response, many bacilli, leonine facies, symmetric glove-and-stocking neuropathy). NPLEX | Microbiology & Immunology | DOK 3

NE-20-G.9 Identify the common causes of bacterial meningitis by age: group B streptococcus, E. coli, and Listeria in neonates; S. pneumoniae and N. meningitidis in children and adults; and S. pneumoniae, N. meningitidis, and Listeria in older adults. NPLEX + USMLE | Microbiology & Immunology | DOK 2

NE-20-G.10 Describe the findings and complications of meningitis (fever, headache, neck stiffness, photophobia, petechial rash in meningococcemia, sensorineural hearing loss, hydrocephalus, seizures), including tuberculous basilar meningitis with cranial nerve palsies. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.11 Describe neurosyphilis forms in tertiary syphilis: meningovascular disease, general paresis, and tabes dorsalis (dorsal column and root degeneration with sensory ataxia, positive Romberg sign, Argyll Robertson pupils, and Charcot joints), with CSF VDRL testing. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.12 Explain poliomyelitis as fecal-oral enterovirus infection that spreads to and destroys anterior horn motor neurons, causing asymmetric flaccid paralysis, fasciculations, and areflexia without sensory loss, and later post-polio syndrome. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.13 Explain prion disease as conversion of normal alpha-helical PrP into protease-resistant beta-pleated PrP (sporadic, inherited, or acquired, including variant CJD from bovine prions), causing rapidly progressive dementia, myoclonus, and spongiform change without inflammation. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.14 Explain PML as JC virus reactivation in severe immunosuppression (AIDS, natalizumab, rituximab), causing lytic oligodendrocyte infection with multifocal white matter demyelination and rapidly progressive deficits. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.15 Describe subacute sclerosing panencephalitis as a persistent defective measles virus infection appearing years after early childhood measles, causing progressive cognitive decline and myoclonus with high CSF anti-measles antibody levels. NPLEX | Pathology | DOK 3

NE-20-G.16 Describe tetanus after spores enter a contaminated wound or the umbilical stump in neonates, causing trismus, risus sardonicus, opisthotonus, and spastic paralysis, and explain prevention by toxoid immunization. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.17 Explain VZV latency in dorsal root and cranial nerve ganglia and reactivation as dermatomal herpes zoster, including herpes zoster ophthalmicus, Ramsay Hunt syndrome (facial palsy with ear vesicles), and postherpetic neuralgia. NPLEX + USMLE | Pathology | DOK 3

NE-20-G.18 Describe Lyme neuroborreliosis (facial nerve palsy, often bilateral, lymphocytic meningitis, radiculopathy) and leptospirosis acquired from water contaminated by animal urine as spirochetal causes of meningitis. USMLE extension | Pathology | DOK 2

NE-20-H CNS trauma

1 Traumatic brain injury (TBI) (concussion, contusion, hemorrhage, hematoma)

2 Spinal cord compression/transection

NE-20-H.1 Describe concussion as trauma-induced transient neurologic dysfunction without structural imaging changes, postconcussion syndrome, and chronic traumatic encephalopathy (dementia pugilistica) with tau deposits after repeated head injury. NPLEX + USMLE | Pathology | DOK 2

NE-20-H.2 Explain coup and contrecoup contusions at the frontal and temporal poles and diffuse axonal injury from acceleration-deceleration shearing, which causes axonal swellings and coma without a mass lesion. NPLEX + USMLE | Pathology | DOK 3

NE-20-H.3 Predict the progression of an expanding traumatic hematoma: rising intracranial pressure, uncal herniation with an ipsilateral dilated pupil and contralateral hemiparesis, secondary Duret hemorrhages in the brainstem, and death without decompression. NPLEX + USMLE | Pathology | DOK 3

NE-20-H.4 Explain the phases of spinal cord transection, from spinal shock (flaccid paralysis, areflexia, neurogenic hypotension) to spastic paralysis with hyperreflexia, and predict ventilator-dependent quadriplegia at or above C3 (phrenic roots C3 to C5) versus paraplegia from thoracic injury. NPLEX + USMLE | Pathology | DOK 3

NE-20-H.5 Distinguish spinal cord compression (back pain, sensory level, upper motor neuron signs) from cauda equina syndrome (saddle anesthesia, urinary retention, lost anal reflex, lower motor neuron signs), caused by metastasis, epidural abscess, or central disc herniation. NPLEX + USMLE | Pathology | DOK 3

NE-20-I PNS trauma and compression

1 Bell palsy **2** Carpal tunnel syndrome (CTS) **3** Disc herniation **4** Nerve root entrapment **5** Sciatica **6** Thoracic outlet syndrome (TOS) **7** Trigeminal neuralgia

NE-20-I.1 Explain Bell palsy as idiopathic lower motor neuron CN VII palsy linked to HSV reactivation, diabetes, pregnancy, and Lyme disease, with whole-face ipsilateral weakness, hyperacusis, loss of anterior tongue taste, and reduced tearing. NPLEX + USMLE Pathology DOK 3

NE-20-I.2 Explain carpal tunnel syndrome as median nerve compression under the flexor retinaculum (risks: pregnancy, hypothyroidism, rheumatoid arthritis, diabetes, dialysis amyloid), with nocturnal thumb-to-ring finger paresthesia, thenar atrophy, Tinel and Phalen signs, and spared palm sensation. NPLEX + USMLE Pathology DOK 3

NE-20-I.3 Explain disc herniation as nucleus pulposus protruding through a torn annulus fibrosus, usually posterolaterally at L4 to L5 or L5 to S1, and predict that it compresses the lower root (for example, an L4 to L5 disc affects L5). NPLEX Anatomy DOK 3

NE-20-I.4 Explain radiculopathy from nerve root entrapment by disc, osteophytes, or foraminal narrowing, and predict the dermatomal pain, weakness, and reflex loss of C6, C7, L4, L5, and S1 root lesions. NPLEX Pathology DOK 3

NE-20-I.5 Describe sciatica as radiating pain from the buttock down the posterior thigh below the knee, usually from L5 or S1 root compression by disc herniation, reproduced by a straight leg raise test. NPLEX Pathology DOK 2

NE-20-I.6 Explain thoracic outlet syndrome as compression of the lower brachial plexus trunk (C8 to T1), subclavian artery, or vein by a cervical rib or scalene muscles, with ulnar-side hand paresthesia, intrinsic hand weakness, and vascular symptoms. NPLEX Pathology DOK 3

NE-20-I.7 Explain trigeminal neuralgia as paroxysmal unilateral lancinating facial pain in the V2 or V3 distribution triggered by touch or chewing, usually from vascular compression of the nerve root, and as a sign of MS in younger patients. NPLEX + USMLE Pathology DOK 3

NE-20-I.8 Predict the findings of cubital tunnel syndrome (ulnar nerve at the elbow, fourth and fifth finger paresthesia, intrinsic weakness), tarsal tunnel syndrome (tibial nerve behind the medial malleolus, sole paresthesia), and common fibular compression at the fibular neck (foot drop). USMLE extension Anatomy DOK 3

NE-20-I.9 Describe lumbar spinal stenosis causing neurogenic claudication (leg pain worse with standing and extension, relieved by sitting or flexion) and cervical stenosis causing myelopathy with upper motor neuron signs. USMLE extension Pathology DOK 3

NE-20-I.10 Describe complex regional pain syndrome (burning pain, allodynia, swelling, and skin and temperature changes after limb injury, with type II following a nerve injury) and phantom limb pain after amputation. USMLE extension Pathology DOK 2

NE-20-J Diseases of the special senses

1 Blepharitis **2** Cataracts **3** Conjunctivitis **4** Glaucoma **5** Iritis/keratitis **6** Macular degeneration **7** Uveitis **8** Ménière disease (idiopathic endolymphatic hydrops) **9** Otitis **10** Vestibular neuritis and labyrinthitis

NE-20-J.1 Describe blepharitis (anterior from staphylococcal or seborrheic disease, posterior from meibomian gland dysfunction), and distinguish a painful hordeolum (acute gland infection, usually S. aureus) from a painless chalazion (meibomian lipogranuloma). NPLEX + USMLE Pathology DOK 2

NE-20-J.2 Explain cataract as lens protein opacification, with risks of age, diabetes (sorbitol), galactosemia (galactitol), corticosteroids, UV light, smoking, and congenital rubella, causing painless progressive blurred vision, glare, and loss of the red reflex. NPLEX + USMLE Pathology DOK 3

NE-20-J.3 Distinguish viral conjunctivitis (adenovirus, watery discharge, preauricular node), bacterial conjunctivitis (purulent; hyperacute with N. gonorrhoeae), chlamydial conjunctivitis and trachoma, and allergic conjunctivitis (type I hypersensitivity, itching). NPLEX + USMLE Microbiology & Immunology DOK 3

NE-20-J.4 Explain glaucoma as optic neuropathy with disc cupping, and contrast open-angle glaucoma (impaired trabecular outflow, painless peripheral vision loss) with acute angle-closure glaucoma (painful red eye, fixed mid-dilated pupil, halos, nausea). NPLEX + USMLE Pathology DOK 3

NE-20-J.5 Describe keratitis (HSV dendritic ulcer; Pseudomonas and Acanthamoeba in contact lens wearers; corneal ulcer risk) and iritis or anterior uveitis (ciliary flush, photophobia, small irregular pupil). NPLEX + USMLE Pathology DOK 3

NE-20-J.6 Distinguish dry macular degeneration (drusen, gradual central vision loss) from wet macular degeneration (VEGF-driven choroidal neovascularization, rapid central loss, distortion), with age and smoking as risk factors. NPLEX + USMLE Pathology DOK 3

NE-20-J.7 Relate anterior uveitis to HLA-B27 spondyloarthropathies, sarcoidosis, and juvenile idiopathic arthritis, and posterior uveitis and chorioretinitis to toxoplasmosis and CMV retinitis in AIDS. NPLEX + USMLE Pathology DOK 3

NE-20-J.8 Explain Ménière disease as endolymphatic hydrops causing episodic vertigo lasting hours, fluctuating low-frequency sensorineural hearing loss, tinnitus, and aural fullness. NPLEX + USMLE Pathology DOK 3

NE-20-J.9 Explain acute otitis media (S. pneumoniae, nontypeable H. influenzae, Moraxella catarrhalis) from eustachian tube dysfunction in children, otitis media with effusion, and complications including perforation, mastoiditis, cholesteatoma, and conductive hearing loss. NPLEX + USMLE Pathology DOK 3

NE-20-J.10 Distinguish otitis externa (Pseudomonas or S. aureus, pain with tragal pressure) from malignant otitis externa (Pseudomonas skull base osteomyelitis in older diabetics, cranial nerve palsies) and auricular perichondritis after cartilage piercing. NPLEX + USMLE Pathology DOK 3

NE-20-J.11 Distinguish vestibular neuritis (acute continuous vertigo for days with normal hearing), labyrinthitis (vertigo with hearing loss), and benign paroxysmal positional vertigo (otoconia in the posterior canal, brief positional vertigo, positive Dix-Hallpike test). NPLEX + USMLE Pathology DOK 3

NE-20-J.12 Distinguish central retinal artery occlusion (sudden painless monocular loss, pale retina, cherry-red spot), amaurosis fugax from carotid emboli, and central retinal vein occlusion (retinal hemorrhages and venous engorgement). USMLE extension Pathology DOK 3

NE-20-J.13 Identify findings of hypertensive retinopathy (arteriolar narrowing, AV nicking, flame hemorrhages, cotton wool spots), diabetic retinopathy (microaneurysms, hard exudates, neovascularization), and papilledema from raised intracranial pressure. USMLE extension Pathology DOK 2

NE-20-J.14 Describe retinal detachment (flashes, floaters, curtain-like vision loss, myopia as a risk) and retinitis pigmentosa (inherited photoreceptor degeneration with night blindness, peripheral vision loss, and bone spicule pigment). USMLE extension Pathology DOK 2

NE-20-J.15 Explain myopia, hyperopia, astigmatism, and presbyopia (age-related loss of lens elasticity), and relate childhood strabismus to amblyopia from cortical suppression of the deviating eye. USMLE extension Physiology DOK 2

NE-20-J.16 Distinguish conductive hearing loss (cerumen impaction, tympanic membrane perforation, barotrauma, eustachian tube dysfunction, otosclerosis with stapes fixation, cholesteatoma) from sensorineural loss (presbycusis, noise-induced hair cell damage), and relate both to tinnitus. USMLE extension Pathology DOK 3

NE-20-J.17 Distinguish preseptal (periorbital) cellulitis from orbital cellulitis (pain with eye movement, proptosis, ophthalmoplegia, usually from ethmoid sinusitis), and describe dacryocystitis and endophthalmitis. USMLE extension Pathology DOK 3

NE-20-J.18 Describe pterygium as UV-related fibrovascular conjunctival growth onto the cornea, and lens dislocation in Marfan syndrome (usually upward) and homocystinuria (usually downward). USMLE extension Pathology DOK 2

NE-20-J.19 Describe eye injuries: corneal abrasion and foreign body (fluorescein uptake), hyphema, subconjunctival hemorrhage, orbital floor fracture with inferior rectus entrapment limiting upward gaze, and alkali burns causing liquefactive necrosis. USMLE extension Pathology DOK 2

HOW THE BODY WORKS

RE-1 Reproductive, placental, and breast embryology

Describe embryological development of male and female reproductive organs, placenta, and breast.

RE-1.1 Explain how SRY on the Y chromosome drives the bipotential gonad to form testes with Sertoli cells, and predict the gonadal phenotype of a 46,XY embryo with an SRY deletion or a 46,XX embryo with an SRY translocation. NPLEX · Anatomy · DOK 3

RE-1.2 Relate Sertoli cell anti-Mullerian hormone, Leydig cell testosterone, and dihydrotestosterone (DHT) to Mullerian duct regression, Wolffian duct development, and masculinization of the external genitalia and prostate. NPLEX · Anatomy · DOK 2

RE-1.3 Trace the paramesonephric (Mullerian) ducts to the uterine tubes, uterus, cervix, and upper vagina, and explain how failed fusion or failed septum resorption produces uterus didelphys, bicornuate uterus, and septate uterus. NPLEX + USMLE · Anatomy · DOK 3

RE-1.4 Match the male and female derivatives of the genital tubercle (glans penis, clitoris), urogenital folds (ventral penile shaft and penile urethra, labia minora), and labioscrotal swellings (scrotum, labia majora). NPLEX · Anatomy · DOK 1

RE-1.5 Describe testicular descent guided by the gubernaculum through the inguinal canal, and explain how a patent processus vaginalis leads to an indirect inguinal hernia or a communicating hydrocele. NPLEX + USMLE · Anatomy · DOK 3

RE-1.6 Describe placental formation from the fetal chorion frondosum (cytotrophoblast and syncytiotrophoblast) and maternal decidua basalis, and identify the umbilical cord vessels (two umbilical arteries, one umbilical vein) in Wharton jelly. NPLEX · Anatomy · DOK 2

RE-1.7 Describe breast development from the ectodermal mammary ridge (milk line), and explain how incomplete regression of the ridge produces supernumerary nipples or breast tissue between the axilla and the groin. NPLEX · Anatomy · DOK 2

RE-2 Reproductive gross and microscopic anatomy

Describe gross and microscopic anatomy of male and female reproductive organs and breast.

RE-2.1 Distinguish Sertoli cells (tight junctions forming the blood-testis barrier, androgen-binding protein, inhibin B) from interstitial Leydig cells (testosterone) on testis histology. NPLEX · Anatomy · DOK 2

RE-2.2 Trace sperm from the seminiferous tubules through the rete testis, efferent ductules, epididymis, ductus deferens, ejaculatory duct, and prostatic, membranous, and spongy urethra. NPLEX · Anatomy · DOK 1

RE-2.3 Identify the epithelium of each female tract segment (ciliated simple columnar uterine tube, simple columnar endometrium and endocervix, stratified squamous ectocervix and vagina) and locate the transformation zone where cervical dysplasia arises. NPLEX + USMLE · Anatomy · DOK 2

RE-2.4 Distinguish primordial, primary, secondary (antral), and mature (Graafian) follicles and the corpus luteum by their oocyte, granulosa cell, and theca cell features. NPLEX · Anatomy · DOK 2

RE-2.5 Describe the breast terminal duct lobular unit, with luminal epithelial cells over a myoepithelial cell layer, and relate it to the origin of most breast carcinomas and fibroadenomas. NPLEX + USMLE · Anatomy · DOK 2

RE-2.6 Describe the peripheral, transition, and central zones of the prostate, and relate benign prostatic hyperplasia to the transition zone and most prostate carcinomas to the peripheral zone. NPLEX + USMLE · Anatomy · DOK 3

RE-3 Gametogenesis, implantation, and embryogenesis

Explain developmental processes related to gametogenesis, implantation, and embryogenesis.

RE-3.1 Compare spermatogenesis and oogenesis, including oocyte arrest in prophase I until ovulation and in metaphase II until fertilization, and the number of functional gametes produced from each meiosis. NPLEX · Physiology · DOK 2

RE-3.2 Describe spermiogenesis (acrosome from the Golgi apparatus, flagellum from a centriole, shedding of cytoplasm), and explain capacitation in the female tract and the acrosome reaction required for fertilization. NPLEX · Physiology · DOK 2

RE-3.3 Explain fertilization in the ampulla of the uterine tube, including the cortical reaction that blocks polyspermy and completion of the second meiotic division. NPLEX · Physiology · DOK 2

RE-3.4 Trace the conceptus from zygote to blastocyst, and explain implantation into the endometrium about day 6 with syncytiotrophoblast invasion and hCG secretion that maintains the corpus luteum. NPLEX · Anatomy · DOK 2

RE-3.5 Predict the consequences of abnormal implantation, including tubal ectopic pregnancy (most often ampullary, with prior salpingitis as a risk factor) and placenta previa over the internal cervical os. NPLEX + USMLE · Pathology · DOK 3

RE-3.6 Describe formation of the bilaminar disc in week 2 and the trilaminar disc by gastrulation in week 3, and explain why weeks 3 to 8 are the period of greatest teratogen sensitivity. NPLEX + USMLE · Anatomy · DOK 2

RE-4 Reproductive organ location and boundaries

Describe location, structure, and boundaries of male and female reproductive systems and breast.

RE-4.1 Describe the broad ligament and its parts (mesometrium, mesosalpinx, mesovarium), and identify the rectouterine pouch as the lowest peritoneal recess where blood or pus collects in a standing woman. NPLEX · Anatomy · DOK 2

RE-4.2 Identify the suspensory (infundibulopelvic) ligament carrying the ovarian vessels, the ovarian ligament, the round ligament through the inguinal canal, the cardinal ligament carrying the uterine vessels, and the uterosacral ligaments. NPLEX · Anatomy · DOK 1

RE-4.3 Describe uterine and vaginal support by the levator ani, perineal body, and cardinal and uterosacral ligaments, and relate weakening of these supports to uterine prolapse and pelvic relaxation. NPLEX + USMLE · Anatomy · DOK 2

RE-4.4 List the layers of the scrotum and spermatic cord (skin, dartos, external spermatic fascia, cremaster, internal spermatic fascia, tunica vaginalis) and the cord contents, including the ductus deferens, testicular artery, and pampiniform plexus. NPLEX · Anatomy · DOK 1

RE-4.5 Describe the breast over the pectoralis major from about rib 2 to rib 6 with an axillary tail, and explain skin dimpling when carcinoma involves the suspensory (Cooper) ligaments. NPLEX + USMLE · Anatomy · DOK 2

RE-4.6 Locate the paired corpora cavernosa, the corpus spongiosum surrounding the urethra, and the tunica albuginea, and relate rupture of the tunica albuginea to penile fracture. NPLEX + USMLE · Anatomy · DOK 2

RE-4.7 Describe the perineum, bounded by the pubic symphysis, ischiopubic rami, ischial tuberosities, sacrotuberous ligaments, and coccyx and divided into urogenital and anal triangles, and locate the prostate below the bladder and in front of the rectum for digital rectal exam. NPLEX · Anatomy · DOK 2

RE-5 Reproductive innervation and blood supply

Describe innervation and pathway of blood supply in reproductive organs and breast.

RE-5.1 Trace the gonadal arteries from the abdominal aorta and the gonadal veins (left to the left renal vein, right to the inferior vena cava), and explain why varicocele is more common on the left. NPLEX + USMLE · Anatomy · DOK 3

RE-5.2 Trace the uterine artery from the anterior division of the internal iliac artery, and explain why the ureter, passing beneath the uterine artery near the cervix, is at risk during ligation of these vessels. NPLEX · Anatomy · DOK 2

RE-5.3 Compare lymph drainage of the testis and ovary (para-aortic nodes), uterus, cervix, and proximal vagina (internal and external iliac nodes), and scrotum, vulva, and distal vagina (superficial inguinal nodes), and predict nodal spread of tumors. NPLEX + USMLE · Anatomy · DOK 3

RE-5.4 Describe the nerves of the male sexual response: erection by parasympathetic pelvic splanchnic nerves (S2 to S4), emission by sympathetic hypogastric nerves, and ejaculation by the somatic pudendal nerve. NPLEX + USMLE · Anatomy · DOK 2

RE-5.5 Trace the pudendal nerve (S2 to S4) out the greater sciatic foramen, around the ischial spine, and through the lesser sciatic foramen to the perineum, and identify the ischial spine as its landmark. NPLEX · Anatomy · DOK 2

RE-5.6 Describe breast blood supply from the internal thoracic, lateral thoracic, and thoracoacromial arteries, and lymph drainage mostly to axillary nodes with medial quadrants also draining to parasternal nodes. NPLEX + USMLE · Anatomy · DOK 2

RE-6 Reproductive processes and lactation

Explain mechanisms, function, regulation, and factors affecting reproductive processes and lactation.

RE-6.1 Relate the follicular and luteal phases of the ovarian cycle to the proliferative, secretory, and menstrual phases of the endometrium, and explain why the luteal phase stays near 14 days while cycle length varies. NPLEX | Physiology | DOK 3

RE-6.2 Explain erection as nitric oxide release raising cGMP in cavernosal smooth muscle, causing relaxation, arterial inflow, and compression of venous outflow against the tunica albuginea. NPLEX + USMLE | Physiology | DOK 2

RE-6.3 Explain how hCG maintains the corpus luteum until the placenta takes over progesterone production near the end of the first trimester, and why placental estriol synthesis depends on fetal adrenal DHEA-S. NPLEX | Physiology | DOK 2

RE-6.4 Predict the maternal adaptations of pregnancy: plasma volume rising more than red cell mass (dilutional anemia), higher cardiac output, lower systemic vascular resistance, higher GFR, respiratory alkalosis, and hypercoagulability. NPLEX + USMLE | Physiology | DOK 3

RE-6.5 Explain how human placental lactogen and other placental hormones increase maternal insulin resistance, and relate this to the development of gestational diabetes. NPLEX + USMLE | Physiology | DOK 3

RE-6.6 Explain lactation: alveolar development under estrogen, progesterone, and prolactin, milk secretion after placental progesterone withdrawal, suckling-driven prolactin (synthesis) and oxytocin (myoepithelial ejection), and suckling suppression of GnRH. NPLEX + USMLE | Physiology | DOK 3

RE-6.7 Explain the positive feedback of parturition, in which cervical stretch increases oxytocin release (Ferguson reflex) and oxytocin and prostaglandins increase myometrial contraction. NPLEX + USMLE | Physiology | DOK 2

RE-7 Reproductive hormones

Explain composition, function, effects, transport, and regulation of reproductive hormones.

RE-7.1 Explain how pulsatile GnRH stimulates LH and FSH release, and predict why continuous GnRH exposure suppresses gonadotropin secretion. NPLEX | Physiology | DOK 2

RE-7.2 Explain the male axis: LH stimulates Leydig cell testosterone, FSH stimulates Sertoli cell androgen-binding protein and inhibin B, with negative feedback by testosterone on LH and by inhibin B on FSH. NPLEX | Physiology | DOK 2

RE-7.3 Explain the two-cell model of ovarian steroidogenesis (LH drives theca cell androgen, FSH drives granulosa cell aromatase), and how sustained high estradiol switches to positive feedback to trigger the LH surge. NPLEX | Physiology | DOK 3

RE-7.4 Describe sex steroid transport bound to SHBG and albumin, and predict how SHBG changes (raised by estrogen, lowered by androgens, insulin resistance, and obesity) alter free testosterone. NPLEX | Physiology | DOK 3

RE-7.5 Distinguish DHT actions (external genitalia, prostate, male-pattern hair loss) from testosterone actions (Wolffian ducts, spermatogenesis, muscle mass, libido), and describe estrogen and progesterone effects on target tissues. NPLEX | Physiology | DOK 2

RE-7.6 Describe puberty as reactivation of pulsatile GnRH secretion, beginning with sleep-associated LH pulses, and list the usual female sequence of thelarche, pubarche, growth spurt, and menarche. NPLEX | Physiology | DOK 2

RE-7.7 Explain prolactin regulation by tonic hypothalamic dopamine inhibition, and predict how hyperprolactinemia suppresses GnRH to cause amenorrhea, galactorrhea, and hypogonadism. NPLEX + USMLE | Physiology | DOK 3

RE-8 Reproductive hormone biochemistry

Explain biochemistry of synthesis and degradation of hormones and other secretions involved in reproductive function and pathology.

RE-8.1 Trace sex steroid synthesis from cholesterol to pregnenolone (cholesterol side-chain cleavage), then to progesterone, androstenedione, testosterone, and estradiol, naming 3-beta-hydroxysteroid dehydrogenase, 17-alpha-hydroxylase, 17,20-lyase, 17-beta-hydroxysteroid dehydrogenase, and aromatase. NPLEX | Biochemistry & Genetics | DOK 2

RE-8.2 Explain the conversion of testosterone to DHT by 5-alpha-reductase in genital skin and prostate, and predict ambiguous genitalia at birth with virilization at puberty in 5-alpha-reductase deficiency. NPLEX | Biochemistry & Genetics | DOK 3

RE-8.3 Explain peripheral aromatization of androstenedione to estrone in adipose tissue, and relate obesity to unopposed estrogen, endometrial hyperplasia, and gynecomastia. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-8.4 Describe hCG, LH, FSH, and TSH as glycoproteins sharing a common alpha subunit with hormone-specific beta subunits, and explain why very high hCG in molar pregnancy can cause hyperthyroidism. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-8.5 Describe hepatic inactivation of sex steroids by conjugation to glucuronides and sulfates for excretion, and explain why cirrhosis causes gynecomastia, spider angiomas, palmar erythema, and testicular atrophy. NPLEX | Biochemistry & Genetics | DOK 3

RE-8.6 Identify the seminal vesicle (fructose) and prostatic (PSA, a serine protease that liquefies the semen coagulum; citrate; zinc) contributions to semen, and explain why PSA rises in prostatitis and BPH as well as carcinoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

RE-9 Reproductive biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to reproductive function and pathology.

RE-9.1 Explain the role of folate in thymidylate and purine synthesis, and relate maternal folate deficiency to neural tube defects, which arise when the neural tube fails to close by about day 28. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-9.2 Explain the higher iron requirement of pregnancy (expanded red cell mass, fetal and placental needs), and distinguish iron deficiency from physiologic dilutional anemia using ferritin and MCV. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-9.3 Describe lactose synthesis by lactose synthase (galactosyltransferase with alpha-lactalbumin), and compare human milk content of lactose, whey proteins, and secretory IgA with its low vitamin D and vitamin K. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

RE-9.4 Explain placental glucose transfer by facilitated diffusion while maternal insulin does not cross, and predict fetal hyperinsulinemia, macrosomia, and neonatal hypoglycemia from maternal hyperglycemia. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-9.5 Relate zinc deficiency to hypogonadism and delayed sexual maturation, and excess vitamin A (including retinoids) to teratogenic craniofacial, cardiac, and central nervous system defects. NPLEX | Biochemistry & Genetics | DOK 2

RE-9.6 Explain how low energy availability and low leptin in athletes or people with very low body fat suppress GnRH pulsatility, causing hypothalamic amenorrhea and loss of bone density. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-9.7 Explain how the gonads and placenta obtain cholesterol for steroid synthesis mainly by LDL receptor-mediated uptake, and identify StAR-mediated transfer of cholesterol into mitochondria as the rate-limiting step in gonadal steroidogenesis. NPLEX | Biochemistry & Genetics | DOK 2

RE-10 Reproductive genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying reproductive disease.

RE-10.1 Explain how meiotic nondisjunction produces 47,XXY and 45,X karyotypes, and how nondisjunction after fertilization produces mosaicism with milder phenotypes. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

RE-10.2 Explain X inactivation by XIST with Barr body formation, and predict the number of Barr bodies in 46,XX, 47,XXY, and 45,X cells. NPLEX | Biochemistry & Genetics | DOK 2

RE-10.3 Explain how high-risk HPV E6 promotes p53 degradation and E7 binds and inactivates Rb to release E2F, removing cell cycle checkpoints in cervical epithelium. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-10.4 Relate autosomal dominant BRCA1 and BRCA2 mutations (homologous recombination DNA repair genes) to hereditary breast and ovarian cancer, and BRCA2 to male breast and prostate cancer. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

RE-10.5 Relate Lynch syndrome (germline mismatch repair gene mutations, microsatellite instability) to endometrial and ovarian carcinoma as well as colorectal carcinoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

RE-10.6 Classify breast carcinomas by estrogen receptor, progesterone receptor, and HER2 (ERBB2 amplification) status, and relate triple-negative basal-like tumors to BRCA1 mutation. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

RE-11 Immune responses in reproductive disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

RE-11.1 Explain testicular immune privilege (Sertoli cell blood-testis barrier, local immunosuppression), predict antisperm antibodies after trauma, vasectomy, or mumps orchitis, and describe autoimmune orchitis as a cause of hypogonadism. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-11.2 Explain maternal tolerance of the semiallogeneic fetus, including trophoblast expression of nonclassical HLA-G in place of HLA-A and HLA-B, which limits maternal T cell and NK cell attack. NPLEX | Microbiology & Immunology | DOK 2

RE-11.3 Explain passive immunity of the newborn through FcRn-mediated placental transfer of maternal IgG (not IgM) and secretory IgA in breast milk. NPLEX | Microbiology & Immunology | DOK 2

RE-11.4 Explain Rh alloimmunization, in which an Rh-negative mother forms anti-D IgG after fetomaternal hemorrhage, causing type II hypersensitivity hemolysis and possible hydrops fetalis in a later Rh-positive fetus. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-11.5 Describe vaginal defenses (lactobacilli producing lactic acid to keep pH at or below 4.5, secretory IgA), and predict overgrowth of Gardnerella and anaerobes when lactobacilli are lost. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-11.6 Explain how HPV vaccines made of L1 capsid virus-like particles induce neutralizing antibodies, and how cell-mediated immunity clears most HPV infections and low-grade lesions, so that immunosuppression such as HIV raises cervical cancer risk. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-11.7 Explain how the inflammatory response to repeated or untreated chlamydial and gonococcal salpingitis causes tubal scarring, increasing the risk of infertility and ectopic pregnancy. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-12 Reproductive pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

RE-12.1 Describe Neisseria gonorrhoeae as an oxidase-positive gram-negative diplococcus that ferments glucose but not maltose, with pili showing antigenic variation, IgA protease, and lipooligosaccharide, cultured on Thayer-Martin agar. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-12.2 Describe Chlamydia trachomatis as an obligate intracellular bacterium with an infectious elementary body and replicating reticulate body, and match serovars A to C (trachoma), D to K (genital and neonatal), and L1 to L3 (LGV). NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-12.3 Describe Treponema pallidum as a thin spirochete that cannot be cultured on standard media, seen by dark-field microscopy, and transmitted sexually and transplacentally. NPLEX + USMLE | Microbiology & Immunology | DOK 1

RE-12.4 Compare HSV-2 (enveloped dsDNA herpesvirus with latency in sacral sensory ganglia) with HPV (nonenveloped dsDNA papillomavirus), distinguishing low-risk types 6 and 11 from high-risk types 16 and 18. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-12.5 Identify Trichomonas vaginalis (motile flagellated protozoan on wet mount), Candida albicans (yeast with pseudohyphae on KOH preparation), and Gardnerella vaginalis (clue cells) by their microscopic features. NPLEX + USMLE | Microbiology & Immunology | DOK 1

RE-12.6 Describe Haemophilus ducreyi as a gram-negative coccobacillus, and explain how Staphylococcus aureus TSST-1 acts as a superantigen by binding MHC class II and the T cell receptor outside the antigen groove. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-12.7 Describe Streptococcus agalactiae (group B, beta-hemolytic, CAMP positive, bacitracin resistant) colonizing the vagina, and identify the TORCH organisms transmitted across the placenta or during delivery. NPLEX + USMLE | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

RE-13-A Reproductive hormone (endocrine) conditions

1 Amenorrhea **2** Anovulation **3** Dysfunctional uterine bleeding **4** Menopause/perimenopause **5** Ovarian insufficiency/failure

RE-13-A.1 Distinguish primary amenorrhea (no menarche by age 15, or by 13 without breast development) from secondary amenorrhea (no menses for 3 months after regular cycles), naming pregnancy as the most common cause of secondary amenorrhea. NPLEX + USMLE | Pathology | DOK 2

RE-13-A.2 Interpret FSH, LH, prolactin, TSH, and estradiol levels to localize amenorrhea to the hypothalamus (low FSH and LH), pituitary, ovary (high FSH), or outflow tract, such as Asherman syndrome. NPLEX + USMLE | Pathology | DOK 3

RE-13-A.3 Explain anovulation from absent LH surge or disordered follicle development (PCOS, hyperprolactinemia, thyroid disease, low energy availability), and predict unopposed estrogen effects of irregular heavy bleeding and endometrial hyperplasia. NPLEX + USMLE | Pathology | DOK 3

RE-13-A.4 Explain anovulatory bleeding (formerly dysfunctional uterine bleeding, now abnormal uterine bleeding from ovulatory dysfunction in the PALM-COEIN system) as estrogen-driven proliferation without progesterone-induced secretory change, and distinguish it from structural causes such as polyps, adenomyosis, leiomyoma, and malignancy. NPLEX + USMLE | Pathology | DOK 3

RE-13-A.5 Explain menopause as follicle depletion with loss of estradiol and inhibin feedback, raising FSH and LH, and relate estrogen loss to vasomotor symptoms, atrophic vaginitis, osteoporosis, and higher cardiovascular risk. NPLEX + USMLE | Pathology | DOK 3

RE-13-A.6 Describe primary ovarian insufficiency (ovarian failure before age 40) with causes including Turner syndrome, FMR1 premutation, autoimmune oophoritis, and chemotherapy or radiation, and its high FSH and low estradiol profile. NPLEX + USMLE | Pathology | DOK 3

RE-13-A.7 Distinguish hirsutism from virilization, and localize androgen excess to the ovary (PCOS, Sertoli-Leydig cell tumor) or adrenal (congenital adrenal hyperplasia, adrenal tumor) using testosterone and DHEA-S levels. USMLE extension | Pathology | DOK 3

RE-13-A.8 Explain primary dysmenorrhea as prostaglandin-driven myometrial contraction and ischemia, and mittelschmerz as midcycle pain at ovulation from follicle rupture. USMLE extension | Physiology | DOK 2

RE-13-A.9 Classify female infertility as ovulatory (PCOS, hyperprolactinemia, ovarian insufficiency), tubal (prior PID), or uterine (Asherman syndrome, submucosal leiomyoma), and explain the mechanism of each. USMLE extension | Pathology | DOK 3

RE-13-B Inflammatory diseases of the reproductive tract

1 Balanitis **2** Cervicitis **3** Endometriosis **4** Endometritis **5** Orchitis **6** Pelvic inflammatory disease **7** Salpingitis **8** Vaginitis (candidal)

RE-13-B.1 Describe balanitis as inflammation of the glans penis, most often from Candida or bacteria, with risk factors of an uncircumcised foreskin, poor hygiene, and diabetes mellitus, and relate recurrent balanitis to phimosis. NPLEX + USMLE | Pathology | DOK 2

RE-13-B.2 Identify Chlamydia trachomatis and Neisseria gonorrhoeae as the main causes of mucopurulent cervicitis, and distinguish it from the punctate hemorrhagic strawberry cervix of trichomoniasis and the ulcerative cervicitis of HSV. NPLEX + USMLE | Pathology | DOK 3

RE-13-B.3 Explain endometriosis as endometrial glands and stroma outside the uterus (theories of retrograde menstruation, coelomic metaplasia, vascular or lymphatic spread), and relate ovarian chocolate cysts and adhesions to dysmenorrhea, dyspareunia, and infertility. NPLEX + USMLE | Pathology | DOK 3

RE-13-B.4 Distinguish adenomyosis (endometrial glands and stroma within the myometrium, causing a uniformly enlarged, tender uterus with dysmenorrhea and heavy bleeding) from endometriosis and leiomyoma. USMLE extension · Pathology · DOK 3

RE-13-B.5 Distinguish acute endometritis (polymicrobial infection after delivery, miscarriage, or retained products, with fever and uterine tenderness) from chronic endometritis, identified by plasma cells in the endometrial stroma. NPLEX + USMLE · Pathology · DOK 3

RE-13-B.6 Distinguish mumps orchitis (postpubertal, risk of testicular atrophy and infertility) from epididymo-orchitis caused by C. trachomatis or N. gonorrhoeae in sexually active young men and by E. coli in older men. NPLEX + USMLE · Pathology · DOK 3

RE-13-B.7 Explain pelvic inflammatory disease as ascending infection by C. trachomatis, N. gonorrhoeae, and vaginal flora, and identify cervical motion tenderness, adnexal tenderness, fever, and purulent cervical discharge. NPLEX + USMLE · Pathology · DOK 3

RE-13-B.8 Predict the complications of salpingitis, including hydrosalpinx, tubo-ovarian abscess, tubal infertility, ectopic pregnancy, and Fitz-Hugh-Curtis perihepatitis with right upper quadrant pain and violin-string adhesions. NPLEX + USMLE · Pathology · DOK 3

RE-13-B.9 Relate candidal vaginitis to risk factors (diabetes, antibiotic use, pregnancy, immunosuppression), and identify thick white curd-like discharge, vulvar itching, normal vaginal pH, and pseudohyphae on KOH preparation. NPLEX + USMLE · Pathology · DOK 3

RE-13-B.10 Describe lichen sclerosus as thinned epidermis with dermal fibrosis forming white parchment-like vulvar plaques, mostly in postmenopausal women, and relate it to a small increase in vulvar squamous cell carcinoma risk. USMLE extension · Pathology · DOK 2

RE-13-C Congenital and genetic reproductive diseases

1 Cryptorchidism **2** Epispadias **3** Fragile X syndrome **4** Hypospadias **5** Imperforate hymen **6** Klinefelter syndrome **7** Paraphimosis **8** Phimosis **9** Pseudohermaphroditism **10** Septate vagina and uterus **11** Turner syndrome

RE-13-C.1 Describe cryptorchidism (most often inguinal, more common in preterm infants), and relate it to impaired spermatogenesis from higher temperature, reduced inhibin B with raised FSH, and increased germ cell tumor risk. NPLEX + USMLE · Pathology · DOK 3

RE-13-C.2 Compare hypospadias (ventral urethral opening from failed fusion of the urethral folds, associated with cryptorchidism and inguinal hernia) with epispadias (dorsal opening from abnormal genital tubercle position, associated with bladder exstrophy). NPLEX + USMLE · Pathology · DOK 3

RE-13-C.3 Explain fragile X syndrome as an FMR1 CGG repeat expansion with hypermethylation and silencing, showing anticipation through maternal transmission, and identify intellectual disability, postpubertal macroorchidism, long face, large jaw, and large ears. NPLEX + USMLE · Biochemistry & Genetics · DOK 3

RE-13-C.4 Explain imperforate hymen as failure of the hymen to perforate during development, and predict primary amenorrhea with cyclic pelvic pain, a bulging bluish membrane, and hematocolpos at menarche. NPLEX + USMLE · Pathology · DOK 3

RE-13-C.5 Explain Klinefelter syndrome (47,XXY) as seminiferous tubule dysgenesis (low inhibin B, high FSH) with Leydig cell dysfunction (low testosterone, high LH), and identify small firm testes, tall stature, gynecomastia, and infertility. NPLEX + USMLE · Biochemistry & Genetics · DOK 3

RE-13-C.6 Distinguish phimosis (foreskin that cannot be retracted over the glans, physiologic in infancy or acquired from scarring) from paraphimosis (retracted foreskin trapped behind the glans, causing edema and ischemia). NPLEX + USMLE · Pathology · DOK 2

RE-13-C.7 Distinguish 46,XX disorders of sex development (DSD), formerly female pseudohermaphroditism (ovaries with virilized genitalia from 21-hydroxylase deficiency or excess maternal androgen), from 46,XY DSD, formerly male pseudohermaphroditism (testes with undervirilized genitalia from androgen insensitivity or 5-alpha-reductase deficiency). NPLEX + USMLE · Pathology · DOK 3

RE-13-C.8 Explain complete androgen insensitivity (X-linked androgen receptor defect) with female external genitalia, blind vaginal pouch, absent uterus and tubes, undescended testes, sparse pubic hair, and raised testosterone, estrogen, and LH. NPLEX + USMLE · Pathology · DOK 3

RE-13-C.9 Distinguish Mullerian agenesis (46,XX, normal ovaries and pubic hair, absent uterus and upper vagina) from complete androgen insensitivity as causes of primary amenorrhea with a blind vaginal pouch. USMLE extension · Pathology · DOK 3

RE-13-C.10 Explain septate uterus as failed resorption of the midline septum after Mullerian duct fusion and a cause of recurrent pregnancy loss, and describe longitudinal and transverse vaginal septa from abnormal fusion or canalization. NPLEX Anatomy DOK 3

RE-13-C.11 Explain Turner syndrome (45,X or mosaic) as ovarian dysgenesis with streak ovaries, and predict primary amenorrhea, absent breast development, low estradiol, and high FSH and LH. NPLEX + USMLE Biochemistry & Genetics DOK 3

RE-13-C.12 Identify the associated findings of Turner syndrome: short stature (SHOX loss), webbed neck from fetal cystic hygroma, lymphedema of hands and feet, bicuspid aortic valve, coarctation of the aorta, and horseshoe kidney. NPLEX + USMLE Biochemistry & Genetics DOK 2

RE-13-D Benign conditions of the penis and scrotum

1 Erectile dysfunction **2** Hematocele **3** Hydrocele **4** Spermatocele **5** Varicocele

RE-13-D.1 Classify erectile dysfunction as vascular (atherosclerosis), neurogenic (diabetes, spinal cord injury, pelvic surgery), endocrine (hypogonadism, hyperprolactinemia), drug-induced, or psychogenic, and use preserved nocturnal erections to identify a psychogenic cause. NPLEX + USMLE Pathology DOK 3

RE-13-D.2 Describe hematocele as blood within the tunica vaginalis after trauma, torsion, or tumor, and distinguish it from hydrocele by its failure to transilluminate. NPLEX Pathology DOK 2

RE-13-D.3 Distinguish a communicating congenital hydrocele (patent processus vaginalis) from a noncommunicating hydrocele (infection, trauma, tumor), and identify a smooth scrotal fluid collection around the testis that transilluminates. NPLEX + USMLE Pathology DOK 3

RE-13-D.4 Describe spermatocele as a cystic dilation of the epididymis containing sperm, felt as a transilluminating mass separate from the testis, and compare it with a simple epididymal cyst. NPLEX + USMLE Pathology DOK 2

RE-13-D.5 Explain varicocele as dilation of the pampiniform plexus, felt as a bag of worms that enlarges on standing and does not transilluminate, and relate it to infertility and, when new on the left, renal cell carcinoma. NPLEX + USMLE Pathology DOK 3

RE-13-D.6 Explain testicular torsion as twisting of the spermatic cord (bell-clapper deformity) causing venous then arterial occlusion, and identify sudden severe pain, a high-riding testis, and an absent cremasteric reflex. USMLE extension Pathology DOK 3

RE-13-D.7 Describe priapism as a painful erection lasting over 4 hours, with sickle cell disease and drugs as causes of the ischemic type, and predict cavernosal fibrosis and erectile dysfunction if it persists. USMLE extension Pathology DOK 2

RE-13-D.8 Describe Peyronie disease as a fibrous plaque of the tunica albuginea causing penile curvature and painful erection. USMLE extension Pathology DOK 1

RE-13-D.9 Classify male infertility as pretesticular (hypogonadotropic hypogonadism), testicular (Klinefelter syndrome, cryptorchidism, varicocele, mumps orchitis), or post-testicular (obstruction, congenital bilateral absence of the vas deferens in cystic fibrosis). USMLE extension Pathology DOK 3

RE-13-E Conditions of the breast

1 Diffuse cystic mastopathy (fibrocystic breast disease) **2** Galactocele **3** Mammary duct ectasia **4** Mastitis **5** Traumatic fat necrosis **6** Benign and malignant neoplasms (fibroadenoma, lobular carcinoma, ductal carcinoma, Paget disease of the breast)

RE-13-E.1 Distinguish nonproliferative fibrocystic change (cysts, apocrine metaplasia, fibrosis; no added cancer risk) from proliferative change without atypia (slightly raised risk) and atypical hyperplasia (moderately raised risk). NPLEX + USMLE Pathology DOK 3

RE-13-E.2 Describe galactocele as a milk-filled cystic dilation of a duct during lactation or weaning, and predict that it may rupture and cause local inflammation. NPLEX Pathology DOK 2

RE-13-E.3 Describe mammary duct ectasia as dilated subareolar ducts with thick secretions and periductal chronic inflammation rich in plasma cells, in older multiparous women, and explain why nipple retraction can mimic carcinoma. NPLEX + USMLE Pathology DOK 3

RE-13-E.4 Explain acute mastitis as Staphylococcus aureus infection entering through nipple fissures during breastfeeding, with a red, painful, warm breast, fever, and possible abscess. NPLEX + USMLE Pathology DOK 2

RE-13-E.5 Describe periductal mastitis in smokers as squamous metaplasia of lactiferous ducts with keratin plugging, producing a painful subareolar mass or abscess. USMLE extension · Pathology · DOK 2

RE-13-E.6 Describe traumatic fat necrosis as nonenzymatic necrosis of breast fat after trauma or surgery, with foamy macrophages, giant cells, and calcification, forming a firm mass that can mimic carcinoma on exam and mammography. NPLEX + USMLE · Pathology · DOK 2

RE-13-E.7 Describe fibroadenoma as the most common benign breast tumor in young women, a small, firm, mobile mass of proliferating stroma and ducts that enlarges with estrogen in pregnancy. NPLEX + USMLE · Pathology · DOK 2

RE-13-E.8 Distinguish phyllodes tumor (leaf-like projections of overgrown, hypercellular stroma, usually in older women, with potential to recur or metastasize) from fibroadenoma. USMLE extension · Pathology · DOK 3

RE-13-E.9 Describe intraductal papilloma as a benign subareolar papillary growth in a large duct and identify it as a common cause of bloody nipple discharge. USMLE extension · Pathology · DOK 2

RE-13-E.10 Relate breast carcinoma risk to age, prolonged estrogen exposure (early menarche, late menopause, nulliparity, late first pregnancy, postmenopausal obesity), atypical hyperplasia, and family history, and identify the upper outer quadrant as the most common site. NPLEX + USMLE · Pathology · DOK 3

RE-13-E.11 Distinguish ductal carcinoma in situ (confined by basement membrane, mammographic microcalcifications, comedo necrosis) from invasive ductal carcinoma (most common type, firm stellate mass with desmoplastic stroma). NPLEX + USMLE · Pathology · DOK 3

RE-13-E.12 Explain lobular carcinoma as loss of E-cadherin (CDH1) causing discohesive cells that invade in single-file lines, and relate it to frequent bilateral and multifocal disease. NPLEX + USMLE · Pathology · DOK 3

RE-13-E.13 Describe Paget disease of the breast as eczematous nipple and areolar change from large Paget cells with clear cytoplasm in the epidermis, nearly always with underlying ductal carcinoma. NPLEX + USMLE · Pathology · DOK 3

RE-13-E.14 Distinguish inflammatory breast carcinoma (tumor emboli in dermal lymphatics causing erythema, edema, and peau d'orange) from acute mastitis. USMLE extension · Pathology · DOK 3

RE-13-E.15 Distinguish gynecomastia (benign ductal and stromal proliferation from a raised estrogen to androgen ratio) from male breast carcinoma, and relate male breast carcinoma to BRCA2 mutation and Klinefelter syndrome. USMLE extension · Pathology · DOK 3

RE-13-F Infectious diseases and hyperplasia of the prostate

1 Benign prostatic hyperplasia **2** Prostatitis

RE-13-F.1 Explain benign prostatic hyperplasia as DHT-dependent nodular hyperplasia of glands and stroma in the transition (periurethral) zone of older men, and state that it is not a precursor of carcinoma. NPLEX + USMLE · Pathology · DOK 2

RE-13-F.2 Predict the urinary findings and complications of benign prostatic hyperplasia: hesitancy, weak stream, frequency, nocturia, urinary retention, bladder wall hypertrophy, infection, and hydronephrosis. NPLEX + USMLE · Pathology · DOK 3

RE-13-F.3 Identify E. coli and other gram-negative rods as the usual cause of acute bacterial prostatitis in older men (C. trachomatis and N. gonorrhoeae in younger men), with fever, chills, dysuria, and a tender, boggy prostate. NPLEX + USMLE · Microbiology & Immunology · DOK 3

RE-13-F.4 Distinguish acute bacterial prostatitis from chronic bacterial prostatitis (recurrent urinary tract infection with the same organism) and chronic nonbacterial prostatitis (chronic pelvic pain without bacteria). NPLEX + USMLE · Pathology · DOK 3

RE-13-G Conditions of the ovary

1 Ovarian cysts **2** Paraovarian cysts **3** Polycystic ovary syndrome (PCOS) **4** Tubo-ovarian cysts

RE-13-G.1 Distinguish follicular cysts (unruptured follicle, may secrete estrogen), corpus luteum cysts (may hemorrhage into the peritoneum), and bilateral theca lutein cysts from high hCG in molar or multiple pregnancy. NPLEX + USMLE · Pathology · DOK 3

RE-13-G.2 Explain ovarian torsion as twisting of the ovary on its ligaments, usually when enlarged by a cyst or tumor, with venous then arterial occlusion causing sudden severe pelvic pain and possible infarction. USMLE extension · Pathology · DOK 3

RE-13-G.3 Describe paraovarian cysts as thin-walled cysts in the broad ligament arising from Mullerian or Wolffian remnants, separate from the ovary, and predict torsion or rupture as complications. NPLEX Pathology DOK 2

RE-13-G.4 Explain PCOS pathogenesis: raised LH to FSH ratio driving theca androgen, insulin resistance raising ovarian androgen and lowering SHBG, and arrested follicles forming multiple small peripheral ovarian cysts. NPLEX + USMLE Pathology DOK 3

RE-13-G.5 Identify the findings and complications of PCOS: oligomenorrhea, hirsutism, acne, obesity, acanthosis nigricans, infertility, type 2 diabetes, dyslipidemia, and endometrial hyperplasia and carcinoma from unopposed estrogen. NPLEX + USMLE Pathology DOK 3

RE-13-G.6 Describe tubo-ovarian cysts and abscesses as sequelae of PID in which an inflamed or fluid-filled tube adheres to the ovary, and predict rupture with peritonitis and later infertility. NPLEX + USMLE Pathology DOK 3

RE-13-H Diseases of the placenta

1 Choriocarcinoma **2** Hydatidiform mole **3** Invasive mole **4** Preeclampsia

RE-13-H.1 Compare complete mole (usually 46,XX, all paternal from an empty ovum, no fetal parts, diffuse villous swelling, p57 negative) with partial mole (triploid, fetal parts, focal swelling, p57 positive). NPLEX + USMLE Pathology DOK 3

RE-13-H.2 Predict the findings of complete hydatidiform mole: vaginal bleeding, uterus large for dates, very high hCG, hyperemesis, theca lutein cysts, hyperthyroidism, preeclampsia before 20 weeks, and higher choriocarcinoma risk. NPLEX + USMLE Pathology DOK 3

RE-13-H.3 Describe invasive mole as hydropic villi penetrating the myometrium, which can cause uterine perforation and hemorrhage and persistent hCG elevation after evacuation. NPLEX + USMLE Pathology DOK 2

RE-13-H.4 Describe gestational choriocarcinoma as malignant cytotrophoblast and syncytiotrophoblast without villi, arising after a mole, abortion, or normal pregnancy, with very high hCG and early hematogenous spread to lung, vagina, and brain. NPLEX + USMLE Pathology DOK 3

RE-13-H.5 Explain preeclampsia as abnormal placentation with failed spiral artery remodeling, placental ischemia, and release of antiangiogenic factors (sFlt-1, soluble endoglin) that cause maternal endothelial dysfunction. NPLEX + USMLE Pathology DOK 3

RE-13-H.6 Identify preeclampsia as new hypertension (140/90 mm Hg or higher) after 20 weeks with proteinuria or end-organ damage, its risk factors (nulliparity, chronic hypertension, diabetes, multiple gestation), and its complications of eclampsia, HELLP syndrome, placental abruption, and fetal growth restriction. NPLEX + USMLE Pathology DOK 3

RE-13-H.7 Distinguish placental abruption (premature separation with painful bleeding, linked to hypertension, cocaine, and trauma, risking DIC) from placenta previa (painless third-trimester bleeding). USMLE extension Pathology DOK 3

RE-13-H.8 Explain placenta accreta as villi attaching to or invading the myometrium through a deficient decidua basalis, with prior cesarean delivery and placenta previa as major risk factors. USMLE extension Pathology DOK 3

RE-13-H.9 Relate oligohydramnios to reduced fetal urine output (bilateral renal agenesis, urinary obstruction) with Potter sequence, and polyhydramnios to impaired fetal swallowing (esophageal atresia, anencephaly) or maternal diabetes. USMLE extension Pathology DOK 3

RE-13-I Benign conditions of the vagina and vulva

1 Bartholin cysts **2** Cystocele **3** Rectocele **4** Urethrocele

RE-13-I.1 Describe a Bartholin cyst as obstruction of the Bartholin gland duct, forming a mass in the posterolateral vulvar vestibule that may become an abscess, and identify its occurrence in reproductive-age women. NPLEX + USMLE Pathology DOK 2

RE-13-I.2 Explain cystocele, urethrocele, and rectocele as herniation through weakened pelvic floor and endopelvic fascia, with risk factors of vaginal childbirth, aging, estrogen loss, obesity, and chronic straining or cough. NPLEX + USMLE Pathology DOK 3

RE-13-I.3 Distinguish cystocele (bladder bulging into the anterior vaginal wall, incomplete emptying), urethrocele (urethra into the distal anterior wall, stress incontinence), and rectocele (rectum into the posterior wall, difficult defecation). NPLEX + USMLE Pathology DOK 3

RE-13-J Reproductive dysplasia and neoplasms

1 Cervical intraepithelial neoplasia (CIN) **2** Endometrial hyperplasia **3** Fibroids (leiomyoma) **4** Invasive carcinoma of cervix **5** Leiomyosarcomas **6** Prostate carcinoma **7** Tumors of ovary **8** Squamous cell carcinoma of penis **9** Testicular tumors **10** Vaginal carcinoma **11** Vulvar carcinoma **12** Vulvar intraepithelial neoplasia

RE-13-J.1 Explain cervical intraepithelial neoplasia as high-risk HPV infection of the transformation zone with koilocytes, graded CIN 1 to 3 by the thickness of epithelium replaced by atypical immature cells (CIN 1 as LSIL, CIN 2 and 3 as HSIL). NPLEX + USMLE Pathology DOK 3

RE-13-J.2 Relate cervical dysplasia risk to persistent HPV 16 or 18 infection, many sexual partners, early sexual activity, smoking, and immunodeficiency such as HIV, and predict that most CIN 1 regresses. NPLEX + USMLE Pathology DOK 3

RE-13-J.3 Describe invasive cervical carcinoma (mostly squamous cell, some adenocarcinoma) with postcoital bleeding, and explain how lateral spread can obstruct the ureters, causing hydronephrosis and renal failure. NPLEX + USMLE Pathology DOK 3

RE-13-J.4 Relate endometrial hyperplasia to unopposed estrogen (obesity, anovulation, PCOS, estrogen-only therapy, granulosa cell tumor) and PTEN loss, and identify atypical hyperplasia as the precursor of endometrioid carcinoma. NPLEX + USMLE Pathology DOK 3

RE-13-J.5 Compare endometrioid endometrial carcinoma (estrogen-related, arising from hyperplasia, PTEN and Lynch syndrome mutations) with serous carcinoma (older women, atrophic endometrium, TP53 mutation, aggressive), both presenting with postmenopausal bleeding. USMLE extension Pathology DOK 3

RE-13-J.6 Describe endometrial and endocervical polyps as benign localized overgrowths of glands and stroma that cause abnormal or intermenstrual bleeding. USMLE extension Pathology DOK 1

RE-13-J.7 Describe leiomyoma as a benign, estrogen-sensitive, well-circumscribed smooth muscle tumor with a whorled pattern, more common in Black women, causing heavy bleeding, pelvic pressure, anemia, and infertility. NPLEX + USMLE Pathology DOK 3

RE-13-J.8 Distinguish leiomyosarcoma (arising de novo in older women, a single large soft mass with necrosis, nuclear atypia, and high mitotic rate) from leiomyoma, which does not transform into it. NPLEX + USMLE Pathology DOK 3

RE-13-J.9 Relate prostate adenocarcinoma to age, Black race, family history, and BRCA2, and explain its origin in the peripheral zone, Gleason grading by gland pattern, and raised PSA. NPLEX + USMLE Pathology DOK 3

RE-13-J.10 Predict osteoblastic bone metastases of prostate carcinoma to the lumbar spine and pelvis through the vertebral venous plexus, with bone pain and raised alkaline phosphatase. NPLEX + USMLE Pathology DOK 3

RE-13-J.11 Classify ovarian tumors by origin as surface epithelial (serous, mucinous, endometrioid, clear cell, Brenner), germ cell (teratoma, dysgerminoma, yolk sac tumor, choriocarcinoma), or sex cord-stromal (granulosa cell, Sertoli-Leydig cell, fibroma, thecoma). NPLEX + USMLE Pathology DOK 2

RE-13-J.12 Relate high-grade serous ovarian carcinoma to BRCA1, BRCA2, and TP53 mutations and the fallopian tube fimbria, and identify psammoma bodies, raised CA-125, ascites, and peritoneal and omental spread. NPLEX + USMLE Pathology DOK 3

RE-13-J.13 Identify mature cystic teratoma (most common ovarian germ cell tumor), struma ovarii (a monodermal teratoma of thyroid tissue that can cause hyperthyroidism), dysgerminoma (sheets of clear cells, raised LDH), and yolk sac tumor (Schiller-Duval bodies, raised AFP). NPLEX + USMLE Pathology DOK 3

RE-13-J.14 Predict the effects of hormonally active ovarian stromal tumors: granulosa cell tumor (estrogen, Call-Exner bodies, precocious puberty or endometrial hyperplasia), Sertoli-Leydig cell tumor (virilization), and fibroma with ascites and hydrothorax (Meigs syndrome). NPLEX + USMLE Pathology DOK 3

RE-13-J.15 Identify Krukenberg tumor as bilateral ovarian metastasis of signet ring cell adenocarcinoma, most often from the stomach. NPLEX Pathology DOK 1

RE-13-J.16 Relate squamous cell carcinoma of the penis to HPV 16, lack of circumcision, phimosis, and smoking, and identify Bowen disease, erythroplasia of Queyrat, and bowenoid papulosis as carcinoma in situ. NPLEX + USMLE Pathology DOK 3

RE-13-J.17 Classify testicular tumors as germ cell (most, with isochromosome 12p and germ cell neoplasia in situ) or sex cord-stromal, and identify a painless firm mass that does not transilluminate, with cryptorchidism and Klinefelter syndrome as risks. NPLEX + USMLE Pathology DOK 3

RE-13-J.18 Distinguish seminoma (most common, uniform clear cells with lymphocytes, placental alkaline phosphatase, radiosensitive, good prognosis) from embryonal carcinoma, yolk sac tumor (AFP), choriocarcinoma (hCG), and teratoma. NPLEX + USMLE | Pathology | DOK 3

RE-13-J.19 Describe Leydig cell tumor (Reinke crystals, androgen or estrogen excess causing precocious puberty or gynecomastia) and testicular lymphoma as the most common testicular tumor in men over 60. NPLEX + USMLE | Pathology | DOK 2

RE-13-J.20 Distinguish vaginal squamous cell carcinoma (older women, HPV, vaginal intraepithelial neoplasia), clear cell adenocarcinoma (in utero DES exposure), and embryonal rhabdomyosarcoma (sarcoma botryoides, grape-like mass in young girls, desmin positive). NPLEX + USMLE | Pathology | DOK 3

RE-13-J.21 Compare HPV-related vulvar squamous cell carcinoma (younger women, from HPV-related intraepithelial neoplasia) with HPV-independent vulvar carcinoma (older women, lichen sclerosus, differentiated VIN with TP53 mutation). NPLEX + USMLE | Pathology | DOK 3

RE-13-J.22 Distinguish usual (HPV-related) vulvar intraepithelial neoplasia from differentiated VIN, and distinguish extramammary Paget disease of the vulva (intraepithelial adenocarcinoma, usually without underlying carcinoma) from Paget disease of the breast. NPLEX + USMLE | Pathology | DOK 3

RE-13-K Infectious diseases of the genitourinary system (including STDs)

1 Chancroid **2** Bacterial vaginosis (BV) **3** Chlamydia **4** Gonorrhea **5** Herpes simplex virus (HSV) **6** Human papillomavirus (HPV) **7** Nongonococcal urethritis **8** Syphilis **9** Toxic shock syndrome (TSS) **10** Trichomoniasis

RE-13-K.1 Distinguish chancroid (Haemophilus ducreyi; painful ulcer with ragged edges and gray exudate, painful suppurative inguinal nodes) from the painless indurated chancre of primary syphilis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.2 Explain bacterial vaginosis as replacement of lactobacilli by Gardnerella vaginalis and anaerobes, causing thin gray discharge with fishy odor, pH above 4.5, positive amine (whiff) test, clue cells, and few neutrophils. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.3 Use vaginal pH, the KOH whiff test, and wet mount findings to distinguish bacterial vaginosis, trichomoniasis, and candidal vaginitis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.4 Identify genital chlamydial infection as often asymptomatic and the cause of urethritis, cervicitis, epididymitis, PID, reactive arthritis, and neonatal conjunctivitis and pneumonia. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.5 Describe lymphogranuloma venereum (C. trachomatis L1 to L3) as a small painless genital ulcer followed by painful inguinal lymphadenopathy (buboes) and possible proctocolitis. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-13-K.6 Identify gonococcal urethritis with purulent discharge and intracellular gram-negative diplococci, cervicitis, PID, neonatal conjunctivitis, and disseminated infection (septic arthritis, tenosynovitis, pustules), and relate recurrent infection to C5 to C9 deficiency. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.7 Describe genital herpes as painful grouped vesicles that ulcerate, with systemic symptoms in primary infection, recurrence from sacral ganglia, multinucleated giant cells, and neonatal infection acquired during delivery. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.8 Distinguish condylomata acuminata (HPV 6 and 11, koilocytes) from condylomata lata of secondary syphilis, and relate high-risk HPV to cervical, vulvar, vaginal, penile, anal, and oropharyngeal carcinoma. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.9 Identify the causes of nongonococcal urethritis (C. trachomatis most often, also Mycoplasma genitalium, Ureaplasma urealyticum, Trichomonas vaginalis), and relate it to reactive arthritis with conjunctivitis in HLA-B27–positive men. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.10 Describe the stages of syphilis: primary painless chancre, secondary rash on palms and soles with condylomata lata and lymphadenopathy, latency, and tertiary gummas, aortitis of the ascending aorta, and tabes dorsalis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.11 Interpret nontreponemal tests (VDRL, RPR; anticardiolipin antibodies, false positives in pregnancy, SLE, and viral infection) and treponemal tests (FTA-ABS, which stays positive for life) in syphilis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.12 Identify the findings of congenital syphilis: rhinitis (snuffles), rash, saddle nose, Hutchinson teeth, mulberry molars, saber shins, and sensorineural deafness. NPLEX + USMLE | Microbiology & Immunology | DOK 2

RE-13-K.13 Explain toxic shock syndrome as TSST-1 superantigen release by S. aureus (retained tampons, nasal packing, wounds) causing polyclonal T cell cytokine release, fever, hypotension, diffuse rash with later palm and sole desquamation, and multiorgan injury. NPLEX + USMLE | Microbiology & Immunology | DOK 3

RE-13-K.14 Describe trichomoniasis as a sexually transmitted flagellated protozoan infection causing frothy yellow-green malodorous discharge, strawberry cervix, pH above 4.5, and motile trophozoites on wet mount. NPLEX + USMLE | Microbiology & Immunology | DOK 3

Urology Urinary System

10% · 20 questions · Week 6

HOW THE BODY WORKS

UR-1 Urinary embryology

Describe embryological development of urinary system organs.

UR-1.1 Describe the sequence of pronephros, mesonephros, and metanephros, and identify the metanephros, appearing in the fifth week, as the source of the permanent kidney. NPLEX · Anatomy · DOK 1

UR-1.2 Explain how reciprocal induction between the ureteric bud and the metanephric mesenchyme forms the collecting system (collecting ducts, calyces, pelvis, ureter) and the nephron (Bowman capsule to distal convoluted tubule), and predict renal agenesis when induction fails. NPLEX + USMLE · Anatomy · DOK 2

UR-1.3 Describe the ascent of the kidneys from the pelvis to the lumbar region, and explain how fusion of the lower poles produces a horseshoe kidney trapped beneath the inferior mesenteric artery. USMLE extension · Anatomy · DOK 2

UR-1.4 Explain how early splitting of the ureteric bud or formation of two buds produces a bifid or duplex ureter, and relate an ectopic ureteral orifice to obstruction or reflux. USMLE extension · Anatomy · DOK 2

UR-1.5 Trace partition of the cloaca by the urorectal septum into the urogenital sinus and anorectal canal, and predict a patent urachus (urine draining from the umbilicus) or urachal cyst from failed allantois obliteration. NPLEX · Anatomy · DOK 2

UR-1.6 Explain how posterior urethral valves in male infants obstruct urine outflow, and predict bilateral hydronephrosis, a distended bladder, and oligohydramnios. USMLE extension · Anatomy · DOK 3

UR-2 Urinary histology

Describe microscopic anatomy of urinary tract.

UR-2.1 Describe the three layers of the glomerular filtration barrier (fenestrated endothelium, basement membrane with type IV collagen and negatively charged heparan sulfate, podocyte foot processes with slit diaphragms) and predict albuminuria when charge or foot processes are lost. NPLEX + USMLE · Anatomy · DOK 2

UR-2.2 Identify the components of the juxtaglomerular apparatus (renin-secreting juxtaglomerular cells of the afferent arteriole, macula densa of the distal thick ascending limb, extraglomerular mesangial cells) and the supporting and phagocytic roles of intraglomerular mesangial cells. NPLEX · Anatomy · DOK 1

UR-2.3 Distinguish proximal tubule (cuboidal cells with brush border and abundant mitochondria), thin limb (simple squamous), distal tubule (cuboidal without brush border), and collecting duct (principal and intercalated cells) epithelium, and relate high proximal tubule energy demand to ischemic injury. NPLEX · Anatomy · DOK 2

UR-2.4 Distinguish cortical from juxtamedullary nephrons by glomerulus position, loop of Henle length, and vasa recta supply, and relate long loops to urine concentrating ability. NPLEX · Anatomy · DOK 2

UR-2.5 Describe urothelium (transitional epithelium with umbrella cells), the lamina propria, and the detrusor muscle, and explain how the urothelium stretches during bladder filling while remaining impermeable to urine. NPLEX · Anatomy · DOK 1

UR-3 Urinary organs: location and boundaries

Describe location, structure, and boundaries of urinary system.

UR-3.1 Locate the kidneys in the retroperitoneum from about T12 to L3, explain why the right kidney lies lower than the left, and identify posterior relations (diaphragm, 12th rib, psoas major, quadratus lumborum). NPLEX · Anatomy · DOK 1

UR-3.2 Identify the anterior relations of the right kidney (liver, descending duodenum, right colic flexure) and left kidney (stomach, spleen, pancreatic tail, jejunum, left colic flexure). NPLEX · Anatomy · DOK 1

UR-3.3 Describe the renal coverings (fibrous capsule, perirenal fat, renal fascia, pararenal fat) and the anterior to posterior order of hilar structures (renal vein, renal artery, renal pelvis). NPLEX · Anatomy · DOK 1

UR-3.4 Trace the ureter along the psoas, across the iliac vessels at the pelvic brim, into the bladder wall, and predict that stones lodge at the ureteropelvic junction, pelvic brim, and ureterovesical junction. NPLEX + USMLE · Anatomy · DOK 2

UR-3.5 Describe the bladder trigone, the ureteric and internal urethral orifices, and the male urethral segments (prostatic, membranous, spongy), and contrast them with the short female urethra and its higher risk of ascending infection. NPLEX · Anatomy · DOK 2

UR-3.6 Predict the site of male urethral injury from a pelvic fracture (posterior urethra at the membranous segment) versus a straddle injury (bulbous part of the spongy urethra). USMLE extension · Anatomy · DOK 3

UR-4 Urinary system in the abdominopelvic cavity

Describe location, structure, and boundaries of abdominopelvic cavity in relation to urinary system.

UR-4.1 Distinguish the retroperitoneal kidneys and ureters from the extraperitoneal bladder, and explain why dome rupture of a full bladder leaks urine into the peritoneal cavity while pelvic fracture causes extraperitoneal extravasation. USMLE extension · Anatomy · DOK 3

UR-4.2 Describe the peritoneal reflections over the bladder, including the rectovesical pouch in males and the vesicouterine and rectouterine pouches in females. NPLEX · Anatomy · DOK 1

UR-4.3 Explain why ureteral colic radiates from the flank to the groin, relating the pain to visceral afferents that enter the lower thoracic and upper lumbar spinal cord. NPLEX + USMLE · Anatomy · DOK 2

UR-4.4 Relate the posterior position of the kidney against the 12th rib and posterior abdominal wall to costovertebral angle tenderness in pyelonephritis and to the spread of a perinephric abscess within the renal fascia. NPLEX + USMLE · Anatomy · DOK 2

UR-4.5 Explain why the ureter is at risk during hysterectomy, where it passes beneath the uterine artery near the cervix, and predict hydronephrosis or urine leak after ligation or transection. USMLE extension · Anatomy · DOK 3

UR-4.6 Relate pelvic floor and urethral support to stress incontinence and cystocele, and distinguish stress, urge (detrusor overactivity), and overflow incontinence by mechanism. USMLE extension · Anatomy · DOK 2

UR-5 Renal circulation

Describe circulation of blood in urinary system.

UR-5.1 Trace renal blood flow from the renal artery through segmental, interlobar, arcuate, and cortical radiate arteries, afferent arteriole, glomerulus, efferent arteriole, and peritubular capillaries or vasa recta to the renal vein. NPLEX · Anatomy · DOK 1

UR-5.2 Explain why segmental and interlobar arteries behave as end arteries, and predict a wedge-shaped cortical infarct after embolic occlusion. NPLEX + USMLE · Anatomy · DOK 2

UR-5.3 Predict how afferent versus efferent arteriolar constriction or dilation changes renal plasma flow, GFR, and filtration fraction. NPLEX · Physiology · DOK 3

UR-5.4 Explain renal autoregulation of blood flow and GFR by the myogenic response and tubuloglomerular feedback through the macula densa. NPLEX · Physiology · DOK 2

UR-5.5 Explain how slow vasa recta flow and its countercurrent arrangement preserve medullary hypertonicity, and relate the low medullary oxygen tension to papillary necrosis and sickling in the vasa recta. NPLEX + USMLE · Physiology · DOK 2

UR-5.6 Describe the course of the left renal vein between the aorta and superior mesenteric artery and its gonadal tributary, and predict a left varicocele when renal cell carcinoma invades the left renal vein. NPLEX + USMLE · Anatomy · DOK 3

UR-6 Filtration, reabsorption, secretion, and micturition

Explain mechanisms, functions, regulation of, and factors affecting micturition and urinary filtration, reabsorption, and secretion.

UR-6.1 Explain how glomerular capillary hydrostatic and oncotic pressures and Bowman space pressure set GFR, and predict the GFR change with ureteral obstruction, hypoalbuminemia, and dehydration. NPLEX · Physiology · DOK 3

UR-6.2 Calculate clearance (urine concentration times urine flow divided by plasma concentration), GFR from inulin or creatinine, effective renal plasma flow from PAH, and filtration fraction, and interpret clearance above or below GFR as net secretion or reabsorption. NPLEX · Physiology · DOK 3

UR-6.3 Describe proximal tubule reabsorption (about two thirds of filtered sodium and water, nearly all glucose and amino acids, Na+/H+ exchange) and thick ascending limb transport (Na+/K+/2Cl- cotransport, water impermeability, paracellular Mg2+ and Ca2+ reabsorption). NPLEX · Physiology · DOK 2

UR-6.4 Explain how aldosterone acts on principal cells (ENaC, Na+/K+-ATPase, K+ secretion), ADH inserts aquaporin-2 in the collecting duct, and PTH increases distal tubule calcium reabsorption. NPLEX + USMLE Physiology DOK 2

UR-6.5 Explain countercurrent multiplication and ADH-dependent urine concentration, and distinguish central diabetes insipidus, nephrogenic diabetes insipidus, SIADH, and primary polydipsia by serum sodium, urine osmolality, and response to water deprivation and desmopressin. NPLEX + USMLE Physiology DOK 3

UR-6.6 Predict changes in extracellular and intracellular fluid volume and osmolality after isotonic, hypotonic, and hypertonic fluid gain or loss, and apply them to dehydration, hypovolemia, and volume overload. NPLEX + USMLE Physiology DOK 3

UR-6.7 Describe the micturition reflex (pelvic parasympathetic detrusor contraction, sympathetic and pudendal sphincter control, pontine micturition center), and predict a reflex spastic bladder above the sacral cord versus a flaccid overflow bladder at the sacral cord. NPLEX + USMLE Physiology DOK 3

UR-7 Kidney in acid–base and blood pressure control

Describe role of kidney in acid-base balance and regulation of blood pressure.

UR-7.1 Explain how the proximal tubule reclaims filtered bicarbonate using carbonic anhydrase and Na+/H+ exchange, and how the kidney generates new bicarbonate by excreting H+ as titratable acid (mainly phosphate) and as ammonium made from glutamine in the proximal tubule. NPLEX + USMLE Physiology DOK 2

UR-7.2 Interpret arterial blood gases to classify metabolic and respiratory acidosis and alkalosis, judge compensation (Winters formula, acute versus chronic renal compensation), and identify mixed disturbances. NPLEX + USMLE Physiology DOK 3

UR-7.3 Calculate the anion gap and distinguish high anion gap metabolic acidosis (lactic acidosis, ketoacidosis, uremia, methanol, ethylene glycol, salicylates) from normal anion gap acidosis (diarrhea, renal tubular acidosis). NPLEX + USMLE Physiology DOK 3

UR-7.4 Distinguish renal tubular acidosis type 1 (distal, urine pH above 5.5, hypokalemia, calcium phosphate stones), type 2 (proximal bicarbonate wasting, hypokalemia, Fanconi syndrome), and type 4 (hypoaldosteronism, hyperkalemia). NPLEX + USMLE Physiology DOK 3

UR-7.5 Explain how vomiting causes hypochloremic metabolic alkalosis with hypokalemia, and why volume depletion and aldosterone maintain it and produce paradoxical aciduria. NPLEX + USMLE Physiology DOK 3

UR-7.6 Trace renin release (low perfusion pressure at juxtaglomerular cells, low NaCl at the macula densa, beta-1 sympathetic input) through angiotensin II and aldosterone actions, and contrast the opposing effects of atrial natriuretic peptide. NPLEX Physiology DOK 2

UR-7.7 Explain pressure natriuresis as the kidney's long-term control of arterial pressure, and predict hypertension with hypokalemia, metabolic alkalosis, and low renin in primary hyperaldosteronism. NPLEX + USMLE Physiology DOK 3

UR-8 Urinary biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to urinary function and pathology.

UR-8.1 Explain sodium-coupled (SGLT2) glucose reabsorption in the proximal tubule with a transport maximum, and predict glucosuria and osmotic diuresis when plasma glucose exceeds the renal threshold (about 200 mg/dL) in diabetes mellitus. NPLEX Biochemistry & Genetics DOK 3

UR-8.2 Distinguish glomerular proteinuria (albumin), overflow proteinuria (Bence Jones light chains), and tubular proteinuria (failed proximal reabsorption of small filtered proteins). NPLEX + USMLE Biochemistry & Genetics DOK 2

UR-8.3 Describe PTH and low phosphate stimulation of vitamin D 1-alpha-hydroxylation in the proximal tubule, predict serum calcium, phosphate, and PTH in chronic kidney disease, primary hyperparathyroidism, and vitamin D deficiency, and explain refractory hypocalcemia in hypomagnesemia. NPLEX + USMLE Biochemistry & Genetics DOK 3

UR-8.4 Explain potassium shifts into cells (insulin, beta-adrenergic stimulation, alkalosis) and out of cells (acidosis, cell lysis, insulin deficiency, hyperosmolality), and predict ECG findings of hyperkalemia (peaked T waves, wide QRS) and hypokalemia (U waves). NPLEX + USMLE Biochemistry & Genetics DOK 3

UR-8.5 Describe erythropoietin release by renal peritubular interstitial cells in response to hypoxia, and relate it to normocytic anemia in chronic kidney disease and secondary polycythemia in renal cell carcinoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

UR-8.6 Explain how NAD+-dependent alcohol dehydrogenase starts the conversion of ethylene glycol to glycolic and oxalic acids, producing high anion gap metabolic acidosis, an osmolal gap, calcium oxalate crystals in urine, and acute tubular injury. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

UR-8.7 Explain the hyperlipidemia of nephrotic syndrome (increased hepatic lipoprotein synthesis and reduced lipoprotein catabolism) and identify lipiduria as oval fat bodies and fatty casts showing Maltese cross birefringence under polarized light. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

UR-9 Urinary genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying urinary disease.

UR-9.1 Explain how autosomal dominant PKD1 (polycystin-1, chromosome 16) and PKD2 (polycystin-2, chromosome 4) mutations and autosomal recessive PKHD1 (fibrocystin) mutations cause polycystic kidney disease, and predict inheritance in a pedigree. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

UR-9.2 Explain how mutations in the alpha-3, alpha-4, or alpha-5 chains of type IV collagen (most often X-linked COL4A5) cause Alport syndrome, and how heterozygous COL4A3 or COL4A4 mutations cause thin basement membrane disease. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

UR-9.3 Explain two-hit loss of the VHL tumor suppressor on chromosome 3p, with HIF stabilization driving VEGF expression, in sporadic clear cell renal carcinoma and von Hippel-Lindau syndrome (bilateral renal carcinoma, hemangioblastomas, pheochromocytoma). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

UR-9.4 Relate WT1 loss on chromosome 11p13 to Wilms tumor in WAGR and Denys-Drash syndromes, and 11p15 imprinting defects to Wilms tumor in Beckwith-Wiedemann syndrome. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

UR-9.5 Explain how TSC1 or TSC2 mutations in tuberous sclerosis remove hamartin and tuberin inhibition of mTOR growth signaling, and relate this to renal angiomyolipomas (blood vessels, smooth muscle, fat) and renal cysts. USMLE extension | Biochemistry & Genetics | DOK 2

UR-9.6 Explain how an autosomal recessive defect in the proximal tubule transporter for cystine, ornithine, lysine, and arginine causes cystinuria with hexagonal cystine stones and a positive cyanide-nitroprusside test. USMLE extension | Biochemistry & Genetics | DOK 2

UR-9.7 Explain how X-linked alpha-galactosidase A deficiency in Fabry disease causes globotriaosylceramide accumulation, with angiokeratomas, painful neuropathy, hypohidrosis, and progressive renal failure. USMLE extension | Biochemistry & Genetics | DOK 2

UR-10 Immune responses in urinary disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

UR-10.1 Classify glomerular injury as type II hypersensitivity (anti-GBM antibodies to the alpha-3 chain of type IV collagen, linear IgG, with lung hemorrhage in Goodpasture syndrome), type III immune complex disease (granular deposits), or pauci-immune ANCA disease, and name an example of each. NPLEX + USMLE | Microbiology & Immunology | DOK 3

UR-10.2 Interpret serum complement in glomerular disease: low C3 in poststreptococcal and membranoproliferative glomerulonephritis, low C3 and C4 in lupus nephritis, and normal levels in IgA nephropathy, anti-GBM disease, and ANCA vasculitis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

UR-10.3 Distinguish granulomatosis with polyangiitis (PR3–ANCA, necrotizing granulomas of the upper and lower airways, glomerulonephritis) from microscopic polyangiitis (MPO-ANCA, no granulomas), and contrast polyarteritis nodosa, which spares glomeruli and causes renal artery microaneurysms. NPLEX + USMLE | Microbiology & Immunology | DOK 3

UR-10.4 Describe IgA vasculitis in children (IgA immune complex deposition after an upper respiratory infection, palpable purpura on the buttocks and legs, arthralgia, abdominal pain, intussusception) and its mesangial IgA glomerulonephritis. NPLEX + USMLE | Microbiology & Immunology | DOK 2

UR-10.5 Describe urinary tract defenses (one-way urine flow, complete bladder emptying, the flap-valve ureterovesical junction, urothelial shedding) and predict how obstruction, reflux, catheters, and pregnancy increase infection risk. NPLEX + USMLE | Microbiology & Immunology | DOK 2

UR-10.6 Explain how CD8+ T cells recognize renal cell carcinoma antigens on MHC class I and NK cells kill tumor cells that lose MHC class I, and how tumors evade immunity through PD-L1 expression and MHC class I downregulation. NPLEX | Microbiology & Immunology | DOK 2

UR-10.7 Explain how the recombinant hepatitis B surface antigen vaccine induces protective anti-HBs antibodies, interpret anti-HBs alone as vaccine immunity, and relate vaccination to prevention of hepatitis B associated membranous nephropathy. NPLEX | Microbiology & Immunology | DOK 2

UR-11 Urinary pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

UR-11.1 Describe uropathogenic Escherichia coli (gram-negative lactose-fermenting rod from fecal flora) as the most common urinary pathogen, and relate type 1 fimbriae to cystitis and P fimbriae to pyelonephritis. NPLEX + USMLE | Microbiology & Immunology | DOK 2

UR-11.2 Identify Staphylococcus saprophyticus (coagulase-negative, novobiocin-resistant, young sexually active women), Klebsiella (mucoid capsule), Enterococcus, and Pseudomonas (catheter-associated) as other urinary pathogens. NPLEX + USMLE | Microbiology & Immunology | DOK 1

UR-11.3 Explain how Proteus mirabilis urease splits urea to ammonia, raising urine pH and precipitating magnesium ammonium phosphate (struvite) staghorn calculi, and relate swarming motility to its identification. NPLEX + USMLE | Microbiology & Immunology | DOK 2

UR-11.4 Interpret urine dipstick and microscopy in infection: leukocyte esterase for pyuria, nitrite from Enterobacterales nitrate reductase, WBC casts for kidney involvement, and sterile pyuria suggesting urethritis or renal tuberculosis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

UR-11.5 Compare Neisseria gonorrhoeae (gram-negative diplococcus, pili with antigenic variation, IgA protease) and Chlamydia trachomatis serovars D to K (obligate intracellular, elementary and reticulate bodies) as causes of urethritis, and identify nucleic acid amplification as the diagnostic test. NPLEX + USMLE | Microbiology & Immunology | DOK 2

UR-11.6 Relate infections to glomerular disease: nephritogenic group A streptococci to poststreptococcal glomerulonephritis, hepatitis B to membranous nephropathy, hepatitis C to membranoproliferative glomerulonephritis with cryoglobulinemia, and HIV to collapsing FSGS. NPLEX + USMLE | Microbiology & Immunology | DOK 3

UR-11.7 Describe hematogenous renal tuberculosis with caseating granulomas and sterile pyuria, and the Schistosoma haematobium life cycle (snail host, skin-penetrating cercariae, adults in vesical venules, terminal-spined eggs) leading to hematuria and bladder squamous cell carcinoma. NPLEX + USMLE | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

UR-12-A Glomerular diseases

1 Glomerulonephritis **2** Glomerulosclerosis **3** Nephrotic syndromes **4** Renal failure (acute and chronic)

UR-12-A.1 Distinguish nephritic syndrome (hematuria with RBC casts, hypertension, oliguria, azotemia, proteinuria under 3.5 g/day) from nephrotic syndrome (proteinuria over 3.5 g/day, hypoalbuminemia, edema, hyperlipidemia, fatty casts). NPLEX + USMLE | Pathology | DOK 3

UR-12-A.2 Interpret urinary findings to localize disease: RBC casts and dysmorphic RBCs (glomerulonephritis), WBC casts (pyelonephritis, interstitial nephritis), muddy brown granular casts (acute tubular necrosis), fatty casts (nephrotic syndrome), and waxy casts (chronic kidney disease). NPLEX + USMLE | Pathology | DOK 3

UR-12-A.3 Describe poststreptococcal glomerulonephritis: onset weeks after streptococcal pharyngitis or impetigo, cola-colored urine, periorbital edema, hypertension, low C3, granular IgG and C3 deposits, and subepithelial humps on electron microscopy. NPLEX + USMLE | Pathology | DOK 2

UR-12-A.4 Distinguish IgA nephropathy (episodic gross hematuria within days of a mucosal infection, mesangial IgA deposits) from poststreptococcal glomerulonephritis by timing of hematuria after infection and complement level. NPLEX + USMLE | Pathology | DOK 3

UR-12-A.5 Classify rapidly progressive (crescentic) glomerulonephritis by immunofluorescence into anti-GBM (linear), immune complex (granular), and pauci-immune (ANCA) types, and identify crescents of fibrin, parietal epithelial cells, and macrophages. NPLEX + USMLE | Pathology | DOK 3

UR-12-A.6 Describe lupus nephritis, with diffuse proliferative disease (subendothelial wire-loop deposits, nephritic picture) as the most common and most severe class and membranous disease causing nephrotic syndrome, and interpret low complement and anti-dsDNA antibodies. NPLEX + USMLE Pathology DOK 3

UR-12-A.7 Describe membranoproliferative glomerulonephritis (mesangial proliferation, tram-track basement membrane duplication), with immune complex disease linked to hepatitis B and C and dense deposit disease (a C3 glomerulopathy) linked to C3 nephritic factor. NPLEX + USMLE Pathology DOK 2

UR-12-A.8 Explain diabetic nephropathy: hyperfiltration, nonenzymatic glycation, and hyaline arteriolosclerosis of afferent and efferent arterioles lead to basement membrane thickening and mesangial expansion with Kimmelstiel-Wilson nodules, progressing from microalbuminuria to nephrotic proteinuria and renal failure. NPLEX Pathology DOK 3

UR-12-A.9 Describe focal segmental glomerulosclerosis (segmental sclerosis with podocyte injury and poor steroid response) and its secondary causes (HIV, heroin use, sickle cell disease, obesity), including the collapsing variant of HIV-associated nephropathy. NPLEX + USMLE Pathology DOK 2

UR-12-A.10 Explain the consequences of nephrotic syndrome: edema from hypoalbuminemia, hyperlipidemia from increased hepatic lipoprotein synthesis, hypercoagulability from urinary antithrombin III loss (renal vein thrombosis), and infection from immunoglobulin loss. NPLEX + USMLE Pathology DOK 3

UR-12-A.11 Describe minimal change disease in children: selective albuminuria, normal light microscopy and immunofluorescence, podocyte foot process effacement on electron microscopy, excellent corticosteroid response, and association with Hodgkin lymphoma. NPLEX + USMLE Pathology DOK 2

UR-12-A.12 Describe membranous nephropathy in adults: subepithelial immune deposits with spike and dome basement membrane, anti-PLA2R antibodies in primary disease, and secondary causes (lupus, hepatitis B, solid tumors, drugs such as NSAIDs). NPLEX + USMLE Pathology DOK 2

UR-12-A.13 Explain how amyloid deposition in glomeruli causes nephrotic syndrome, and identify AL (light chain) and AA (chronic inflammation) amyloid by apple-green birefringence with Congo red stain under polarized light. NPLEX + USMLE Pathology DOK 2

UR-12-A.14 Classify acute kidney injury as prerenal, intrinsic, or postrenal, and interpret BUN to creatinine ratio (over 20 prerenal), fractional excretion of sodium (under 1 percent prerenal, over 2 percent tubular necrosis), and urine osmolality. NPLEX + USMLE Pathology DOK 3

UR-12-A.15 Identify diabetes and hypertension as the leading causes of chronic kidney disease, and predict its consequences: uremia (pericarditis, encephalopathy, platelet dysfunction), hyperkalemia, metabolic acidosis, volume overload, hypertension, anemia, and renal osteodystrophy. NPLEX + USMLE Pathology DOK 3

UR-12-B Tubulointerstitial disease

1 Tubular necrosis

UR-12-B.1 Explain ischemic and nephrotoxic acute tubular necrosis, with injury concentrated in the proximal tubule and thick ascending limb, tubular cell sloughing that obstructs flow, and muddy brown granular casts. NPLEX + USMLE Pathology DOK 3

UR-12-B.2 Describe the initiation, maintenance (oliguria, hyperkalemia, metabolic acidosis, volume overload), and recovery (polyuria with potassium and sodium loss) phases of acute tubular necrosis. NPLEX + USMLE Pathology DOK 2

UR-12-B.3 Identify nephrotoxic causes of tubular necrosis (myoglobin from rhabdomyolysis, hemoglobin, aminoglycosides, iodinated contrast, cisplatin, ethylene glycol) and predict a dipstick positive for blood without RBCs on microscopy in rhabdomyolysis. NPLEX + USMLE Pathology DOK 3

UR-12-B.4 Describe Fanconi syndrome as generalized proximal tubule dysfunction with glucosuria despite normal plasma glucose, aminoaciduria, phosphaturia, and bicarbonaturia (proximal renal tubular acidosis), and identify causes such as cystinosis, Wilson disease, and multiple myeloma. USMLE extension Pathology DOK 2

UR-12-C Obstructive urinary diseases

1 Hydronephrosis **2** Renal calculi

UR-12-C.1 Identify causes of hydronephrosis by age and sex (posterior urethral valves, ureteropelvic junction obstruction, calculi, pregnancy, prostatic enlargement, pelvic tumors), and explain how persistent obstruction lowers GFR and causes papillary and cortical atrophy. NPLEX + USMLE Pathology DOK 3

UR-12-C.2 Predict that unilateral hydronephrosis may stay silent while bilateral obstruction causes postrenal acute kidney injury with oliguria or anuria, and explain postobstructive diuresis after relief. NPLEX + USMLE Pathology DOK 3

UR-12-C.3 Classify kidney stones by composition and urine pH: calcium oxalate (most common, hypercalciuria), calcium phosphate (alkaline urine), struvite (urease-producing bacteria, staghorn), uric acid (acidic urine, radiolucent, gout, tumor lysis), and cystine (hexagonal crystals). NPLEX + USMLE Pathology DOK 3

UR-12-C.4 Describe renal colic (colicky flank pain radiating to the groin with hematuria) and the complications of calculi, including obstruction, hydronephrosis, and infection above the stone. NPLEX + USMLE Pathology DOK 2

UR-12-D Inflammatory urinary tract diseases

1 Drug-induced nephritis **2** Chronic pyelonephritis

UR-12-D.1 Describe drug-induced acute interstitial nephritis as a hypersensitivity reaction days to weeks after drugs such as penicillins, NSAIDs, proton pump inhibitors, and diuretics, presenting with fever, rash, eosinophilia, pyuria with WBC casts, and rising creatinine. NPLEX + USMLE Pathology DOK 3

UR-12-D.2 Explain chronic analgesic nephropathy, with long-term combination analgesic use causing chronic interstitial nephritis and papillary necrosis, and identify other causes of papillary necrosis (diabetes, sickle cell disease, severe pyelonephritis, obstruction). NPLEX + USMLE Pathology DOK 3

UR-12-D.3 Explain how vesicoureteral reflux or obstruction leads to chronic pyelonephritis, with coarse asymmetric corticomedullary scars over blunted calyces and thyroidization of tubules filled with eosinophilic casts. NPLEX + USMLE Pathology DOK 3

UR-12-D.4 Describe xanthogranulomatous pyelonephritis as a chronic destructive infection, usually with obstructing stones and Proteus, filled with lipid-laden foamy macrophages that can mimic renal cell carcinoma. NPLEX + USMLE Pathology DOK 2

UR-12-D.5 Describe interstitial cystitis (bladder pain syndrome) as chronic pelvic pain with urgency and frequency in the absence of infection, most often in women, with Hunner ulcers in some patients. USMLE extension Pathology DOK 1

UR-12-E Congenital urinary disease

1 Alport syndrome **2** Cystic renal disease **3** Renal agenesis **4** Vesicoureteral reflux (VUR)

UR-12-E.1 Describe Alport syndrome: hematuria progressing to renal failure, sensorineural hearing loss, and anterior lenticonus, with thinning and splitting (basket-weave) of the glomerular basement membrane on electron microscopy, and more severe disease in males with X-linked inheritance. NPLEX + USMLE Pathology DOK 2

UR-12-E.2 Describe autosomal dominant polycystic kidney disease in adults: bilaterally enlarged cystic kidneys, flank pain, hematuria, hypertension, progressive renal failure, hepatic cysts, berry aneurysms with subarachnoid hemorrhage, and mitral valve prolapse. NPLEX + USMLE Pathology DOK 2

UR-12-E.3 Distinguish autosomal recessive polycystic kidney disease (bilateral enlarged kidneys in infancy, Potter sequence, congenital hepatic fibrosis with portal hypertension) from multicystic dysplastic kidney (usually unilateral, noninherited) and acquired cystic disease in long-term dialysis (renal carcinoma risk). NPLEX + USMLE Pathology DOK 3

UR-12-E.4 Explain how bilateral renal agenesis causes oligohydramnios and Potter sequence (pulmonary hypoplasia, flattened facies, limb deformities), while unilateral agenesis or a single kidney leads to compensatory hypertrophy of the other kidney. NPLEX + USMLE Pathology DOK 3

UR-12-E.5 Explain how a short intravesical ureteral segment causes primary vesicoureteral reflux, predict recurrent pyelonephritis and reflux nephropathy scarring, and identify the voiding cystourethrogram as the diagnostic study. NPLEX + USMLE Pathology DOK 3

UR-12-F Urinary vascular diseases

1 Hemolytic uremic syndrome (HUS) **2** Hypertensive nephrosclerosis **3** Renal artery stenosis **4** Renal infarction **5** Sickle cell nephropathy

UR-12-F.1 Explain how Shiga toxin from enterohemorrhagic E. coli O157:H7 damages endothelium in hemolytic uremic syndrome, causing microangiopathic hemolytic anemia with schistocytes, thrombocytopenia, and acute kidney injury after bloody diarrhea in children, and contrast TTP (ADAMTS13 deficiency). NPLEX Pathology DOK 3

UR-12-F.2 Distinguish benign nephrosclerosis (hyaline arteriolosclerosis, symmetrically shrunken kidneys with finely granular surfaces) from malignant hypertension (hyperplastic onion-skin arteriolosclerosis, fibrinoid necrosis, flea-bitten kidney, acute renal failure). NPLEX + USMLE Pathology DOK 3

UR-12-F.3 Distinguish atherosclerotic renal artery stenosis (older adults, ostial or proximal lesion) from fibromuscular dysplasia (young women, string of beads), and explain renin-driven hypertension, an abdominal bruit, and atrophy of the affected kidney. NPLEX + USMLE Pathology DOK 3

UR-12-F.4 Describe renal infarction from emboli (atrial fibrillation, mural thrombus, endocarditis) or arterial thrombosis, producing a pale wedge-shaped coagulative infarct with flank pain and hematuria. NPLEX + USMLE Pathology DOK 2

UR-12-F.5 Explain how the hypoxic, hypertonic, acidic renal medulla promotes sickling in the vasa recta, and predict isosthenuria, hematuria, and papillary necrosis in sickle cell disease and sickle cell trait. NPLEX Pathology DOK 3

UR-12-F.6 Identify causes of renal vein thrombosis (nephrotic syndrome, especially membranous nephropathy; renal cell carcinoma invading the vein; dehydration in infants) and predict flank pain, hematuria, and an enlarged kidney. USMLE extension Pathology DOK 2

UR-12-G Neoplasms of the urinary tract

1 Renal cell carcinoma **2** Nephroblastoma (Wilms tumor)

UR-12-G.1 Describe clear cell renal cell carcinoma: proximal tubule origin, golden-yellow cut surface, lipid- and glycogen-filled clear cells, risk factors (smoking, obesity, hypertension, acquired cystic disease), and spread into the renal vein and inferior vena cava. NPLEX + USMLE Pathology DOK 2

UR-12-G.2 Predict renal cell carcinoma presentations: hematuria, flank pain, and a flank mass, plus paraneoplastic polycythemia (erythropoietin), hypercalcemia (PTHrP), and hypertension (renin), and hematogenous spread to lung and bone. NPLEX + USMLE Pathology DOK 3

UR-12-G.3 Distinguish papillary renal cell carcinoma (trisomy 7 and 17, papillae with foamy macrophages) and chromophobe carcinoma (intercalated cell origin) from clear cell carcinoma, and identify oncocytoma as a benign mitochondria-rich tumor. NPLEX + USMLE Pathology DOK 2

UR-12-G.4 Describe Wilms tumor in children aged 2 to 5 years: a large unilateral flank mass, often with hematuria or hypertension, triphasic histology (blastema, stroma, epithelium), and associated syndromes (WAGR, Denys-Drash, Beckwith-Wiedemann). NPLEX + USMLE Pathology DOK 2

UR-12-G.5 Describe urothelial carcinoma of the bladder and collecting system: painless hematuria, multifocal recurrence, flat carcinoma in situ versus papillary tumors, and risk factors (smoking, aromatic amine dyes, cyclophosphamide, phenacetin). USMLE extension Pathology DOK 2

UR-12-G.6 Relate bladder squamous cell carcinoma to Schistosoma haematobium infection and chronic irritation, and bladder adenocarcinoma to urachal remnants and bladder exstrophy. USMLE extension Pathology DOK 2

UR-12-H Infectious urinary diseases

1 Acute pyelonephritis **2** Cystitis **3** Urethritis

UR-12-H.1 Describe acute pyelonephritis: mostly ascending infection, fever, flank pain, costovertebral angle tenderness, WBC casts, and neutrophilic infiltration of the interstitium and tubules, with complications of papillary necrosis, perinephric abscess, pyonephrosis, and sepsis. NPLEX + USMLE Pathology DOK 3

UR-12-H.2 Identify risk factors for cystitis (female sex, sexual intercourse, pregnancy, catheters, diabetes, obstruction) and distinguish cystitis (dysuria, frequency, urgency, suprapubic pain, no fever) from pyelonephritis. NPLEX + USMLE Pathology DOK 3

UR-12-H.3 Distinguish gonococcal urethritis (purulent discharge, gram-negative intracellular diplococci) from nongonococcal urethritis (Chlamydia trachomatis, scant discharge), and predict complications such as epididymitis, pelvic inflammatory disease, and reactive arthritis. NPLEX + USMLE Pathology DOK 3

Endo Endocrine System

8% · 16 questions · Week 7

HOW THE BODY WORKS

EN-1 Endocrine embryology

Describe embryological development of endocrine system organs.

EN-1.1 Trace thyroid development from an endodermal diverticulum at the foramen cecum of the tongue, its descent along the thyroglossal duct anterior to the hyoid bone, and relate failed descent to a lingual thyroid. NPLEX Anatomy DOK 2

EN-1.2 Identify the pharyngeal pouch origins of the inferior parathyroids (third pouch, with the thymus), superior parathyroids (fourth pouch), and parafollicular C cells (ultimobranchial body, neural crest), and predict ectopic inferior parathyroid locations in the mediastinum. NPLEX Anatomy DOK 2

EN-1.3 Distinguish the origin of the anterior pituitary (Rathke pouch, oral ectoderm) from the posterior pituitary (infundibulum, diencephalic neuroectoderm), and relate Rathke pouch remnants to craniopharyngioma. NPLEX Anatomy DOK 2

EN-1.4 Compare the mesodermal origin of the adrenal cortex with the neural crest origin of the adrenal medulla, and relate the medulla's neural crest origin to pheochromocytoma and neuroblastoma. NPLEX Anatomy DOK 2

EN-1.5 Explain how failed development of the third and fourth pharyngeal pouches in 22q11.2 deletion (DiGeorge) syndrome produces neonatal hypocalcemic tetany together with thymic hypoplasia and T cell deficiency. NPLEX Anatomy DOK 3

EN-1.6 Describe development of the pancreatic islets from foregut endoderm of the dorsal and ventral pancreatic buds. NPLEX Anatomy DOK 1

EN-2 Endocrine histology and derivations

Describe microscopic anatomy and derivations of endocrine organs.

EN-2.1 Identify thyroid follicles (cuboidal follicular cells surrounding colloid) and parafollicular C cells, and relate tall columnar cells with scalloped, scant colloid to an actively stimulated gland, as in Graves disease. NPLEX Anatomy DOK 2

EN-2.2 Distinguish the zona glomerulosa, zona fasciculata, zona reticularis, and medulla of the adrenal gland by histology and principal product (aldosterone, cortisol, adrenal androgens, catecholamines from chromaffin cells). NPLEX Anatomy DOK 1

EN-2.3 Distinguish anterior pituitary acidophils (GH, prolactin), basophils (FSH, LH, ACTH, TSH), and chromophobes from the posterior pituitary, which contains pituicytes and Herring bodies formed by axon terminals of hypothalamic neurons. NPLEX Anatomy DOK 2

EN-2.4 Identify parathyroid chief cells (PTH secretion) and oxyphil cells, and identify pancreatic islet alpha, beta, and delta cells by their hormones (glucagon, insulin, somatostatin). NPLEX Anatomy DOK 1

EN-2.5 Classify endocrine cells by derivation (neural crest: chromaffin cells and C cells; endoderm: thyroid follicular cells, parathyroids, islets; ectoderm: adenohypophysis) and relate neuroendocrine origin to chromogranin and synaptophysin staining of neuroendocrine tumors. NPLEX Anatomy DOK 2

EN-3 Endocrine organ location and structure

Describe location and structure of endocrine organs.

EN-3.1 Locate the thyroid lobes and the isthmus anterior to the second and third tracheal rings, and explain why the gland, enclosed in pretracheal fascia, moves upward on swallowing. NPLEX Anatomy DOK 1

EN-3.2 Describe the relationship of the recurrent laryngeal nerve (tracheoesophageal groove) and external branch of the superior laryngeal nerve (with the superior thyroid artery) to the thyroid, and predict voice changes after injury during thyroidectomy. NPLEX Anatomy DOK 3

EN-3.3 Locate the four parathyroid glands on the posterior surface of the thyroid lobes, and explain why their variable position, especially of the inferior glands, matters in thyroid and parathyroid surgery. NPLEX Anatomy DOK 2

EN-3.4 Locate the pituitary in the sella turcica beneath the diaphragma sellae, relate it to the optic chiasm above and cavernous sinuses laterally, and predict bitemporal hemianopia from suprasellar tumor extension. NPLEX Anatomy DOK 3

EN-3.5 Locate the retroperitoneal adrenal glands within the renal fascia superomedial to the kidneys, contrasting the pyramidal right gland behind the inferior vena cava with the crescent-shaped left gland along the medial border of the left kidney. NPLEX Anatomy DOK 1

EN-4 Endocrine blood supply

Describe location, structure, and circulatory pathways of blood to endocrine organs.

EN-4.1 Trace the thyroid's arterial supply (superior thyroid artery from the external carotid, inferior thyroid artery from the thyrocervical trunk, occasional thyroid ima artery) and venous drainage (superior and middle to internal jugular, inferior to brachiocephalic). NPLEX Anatomy DOK 1

EN-4.2 Trace the hypothalamo-hypophyseal portal system from superior hypophyseal arteries through the median eminence plexus and portal veins to the anterior pituitary, and explain why the anterior lobe is vulnerable to ischemic necrosis. NPLEX Anatomy DOK 2

EN-4.3 Trace the adrenal arteries (superior from inferior phrenic, middle from aorta, inferior from renal) and veins (right to inferior vena cava, left to left renal vein), and explain how cortical blood reaching the medulla induces PNMT for epinephrine synthesis. NPLEX Anatomy DOK 2

EN-4.4 Explain why the parathyroids depend mainly on the inferior thyroid artery, and predict hypocalcemia after thyroidectomy from devascularization or removal of parathyroid tissue. NPLEX Anatomy DOK 3

EN-4.5 Explain why insulin and glucagon draining into the hepatic portal vein reach the liver at high concentration first, and relate this to hepatic first-pass extraction of insulin. NPLEX Physiology DOK 2

EN-5 Endocrine organ function and control

Explain mechanisms, factors affecting, functions, and control of endocrine organs.

EN-5.1 Match each hypothalamic hormone (TRH, CRH, GnRH, GHRH, somatostatin, dopamine) to its anterior pituitary target, and predict the effect of pituitary stalk section (rising prolactin, falling other anterior pituitary hormones, diabetes insipidus). NPLEX Physiology DOK 3

EN-5.2 Explain control of ADH release by plasma osmolality (hypothalamic osmoreceptors) and blood volume (baroreceptors), and its action on V2 receptors inserting aquaporin-2 channels into collecting duct principal cells. NPLEX Physiology DOK 2

EN-5.3 Explain calcium homeostasis: calcium-sensing receptor control of PTH, PTH effects on bone resorption, distal tubule calcium reabsorption, proximal tubule phosphate excretion, and renal 1-alpha-hydroxylase, and the opposing action of calcitonin. NPLEX + USMLE Physiology DOK 2

EN-5.4 Explain regulation of aldosterone by angiotensin II and hyperkalemia (ACTH having a minor role), and its collecting duct actions: sodium reabsorption (ENaC, Na/K-ATPase) and potassium secretion by principal cells, and hydrogen ion secretion by alpha intercalated cells. NPLEX Physiology DOK 2

EN-5.5 Trace glucose-stimulated insulin secretion (GLUT2 entry, glucokinase, rising ATP closing K-ATP channels, depolarization, calcium influx, exocytosis), and explain why oral glucose releases more insulin than intravenous glucose through incretins (GLP-1, GIP). NPLEX Physiology DOK 2

EN-5.6 Explain glucose counterregulation during fasting and hypoglycemia, comparing the rapid effects of glucagon and epinephrine with the slower effects of cortisol and growth hormone. NPLEX Physiology DOK 2

EN-6 Hormone action, transport, and feedback

Explain composition, function, effects, transport, and regulation of endocrine hormones, including feedback mechanisms.

EN-6.1 Classify hormones by signaling pathway: Gs/cAMP (ACTH, TSH, LH, FSH, PTH, glucagon), Gq/IP3–calcium (GnRH, TRH, oxytocin), intrinsic tyrosine kinase (insulin, IGF-1), JAK/STAT (GH, prolactin), and intracellular receptors (steroids, thyroid hormone, calcitriol). NPLEX Physiology DOK 2

EN-6.2 Explain plasma transport of thyroid hormone (thyroxine-binding globulin) and cortisol (cortisol-binding globulin), why only free hormone is active, and predict raised total T4 with normal free T4 when binding globulin rises in pregnancy. NPLEX Physiology DOK 3

EN-6.3 Interpret paired trophic and target hormone levels (for example, high cortisol with low ACTH, or low free T4 with high TSH) to localize an endocrine disorder to the target gland, pituitary, or hypothalamus by negative feedback logic. NPLEX Physiology DOK 3

EN-6.4 Describe actions of thyroid hormone (raised basal metabolic rate and Na/K-ATPase, beta-1 adrenergic receptor upregulation, bone growth, CNS maturation) and its activation by peripheral deiodinase conversion of T4 to T3. NPLEX Physiology DOK 2

EN-6.5 Explain direct GH actions (lipolysis, insulin antagonism) and indirect actions through hepatic IGF-1 (linear growth, protein synthesis), and GH regulation by GHRH, somatostatin, IGF-1 feedback, sleep, exercise, and hypoglycemia. NPLEX Physiology DOK 2

EN-6.6 Explain cortisol's effects (gluconeogenesis, proteolysis, lipolysis, anti-inflammatory and immunosuppressive actions, permissive effect on catecholamine vasoconstriction, reduced collagen synthesis) and the morning peak of ACTH and cortisol secretion. NPLEX Physiology DOK 2

EN-6.7 Explain tonic dopamine inhibition of prolactin secretion and TRH stimulation of prolactin, and predict hypogonadism from hyperprolactinemia through suppression of GnRH. NPLEX + USMLE Physiology DOK 3

EN-7 Endocrine biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to endocrine function and pathology.

EN-7.1 Contrast insulin's effects (GLUT4 uptake in muscle and adipose, glycogen synthesis, glycolysis, lipogenesis, protein synthesis) with glucagon's effects (glycogenolysis, gluconeogenesis, ketogenesis) on fuel metabolism in the fed and fasting states. NPLEX Biochemistry & Genetics DOK 2

EN-7.2 Explain how insulin deficiency with glucagon excess increases lipolysis, hepatic fatty acid oxidation, and conversion of acetyl-CoA to acetoacetate and beta-hydroxybutyrate, producing an anion gap metabolic acidosis. NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-7.3 Relate dietary iodine to thyroid hormone synthesis and selenium to the deiodinases that convert T4 to T3, and explain how iodine deficiency raises TSH and causes goiter. NPLEX Biochemistry & Genetics DOK 2

EN-7.4 Trace vitamin D activation (UVB conversion of 7-dehydrocholesterol in skin, hepatic 25-hydroxylation, renal 1-alpha-hydroxylation stimulated by PTH and low phosphate) and predict rickets, osteomalacia, and secondary hyperparathyroidism from deficiency. NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-7.5 Explain how hypomagnesemia impairs PTH secretion and end-organ PTH action, producing hypocalcemia that does not correct until magnesium is replaced. NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-7.6 Explain how leptin from adipocytes (satiety) and ghrelin from the stomach (hunger) act on the hypothalamic arcuate nucleus to regulate appetite, and relate leptin resistance to common obesity and visceral adiposity to insulin resistance. NPLEX + USMLE Biochemistry & Genetics DOK 2

EN-8 Hormone synthesis and degradation

Explain biochemistry of synthesis and degradation of hormones in the endocrine system.

EN-8.1 Trace thyroid hormone synthesis: iodide uptake by the sodium-iodide symporter, oxidation and organification by thyroid peroxidase onto thyroglobulin tyrosines, coupling of MIT and DIT to T3 and T4, and release by thyroglobulin proteolysis. NPLEX Biochemistry & Genetics DOK 2

EN-8.2 Trace adrenal steroid synthesis from cholesterol (StAR transport, side-chain cleavage) through 17-alpha-hydroxylase, 21-hydroxylase, and 11-beta-hydroxylase to aldosterone, cortisol, and androgens, and predict which precursors accumulate when each enzyme is blocked. NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-8.3 Trace catecholamine synthesis from tyrosine (tyrosine hydroxylase as rate-limiting step, DOPA decarboxylase, dopamine beta-hydroxylase, PNMT) and breakdown by COMT and MAO to metanephrines and vanillylmandelic acid, and relate these metabolites to pheochromocytoma testing. NPLEX + USMLE Biochemistry & Genetics DOK 2

EN-8.4 Describe processing of preproinsulin to proinsulin and then insulin plus C-peptide, and use C-peptide to distinguish endogenous hyperinsulinism (insulinoma, sulfonylurea ingestion) from exogenous insulin as the cause of hypoglycemia. NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-8.5 Describe synthesis of ADH and oxytocin in the supraoptic and paraventricular nuclei, their axonal transport bound to neurophysins, and storage and release from the posterior pituitary. NPLEX Biochemistry & Genetics DOK 1

EN-8.6 Explain how renal 11-beta-hydroxysteroid dehydrogenase type 2 inactivates cortisol to cortisone to protect mineralocorticoid receptors, and predict hypertension with hypokalemia when it is inhibited by licorice (glycyrrhetinic acid). NPLEX + USMLE Biochemistry & Genetics DOK 3

EN-9 Endocrine genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying endocrine disease.

EN-9.1 Contrast loss-of-function mutation of the MEN1 tumor suppressor gene (menin, with a second hit needed in tumor cells) with gain-of-function mutation of the RET proto-oncogene in MEN2, and explain the autosomal dominant inheritance of both syndromes. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

EN-9.2 Explain the autosomal recessive inheritance of congenital adrenal hyperplasia, with CYP21A2 (21-hydroxylase) mutations as the most common cause. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

EN-9.3 Compare the genetics of type 1 diabetes (HLA-DR3 and DR4 association), type 2 diabetes (polygenic, with higher monozygotic twin concordance than type 1), and maturity-onset diabetes of the young (autosomal dominant glucokinase or HNF1A mutations). NPLEX + USMLE | Biochemistry & Genetics | DOK 2

EN-9.4 Relate thyroid cancer types to their characteristic mutations: BRAF V600E and RET/PTC rearrangements in papillary carcinoma, RAS mutations and PAX8–PPARG fusion in follicular carcinoma, and RET mutations in medullary carcinoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

EN-9.5 Distinguish androgen insensitivity syndrome (X-linked androgen receptor defect, 46,XY, female external genitalia, testes, no uterus, high testosterone and LH) from 5-alpha-reductase deficiency (low DHT, ambiguous genitalia that virilize at puberty). USMLE extension | Biochemistry & Genetics | DOK 3

EN-9.6 Distinguish primary hypogonadism (for example, Klinefelter syndrome 47,XXY, with high LH and FSH) from secondary hypogonadism (for example, Kallmann syndrome, failed GnRH neuron migration with anosmia, with low LH and FSH). USMLE extension | Biochemistry & Genetics | DOK 3

EN-9.7 Explain how steroid and thyroid hormones bind intracellular receptors that act as transcription factors at hormone response elements in DNA (the thyroid hormone receptor pairing with the retinoid X receptor), and relate this to their slower onset than peptide hormones. NPLEX | Biochemistry & Genetics | DOK 2

EN-10 Immune responses in endocrine disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

EN-10.1 Explain Graves disease as a type II hypersensitivity in which thyroid-stimulating immunoglobulins activate the TSH receptor, and predict transient neonatal thyrotoxicosis from transplacental IgG. NPLEX + USMLE | Microbiology & Immunology | DOK 3

EN-10.2 Describe the immune mechanisms of Hashimoto thyroiditis (T cell mediated follicular cell destruction, antibodies against thyroid peroxidase and thyroglobulin), relate these antibodies to diagnosis, and relate chronic lymphoid stimulation to risk of thyroid B cell lymphoma. NPLEX + USMLE | Microbiology & Immunology | DOK 2

EN-10.3 Explain T cell mediated beta cell destruction (insulitis) in type 1 diabetes, the use of autoantibodies (GAD65, insulin, IA-2, ZnT8) as markers, and slowly progressive autoimmune diabetes in adults (type 1.5, LADA). NPLEX + USMLE | Microbiology & Immunology | DOK 2

EN-10.4 Contrast autoimmune polyendocrine syndrome type 1 (AIRE mutation, autosomal recessive: chronic mucocutaneous candidiasis, hypoparathyroidism, Addison disease) with type 2 (Addison disease with autoimmune thyroid disease or type 1 diabetes). NPLEX + USMLE | Microbiology & Immunology | DOK 2

EN-10.5 Explain how inflammatory cytokines (IL-1, IL-6, TNF-alpha) activate the hypothalamic-pituitary-adrenal axis, and how cortisol limits inflammation by inhibiting phospholipase A2 and NF-kB and reducing lymphocyte and eosinophil counts. NPLEX | Microbiology & Immunology | DOK 2

EN-10.6 Explain why conjugating meningococcal capsular polysaccharide to a protein carrier turns a T cell independent response into a T cell dependent response with class switching and memory, and relate vaccination to prevention of Waterhouse-Friderichsen syndrome. NPLEX + USMLE | Microbiology & Immunology | DOK 3

EN-11 Endocrine pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

EN-11.1 Describe Neisseria meningitidis (gram-negative diplococcus, polysaccharide capsule, lipooligosaccharide endotoxin, respiratory droplet spread) and explain how endotoxin-driven sepsis and DIC cause adrenal hemorrhage in Waterhouse-Friderichsen syndrome. NPLEX + USMLE | Microbiology & Immunology | DOK 2

EN-11.2 Identify bacteria that cause acute suppurative thyroiditis (Staphylococcus aureus, Streptococcus pyogenes, Streptococcus pneumoniae), and relate recurrent left-sided infection in children to a pyriform sinus fistula. NPLEX | Microbiology & Immunology | DOK 2

EN-11.3 Relate subacute granulomatous thyroiditis to a preceding viral upper respiratory infection, with viruses such as coxsackievirus and mumps implicated. NPLEX | Microbiology & Immunology | DOK 1

EN-11.4 Identify infectious causes of adrenalitis leading to Addison disease, including Mycobacterium tuberculosis (acid-fast, caseating granulomas), Histoplasma capsulatum, and cytomegalovirus in patients with AIDS. NPLEX | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

EN-12-A Diseases of endocrine hyperfunction

1 Hyperadrenalism (Cushing syndrome, Conn syndrome, congenital adrenal hyperplasia)

2 Hyperparathyroidism (primary and secondary) **3** Hyperpituitarism (acromegaly, gigantism)

4 Hyperthyroidism (multinodular goiter, Graves disease)

EN-12-A.1 Distinguish ACTH-dependent Cushing syndrome (pituitary adenoma, ectopic ACTH from small cell lung carcinoma) from ACTH-independent causes (adrenal adenoma or carcinoma, exogenous glucocorticoids) using plasma ACTH and high-dose dexamethasone suppression. NPLEX + USMLE | Pathology | DOK 3

EN-12-A.2 Relate cortisol excess to central obesity, moon facies, purple striae, proximal muscle weakness, hyperglycemia, hypertension, osteoporosis, and poor wound healing, and identify screening tests (24-hour urine free cortisol, late-night salivary cortisol, low-dose dexamethasone suppression). NPLEX + USMLE | Pathology | DOK 2

EN-12-A.3 Explain primary hyperaldosteronism (Conn syndrome from adrenal adenoma, or bilateral adrenal hyperplasia) as hypertension with hypokalemia, metabolic alkalosis, and low renin, and contrast it with secondary hyperaldosteronism with high renin. NPLEX + USMLE | Pathology | DOK 3

EN-12-A.4 Predict findings in 21-hydroxylase deficiency (salt wasting, hyperkalemia, raised 17-hydroxyprogesterone, virilization), 11-beta-hydroxylase deficiency (virilization with hypertension), and 17-alpha-hydroxylase deficiency (hypertension, hypokalemia, absent sex steroids). NPLEX + USMLE | Pathology | DOK 3

EN-12-A.5 Explain primary hyperparathyroidism, most often from a single parathyroid adenoma, and interpret hypercalcemia, hypophosphatemia, and raised PTH with stones, bone pain, abdominal symptoms, and osteitis fibrosa cystica (subperiosteal resorption, brown tumors). NPLEX + USMLE | Pathology | DOK 3

EN-12-A.6 Explain secondary hyperparathyroidism in chronic kidney disease (phosphate retention, low calcitriol) and vitamin D deficiency, interpret low or normal calcium with high PTH, and relate it to renal osteodystrophy and tertiary hyperparathyroidism. NPLEX + USMLE | Pathology | DOK 3

EN-12-A.7 Distinguish gigantism (GH excess before epiphyseal closure) from acromegaly (after closure), and relate acromegaly to coarse facial features, enlarged hands and feet, macroglossia, insulin resistance, and cardiovascular disease. NPLEX + USMLE | Pathology | DOK 2

EN-12-A.8 Interpret raised IGF-1 and failure of GH suppression after an oral glucose load as confirmation of GH excess, usually from a somatotroph pituitary adenoma. NPLEX + USMLE | Pathology | DOK 3

EN-12-A.9 Distinguish Graves disease (diffuse goiter, ophthalmopathy, pretibial myxedema, diffusely increased radioiodine uptake) from toxic multinodular goiter (patchy uptake, no eye or skin signs), both showing low TSH and high free T4. NPLEX + USMLE | Pathology | DOK 3

EN-12-A.10 Relate thyrotoxic findings (weight loss, heat intolerance, tremor, tachycardia, atrial fibrillation, bone loss) to raised metabolic rate and beta-adrenergic sensitivity, and identify thyroid storm as a life-threatening complication. NPLEX + USMLE | Pathology | DOK 2

EN-12-A.11 Explain the syndrome of inappropriate ADH secretion (small cell lung carcinoma, CNS and pulmonary disease, drugs) and interpret euvolemic hyponatremia with inappropriately concentrated urine and high urine sodium. USMLE extension | Pathology | DOK 3

EN-12-A.12 Distinguish familial hypocalciuric hypercalcemia (inactivating calcium-sensing receptor mutation, low urine calcium, mildly raised PTH) from primary hyperparathyroidism. USMLE extension | Pathology | DOK 3

EN-12-A.13 Distinguish central precocious puberty (early GnRH activation, raised LH and FSH) from peripheral precocious puberty (low LH and FSH, for example from congenital adrenal hyperplasia, McCune-Albright syndrome, or sex steroid secreting tumors). USMLE extension | Pathology | DOK 3

EN-12-B Diseases of endocrine hypofunction

1 Diabetes insipidus

2 Hypoadrenalism (Addison disease, primary acute insufficiency, secondary adrenocortical insufficiency)

3 Hypoparathyroidism **4** Hypopituitarism (empty sella syndrome, hypothalamic lesions)

5 Hypothyroidism (iodine deficiency goiter)

EN-12-B.1 Distinguish central diabetes insipidus (ADH deficiency from head trauma, surgery, or tumors) from nephrogenic diabetes insipidus (V2 receptor or aquaporin-2 defects, lithium, hypercalcemia, hypokalemia), both causing polyuria with dilute urine and a tendency to hypernatremia when water intake is limited. NPLEX + USMLE | Pathology | DOK 2

EN-12-B.2 Interpret a water deprivation test with desmopressin, predicting urine concentration in central diabetes insipidus, nephrogenic diabetes insipidus, and primary polydipsia. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.3 Explain primary adrenal insufficiency (Addison disease; autoimmune adrenalitis, tuberculosis, metastases) and relate hypotension, hyponatremia, hyperkalemia, and hyperpigmentation from raised ACTH and POMC-derived MSH to low cortisol and aldosterone. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.4 Identify causes of acute adrenocortical insufficiency (adrenal crisis), including stress in a patient with chronic insufficiency, rapid withdrawal of long-term glucocorticoids, and bilateral adrenal hemorrhage (anticoagulation, DIC, Waterhouse-Friderichsen syndrome), and relate it to shock. NPLEX + USMLE | Pathology | DOK 2

EN-12-B.5 Explain secondary adrenocortical insufficiency (pituitary disease, suppression by exogenous glucocorticoids) and predict why it lacks hyperpigmentation and hyperkalemia because ACTH is low and the renin-angiotensin system preserves aldosterone. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.6 Identify causes of hypoparathyroidism (surgical removal, autoimmune, DiGeorge syndrome, hypomagnesemia) and relate hypocalcemia, hyperphosphatemia, and low PTH to tetany, perioral numbness, Chvostek and Trousseau signs, and QT prolongation. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.7 Distinguish pseudohypoparathyroidism (end-organ PTH resistance from GNAS mutation, high PTH, Albright hereditary osteodystrophy with short fourth and fifth metacarpals) from true hypoparathyroidism with low PTH. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.8 Identify causes of hypopituitarism (nonfunctioning macroadenoma, Sheehan necrosis, pituitary apoplexy, surgery or radiation), and explain primary empty sella syndrome as arachnoid herniation through a defective diaphragma sellae flattening the pituitary. NPLEX + USMLE | Pathology | DOK 2

EN-12-B.9 Predict findings of hypothalamic lesions (craniopharyngioma, infiltrative disease, radiation), including deficiency of releasing hormones with low anterior pituitary output, diabetes insipidus, and hyperprolactinemia from interrupted dopamine delivery. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.10 Explain how iodine deficiency lowers thyroid hormone, raises TSH, and causes diffuse then multinodular goiter, and relate hypothyroidism to cold intolerance, weight gain, bradycardia, myxedema, hyperlipidemia, and delayed reflex relaxation. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.11 Interpret thyroid function tests in primary hypothyroidism (high TSH, low free T4), central hypothyroidism (low or normal TSH, low free T4), and TSH-secreting adenoma (high TSH, high free T4), and relate hypothyroid hyperprolactinemia to menstrual disturbance and infertility. NPLEX + USMLE | Pathology | DOK 3

EN-12-B.12 Identify causes of congenital hypothyroidism (thyroid dysgenesis, dyshormonogenesis, maternal iodine deficiency) and relate them to lethargy, prolonged jaundice, macroglossia, umbilical hernia, and intellectual disability if untreated. USMLE extension | Pathology | DOK 2

EN-12-B.13 Interpret nonthyroidal illness (euthyroid sick) syndrome in severe illness, with low T3, raised reverse T3, and usually normal TSH from reduced peripheral T4 to T3 conversion. USMLE extension | Pathology | DOK 3

EN-12-B.14 Distinguish causes of short stature, including GH deficiency (low IGF-1, delayed bone age), GH receptor defects (Laron syndrome, high GH, low IGF-1), constitutional growth delay (delayed bone age), and familial short stature (normal bone age). USMLE extension | Pathology | DOK 3

EN-12-C Inflammatory endocrine diseases

1 Hashimoto thyroiditis **2** Granulomatous subacute thyroiditis

EN-12-C.1 Explain the pathogenesis of Hashimoto thyroiditis and identify its histology (lymphocytic infiltrate with germinal centers, atrophic follicles, Hürthle cells) and painless goiter in a middle-aged woman. NPLEX + USMLE | Pathology | DOK 2

EN-12-C.2 Predict the course of Hashimoto thyroiditis from possible transient hyperthyroidism to permanent hypothyroidism, its association with other autoimmune diseases, and its increased risk of primary thyroid B cell lymphoma. NPLEX + USMLE Pathology DOK 3

EN-12-C.3 Identify granulomatous subacute (de Quervain) thyroiditis by a painful, tender thyroid after a viral illness, fever, raised ESR, and granulomas with multinucleated giant cells, and predict transient hyperthyroidism with low radioiodine uptake. NPLEX + USMLE Pathology DOK 3

EN-12-C.4 Predict the self-limited course of subacute thyroiditis from a thyrotoxic phase to a transient hypothyroid phase and usual recovery to normal thyroid function. NPLEX + USMLE Pathology DOK 2

EN-12-C.5 Distinguish painless (lymphocytic and postpartum) thyroiditis and Riedel thyroiditis (IgG4–related dense fibrosis producing a rock-hard thyroid fixed to adjacent structures) from Hashimoto and subacute thyroiditis. USMLE extension Pathology DOK 3

EN-12-D Metabolic endocrine disease

1 Diabetes types 1 and 2 (DM)

EN-12-D.1 Compare type 1 diabetes (autoimmune beta cell loss, absolute insulin deficiency, young onset, ketosis prone) with type 2 diabetes (insulin resistance with progressive beta cell dysfunction and islet amyloid, linked to obesity and family history). NPLEX + USMLE Pathology DOK 3

EN-12-D.2 Interpret diagnostic criteria for diabetes (fasting glucose at least 126 mg/dL, 2–hour oral glucose tolerance value at least 200 mg/dL, HbA1c at least 6.5%, or random glucose at least 200 mg/dL with symptoms) and explain HbA1c as a three-month measure. NPLEX + USMLE Pathology DOK 3

EN-12-D.3 Contrast diabetic ketoacidosis (anion gap acidosis, ketones, Kussmaul breathing, raised serum potassium despite total body depletion) with hyperosmolar hyperglycemic state (very high glucose and osmolality, altered mental status, little ketosis). NPLEX + USMLE Pathology DOK 3

EN-12-D.4 Explain how nonenzymatic glycation (advanced glycation end products) and the sorbitol (polyol) pathway produce hyaline arteriolosclerosis, accelerated atherosclerosis, peripheral neuropathy, and cataracts in chronic hyperglycemia. NPLEX + USMLE Pathology DOK 3

EN-12-D.5 Predict chronic diabetic complications by organ: nephropathy (glomerular basement membrane thickening, Kimmelstiel-Wilson nodules, albuminuria), retinopathy, autonomic neuropathy with gastroparesis, and peripheral vascular disease with foot ulcers. NPLEX + USMLE Pathology DOK 3

EN-12-D.6 Identify components of metabolic syndrome (abdominal obesity, high triglycerides, low HDL cholesterol, hypertension, raised fasting glucose), relate them to insulin resistance, and recognize acanthosis nigricans as a skin sign of insulin resistance. USMLE extension Pathology DOK 2

EN-12-E Congenital endocrine disease

1 Thyroglossal duct cyst

EN-12-E.1 Explain how a persistent thyroglossal duct forms a midline neck cyst near the hyoid bone that moves upward with swallowing and tongue protrusion, and identify infection as a complication. NPLEX + USMLE Pathology DOK 2

EN-12-E.2 Distinguish a midline thyroglossal duct cyst from a lateral branchial cleft cyst at the anterior border of the sternocleidomastoid, and explain why a normally located thyroid should be confirmed before removal, since the mass may be the only functioning thyroid tissue. NPLEX + USMLE Pathology DOK 3

EN-12-F Endocrine vascular diseases

1 Postpartum pituitary necrosis (Sheehan necrosis)

EN-12-F.1 Explain Sheehan syndrome as ischemic necrosis of a pregnancy-enlarged anterior pituitary after severe postpartum hemorrhage and hypotension, and explain why the posterior pituitary, with its direct arterial supply, is usually spared. NPLEX + USMLE Pathology DOK 3

EN-12-F.2 Predict the presentation of Sheehan syndrome (failure of lactation, amenorrhea, loss of axillary and pubic hair, hypothyroidism, adrenal insufficiency) from loss of anterior pituitary hormones. NPLEX + USMLE Pathology DOK 3

EN-12-F.3 Explain pituitary apoplexy as sudden hemorrhage or infarction of a pituitary adenoma, and relate severe headache, visual loss, ophthalmoplegia from cavernous sinus compression, and acute adrenal insufficiency to its mechanism. USMLE extension Pathology DOK 3

EN-12-G **Endocrine neoplasms**

1 Adrenal **2** Pancreas (insulinoma) **3** Parathyroid

4 Pituitary (adenoma, non-functioning tumor, craniopharyngioma)

5 Thyroid (adenoma and follicular, papillary and medullary carcinomas, euthyroid goiter)

6 Other neoplasms (multiple endocrine neoplasia types 1 and 2, pheochromocytoma)

EN-12-G.1 Distinguish adrenal cortical adenoma (small, often nonfunctional or cortisol or aldosterone secreting) from adrenocortical carcinoma (large, invasive, often virilizing), and recognize that metastases are more common than primary adrenal cancer. NPLEX + USMLE Pathology DOK 3

EN-12-G.2 Identify neuroblastoma as an adrenal medulla tumor of young children with N-MYC amplification, Homer Wright rosettes, raised urine homovanillic acid and vanillylmandelic acid, and possible opsoclonus-myoclonus. NPLEX + USMLE Pathology DOK 2

EN-12-G.3 Identify insulinoma as a beta cell tumor causing Whipple triad (hypoglycemic symptoms, low glucose, relief with glucose) with inappropriately high insulin and C-peptide, and relate it to MEN1. NPLEX + USMLE Pathology DOK 3

EN-12-G.4 Distinguish causes of hypoglycemia, including insulinoma (high insulin and C-peptide), surreptitious insulin use (high insulin, low C-peptide), sepsis, and liver failure (impaired gluconeogenesis and glycogen stores). NPLEX + USMLE Pathology DOK 3

EN-12-G.5 Identify other pancreatic neuroendocrine tumors by their syndromes: glucagonoma (hyperglycemia, necrolytic migratory erythema, weight loss) and somatostatinoma (diabetes, gallstones, steatorrhea, achlorhydria). USMLE extension Pathology DOK 2

EN-12-G.6 Distinguish parathyroid adenoma (single enlarged gland of chief cells, suppressed remaining glands), hyperplasia (multiple glands, as in MEN syndromes), and the rare carcinoma (marked hypercalcemia, invasion, metastasis). NPLEX + USMLE Pathology DOK 3

EN-12-G.7 Classify pituitary adenomas by size (microadenoma under 1 cm, macroadenoma) and hormone, and relate prolactinoma, the most common type, to galactorrhea, amenorrhea, and infertility in women and decreased libido in men. NPLEX + USMLE Pathology DOK 2

EN-12-G.8 Predict mass effects of a nonfunctioning pituitary macroadenoma: bitemporal hemianopia from chiasm compression, headache, hypopituitarism, and mild hyperprolactinemia from stalk compression. NPLEX + USMLE Pathology DOK 3

EN-12-G.9 Identify craniopharyngioma as a suprasellar tumor from Rathke pouch remnants, often calcified and cystic with oily fluid, causing headache, bitemporal hemianopia, diabetes insipidus, and growth failure in children. NPLEX + USMLE Pathology DOK 2

EN-12-G.10 Distinguish follicular adenoma from follicular carcinoma by capsular or vascular invasion, and explain hematogenous spread of follicular carcinoma to bone and lung. NPLEX + USMLE Pathology DOK 3

EN-12-G.11 Identify papillary carcinoma, the most common thyroid cancer, by Orphan Annie eye nuclei, nuclear grooves, and psammoma bodies, relate it to childhood radiation exposure, and explain lymphatic spread to cervical nodes with good prognosis. NPLEX + USMLE Pathology DOK 2

EN-12-G.12 Identify medullary carcinoma as a C cell tumor secreting calcitonin, with amyloid stroma derived from calcitonin, and relate it to RET mutations and MEN2. NPLEX + USMLE Pathology DOK 2

EN-12-G.13 Distinguish euthyroid (nontoxic) diffuse and multinodular goiter, with normal TSH and compressive symptoms such as dysphagia, from thyroid neoplasms, and relate a cold nodule on radioiodine scan to higher cancer risk. NPLEX + USMLE Pathology DOK 3

EN-12-G.14 Identify anaplastic thyroid carcinoma as a rapidly growing, locally invasive tumor of older adults with TP53 mutations and very poor prognosis. USMLE extension Pathology DOK 2

EN-12-G.15 Match the tumors of MEN1 (parathyroid, pituitary, pancreatic endocrine tumors such as gastrinoma), MEN2A (medullary thyroid carcinoma, pheochromocytoma, parathyroid hyperplasia), and MEN2B (medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, marfanoid habitus). NPLEX + USMLE Pathology DOK 2

EN-12-G.16 Identify pheochromocytoma as a chromaffin cell tumor causing episodic hypertension, headache, palpitations, and sweating, diagnosed by raised plasma or urine metanephrines, and associated with MEN2, von Hippel-Lindau disease, and neurofibromatosis type 1. NPLEX + USMLE Pathology DOK 3

EN-12-G.17 Identify endocrine paraneoplastic syndromes: ACTH and ADH from small cell lung carcinoma, and hypercalcemia of malignancy from PTH-related peptide (squamous cell lung carcinoma, renal cell carcinoma) with suppressed PTH. USMLE extension Pathology DOK 3

EN-12-H Infectious endocrine diseases

1 Infectious thyroiditis 2 Waterhouse-Friderichsen syndrome

EN-12-H.1 Identify acute infectious (suppurative) thyroiditis as a bacterial infection presenting with fever and a painful, tender, sometimes fluctuant thyroid with usually normal thyroid function, and identify abscess as a complication. NPLEX Pathology DOK 2

EN-12-H.2 Explain Waterhouse-Friderichsen syndrome as bilateral adrenal hemorrhage during overwhelming sepsis, classically meningococcemia with DIC, and relate petechial rash, shock, and acute adrenal insufficiency to its mechanism. NPLEX + USMLE Pathology DOK 3

Immuno Immunological System

8% · 16 questions · Week 8

HOW THE BODY WORKS

IM-1 Thymus embryology

Describe embryological development of the thymus.

IM-1.1 Trace thymic development from third pharyngeal pouch endoderm (shared with the inferior parathyroids), with neural crest derived capsule and septa, and its descent into the anterior mediastinum. NPLEX · Anatomy · DOK 2

IM-1.2 Describe colonization of the thymic epithelial framework by bone marrow derived T cell precursors and the formation of cortex, medulla, and Hassall corpuscles during fetal life. NPLEX · Anatomy · DOK 1

IM-1.3 Relate failed third and fourth pharyngeal pouch development in 22q11.2 deletion (DiGeorge) syndrome to absent thymic shadow, T cell deficiency, hypocalcemia from parathyroid hypoplasia or aplasia, and conotruncal heart defects. NPLEX + USMLE · Anatomy · DOK 3

IM-1.4 Describe age-related thymic involution with replacement by adipose tissue, and explain why established peripheral T cell immunity persists after involution or adult thymectomy. NPLEX · Anatomy · DOK 2

IM-1.5 Locate the adult thymus in the anterior and superior mediastinum behind the manubrium, and distinguish thymoma from the other classic anterior mediastinal masses (teratoma, lymphoma, retrosternal thyroid). NPLEX · Anatomy · DOK 2

IM-2 Lymphoid organ histology

Describe microscopic anatomy of lymphoid organs.

IM-2.1 Describe lymph node microanatomy: capsule and subcapsular sinus, cortical B cell follicles with germinal centers, paracortical T cells with high endothelial venules, and medullary cords and sinuses. NPLEX · Anatomy · DOK 1

IM-2.2 Predict which lymph node zone is depleted in DiGeorge syndrome (paracortex) and X-linked agammaglobulinemia (follicles and germinal centers), and which expands in reactive follicular versus paracortical hyperplasia. NPLEX + USMLE · Pathology · DOK 3

IM-2.3 Describe splenic histology: white pulp (periarteriolar lymphoid sheath of T cells, B cell follicles, marginal zone) and red pulp (cords of Billroth and venous sinusoids), and trace blood flow through them. NPLEX · Anatomy · DOK 2

IM-2.4 Distinguish thymic cortex (densely packed immature thymocytes, epithelial reticular cells, blood thymus barrier) from medulla (mature T cells, Hassall corpuscles) on a histologic section. NPLEX · Anatomy · DOK 1

IM-2.5 Describe mucosa associated lymphoid tissue, including tonsils and ileal Peyer patches with antigen-sampling M cells, and explain its role in generating IgA-secreting plasma cells. NPLEX · Anatomy · DOK 2

IM-3 Histocompatibility antigens and disease

Describe structure and function of histocompatibility antigens and their associated diseases.

IM-3.1 Compare MHC class I (HLA-A, -B, -C; heavy chain with beta2-microglobulin; all nucleated cells; presents to CD8 T cells) with class II (HLA-DP, -DQ, -DR; antigen-presenting cells; presents to CD4 T cells). NPLEX · Microbiology & Immunology · DOK 2

IM-3.2 Trace antigen processing for class I (proteasome, TAP transport, loading in the endoplasmic reticulum) and class II (endosomal degradation, invariant chain, CLIP exchange by HLA-DM). NPLEX · Microbiology & Immunology · DOK 2

IM-3.3 Describe the HLA locus on chromosome 6, codominant expression, polymorphism, and haplotype inheritance, and calculate the probability (25%) that a sibling is a full HLA match for transplantation. NPLEX + USMLE · Biochemistry & Genetics · DOK 3

IM-3.4 Relate HLA alleles to disease: B27 (ankylosing spondylitis, reactive and psoriatic arthritis), DQ2 and DQ8 (celiac disease), DR3 and DR4 (type 1 diabetes), DR4 (rheumatoid arthritis), and DR2 (multiple sclerosis, Goodpasture syndrome). NPLEX · Microbiology & Immunology · DOK 2

IM-3.5 Explain how NK cells kill cells that lose MHC class I expression (missing self), and why virus-infected and tumor cells that downregulate class I become NK targets. NPLEX · Microbiology & Immunology · DOK 2

IM-3.6 Explain how preformed anti-HLA antibodies from prior transfusion, pregnancy, or transplant cause hyperacute rejection, and why crossmatching and HLA matching improve graft survival. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-4 Lymphatic drainage

Describe location and drainage patterns of lymphatic vessels.

IM-4.1 Trace lymph from the lower limbs, pelvis, and abdomen through the cisterna chyli and thoracic duct to the left venous angle, and contrast the right lymphatic duct territory (right head, neck, upper limb, thorax). NPLEX Anatomy DOK 2

IM-4.2 Identify the regions drained by the palpable node groups (cervical, supraclavicular, axillary, epitrochlear, superficial inguinal) and predict which group enlarges with an infection of a given site. NPLEX Anatomy DOK 2

IM-4.3 Predict first nodal spread of tumors: lateral breast to axillary and medial breast to parasternal nodes, testis to para-aortic but scrotum to superficial inguinal nodes, and anal canal above the pectinate line to internal iliac but below it to superficial inguinal nodes. NPLEX Anatomy DOK 3

IM-4.4 Relate an enlarged left supraclavicular (Virchow) node to abdominal or thoracic malignancy spreading along the thoracic duct. NPLEX Anatomy DOK 2

IM-4.5 Predict chylothorax after thoracic duct injury and describe the milky, triglyceride-rich pleural fluid containing chylomicrons and lymphocytes. NPLEX Anatomy DOK 3

IM-5 Humoral and cell-mediated immunity

Explain functions of cells, antibodies, and cytokines in humoral and cell-mediated immunity.

IM-5.1 Describe immunoglobulin structure (two heavy and two light chains, Fab with hypervariable antigen-binding regions, Fc determining isotype functions such as complement fixation and Fc receptor binding). NPLEX Microbiology & Immunology DOK 1

IM-5.2 Compare the isotypes: IgM (pentamer, first response, strongest complement activation), IgG (placental transfer, opsonization), secretory IgA (mucosal dimer), IgE (mast cells, helminths), and IgD (naive B cell receptor). NPLEX Microbiology & Immunology DOK 2

IM-5.3 Explain T cell dependent B cell activation, including CD40L to CD40 signaling and cytokines driving isotype switching and affinity maturation in germinal centers, and contrast T independent responses to polysaccharide antigens. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-5.4 Compare Th1 (IL-12 induced; IFN-gamma activates macrophages), Th2 (IL-4, IL-5, IL-13; IgE and eosinophils), Th17 (IL-17; neutrophils against extracellular bacteria and fungi), and regulatory T cells (FOXP3; IL-10, TGF-beta). NPLEX Microbiology & Immunology DOK 2

IM-5.5 Explain how CD8 cytotoxic T cells and NK cells kill target cells through perforin and granzymes and through Fas ligand binding Fas, both producing apoptosis. NPLEX Microbiology & Immunology DOK 2

IM-5.6 Interpret primary versus secondary antibody response curves (lag phase, IgM then IgG, higher titer and affinity on re-exposure), and distinguish active from passive immunity such as maternal IgG. NPLEX Microbiology & Immunology DOK 3

IM-6 Cell and cytokine signaling in injury and infection

Explain pathways of cellular and cytokine signaling in response to injury, infection, and foreign bodies.

IM-6.1 Describe pattern recognition: TLR4 with CD14 binding LPS, TLR5 binding flagellin, endosomal TLR3, 7, 8, and 9 binding nucleic acids, and the NLRP3 inflammasome activating caspase-1 to release IL-1beta in response to DAMPs. NPLEX Microbiology & Immunology DOK 2

IM-6.2 Trace leukocyte extravasation (selectin rolling, integrin adhesion to ICAM-1 induced by TNF and IL-1, PECAM-1 diapedesis, chemotaxis to C5a, LTB4, IL-8) and relate CD18 deficiency to leukocyte adhesion deficiency. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-6.3 Explain the acute phase response: IL-1, IL-6, and TNF-alpha cause fever through hypothalamic PGE2, raise CRP, fibrinogen, serum amyloid A, ferritin, and hepcidin, and lower albumin and transferrin. NPLEX Microbiology & Immunology DOK 2

IM-6.4 Explain opsonization by IgG and C3b and phagolysosome fusion, and relate LYST mutation in Chediak-Higashi syndrome to giant granules, partial albinism, neuropathy, and recurrent pyogenic infections. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-6.5 Describe the foreign body granuloma (macrophages, epithelioid cells, foreign body giant cells around sutures, talc, or splinters) and distinguish it from an immune granuloma driven by Th1 IFN-gamma. NPLEX Pathology DOK 2

IM-6.6 Explain how LPS signaling through TLR4 drives TNF, IL-1, and IL-6 release in sepsis, causing vasodilation, capillary leak, DIC, and organ dysfunction, and apply SIRS criteria (temperature above 38 °C (100.4 °F) or below 36 °C (96.8 °F), pulse, respirations, WBC). NPLEX + USMLE Microbiology & Immunology DOK 3

IM-6.7 Explain how superantigens (S aureus TSST-1, S pyogenes pyrogenic exotoxins) bind MHC class II and TCR V-beta outside the peptide groove, causing polyclonal T cell activation and cytokine storm in toxic shock syndrome. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-7 Complement

Explain structure, function, and pathways of complement compounds.

IM-7.1 Compare activation of the classical (C1q binding IgM or IgG complexes), lectin (mannose-binding lectin on microbial sugars), and alternative (spontaneous C3 hydrolysis on microbial surfaces) pathways converging on C3 convertase. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-7.2 Describe complement effector functions: C3b opsonization, C3a and C5a anaphylatoxins, C5a neutrophil chemotaxis, C5b-9 membrane attack complex lysis, and clearance of immune complexes. NPLEX Microbiology & Immunology DOK 1

IM-7.3 Explain regulation by C1 esterase inhibitor, DAF (CD55), and CD59, and predict hereditary angioedema (bradykinin-mediated swelling without urticaria, low C4) and paroxysmal nocturnal hemoglobinuria (PIGA mutation, intravascular hemolysis). NPLEX + USMLE Microbiology & Immunology DOK 3

IM-7.4 Predict the disease from each deficiency: C1q, C2, or C4 (SLE-like immune complex disease), C3 (severe pyogenic infections, immune complex glomerulonephritis), C5 to C9 (recurrent Neisseria), and MBL (childhood infections). NPLEX + USMLE Microbiology & Immunology DOK 3

IM-7.5 Interpret serum complement: low C3 and C4 in active SLE, low C3 with normal C4 in poststreptococcal glomerulonephritis, and low C4 with normal C3 in hereditary angioedema. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-8 Lymphatic organ function

Explain mechanisms, factors affecting, functions, and control of lymphatic organs.

IM-8.1 Explain thymic T cell maturation: double negative to double positive thymocytes, positive selection on cortical epithelial MHC, and negative selection in the medulla by AIRE-driven expression of tissue antigens. NPLEX Microbiology & Immunology DOK 2

IM-8.2 Explain splenic functions: removal of aged RBCs and inclusions, clearance of opsonized and encapsulated bacteria by marginal zone B cells and macrophages, IgM production, and platelet storage. NPLEX Physiology DOK 2

IM-8.3 Predict findings after splenectomy or functional asplenia (sickle cell disease): Howell-Jolly bodies, target cells, thrombocytosis, and sepsis from S pneumoniae, H influenzae type b, and N meningitidis. NPLEX Physiology DOK 3

IM-8.4 Explain naive lymphocyte homing into lymph nodes through high endothelial venules (L-selectin, CCR7), meeting antigen carried by dendritic cells in afferent lymph, and exit through efferent lymph. NPLEX Physiology DOK 2

IM-8.5 Predict the effect of high cortisol (stress or Cushing syndrome) on the blood count and lymphoid organs: lymphopenia, eosinopenia, neutrophilia from demargination, and thymic atrophy. NPLEX Physiology DOK 3

IM-9 Lymph composition and transport

Explain composition, function, and transport of lymphatic fluid.

IM-9.1 Explain lymph formation from net capillary filtration by Starling forces, with about 2 to 3 L of lymph returned to the blood each day. NPLEX Physiology DOK 2

IM-9.2 Describe lymph composition (interstitial fluid with filtered plasma protein, lymphocytes, and chylomicrons in intestinal chyle) and explain why lymph is the only route returning filtered protein to the blood. NPLEX Physiology DOK 2

IM-9.3 Explain lymph propulsion by intrinsic lymphangion smooth muscle contraction, one-way valves, skeletal muscle pumping, and respiratory pressure changes, and why rising interstitial pressure increases lymph flow. NPLEX Physiology DOK 2

IM-9.4 Predict lymphedema (protein-rich edema that becomes nonpitting with fibrosis) from axillary dissection or radiation, Wuchereria bancrofti filariasis, Milroy disease (FLT4), and Turner syndrome (cystic hygroma). NPLEX | Pathology | DOK 3

IM-10 Immune biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors, and biochemical processes and associated constituents involved in immunological function and pathology.

IM-10.1 Trace the respiratory burst (NADPH from the pentose phosphate pathway, NADPH oxidase to superoxide, superoxide dismutase to H2O2, myeloperoxidase to HOCl) and relate NADPH oxidase defects to chronic granulomatous disease. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

IM-10.2 Predict chronic granulomatous disease infections with catalase-positive organisms (S aureus, Aspergillus, Burkholderia cepacia, Serratia, Nocardia) and interpret an abnormal dihydrorhodamine or nitroblue tetrazolium test. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-10.3 Explain how adenosine deaminase deficiency causes dATP accumulation that inhibits ribonucleotide reductase and kills lymphocytes, producing autosomal recessive SCID. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-10.4 Compare arachidonic acid products: COX-derived PGE2 (fever, pain, vasodilation) and thromboxane, and 5-lipoxygenase-derived LTB4 (neutrophil chemotaxis) and LTC4, LTD4, LTE4 (bronchoconstriction, vascular permeability). NPLEX | Biochemistry & Genetics | DOK 2

IM-10.5 Explain how granuloma macrophages in sarcoidosis express 1-alpha-hydroxylase, producing calcitriol and hypercalcemia, and how vitamin D induces macrophage cathelicidin. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-10.6 Relate zinc deficiency (thymic atrophy, impaired T cell function, acrodermatitis enteropathica) and vitamin A deficiency (mucosal squamous metaplasia, more severe measles) to infection risk. NPLEX | Biochemistry & Genetics | DOK 2

IM-10.7 Explain why protein-energy malnutrition (marasmus, kwashiorkor) is the most common cause of acquired immunodeficiency worldwide, with thymic atrophy and impaired cell-mediated immunity. NPLEX | Biochemistry & Genetics | DOK 2

IM-11 Lymph biochemistry

Explain biochemistry of synthesis and degradation of lymphatic fluid and its components.

IM-11.1 Explain chylomicron assembly in enterocytes (triglyceride resynthesis, apoB-48, microsomal triglyceride transfer protein) and their entry into lacteals and thoracic duct lymph, bypassing the liver. NPLEX | Biochemistry & Genetics | DOK 2

IM-11.2 Trace chylomicron breakdown after entering the blood: apoC-II activation of endothelial lipoprotein lipase, fatty acid release to muscle and adipose tissue, and apoE-mediated hepatic remnant uptake. NPLEX | Biochemistry & Genetics | DOK 2

IM-11.3 Predict the findings of abetalipoproteinemia (MTP deficiency): absent chylomicrons and apoB lipoproteins, fat malabsorption, acanthocytes, and fat-soluble vitamin deficiency. NPLEX | Biochemistry & Genetics | DOK 3

IM-11.4 Explain why hepatic and intestinal lymph carry more protein than lymph from skeletal muscle, relating it to sinusoidal and fenestrated versus continuous capillary endothelium. NPLEX | Physiology | DOK 2

IM-12 Immunogenetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying immunological disease.

IM-12.1 Explain V(D)J recombination by RAG1 and RAG2 and junctional diversity from TdT, and predict that RAG deficiency causes SCID with absent T and B cells but preserved NK cells. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-12.2 Classify SCID genotypes by lymphocyte profile: X-linked IL2RG common gamma chain (T and NK absent, B present), ADA deficiency (all absent), and RAG deficiency, with low T cell receptor excision circles. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-12.3 Relate X-linked WAS gene mutation in Wiskott-Aldrich syndrome to thrombocytopenia with small platelets, eczema, recurrent infections, low IgM with high IgA and IgE, and calculate risk for sons of carriers. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IM-12.4 Explain how X-linked CD40 ligand deficiency blocks isotype switching, producing hyper-IgM syndrome with normal or high IgM, low IgG, IgA, and IgE, and Pneumocystis and Cryptosporidium infections. NPLEX + USMLE Biochemistry & Genetics DOK 3

IM-12.5 Relate AIRE mutation to autoimmune polyglandular syndrome type 1 (chronic mucocutaneous candidiasis, hypoparathyroidism, adrenal insufficiency) and FOXP3 mutation to IPEX syndrome. NPLEX + USMLE Biochemistry & Genetics DOK 3

IM-12.6 Explain how Fas-mediated apoptosis ends lymphocyte clonal expansion and removes self-reactive lymphocytes, and predict lymphadenopathy, splenomegaly, and autoimmune cytopenias in autoimmune lymphoproliferative syndrome from FAS mutation. NPLEX Biochemistry & Genetics DOK 3

IM-13 Innate and adaptive immune activation

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

IM-13.1 Compare innate immunity (germline pattern recognition receptors, rapid, no memory; neutrophils, macrophages, NK cells, complement) with adaptive immunity (rearranged receptors, clonal expansion, memory). NPLEX Microbiology & Immunology DOK 2

IM-13.2 Explain two-signal T cell activation (TCR binding peptide on MHC plus CD28 binding B7), anergy when costimulation is absent, and inhibition through CTLA-4 and PD-1. NPLEX Microbiology & Immunology DOK 2

IM-13.3 Explain antitumor immunity by CD8 T cells recognizing tumor antigens on MHC class I and by NK cells, and tumor escape through loss of MHC class I and PD-L1 expression. NPLEX Microbiology & Immunology DOK 2

IM-13.4 Compare live attenuated vaccines (MMR, varicella, yellow fever; strong cellular and humoral immunity, avoided in immunodeficiency) with inactivated, subunit, toxoid, and protein-conjugated polysaccharide vaccines. NPLEX Microbiology & Immunology DOK 2

IM-13.5 Explain mechanisms of autoimmunity: failed central or peripheral tolerance, HLA susceptibility, and molecular mimicry (group A strep M protein in rheumatic fever, Campylobacter jejuni in Guillain-Barre syndrome). NPLEX + USMLE Microbiology & Immunology DOK 2

IM-13.6 Compare the four hypersensitivity types by effector (IgE and mast cells; IgG or IgM against cell or matrix antigens; immune complexes; T cells) and time course, and classify a clinical reaction. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-14 Systemic pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

IM-14.1 Describe parvovirus B19 (small nonenveloped ssDNA virus) tropism for erythroid progenitors through the P antigen, respiratory and transplacental spread, and why infection halts red cell production. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.2 Describe Haemophilus influenzae type b: gram-negative coccobacillus, polyribosylribitol phosphate capsule as the key virulence factor, IgA protease, respiratory spread, and growth requiring factors X and V on chocolate agar. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.3 Describe influenza virus (orthomyxovirus with segmented negative-sense RNA, hemagglutinin binding sialic acid, neuraminidase releasing virions) and distinguish antigenic drift (point mutations) from shift (reassortment causing pandemics). NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.4 Describe measles and mumps viruses (enveloped nonsegmented negative-sense RNA paramyxoviruses whose F protein forms syncytia, such as Warthin-Finkeldey cells in measles) and rubella virus (enveloped positive-sense RNA, formerly classed as a togavirus), all spread by respiratory droplets. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.5 Describe the herpesviruses behind mononucleosis and roseola (enveloped dsDNA with lifelong latency): EBV entering B cells through CD21, and HHV-6 and HHV-7 infecting CD4 T cells, all spread by saliva. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.6 Describe Streptococcus pyogenes (catalase-negative, beta-hemolytic, bacitracin-sensitive gram-positive cocci in chains), M protein as antiphagocytic and mimicry antigen, and pyrogenic exotoxins as superantigens causing scarlet fever. NPLEX + USMLE Microbiology & Immunology DOK 2

IM-14.7 Trace the Toxoplasma gondii life cycle: sexual stage in the cat (definitive host), oocysts in cat feces, tissue cysts with bradyzoites in undercooked meat, and transplacental tachyzoites. NPLEX + USMLE Microbiology & Immunology DOK 2

DISEASES AND CONDITIONS

IM-15-A Congenital immunodeficiency diseases

1 Common variable immunodeficiency **2** DiGeorge syndrome **3** Selective IgA deficiency **4** Severe combined immunodeficiency **5** X-linked agammaglobulinemia

IM-15-A.1 Explain common variable immunodeficiency (failed B cell differentiation into plasma cells with usually normal B cell counts), its frequent onset in adulthood, low IgG and IgA, sinopulmonary and Giardia infections, and autoimmune and lymphoma risk. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-A.2 Identify DiGeorge syndrome (22q11.2 deletion, TBX1) by cardiac defects (truncus arteriosus, tetralogy of Fallot), abnormal facies, thymic hypoplasia with viral and fungal infections, cleft palate, and hypocalcemic tetany. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-A.3 Describe selective IgA deficiency, the most common primary immunodeficiency: often asymptomatic, sinopulmonary and Giardia infections, celiac and autoimmune associations, anaphylaxis to IgA-containing blood products, and falsely negative IgA serologies. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-A.4 Identify SCID in an infant with failure to thrive, chronic diarrhea, thrush, Pneumocystis pneumonia, and severe viral infections, with absent thymic shadow, and predict harm from live vaccines and nonirradiated transfusions. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-A.5 Explain how BTK mutation in X-linked agammaglobulinemia blocks pre-B cell maturation, leaving absent B cells, very low immunoglobulins, scant tonsils, and infections after 6 months with encapsulated bacteria, enteroviruses, and Giardia. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-A.6 Predict the defective arm from the infection pattern: B cell (encapsulated bacteria, enteroviruses), T cell (viruses, fungi, Pneumocystis), phagocyte (S aureus, Aspergillus), terminal complement (Neisseria), and choose the first test (immunoglobulin levels, flow cytometry, DHR, CH50). NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-B Acquired immunodeficiency diseases

1 Drug-induced immunodeficiencies

2 Human immunodeficiency virus (HIV)/acquired immune deficiency syndrome (AIDS)

IM-15-B.1 Predict the infections caused by drug-induced immunodeficiency: glucocorticoids and calcineurin inhibitors (T cell suppression, Pneumocystis), TNF-alpha inhibitors (latent tuberculosis reactivation), rituximab (B cell depletion, hepatitis B reactivation), and chemotherapy neutropenia. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-B.2 Describe HIV structure and entry: retrovirus with gag (p24 capsid), pol (reverse transcriptase, integrase, protease), and env (gp120 binding CD4 with CCR5 or CXCR4, gp41 fusion), and why CCR5 delta32 homozygotes resist infection. NPLEX + USMLE | Microbiology & Immunology | DOK 2

IM-15-B.3 Trace the natural history of HIV: transmission routes, acute retroviral syndrome, clinical latency with ongoing lymphoid replication, CD4 decline, and AIDS defined by CD4 below 200 cells per mm3 or an AIDS-defining illness. NPLEX + USMLE | Microbiology & Immunology | DOK 2

IM-15-B.4 Interpret HIV testing: fourth generation antigen and antibody immunoassay followed by HIV-1 and HIV-2 differentiation, CD4 count for staging, viral load for replication, and why infants of infected mothers need nucleic acid testing rather than antibody tests. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-B.5 Predict opportunistic infections by CD4 count: oral thrush and oral hairy leukoplakia earlier, Pneumocystis pneumonia below 200, Toxoplasma encephalitis below 100, and CMV retinitis and Mycobacterium avium complex below 50. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-B.6 Relate noninfectious AIDS complications to their causes: Kaposi sarcoma (HHV-8), primary CNS lymphoma (EBV), HIV-associated nephropathy (collapsing FSGS), neurocognitive disorder, peripheral neuropathy, and immune reconstitution inflammatory syndrome. NPLEX + USMLE | Pathology | DOK 3

IM-15-C Diseases of hypersensitivity

1 Type I (anaphylaxis)

2 Type II (autoimmune hemolytic anemia, Goodpasture syndrome, myasthenia gravis [MG])

3 Type III (systemic lupus erythematosus [SLE], polyarteritis nodosa, poststreptococcal glomerulonephritis, rheumatoid arthritis)

4 Type IV (granulomatous inflammation, transplant rejection)

IM-15-C.1 Explain type I hypersensitivity: Th2 and IL-4 driven IgE sensitization, mast cell FcepsilonRI cross-linking on re-exposure, immediate histamine and tryptase release, and a late phase of leukotrienes and eosinophils. NPLEX + USMLE | Microbiology & Immunology | DOK 2

IM-15-C.2 Identify anaphylaxis from common triggers (foods such as peanut, insect stings, penicillin, latex) by urticaria, angioedema, bronchospasm, and hypotension, and interpret an elevated serum tryptase. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-C.3 Compare warm autoimmune hemolytic anemia (IgG, extravascular splenic hemolysis, spherocytes; SLE, CLL) with cold agglutinin disease (IgM fixing complement; Mycoplasma pneumoniae, EBV; acrocyanosis), both with positive direct Coombs test. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.4 Explain Goodpasture syndrome (anti-GBM disease; antibodies to the alpha3 chain of type IV collagen) causing hemoptysis and rapidly progressive crescentic glomerulonephritis with linear IgG on immunofluorescence. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.5 Explain myasthenia gravis as noncytotoxic type II hypersensitivity (antibodies to the acetylcholine receptor or MuSK), with fatigable ptosis, diplopia, and proximal weakness, and its association with thymic hyperplasia and thymoma. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-C.6 Explain SLE pathogenesis (impaired clearance of apoptotic nuclear material, immune complexes) and interpret ANA (sensitive), anti-dsDNA and anti-Smith (specific), anti-histone (drug-induced), and low C3 and C4 during flares. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-C.7 Identify SLE manifestations in a young woman (malar rash, photosensitivity, nonerosive arthritis, serositis, cytopenias, lupus nephritis with wire loops, Libman-Sacks endocarditis) and antiphospholipid syndrome thrombosis and pregnancy loss. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.8 Describe polyarteritis nodosa: immune complex medium-vessel vasculitis linked to hepatitis B, ANCA-negative, sparing the lungs, with transmural fibrinoid necrosis at different stages and renal and mesenteric microaneurysms. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.9 Explain poststreptococcal glomerulonephritis 1 to 4 weeks after group A strep pharyngitis or skin infection: nephritic syndrome, low C3, elevated ASO or anti-DNase B, subepithelial humps, and granular immunofluorescence. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.10 Describe rheumatoid arthritis pathogenesis (HLA-DR4, citrullinated peptide autoimmunity, synovial pannus) and findings: symmetric MCP, PIP, and wrist erosions sparing DIP joints, rheumatoid factor, anti-CCP, nodules, and Felty syndrome. NPLEX + USMLE | Pathology | DOK 3

IM-15-C.11 Explain type IV granulomatous inflammation (Th1 IFN-gamma activating macrophages, TNF maintaining granulomas), distinguishing caseating tuberculosis granulomas from noncaseating ones, and interpret a tuberculin skin test read at 48 to 72 hours. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-C.12 Identify sarcoidosis by noncaseating granulomas, bilateral hilar lymphadenopathy, elevated ACE, hypercalcemia, uveitis, and erythema nodosum, and recognize Lofgren syndrome (erythema nodosum, hilar adenopathy, arthritis). NPLEX + USMLE | Pathology | DOK 3

IM-15-C.13 Compare hyperacute (preformed antibody, minutes), acute (T cell or antibody mediated, weeks to months), and chronic (graft arteriosclerosis, bronchiolitis obliterans, months to years) rejection, and distinguish graft-versus-host disease (skin, liver, gut). NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-C.14 Explain serum sickness (fever, urticaria, arthralgia, proteinuria, low complement about 1 to 2 weeks after foreign protein exposure) and the local Arthus reaction as systemic and local type III reactions. USMLE extension | Microbiology & Immunology | DOK 3

IM-15-C.15 Classify drug hypersensitivity examples by mechanism: penicillin anaphylaxis (type I), penicillin or methyldopa hemolytic anemia (type II), serum sickness from chimeric antibodies (type III), and contact dermatitis from topical neomycin (type IV). USMLE extension | Microbiology & Immunology | DOK 3

IM-15-D Other autoimmune diseases

1 Progressive systemic sclerosis (scleroderma) **2** Rheumatic fever **3** Sjögren syndrome

IM-15-D.1 Explain systemic sclerosis pathogenesis (endothelial injury, autoimmunity, TGF-beta driven fibroblast collagen deposition) causing skin thickening, Raynaud phenomenon, esophageal dysmotility, and interstitial lung disease. NPLEX + USMLE | Pathology | DOK 3

IM-15-D.2 Distinguish diffuse systemic sclerosis (anti-Scl-70, early visceral disease, scleroderma renal crisis) from limited CREST disease (anticentromere; calcinosis, Raynaud, esophageal dysmotility, sclerodactyly, telangiectasia; pulmonary hypertension). NPLEX + USMLE | Pathology | DOK 3

IM-15-D.3 Explain acute rheumatic fever as type II hypersensitivity from M protein mimicry of cardiac proteins 2 to 4 weeks after untreated group A strep pharyngitis, with Aschoff bodies and Anitschkow cells in pancarditis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-D.4 Apply the Jones major criteria (migratory polyarthritis, carditis, subcutaneous nodules, erythema marginatum, Sydenham chorea) with evidence of strep infection, and predict chronic mitral stenosis. NPLEX + USMLE | Pathology | DOK 3

IM-15-D.5 Explain Sjogren syndrome (lymphocytic destruction of salivary and lacrimal glands) causing dry mouth and keratoconjunctivitis sicca, with anti-Ro (SSA) and anti-La (SSB), RA association, and marginal zone lymphoma risk. NPLEX + USMLE | Pathology | DOK 3

IM-15-D.6 Predict congenital heart block and neonatal lupus in infants of mothers with anti-Ro (SSA) antibodies from Sjogren syndrome or SLE, through transplacental IgG. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IM-15-D.7 Identify mixed connective tissue disease by overlapping features of SLE, systemic sclerosis, and polymyositis with high-titer anti-U1 RNP antibodies. USMLE extension | Pathology | DOK 2

IM-15-E Diseases of amyloids

1 Primary amyloidosis **2** Secondary amyloidosis

IM-15-E.1 Describe amyloid as misfolded proteins in beta-pleated sheets deposited extracellularly, showing apple-green birefringence with Congo red under polarized light, and diagnosed by abdominal fat pad or organ biopsy. NPLEX + USMLE | Pathology | DOK 2

IM-15-E.2 Explain primary (AL) amyloidosis from monoclonal plasma cell light chains (a plasma cell dyscrasia, sometimes with multiple myeloma) and predict restrictive cardiomyopathy, nephrotic syndrome, macroglossia, periorbital purpura, and carpal tunnel neuropathy. NPLEX + USMLE | Pathology | DOK 3

IM-15-E.3 Explain secondary (AA) amyloidosis from serum amyloid A in chronic inflammation (rheumatoid arthritis, inflammatory bowel disease, chronic infections, familial Mediterranean fever) and predict renal involvement with nephrotic syndrome. NPLEX + USMLE | Pathology | DOK 3

IM-15-E.4 Match other amyloid proteins to their settings: beta2-microglobulin (long-term dialysis), transthyretin (wild-type age-related cardiac and hereditary), A-beta (Alzheimer disease), calcitonin (medullary thyroid carcinoma), and amylin (type 2 diabetes islets). USMLE extension | Pathology | DOK 2

IM-15-F Systemic infectious diseases

1 Erythema infectiosum (fifth disease) **2** Haemophilus influenzae type b **3** Influenza **4** Measles **5** Mononucleosis **6** Mumps **7** Roseola infantum **8** Rubella **9** Scarlet fever **10** Toxoplasmosis

IM-15-F.1 Identify erythema infectiosum (slapped cheek then lacy reticular rash in children, arthropathy in adults) and predict parvovirus B19 aplastic crisis in hemolytic anemias and hydrops fetalis in pregnancy. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.2 Identify invasive Haemophilus influenzae type b disease in unvaccinated young children (epiglottitis with drooling and stridor, meningitis, bacteremia, pneumonia), and contrast nontypeable strains causing otitis media and sinusitis. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.3 Identify influenza by abrupt fever, myalgia, headache, and cough, and predict complications: primary viral pneumonia, secondary bacterial pneumonia (S aureus, S pneumoniae), and Reye syndrome with aspirin use in children. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.4 Identify measles by cough, coryza, conjunctivitis, Koplik spots, and a rash spreading from the head downward, and predict pneumonia, encephalitis, subacute sclerosing panencephalitis, and worse outcomes with vitamin A deficiency. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.5 Identify infectious mononucleosis in an adolescent (fever, pharyngitis, posterior cervical lymphadenopathy, splenomegaly) by atypical CD8 T lymphocytes and heterophile antibodies, and predict rash after amoxicillin. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.6 Predict complications of EBV infection: splenic rupture, hepatitis, autoimmune hemolytic anemia, and associated malignancies (Burkitt lymphoma, Hodgkin lymphoma, nasopharyngeal carcinoma, lymphoma in immunosuppression). NPLEX + USMLE Pathology DOK 3

IM-15-F.7 Identify mumps by parotitis, and predict orchitis in postpubertal males, aseptic meningitis, and pancreatitis, with elevated serum amylase. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.8 Identify roseola infantum (HHV-6, less often HHV-7) in a child under 2 years: several days of high fever, then a truncal rash as the fever breaks, with risk of febrile seizures. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.9 Identify rubella (enveloped positive-sense RNA virus) by mild face-to-trunk rash, postauricular lymphadenopathy, and adult arthralgia, and predict congenital rubella (cataracts, sensorineural deafness, PDA, blueberry muffin rash) after first trimester infection. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.10 Identify scarlet fever after strep pharyngitis (sandpaper rash, circumoral pallor, strawberry tongue, Pastia lines, later desquamation) and predict rheumatic fever and poststreptococcal glomerulonephritis. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.11 Predict toxoplasmosis presentations: mono-like lymphadenopathy in healthy hosts, congenital chorioretinitis, hydrocephalus, and diffuse intracranial calcifications, and multiple ring-enhancing brain lesions in AIDS with CD4 below 100. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.12 Distinguish the childhood exanthems by rash pattern and timing relative to fever: measles, rubella, erythema infectiosum, roseola, and scarlet fever. NPLEX + USMLE Microbiology & Immunology DOK 3

IM-15-F.13 Identify CMV infection: heterophile-negative mononucleosis, owl's eye inclusions, congenital infection with periventricular calcifications and hearing loss, and retinitis, colitis, or pneumonitis in transplant and AIDS patients. USMLE extension Microbiology & Immunology DOK 3

Pulm Pulmonary System and Upper Respiratory Tract

8% · 16 questions · Week 9

HOW THE BODY WORKS

PU-1 Respiratory embryology

Describe embryological development of respiratory tract.

PU-1.1 Describe how the respiratory diverticulum (lung bud) arises from ventral foregut endoderm in week 4, and distinguish endodermal airway epithelium and glands from the splanchnic mesoderm that forms cartilage, smooth muscle, and connective tissue. NPLEX · Anatomy · DOK 1

PU-1.2 Explain how the tracheoesophageal ridges fuse to form the tracheoesophageal septum, and relate faulty septation to esophageal atresia with a distal tracheoesophageal fistula. NPLEX + USMLE · Anatomy · DOK 2

PU-1.3 Sequence the pseudoglandular, canalicular, saccular, and alveolar stages of lung development, and explain why survival with support first becomes possible near the end of the canalicular stage, once respiratory bronchioles and capillaries appear (about weeks 24 to 26). NPLEX · Anatomy · DOK 2

PU-1.4 Explain the timing of surfactant production by type II pneumocytes (beginning in the second trimester and reaching mature levels by about week 35), and interpret an amniotic lecithin to sphingomyelin ratio above 2 as a sign of fetal lung maturity. NPLEX · Anatomy · DOK 2

PU-1.5 Explain how oligohydramnios (for example from bilateral renal agenesis) and congenital diaphragmatic hernia cause pulmonary hypoplasia, and predict respiratory failure at birth. USMLE extension · Anatomy · DOK 3

PU-2 Respiratory histology

Describe microscopic anatomy of respiratory tract.

PU-2.1 Trace the change in airway epithelium from pseudostratified ciliated columnar with goblet cells in the trachea and bronchi, to simple cuboidal with club cells in terminal bronchioles, to type I and type II pneumocytes in alveoli. NPLEX · Anatomy · DOK 1

PU-2.2 Distinguish bronchi from bronchioles by the presence of cartilage plates, submucosal glands, and goblet cells, and explain why bronchioles depend on smooth muscle tone and radial traction from surrounding alveoli to stay open. NPLEX · Anatomy · DOK 2

PU-2.3 Describe the functions of type I pneumocytes (thin gas exchange surface), type II pneumocytes (surfactant secretion and regeneration of type I cells), club cells (secretion, cytochrome P450 detoxification, progenitors), and alveolar macrophages. NPLEX · Anatomy · DOK 2

PU-2.4 Explain mucociliary clearance by cilia with a 9 plus 2 microtubule axoneme and dynein arms, and predict how smoking-induced squamous metaplasia or ciliary dysfunction leads to retained mucus and recurrent infection. NPLEX · Anatomy · DOK 3

PU-2.5 Describe the layers of the blood-air barrier (type I pneumocyte, fused basement membranes, capillary endothelium), and relate interstitial fibrosis or edema that thickens it to impaired diffusion. NPLEX · Anatomy · DOK 2

PU-2.6 Identify the nonkeratinized stratified squamous epithelium of the true vocal folds and oropharynx, and relate it to squamous cell carcinoma as the main malignant tumor of the larynx. NPLEX · Anatomy · DOK 1

PU-3 Upper respiratory tract

Describe boundaries and anatomical structures associated with upper respiratory tract organs.

PU-3.1 Describe the nasal conchae and meatuses and identify sinus drainage: frontal, maxillary, and anterior ethmoid into the middle meatus; posterior ethmoid into the superior meatus; sphenoid into the sphenoethmoidal recess; nasolacrimal duct into the inferior meatus. NPLEX · Anatomy · DOK 1

PU-3.2 Explain why the maxillary sinus is the sinus most often infected (ostium high on its medial wall, poor gravity drainage), and relate the closeness of maxillary molar roots to its floor to odontogenic sinusitis. NPLEX + USMLE · Anatomy · DOK 2

PU-3.3 Locate Kiesselbach area on the anterior nasal septum, name the arteries that anastomose there (sphenopalatine, anterior ethmoidal, greater palatine, superior labial), and contrast it with posterior epistaxis from sphenopalatine branches. NPLEX + USMLE · Anatomy · DOK 2

PU-3.4 Describe the nasopharynx, oropharynx, and laryngopharynx, the tonsils of Waldeyer ring, and the pharyngotympanic tube opening, and relate adenoid enlargement to mouth breathing, snoring, and otitis media. NPLEX · Anatomy · DOK 2

PU-3.5 Identify the laryngeal cartilages and intrinsic muscles, state that the posterior cricoarytenoid is the only abductor of the vocal folds, and predict hoarseness from recurrent laryngeal nerve injury, noting that the left nerve loops under the aortic arch. NPLEX · Anatomy · DOK 3

PU-3.6 Distinguish laryngeal innervation by the internal laryngeal nerve (sensation above the vocal folds), external laryngeal nerve (cricothyroid), and recurrent laryngeal nerve (other intrinsic muscles, sensation below the folds), and relate supraglottic sensory loss to aspiration. NPLEX · Anatomy · DOK 2

PU-4 Thorax, pleura, lungs, and mediastinum

Describe boundaries and components of thorax in relation to pleura, lungs, heart, and mediastinum.

PU-4.1 Describe the surface projections of the lungs (ribs 6, 8, and 10) and pleura (ribs 8, 10, and 12) at the midclavicular, midaxillary, and paravertebral lines, and explain why a thoracentesis needle passes just above a rib. NPLEX · Anatomy · DOK 2

PU-4.2 Compare the right lung (three lobes, horizontal and oblique fissures) with the left (two lobes, lingula, cardiac notch), and predict where aspirated material lodges when upright (right lower lobe) or supine (superior segment of right lower lobe). NPLEX + USMLE · Anatomy · DOK 3

PU-4.3 Identify the relationships of the pulmonary artery, main bronchus, and pulmonary veins at each hilum, noting that the right pulmonary artery lies anterior and the left lies superior to its bronchus. NPLEX · Anatomy · DOK 1

PU-4.4 Describe the superior, anterior, middle, and posterior mediastinum and their contents, and predict typical masses in each, such as thymoma, teratoma, lymphoma, and retrosternal thyroid anteriorly and neurogenic tumors posteriorly. NPLEX + USMLE · Anatomy · DOK 2

PU-4.5 Compare the pain-sensitive parietal pleura (intercostal and phrenic nerves) with the visceral pleura (autonomic supply), and explain why irritation of the central diaphragmatic pleura refers pain to the shoulder (C3 to C5). NPLEX + USMLE · Anatomy · DOK 2

PU-4.6 Identify the diaphragmatic openings (inferior vena cava at T8, esophagus and vagal trunks at T10, aorta, thoracic duct, and azygos vein at T12) and the phrenic nerve supply of the diaphragm. NPLEX · Anatomy · DOK 1

PU-4.7 Trace the thoracic duct from the cisterna chyli through the aortic hiatus to the junction of the left subclavian and left internal jugular veins, and relate its injury to chylothorax. NPLEX + USMLE · Anatomy · DOK 2

PU-5 Pulmonary blood flow and airflow

Describe circulation of blood and flow of air in lungs.

PU-5.1 Compare the low-pressure, low-resistance pulmonary circulation with the systemic bronchial circulation, and explain why dual blood supply makes pulmonary infarction uncommon after embolism unless bronchial flow is also compromised. NPLEX + USMLE · Anatomy · DOK 2

PU-5.2 Explain hypoxic pulmonary vasoconstriction, contrast it with hypoxic vasodilation in systemic arterioles, and relate chronic alveolar hypoxia to pulmonary hypertension and right ventricular hypertrophy (cor pulmonale). NPLEX + USMLE · Physiology · DOK 3

PU-5.3 Describe West zones 1, 2, and 3 by the relationship of alveolar, arterial, and venous pressures, and predict that ventilation-perfusion ratio is highest at the apex, explaining why reactivation tuberculosis favors the apices. NPLEX + USMLE · Physiology · DOK 3

PU-5.4 Trace airflow through the conducting zone (trachea to terminal bronchioles, anatomic dead space about 150 mL) and the respiratory zone (respiratory bronchioles, alveolar ducts, alveoli), and explain why medium-sized bronchi contribute the most airway resistance. NPLEX · Anatomy · DOK 2

PU-5.5 Apply Poiseuille law to show that airway resistance varies inversely with the fourth power of the radius, and predict how bronchoconstriction, lung volume, and gas density change resistance and airflow. NPLEX · Physiology · DOK 3

PU-5.6 Explain why pulmonary vascular resistance is lowest at functional residual capacity and rises at low lung volumes (narrowed extra-alveolar vessels) and high lung volumes (compressed alveolar vessels). NPLEX · Physiology · DOK 2

PU-6 Ventilation

Explain mechanisms, functions, regulation, and factors affecting ventilation.

PU-6.1 Identify tidal volume, inspiratory and expiratory reserve volumes, residual volume, functional residual capacity, vital capacity, and total lung capacity, and explain why residual volume requires helium dilution or body plethysmography rather than simple spirometry. NPLEX · Physiology · DOK 1

PU-6.2 Calculate minute and alveolar ventilation, and physiologic dead space with the Bohr equation, and predict why rapid shallow breathing lowers alveolar ventilation at the same minute ventilation. NPLEX + USMLE Physiology DOK 3

PU-6.3 Explain lung and chest wall compliance, elastic recoil, the negative intrapleural pressure at functional residual capacity, and how surfactant lowers surface tension to stabilize small alveoli according to the law of Laplace. NPLEX Physiology DOK 2

PU-6.4 Interpret spirometry and flow-volume loops to distinguish obstructive disease (reduced FEV1/FVC, normal or increased TLC) from restrictive disease (normal or increased FEV1/FVC, reduced TLC) and fixed upper airway obstruction. NPLEX + USMLE Physiology DOK 3

PU-6.5 Describe medullary control of breathing (dorsal respiratory group for inspiration, ventral group for forced breathing), the pontine pneumotaxic and apneustic centers, and the Hering-Breuer inflation reflex. NPLEX Physiology DOK 1

PU-6.6 Compare central chemoreceptors, which respond to CSF hydrogen ion generated from arterial CO2, with carotid and aortic body chemoreceptors, which respond mainly to PaO2 below about 60 mmHg, and predict the ventilatory response to hypercapnia and hypoxemia. NPLEX + USMLE Physiology DOK 3

PU-6.7 Distinguish obstructive sleep apnea (pharyngeal collapse despite respiratory effort), central sleep apnea (absent effort), and obesity hypoventilation syndrome (daytime PaCO2 above 45 mmHg with BMI 30 or more), and predict pulmonary hypertension and secondary polycythemia. USMLE extension Physiology DOK 3

PU-7 Gas exchange and tissue perfusion

Explain mechanisms, functions, regulation, and factors affecting gas exchange and tissue perfusion.

PU-7.1 Calculate PAO2 with the alveolar gas equation (PAO2 = FiO2 times (Patm minus 47) minus PaCO2/0.8) and the A-a gradient, and interpret a normal versus a widened gradient. NPLEX + USMLE Physiology DOK 3

PU-7.2 Classify causes of hypoxemia (low inspired O2, hypoventilation, diffusion limitation, ventilation-perfusion mismatch, right-to-left shunt), and predict which have a normal A-a gradient and which fail to correct with supplemental oxygen. NPLEX + USMLE Physiology DOK 3

PU-7.3 Contrast dead space (V/Q approaching infinity, as in pulmonary embolism) with shunt (V/Q of zero, as in airway obstruction or alveoli filled by pneumonia or edema), and predict the effect of each on PaO2 and PaCO2. NPLEX + USMLE Physiology DOK 3

PU-7.4 Apply Fick law of diffusion, distinguish perfusion-limited exchange (N2O, O2 at rest) from diffusion-limited exchange (CO, O2 in fibrosis or strenuous exercise at altitude), and interpret reduced DLCO in emphysema and pulmonary fibrosis. NPLEX + USMLE Physiology DOK 3

PU-7.5 Distinguish hypoxemia from tissue hypoxia, and classify hypoxia as hypoxemic, anemic, circulatory, or histotoxic (cyanide), predicting PaO2, SaO2, and O2 content in each. NPLEX + USMLE Physiology DOK 3

PU-7.6 Explain acclimatization to high altitude (hyperventilation with respiratory alkalosis, renal bicarbonate excretion, increased erythropoietin and 2,3-BPG, hypoxic pulmonary vasoconstriction) and relate it to acute mountain sickness, high altitude pulmonary edema, and cerebral edema. USMLE extension Physiology DOK 3

PU-8 Blood gas transport

Explain transport and regulation of blood gases.

PU-8.1 Calculate arterial O2 content (1.34 times Hb times SaO2, plus 0.003 times PaO2), and predict how anemia, polycythemia, and carbon monoxide poisoning change PaO2, SaO2, and O2 content. NPLEX Physiology DOK 3

PU-8.2 Interpret the oxyhemoglobin dissociation curve, explain cooperative binding between T and R states, and name factors that shift it right (increased H+, CO2, temperature, 2,3-BPG) or left (fetal hemoglobin, CO, methemoglobin). NPLEX Physiology DOK 2

PU-8.3 Describe CO2 transport as bicarbonate (about 70 percent), carbaminohemoglobin, and dissolved CO2, including red cell carbonic anhydrase and the chloride shift. NPLEX Physiology DOK 2

PU-8.4 Explain the Bohr effect in tissues and the Haldane effect in the lungs, and relate them to O2 unloading and CO2 loading and unloading. NPLEX Physiology DOK 2

PU-8.5 Explain how carbon monoxide (about 200 times the affinity of O2, left shift) and methemoglobin (ferric iron, chocolate-colored blood) impair O2 delivery, and predict why pulse oximetry misleads while PaO2 stays normal. NPLEX + USMLE Physiology DOK 3

PU-8.6 Interpret arterial blood gases in respiratory acidosis and alkalosis, applying expected renal compensation (bicarbonate rises about 1 mEq/L acutely and 3 to 4 mEq/L chronically per 10 mmHg rise in PaCO2), and identify mixed disturbances. NPLEX + USMLE | Physiology | DOK 3

PU-9 Pulmonary biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to pulmonary function and pathology.

PU-9.1 Describe surfactant composition (dipalmitoylphosphatidylcholine as the main lipid, phosphatidylglycerol, surfactant proteins), and explain why fetal cortisol and thyroid hormone promote its synthesis while fetal hyperinsulinemia in maternal diabetes delays it. NPLEX | Biochemistry & Genetics | DOK 2

PU-9.2 Explain the protease and antiprotease balance in the lung, in which alpha-1 antitrypsin inhibits neutrophil elastase, and predict how oxidants in cigarette smoke and recruited neutrophils tip it toward alveolar wall destruction. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

PU-9.3 Relate elastin (cross-linked by copper-dependent lysyl oxidase) to lung recoil, and contrast increased compliance from elastin loss in emphysema with decreased compliance from collagen deposition in fibrosis. NPLEX | Biochemistry & Genetics | DOK 2

PU-9.4 Explain how vitamin A deficiency causes squamous metaplasia of respiratory epithelium and increases respiratory infections, and why pancreatic insufficiency in cystic fibrosis causes deficiency of fat-soluble vitamins A, D, E, and K. NPLEX | Biochemistry & Genetics | DOK 2

PU-9.5 Explain how high inspired oxygen concentrations and cigarette smoke generate reactive oxygen species that injure the lung, and describe protection by superoxide dismutase, catalase, and selenium-dependent glutathione peroxidase. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

PU-9.6 Describe airway mucus as heavily O-glycosylated mucin glycoproteins secreted by goblet cells and submucosal glands, and relate mucous gland hyperplasia in chronic bronchitis and dehydrated mucus in cystic fibrosis to impaired clearance. NPLEX | Biochemistry & Genetics | DOK 2

PU-10 Pulmonary genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying pulmonary disease.

PU-10.1 Describe cystic fibrosis as an autosomal recessive CFTR defect on chromosome 7, and explain how the common F508 deletion causes misfolding and proteasomal degradation, reducing chloride secretion and leaving dehydrated, thick airway mucus. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

PU-10.2 Explain alpha-1 antitrypsin deficiency (codominant SERPINA1 alleles, PiZZ most severe), and predict panacinar lower lobe emphysema plus cirrhosis with PAS-positive globules from misfolded protein retained in hepatocytes. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

PU-10.3 Relate dynein arm defects in primary ciliary dyskinesia (autosomal recessive) to bronchiectasis, chronic sinusitis, and infertility, and explain situs inversus in Kartagener syndrome. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

PU-10.4 Describe how p53 and Rb restrain the G1 to S transition, and explain how tobacco carcinogens such as benzo[a]pyrene form DNA adducts that cause mutations, including in TP53. NPLEX | Biochemistry & Genetics | DOK 2

PU-10.5 Identify common driver alterations in lung cancer (EGFR mutations, ALK rearrangements, and KRAS mutations in adenocarcinoma; TP53 and RB1 loss in small cell carcinoma), and relate EGFR mutations to adenocarcinoma in nonsmokers. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

PU-11 Energy production and the respiratory system

Explain biochemistry of energy production and utilization as it relates to respiratory system.

PU-11.1 Relate O2 as the final electron acceptor at cytochrome c oxidase (complex IV) to ATP production, and predict the shift to anaerobic glycolysis with lactic acidosis when tissue O2 delivery fails. NPLEX | Biochemistry & Genetics | DOK 2

PU-11.2 Explain how cyanide from burning plastics and carbon monoxide in smoke inhalation inhibit complex IV, and predict lactic acidosis with a normal PaO2 and, in cyanide poisoning, a high venous O2 saturation. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

PU-11.3 Calculate the respiratory quotient (CO2 produced divided by O2 consumed) for carbohydrate (1.0), fat (about 0.7), and protein (about 0.8), and predict how fuel mix changes CO2 production and ventilatory demand. NPLEX | Biochemistry & Genetics | DOK 3

PU-11.4 Explain how 2,3-BPG made in the red cell glycolytic shunt lowers hemoglobin O2 affinity, and predict how high altitude and pyruvate kinase deficiency increase 2,3-BPG and O2 unloading. NPLEX | Biochemistry & Genetics | DOK 3

PU-11.5 Relate the increased work of breathing in severe emphysema to higher resting energy expenditure and the weight loss seen in these patients. NPLEX | Biochemistry & Genetics | DOK 2

PU-12 Immune responses in pulmonary disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

PU-12.1 Describe innate lung defenses (nasal filtration, mucociliary escalator, cough, secretory IgA, alveolar macrophages, surfactant proteins A and D acting as opsonins), and relate their impairment by smoking, alcohol, and ciliary defects to pneumonia risk. NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-12.2 Explain type I hypersensitivity in allergic asthma and rhinitis, including Th2 cytokines (IL-4 for IgE class switching, IL-5 for eosinophils, IL-13 for mucus), mast cell degranulation, and the early and late phase responses. NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-12.3 Explain granuloma formation in tuberculosis and sarcoidosis (Th1 cells, IFN-gamma activation of macrophages, TNF-alpha maintenance of granulomas), and predict tuberculosis reactivation when TNF-alpha is blocked. NPLEX + USMLE | Microbiology & Immunology | DOK 3

PU-12.4 Distinguish hypersensitivity pneumonitis (type III and IV reactions to inhaled organic antigens such as thermophilic actinomycetes in farmer lung, with poorly formed interstitial granulomas) from asthma. USMLE extension | Microbiology & Immunology | DOK 3

PU-12.5 Contrast Goodpasture syndrome (type II hypersensitivity against type IV collagen, pulmonary hemorrhage, glomerulonephritis) with granulomatosis with polyangiitis (PR3-ANCA, necrotizing granulomatous vasculitis of upper and lower airways and kidney). USMLE extension | Microbiology & Immunology | DOK 3

PU-12.6 Explain how toxoid vaccines (diphtheria, tetanus), acellular pertussis vaccine, and polysaccharide-protein conjugate vaccines (Haemophilus influenzae type b, pneumococcus) induce protection, contrasting T-cell independent polysaccharide responses with T-cell dependent conjugate responses. NPLEX | Microbiology & Immunology | DOK 2

PU-12.7 Explain how lung cancers evade immunity by reducing MHC class I and expressing PD-L1, and relate antibodies to shared tumor and neural antigens (anti-Hu, anti-voltage-gated calcium channel) to paraneoplastic neurologic syndromes. NPLEX + USMLE | Microbiology & Immunology | DOK 3

PU-13 Pulmonary pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

PU-13.1 Describe Streptococcus pneumoniae (gram-positive lancet-shaped diplococci, alpha-hemolytic, optochin sensitive, bile soluble, polysaccharide capsule, IgA protease) as the most common cause of community-acquired and lobar pneumonia. NPLEX + USMLE | Microbiology & Immunology | DOK 1

PU-13.2 Compare gram-negative respiratory pathogens: Haemophilus influenzae (needs factors X and V), Klebsiella pneumoniae (mucoid capsule, alcoholics), Pseudomonas aeruginosa (oxidase positive, cystic fibrosis and ventilated patients), and Legionella (charcoal yeast extract agar, contaminated water). NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-13.3 Explain the toxins of Corynebacterium diphtheriae (prophage-encoded, ADP-ribosylates EF-2) and Bordetella pertussis (ADP-ribosylates Gi, raising cAMP), and link each to its clinical picture. NPLEX + USMLE | Microbiology & Immunology | DOK 3

PU-13.4 Describe Mycobacterium tuberculosis (acid-fast mycolic acid wall, cord factor, sulfatides), its spread by airborne droplet nuclei, and its survival in macrophages by blocking phagolysosome fusion. NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-13.5 Compare influenza (segmented negative-sense RNA, hemagglutinin and neuraminidase, antigenic drift and shift), respiratory syncytial virus (a pneumovirus, formerly classed as a paramyxovirus, whose F protein forms syncytia), and coronaviruses (enveloped positive-sense RNA; SARS-CoV-2 spike binds ACE2). NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-13.6 Distinguish the dimorphic fungi by geography and tissue form: Histoplasma (Ohio and Mississippi valleys, bird and bat droppings, small yeasts in macrophages), Blastomyces (Great Lakes and Ohio and Mississippi basins, broad-based budding yeast), and Coccidioides (southwestern United States, spherules with endospores). NPLEX + USMLE | Microbiology & Immunology | DOK 2

PU-13.7 Identify Aspergillus (septate hyphae branching at acute angles) and Pneumocystis jirovecii (cysts on methenamine silver stain), and relate each to the host defect that allows infection (neutropenia, CD4 count below 200 cells/mm3). NPLEX + USMLE | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

PU-14-A Restrictive pulmonary diseases

1 Respiratory distress syndrome **2** Idiopathic pulmonary fibrosis **3** Pneumoconiosis **4** Sarcoidosis

PU-14-A.1 Explain neonatal respiratory distress syndrome (surfactant deficiency from prematurity, maternal diabetes, or cesarean delivery) leading to atelectasis and hyaline membranes, and predict patent ductus arteriosus, plus bronchopulmonary dysplasia and retinopathy of prematurity from oxygen therapy. NPLEX | Pathology | DOK 3

PU-14-A.2 Explain acute respiratory distress syndrome as diffuse alveolar damage after sepsis, pneumonia, aspiration, trauma, or pancreatitis, with neutrophil-mediated capillary leak, hyaline membranes, bilateral infiltrates, and hypoxemia (PaO2/FiO2 of 300 or less) not explained by heart failure. NPLEX + USMLE | Pathology | DOK 3

PU-14-A.3 Describe idiopathic pulmonary fibrosis (repeated alveolar epithelial injury with TGF-beta driven fibrosis), its usual interstitial pneumonia pattern (patchy subpleural and basal fibrosis, fibroblastic foci, honeycombing), clubbing, and restrictive spirometry with reduced DLCO. NPLEX + USMLE | Pathology | DOK 3

PU-14-A.4 Compare pneumoconioses: coal workers (upper lobes, progressive massive fibrosis), silicosis (upper lobes, eggshell calcified hilar nodes, tuberculosis risk), asbestosis (lower lobes, ferruginous bodies), and berylliosis (noncaseating granulomas, aerospace work). NPLEX + USMLE | Pathology | DOK 3

PU-14-A.5 Describe sarcoidosis (noncaseating granulomas, bilateral hilar lymphadenopathy, elevated ACE, hypercalcemia from macrophage 1–alpha-hydroxylase), its higher incidence in Black women, and Lofgren syndrome (erythema nodosum, hilar lymphadenopathy, arthritis, fever). NPLEX + USMLE | Pathology | DOK 3

PU-14-A.6 Distinguish intrinsic restrictive disease (reduced DLCO) from extrapulmonary restriction caused by obesity, kyphoscoliosis, or neuromuscular weakness (usually normal DLCO and a normal A-a gradient with hypoventilation). NPLEX + USMLE | Physiology | DOK 3

PU-14-A.7 Distinguish desquamative interstitial pneumonia (smokers, macrophages filling alveoli), nonspecific interstitial pneumonia (uniform interstitial inflammation and fibrosis), and cryptogenic organizing pneumonia (polypoid fibroblast plugs, Masson bodies). USMLE extension | Pathology | DOK 3

PU-14-A.8 Identify occupational causes of lung disease: byssinosis (cotton dust), bagassosis (moldy sugarcane, hypersensitivity pneumonitis), and silo-filler disease (nitrogen dioxide causing bronchiolitis and pulmonary edema). USMLE extension | Pathology | DOK 1

PU-14-A.9 Identify bleomycin, amiodarone, busulfan, and methotrexate as causes of drug-induced pulmonary fibrosis to be distinguished from idiopathic pulmonary fibrosis. USMLE extension | Pathology | DOK 1

PU-14-B Obstructive pulmonary diseases

1 Asthma **2** Bronchiectasis **3** Chronic bronchitis **4** Emphysema

PU-14-B.1 Explain asthma pathogenesis (atopic Th2 inflammation and nonatopic triggers such as viral infection, cold air, exercise, and aspirin), and identify Curschmann spirals, Charcot-Leyden crystals, goblet cell hyperplasia, basement membrane thickening, and smooth muscle hypertrophy. NPLEX + USMLE | Pathology | DOK 3

PU-14-B.2 Predict findings in acute asthma (wheezing, prolonged expiration, pulsus paradoxus, reversible obstruction on spirometry), and interpret a normal or rising PaCO2 after early respiratory alkalosis as a sign of tiring and impending respiratory failure. NPLEX + USMLE | Pathology | DOK 3

PU-14-B.3 Explain bronchiectasis as permanent airway dilation from obstruction plus necrotizing infection, identify causes (cystic fibrosis, primary ciliary dyskinesia, allergic bronchopulmonary aspergillosis, tumor or foreign body), and predict copious purulent sputum, hemoptysis, clubbing, and amyloidosis. NPLEX + USMLE | Pathology | DOK 3

PU-14-B.4 Describe chronic bronchitis (productive cough for at least 3 months in each of 2 consecutive years), mucous gland hyperplasia with a Reid index above 0.5, and the hypoxemia, hypercapnia, cyanosis, and cor pulmonale of the blue bloater. NPLEX + USMLE | Pathology | DOK 3

PU-14-B.5 Compare centriacinar emphysema (smoking, upper lobes), panacinar emphysema (alpha-1 antitrypsin deficiency, lower lobes), and paraseptal emphysema (subpleural bullae, spontaneous pneumothorax in young adults), and predict barrel chest, air trapping, and pursed-lip breathing. NPLEX + USMLE Pathology DOK 3

PU-14-B.6 Interpret COPD testing (FEV1/FVC below 0.7, increased TLC and residual volume), and contrast DLCO and blood gases in predominant emphysema (low DLCO, near normal gases until late) with predominant chronic bronchitis (normal DLCO, early hypoxemia and hypercapnia). NPLEX + USMLE Physiology DOK 3

PU-14-C Diseases of the upper respiratory tract

1 Bronchitis **2** Epiglottitis **3** Laryngitis **4** Rhinitis **5** Sinusitis **6** Pharyngitis **7** Tonsillitis

PU-14-C.1 Explain acute bronchitis as a mostly viral, self-limited inflammation of the large airways causing cough for up to about 3 weeks, and distinguish it from pneumonia (no consolidation) and from chronic bronchitis. NPLEX + USMLE Pathology DOK 2

PU-14-C.2 Describe epiglottitis (classically Haemophilus influenzae type b in unvaccinated children) with sudden fever, drooling, dysphagia, muffled voice, tripod posture, and a thumb sign on lateral neck radiograph, and predict sudden airway obstruction. NPLEX + USMLE Pathology DOK 3

PU-14-C.3 Distinguish acute laryngitis (usually viral, hoarseness) from croup (parainfluenza virus laryngotracheobronchitis, barking cough, inspiratory stridor, steeple sign) and bacterial tracheitis (toxic child, purulent tracheal secretions, often Staphylococcus aureus). NPLEX + USMLE Pathology DOK 3

PU-14-C.4 Compare allergic rhinitis (IgE, eosinophils, pale boggy mucosa, seasonal or perennial) with infectious rhinitis (rhinovirus, an acid-labile picornavirus binding ICAM-1, as the main cause of the common cold) and nonallergic vasomotor rhinitis. NPLEX + USMLE Pathology DOK 2

PU-14-C.5 Explain sinusitis from ostial obstruction (mostly viral; bacterial Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis), and predict complications such as orbital cellulitis, cavernous sinus thrombosis, and mucormycosis in diabetic ketoacidosis. NPLEX + USMLE Pathology DOK 3

PU-14-C.6 Distinguish Streptococcus pyogenes pharyngitis (beta-hemolytic, bacitracin sensitive, M protein; fever, tonsillar exudate, tender anterior nodes, no cough) from viral pharyngitis, and predict rheumatic fever (molecular mimicry) and poststreptococcal glomerulonephritis. NPLEX + USMLE Pathology DOK 3

PU-14-C.7 Distinguish streptococcal tonsillitis from infectious mononucleosis (EBV, exudative tonsillitis, posterior cervical nodes, atypical lymphocytes, heterophile antibodies, splenomegaly), and predict peritonsillar abscess (uvular deviation, trismus, muffled voice). NPLEX + USMLE Pathology DOK 3

PU-14-D Disorders of the pleural cavity and lung expansion

1 Chylothorax **2** Hemothorax **3** Obstructive atelectasis **4** Pleural fibrosis (asbestosis) **5** Pneumothorax

PU-14-D.1 Explain chylothorax from thoracic duct disruption (trauma, surgery, lymphoma), and identify milky pleural fluid with triglycerides above 110 mg/dL and chylomicrons. NPLEX + USMLE Pathology DOK 2

PU-14-D.2 Describe hemothorax from chest trauma, vascular rupture, or malignancy, define it by a pleural fluid hematocrit at least half the blood hematocrit, and predict hypovolemic shock and later fibrothorax. NPLEX + USMLE Pathology DOK 3

PU-14-D.3 Explain obstructive (resorption) atelectasis from absorption of trapped alveolar air beyond a mucus plug, tumor, or foreign body, predict mediastinal shift toward the collapse, and contrast compression atelectasis, which shifts the mediastinum away. NPLEX + USMLE Pathology DOK 3

PU-14-D.4 Describe asbestos-related pleural disease (calcified parietal pleural plaques, effusion, diffuse pleural fibrosis), ferruginous bodies, and the increased risk of bronchogenic carcinoma (synergistic with smoking) and mesothelioma. NPLEX + USMLE Pathology DOK 3

PU-14-D.5 Compare primary spontaneous pneumothorax (tall thin young men, ruptured apical blebs), secondary pneumothorax (COPD, cystic fibrosis), and tension pneumothorax (tracheal deviation away, hypotension, distended neck veins), and predict hyperresonance with absent breath sounds. NPLEX + USMLE Pathology DOK 3

PU-14-D.6 Classify pleural effusions with Light criteria (exudate if pleural to serum protein above 0.5, LDH ratio above 0.6, or LDH above two thirds of the serum upper limit), and identify transudative (heart failure, cirrhosis, nephrotic syndrome) versus exudative causes. USMLE extension Pathology DOK 3

PU-14-D.7 Identify empyema (purulent exudate after pneumonia), mediastinitis (esophageal perforation, sternotomy, inhalational anthrax), pneumomediastinum (subcutaneous crepitus, Hamman crunch), and costochondritis (reproducible chest wall tenderness). USMLE extension Pathology DOK 2

PU-14-D.8 Predict findings of chest trauma: flail chest (paradoxical segment motion), pulmonary contusion (hemorrhage with worsening hypoxemia), open pneumothorax from a penetrating wound, and diaphragmatic rupture (bowel in the left hemithorax). USMLE extension · Pathology · DOK 3

PU-14-E Pulmonary vascular disease

1 Pulmonary edema **2** Pulmonary emboli **3** Pulmonary infarction

PU-14-E.1 Explain pulmonary edema using Starling forces, contrasting cardiogenic edema (raised capillary hydrostatic pressure, wedge pressure above 18 mmHg, hemosiderin-laden macrophages) with noncardiogenic edema from capillary injury, low oncotic pressure, or lymphatic obstruction. NPLEX + USMLE · Pathology · DOK 3

PU-14-E.2 Explain pulmonary embolism from deep vein thrombosis (Virchow triad), and predict sudden dyspnea, pleuritic pain, tachycardia, hypoxemia with respiratory alkalosis, a widened A-a gradient, raised D-dimer, and acute right heart strain with saddle emboli. NPLEX + USMLE · Pathology · DOK 3

PU-14-E.3 Distinguish fat embolism (long bone fracture, hypoxemia, neurologic changes, petechial rash within 24 to 72 hours), air embolism (venous catheters, diving decompression), and amniotic fluid embolism (sudden collapse and DIC during delivery). USMLE extension · Pathology · DOK 3

PU-14-E.4 Describe pulmonary infarction as a wedge-shaped hemorrhagic infarct based on the pleura, occurring when an embolus blocks a medium or small artery and bronchial flow is reduced (as in heart failure), and predict hemoptysis and pleuritic pain. NPLEX + USMLE · Pathology · DOK 3

PU-14-E.5 Classify pulmonary hypertension (pulmonary arterial including BMPR2 mutation, left heart disease, lung disease or hypoxia, chronic thromboembolism), and describe medial hypertrophy, intimal fibrosis, plexiform lesions, and progression to cor pulmonale. USMLE extension · Pathology · DOK 3

PU-14-E.6 Describe pulmonary arteriovenous malformations in hereditary hemorrhagic telangiectasia (autosomal dominant), and predict hypoxemia from right-to-left shunting, hemoptysis, and paradoxical embolic stroke or brain abscess. USMLE extension · Pathology · DOK 3

PU-14-F Neoplasms of the pulmonary system and upper respiratory tract

1 Adenocarcinomas **2** Bronchial carcinoid **3** Esophageal **4** Laryngeal **5** Mesothelioma **6** Nasopharyngeal carcinoma **7** Polyps **8** Small-cell carcinoma **9** Non-small-cell carcinoma

PU-14-F.1 Describe adenocarcinoma as the most common lung cancer overall and in nonsmokers and women, typically peripheral and mucin producing, and identify adenocarcinoma in situ by lepidic growth along alveolar walls without invasion. NPLEX + USMLE · Pathology · DOK 3

PU-14-F.2 Describe bronchial carcinoid as a low-grade neuroendocrine tumor (nests of uniform cells, chromogranin positive, often a central polypoid mass, not linked to smoking) that may obstruct a bronchus and rarely produces carcinoid syndrome. NPLEX + USMLE · Pathology · DOK 2

PU-14-F.3 Compare esophageal squamous cell carcinoma (upper and middle third; smoking, alcohol, very hot drinks, achalasia, nitrosamines) with adenocarcinoma (distal third; GERD, Barrett esophagus, obesity), and predict progressive dysphagia, weight loss, and tracheoesophageal fistula. NPLEX · Pathology · DOK 3

PU-14-F.4 Describe laryngeal squamous cell carcinoma (smoking and alcohol, early hoarseness when glottic), and contrast it with benign vocal fold nodules from voice overuse and juvenile laryngeal papillomatosis from HPV types 6 and 11. NPLEX + USMLE · Pathology · DOK 3

PU-14-F.5 Describe malignant mesothelioma (asbestos exposure decades earlier, no smoking link, recurrent hemorrhagic effusion, tumor encasing the lung, calretinin and cytokeratin positive, long slender microvilli), and distinguish it from metastatic pleural adenocarcinoma. NPLEX + USMLE · Pathology · DOK 3

PU-14-F.6 Describe nasopharyngeal carcinoma (EBV association, higher incidence in southern China and parts of Africa, nonkeratinizing or undifferentiated histology), and predict cervical lymph node metastasis and unilateral serous otitis media from eustachian tube blockage. NPLEX + USMLE · Pathology · DOK 3

PU-14-F.7 Explain nasal polyps (chronic rhinosinusitis, allergy, aspirin-exacerbated respiratory disease with asthma, cystic fibrosis in a child) and vocal fold polyps (voice strain, smoking), and predict nasal obstruction, anosmia, and hoarseness. NPLEX + USMLE · Pathology · DOK 2

PU-14-F.8 Describe small cell carcinoma (central, strongly linked to smoking, small neuroendocrine cells with scant cytoplasm and nuclear molding, chromogranin and synaptophysin positive), and predict early metastasis and ACTH, ADH, and Lambert-Eaton paraneoplastic syndromes. NPLEX + USMLE · Pathology · DOK 3

PU-14-F.9 Compare non-small-cell carcinomas: squamous cell (central, smoking, keratin pearls, intercellular bridges, cavitation, PTHrP hypercalcemia) and large cell (peripheral, poorly differentiated, poor prognosis), alongside adenocarcinoma. NPLEX + USMLE | Pathology | DOK 3

PU-14-F.10 Match lung cancer paraneoplastic syndromes to type: PTHrP hypercalcemia (squamous cell), ACTH Cushing syndrome and SIADH (small cell), Lambert-Eaton myasthenic syndrome (small cell), and hypertrophic pulmonary osteoarthropathy with clubbing (especially adenocarcinoma). NPLEX + USMLE | Pathology | DOK 3

PU-14-F.11 Predict local effects of lung cancer spread: superior sulcus (Pancoast) tumor with Horner syndrome and brachial plexus deficits, superior vena cava syndrome, hoarseness from recurrent laryngeal nerve invasion, and diaphragm elevation from phrenic nerve invasion. NPLEX + USMLE | Pathology | DOK 3

PU-14-F.12 Distinguish metastases to the lung (multiple well-circumscribed nodules, more common than primary lung cancer) from a benign hamartoma (cartilage and fat, popcorn calcification) when evaluating a solitary pulmonary nodule. USMLE extension | Pathology | DOK 3

PU-14-F.13 Relate HPV type 16 to oropharyngeal squamous cell carcinoma of the tonsil and base of tongue, and contrast it with tobacco and alcohol related squamous carcinoma of the oral cavity and pharynx. USMLE extension | Pathology | DOK 2

PU-14-G Congenital pulmonary diseases

1 Cystic fibrosis (CF) **2** Tracheoesophageal fistula

PU-14-G.1 Predict the clinical picture of cystic fibrosis: meconium ileus, pancreatic insufficiency with steatorrhea, chronic Pseudomonas and Staphylococcus aureus infection, bronchiectasis, nasal polyps, male infertility from absent vas deferens, and sweat chloride of 60 mmol/L or higher. NPLEX + USMLE | Pathology | DOK 3

PU-14-G.2 Explain why sweat is salty in cystic fibrosis (failed chloride reabsorption in the sweat duct) while airway surface liquid is depleted (failed chloride secretion with excess sodium absorption through ENaC). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

PU-14-G.3 Describe tracheoesophageal fistula, most often esophageal atresia with a distal fistula, and predict polyhydramnios, drooling, choking and cyanosis with feeds, a gastric air bubble, and failure to pass a nasogastric tube, noting the VACTERL association. NPLEX + USMLE | Pathology | DOK 3

PU-14-G.4 Describe congenital diaphragmatic hernia (usually left posterolateral, failed pleuroperitoneal fold closure) with abdominal contents in the chest, pulmonary hypoplasia, and a scaphoid abdomen at birth. USMLE extension | Anatomy | DOK 3

PU-14-G.5 Distinguish pulmonary sequestration (nonfunctioning lung tissue with systemic arterial supply from the aorta and no bronchial connection) from bronchogenic cyst (abnormal foregut budding, fluid-filled, usually mediastinal), and predict recurrent infection. USMLE extension | Anatomy | DOK 3

PU-14-H Infectious diseases of the pulmonary system and upper respiratory tract

1 Atypical pneumonia **2** Bronchopneumonia **3** Diphtheria **4** Fungal pneumonia **5** Lobar pneumonia **6** Lung abscess **7** Pertussis **8** Respiratory syncytial virus (RSV) **9** Tuberculosis (TB) **10** Coronaviruses

PU-14-H.1 Compare lobar pneumonia (Streptococcus pneumoniae, Klebsiella; congestion, red hepatization, gray hepatization, resolution; dullness, egophony, increased fremitus) with bronchopneumonia (Staphylococcus aureus, Haemophilus influenzae, Pseudomonas; patchy, often multilobar, centered on bronchioles). NPLEX + USMLE | Pathology | DOK 3

PU-14-H.2 Describe atypical pneumonia (Mycoplasma pneumoniae, which lacks a cell wall, with cold agglutinins in young adults in close quarters; Chlamydia pneumoniae; Legionella with hyponatremia and diarrhea; viruses), and predict patchy interstitial infiltrates that look worse than the patient. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.3 Describe diphtheria (gray adherent pseudomembrane that bleeds when scraped, bull neck lymphadenopathy, tellurite agar, Elek test) and predict toxin-mediated myocarditis, heart block, and cranial and peripheral neuropathy. NPLEX | Pathology | DOK 3

PU-14-H.4 Describe endemic fungal pneumonia (histoplasmosis, blastomycosis, coccidioidomycosis) in healthy hosts, and predict dissemination to skin, bone, and meninges, and granulomas mimicking tuberculosis. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.5 Distinguish Pneumocystis pneumonia (CD4 below 200, bilateral ground-glass infiltrates, exertional hypoxemia, high LDH) from aspergillosis (aspergilloma in an old cavity, angioinvasive disease in neutropenia, allergic bronchopulmonary aspergillosis in asthma and cystic fibrosis). NPLEX + USMLE | Pathology | DOK 3

PU-14-H.6 Explain lung abscess from aspiration of oropharyngeal anaerobes (Bacteroides, Fusobacterium, Peptostreptococcus) in alcoholism, seizures, or dysphagia, and predict an air-fluid level in dependent segments and foul-smelling sputum; identify Staphylococcus aureus and Klebsiella as other causes. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.7 Describe the catarrhal, paroxysmal (inspiratory whoop, posttussive vomiting), and convalescent stages of pertussis, its marked lymphocytosis, and apnea and pneumonia in young infants. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.8 Describe respiratory syncytial virus as the most common cause of bronchiolitis and pneumonia in infants (winter epidemics, wheezing, tachypnea), and identify prematurity and congenital heart or lung disease as risk factors for severe disease. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.9 Trace primary tuberculosis from the Ghon focus and hilar node (Ghon complex) with caseating granulomas to latent infection, and interpret tuberculin skin test and interferon-gamma release assay results. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.10 Predict secondary (reactivation) tuberculosis in the apices with cavitation, fever, night sweats, weight loss, and hemoptysis, and miliary spread to kidneys, meninges, and vertebrae (Pott disease), especially with HIV or immunosuppression. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.11 Describe coronavirus disease: common cold strains, and SARS, MERS, and COVID-19 (SARS-CoV-2 entry via ACE2 on airway and alveolar cells, loss of smell, diffuse alveolar damage, thrombosis), and identify older age and comorbidity as risk factors. NPLEX + USMLE | Pathology | DOK 3

PU-14-H.12 Identify likely organisms in hospital-acquired and ventilator-associated pneumonia (Pseudomonas, MRSA, enteric gram-negative rods), and distinguish chemical aspiration pneumonitis from bacterial aspiration pneumonia. USMLE extension | Pathology | DOK 3

PU-14-H.13 Explain influenza pneumonia and predict secondary bacterial pneumonia with Staphylococcus aureus or Streptococcus pneumoniae, and identify avian influenza as a source of novel strains through reassortment. USMLE extension | Pathology | DOK 3

PU-14-H.14 Identify zoonotic and bioterrorism causes of pneumonia: inhalational anthrax (hemorrhagic mediastinitis, widened mediastinum), tularemia (rabbits, ticks), Q fever (Coxiella from farm animal birth products), and melioidosis (Burkholderia pseudomallei, Southeast Asia). USMLE extension | Microbiology & Immunology | DOK 2

Heme Hematopoietic System

6% · 12 questions · Week 10

HOW THE BODY WORKS

HE-1 Blood cell histology, origin, and maturation

Describe microscopic anatomy, origins, and maturation of blood cells.

HE-1.1 Identify erythrocytes, neutrophils, bands, eosinophils, basophils, lymphocytes, monocytes, and platelets on a Wright-stained peripheral smear by cell size, nuclear shape, and granule color. NPLEX · Anatomy · DOK 1

HE-1.2 Trace hematopoiesis from the hematopoietic stem cell through common myeloid and common lymphoid progenitors to each mature blood cell lineage. NPLEX · Anatomy · DOK 2

HE-1.3 Describe the sequence of hematopoietic sites in development (yolk sac, then liver and spleen, then bone marrow) and relate marrow fibrosis and ineffective erythropoiesis (beta thalassemia major) after birth to extramedullary hematopoiesis and splenomegaly. NPLEX + USMLE · Anatomy · DOK 2

HE-1.4 Describe erythroid maturation from proerythroblast through reticulocyte to mature erythrocyte, including nuclear condensation and extrusion, hemoglobin accumulation, and loss of residual ribosomal RNA, and explain what an elevated reticulocyte count indicates about marrow response. NPLEX + USMLE · Anatomy · DOK 2

HE-1.5 Describe granulocyte maturation from myeloblast through promyelocyte, myelocyte, metamyelocyte, and band to segmented neutrophil, and identify a left shift on a peripheral smear. NPLEX · Anatomy · DOK 2

HE-1.6 Describe platelet formation by fragmentation of megakaryocyte cytoplasm and identify platelet granule contents: alpha granules (von Willebrand factor, fibrinogen) and dense granules (ADP, calcium, serotonin). NPLEX · Anatomy · DOK 1

HE-1.7 Locate B cell and T cell zones in the lymph node (cortical follicles, paracortex, medullary cords and sinuses) and the spleen (white pulp, periarteriolar lymphoid sheath, marginal zone, red pulp cords and sinusoids). NPLEX · Anatomy · DOK 1

HE-2 Blood cells and plasma

Describe composition and explain function and regulation of blood cells and plasma.

HE-2.1 Describe the composition of plasma (water, albumin, globulins, fibrinogen, electrolytes) and distinguish plasma from serum. NPLEX · Physiology · DOK 1

HE-2.2 Explain the roles of albumin in plasma oncotic pressure and in transport of bilirubin, fatty acids, and calcium, and predict edema from hypoalbuminemia using Starling forces. NPLEX · Physiology · DOK 2

HE-2.3 Describe red cell features that support gas transport (biconcave shape, no nucleus or mitochondria, carbonic anhydrase converting CO2 to bicarbonate, chloride shift) and explain their functional advantages. NPLEX · Physiology · DOK 2

HE-2.4 Explain how erythropoietin from renal peritubular interstitial cells responds to tissue hypoxia, and predict erythropoietin levels in chronic kidney disease, high altitude, and polycythemia vera. NPLEX + USMLE · Physiology · DOK 3

HE-2.5 Interpret a complete blood count with hemoglobin, hematocrit, MCV, MCHC, RDW, and reticulocyte count to classify an anemia as microcytic, normocytic, or macrocytic and as underproduction or hemolytic. NPLEX + USMLE · Physiology · DOK 3

HE-2.6 Explain the functions of neutrophils, eosinophils, basophils, monocytes, and lymphocytes, and relate neutrophilia, eosinophilia, and lymphocytosis to bacterial infection, parasitic or allergic disease, and viral infection. NPLEX + USMLE · Physiology · DOK 2

HE-2.7 Describe ABO and Rh antigens and naturally occurring IgM anti-A and anti-B antibodies, and predict compatible donors for red cell and plasma transfusion. NPLEX + USMLE · Physiology · DOK 2

HE-3 Blood cell synthesis and breakdown

Describe synthesis and degradation of blood cells.

HE-3.1 Describe the red cell life span of about 120 days and the removal of senescent red cells by macrophages of the spleen and liver. NPLEX · Physiology · DOK 1

HE-3.2 Trace heme catabolism from heme to biliverdin (heme oxygenase) and unconjugated bilirubin (biliverdin reductase), albumin transport, hepatic conjugation by UDP-glucuronosyltransferase, and excretion as urobilinogen, urobilin, and stercobilin. NPLEX + USMLE · Biochemistry & Genetics · DOK 2

HE-3.3 Distinguish intravascular from extravascular hemolysis using haptoglobin, LDH, unconjugated bilirubin, hemoglobinuria, hemosiderinuria, and smear findings (schistocytes versus spherocytes). NPLEX + USMLE Pathology DOK 3

HE-3.4 Explain iron recycling and absorption: duodenal uptake of ferrous iron by DMT1, export through ferroportin, transferrin transport, ferritin storage, and hepcidin inhibition of ferroportin. NPLEX + USMLE Biochemistry & Genetics DOK 2

HE-3.5 Relate the short circulating life of neutrophils (hours) and platelets (about 7 to 10 days) compared with red cells to explain why neutropenia and thrombocytopenia appear before anemia after sudden marrow suppression. NPLEX + USMLE Physiology DOK 3

HE-4 Hematopoiesis and hemostasis

Explain mechanisms and factors affecting hematopoiesis and hemostasis.

HE-4.1 Identify the growth factors that drive each lineage: erythropoietin for red cells, G-CSF and GM-CSF for granulocytes, thrombopoietin for megakaryocytes, and IL-5 for eosinophils. NPLEX Physiology DOK 1

HE-4.2 Describe primary hemostasis: vasoconstriction, von Willebrand factor bridging exposed collagen to platelet GpIb, platelet degranulation with ADP and thromboxane A2 release, and fibrinogen linking GpIIb/IIIa receptors during aggregation. NPLEX + USMLE Physiology DOK 2

HE-4.3 Trace secondary hemostasis through the tissue factor (extrinsic) and contact (intrinsic) pathways to the common pathway, thrombin generation, and fibrin cross-linking by factor XIIIa, and relate PT and PTT to each pathway. NPLEX + USMLE Physiology DOK 2

HE-4.4 Explain how antithrombin, thrombomodulin with proteins C and S, and tissue factor pathway inhibitor limit clotting, and how plasmin from tPA-activated plasminogen produces fibrin degradation products and D-dimer. NPLEX + USMLE Physiology DOK 2

HE-4.5 Interpret platelet count, bleeding time, PT/INR, PTT, fibrinogen, and a mixing study to localize a bleeding disorder to platelets, a specific coagulation pathway, a factor inhibitor, or consumption. NPLEX + USMLE Physiology DOK 3

HE-4.6 Describe Virchow triad (stasis, endothelial injury, hypercoagulability) and relate immobility, surgery, pregnancy, malignancy, and estrogen use to venous thrombosis risk. NPLEX + USMLE Pathology DOK 2

HE-5 Hemoglobin and hematologic biochemistry

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to hematopoietic function, hemostasis, and hemoglobin function, formation, and pathology.

HE-5.1 Trace heme synthesis from glycine and succinyl-CoA through ALA synthase (rate limiting, pyridoxine dependent), ALA dehydratase, and ferrochelatase, locating mitochondrial and cytosolic steps and explaining heme feedback on ALA synthase. NPLEX + USMLE Biochemistry & Genetics DOK 2

HE-5.2 Distinguish acute intermittent porphyria (porphobilinogen deaminase deficiency, abdominal pain, neuropsychiatric symptoms, high urine porphobilinogen) from porphyria cutanea tarda (uroporphyrinogen decarboxylase deficiency, blistering photosensitivity, hepatitis C association). NPLEX + USMLE Biochemistry & Genetics DOK 3

HE-5.3 Describe the globin chains of hemoglobin A, A2, and F, explain cooperative oxygen binding, and predict shifts of the oxygen dissociation curve with 2,3-BPG, H+, CO2, temperature, fetal hemoglobin, and carbon monoxide. NPLEX + USMLE Biochemistry & Genetics DOK 3

HE-5.4 Explain how ferric iron in methemoglobin and carbon monoxide binding reduce oxygen delivery despite a normal PaO2, and identify congenital (cytochrome b5 reductase deficiency) and acquired (nitrites, dapsone, benzocaine) methemoglobinemia. NPLEX + USMLE Biochemistry & Genetics DOK 3

HE-5.5 Explain why red cells depend on glycolysis and the pentose phosphate pathway, and relate pyruvate kinase deficiency to ATP loss and G6PD deficiency to NADPH and reduced glutathione loss with oxidative hemolysis. NPLEX + USMLE Biochemistry & Genetics DOK 3

HE-5.6 Explain the roles of vitamin B12 and folate in thymidine synthesis, methionine synthase, and methylmalonyl-CoA mutase, and use methylmalonic acid and homocysteine levels to distinguish B12 from folate deficiency. NPLEX + USMLE Biochemistry & Genetics DOK 3

HE-5.7 Explain vitamin K dependent gamma-carboxylation of factors II, VII, IX, and X and proteins C and S, and why these carboxylated residues need calcium to bind phosphatidylserine on activated platelet membranes. NPLEX + USMLE Biochemistry & Genetics DOK 2

HE-6 Hematologic genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying hematopoietic disease.

HE-6.1 Explain how a glutamate to valine substitution at position 6 of beta globin causes HbS polymerization when deoxygenated, and predict how fetal hemoglobin, dehydration, acidosis, and hypoxia change sickling. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

HE-6.2 Relate the number of deleted alpha globin genes (one to four) to silent carrier, alpha thalassemia trait, HbH disease, and Hb Barts hydrops fetalis, and contrast beta thalassemia caused by point mutations (often splice site or promoter). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

HE-6.3 Identify the inheritance of hemophilia A and B and G6PD deficiency (X-linked recessive), sickle cell disease and thalassemias (autosomal recessive), and most hereditary spherocytosis and von Willebrand disease (autosomal dominant), and predict offspring risk. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

HE-6.4 Explain how t(9;22) forms the BCR-ABL1 fusion gene with constitutive tyrosine kinase activity in CML, and how the JAK2 V617F mutation drives growth factor independent proliferation in polycythemia vera. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

HE-6.5 Relate characteristic translocations to the genes they deregulate: t(8;14) MYC in Burkitt lymphoma, t(14;18) BCL2 (blocked apoptosis) in follicular lymphoma, t(11;14) cyclin D1 in mantle cell lymphoma, and t(15;17) PML-RARA (blocked differentiation) in acute promyelocytic leukemia. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

HE-6.6 Explain how defective DNA crosslink repair in Fanconi anemia causes congenital marrow failure with short stature, thumb and radial anomalies, and increased risk of AML. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

HE-6.7 Explain how factor V Leiden (resistance to activated protein C) and the prothrombin G20210A mutation (increased prothrombin levels) raise venous thrombosis risk. USMLE extension | Biochemistry & Genetics | DOK 2

HE-7 Immune responses in hematologic disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

HE-7.1 Distinguish warm autoimmune hemolytic anemia (IgG, extravascular, spherocytes; SLE, CLL) from cold agglutinin disease (IgM with complement; Mycoplasma pneumoniae, EBV), and interpret a positive direct Coombs test. NPLEX + USMLE | Microbiology & Immunology | DOK 3

HE-7.2 Classify hematologic immune reactions by hypersensitivity type: acute hemolytic transfusion reaction, autoimmune hemolysis, and hemolytic disease of the newborn (type II), and cryoglobulinemia with immune complex vasculitis (type III). NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-7.3 Distinguish acute hemolytic, delayed hemolytic, febrile nonhemolytic, and anaphylactic (IgA deficiency) transfusion reactions, septic reactions from contaminated units, and TRALI by mechanism, timing, and key findings. NPLEX + USMLE | Microbiology & Immunology | DOK 3

HE-7.4 Explain how IgG antibodies against heparin and platelet factor 4 complexes activate platelets in heparin-induced thrombocytopenia, producing a falling platelet count with arterial and venous thrombosis. USMLE extension | Microbiology & Immunology | DOK 2

HE-7.5 Explain how antiphospholipid antibodies (lupus anticoagulant, anticardiolipin, anti-beta2 glycoprotein I) cause arterial and venous thrombosis and pregnancy loss despite prolonging the PTT in vitro. USMLE extension | Microbiology & Immunology | DOK 3

HE-7.6 Explain how Epstein-Barr virus enters B cells through CD21, produces atypical CD8 T cell lymphocytosis in mononucleosis, and contributes to Burkitt lymphoma, Hodgkin lymphoma, and lymphomas in immunosuppressed patients. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-7.7 Explain why asplenia (surgical or sickle cell autosplenectomy) raises risk of sepsis from encapsulated organisms (Streptococcus pneumoniae, Haemophilus influenzae type b, Neisseria meningitidis) and why vaccination against them is protective. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-8 Blood-borne pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

HE-8.1 Trace the Plasmodium life cycle from sporozoite inoculation by the female Anopheles mosquito through liver schizogony, the erythrocytic cycle, and gametocytes, and explain relapse from hypnozoites of P. vivax and P. ovale. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-8.2 Explain the virulence of Plasmodium falciparum: invasion of red cells of all ages, high parasitemia, PfEMP1–mediated adherence of infected red cells to endothelium causing microvascular obstruction, and immune evasion by switching var genes that encode PfEMP1. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-8.3 Explain how sickle cell trait, G6PD deficiency, and thalassemia trait reduce severe malaria, and why Duffy antigen negative red cells resist Plasmodium vivax invasion. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-8.4 Describe Babesia microti transmission by Ixodes scapularis ticks and by transfusion, its intraerythrocytic ring forms and Maltese cross tetrads, and coinfection with Borrelia burgdorferi. NPLEX + USMLE | Microbiology & Immunology | DOK 2

HE-8.5 Describe the Schistosoma life cycle: freshwater snail intermediate host, cercarial skin penetration, adult worms in mesenteric (S. mansoni, S. japonicum) or vesical (S. haematobium) venules, and egg deposition. NPLEX | Microbiology & Immunology | DOK 2

HE-8.6 Explain how parvovirus B19, a nonenveloped single-stranded DNA virus, infects erythroid progenitors through the P antigen, causing transient aplastic crisis in chronic hemolytic anemias and hydrops fetalis in fetal infection. USMLE extension | Microbiology & Immunology | DOK 2

DISEASES AND CONDITIONS

HE-9-A Diseases of blood cell production

1 Anemias (macrocytic, microcytic, aplastic) **2** Polycythemia (vera, secondary)

HE-9-A.1 Identify causes of iron deficiency anemia (chronic gastrointestinal or menstrual blood loss, poor intake or absorption, pregnancy) and predict its profile: low ferritin, high TIBC, low transferrin saturation, high RDW, and microcytic hypochromic cells. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.2 Distinguish iron deficiency, anemia of chronic disease (IL-6 driven hepcidin, high ferritin, low TIBC), thalassemia trait (normal iron studies, target cells), and sideroblastic anemia (high iron, ringed sideroblasts) using iron studies and smear. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.3 Explain how lead inhibition of ALA dehydratase and ferrochelatase, alcohol, pyridoxine deficiency, and X-linked ALAS2 mutations cause sideroblastic anemia, and identify basophilic stippling and high zinc protoporphyrin in lead poisoning. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.4 Explain how pernicious anemia (antibodies to parietal cells and intrinsic factor), ileal resection, and the fish tapeworm Dibothriocephalus (formerly Diphyllobothrium) latus cause B12 deficiency with megaloblastic anemia, hypersegmented neutrophils, and subacute combined degeneration of the spinal cord. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.5 Identify causes of folate deficiency (poor intake, alcohol use, pregnancy, chronic hemolysis, methotrexate) and distinguish megaloblastic anemia from nonmegaloblastic macrocytosis due to liver disease, alcohol, and reticulocytosis. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.6 Explain aplastic anemia as stem cell failure from autoimmune T cell injury, drugs, benzene, radiation, viral hepatitis, or Fanconi anemia, and identify pancytopenia with a hypocellular fatty marrow and no splenomegaly. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.7 Distinguish polycythemia vera (JAK2 V617F, low erythropoietin, aquagenic pruritus, splenomegaly) from secondary polycythemia with appropriately high erythropoietin (hypoxia, altitude, smoking) or inappropriately high erythropoietin (renal cell or hepatocellular carcinoma), and from relative polycythemia. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.8 Predict complications of polycythemia vera from hyperviscosity and thrombosis (stroke, Budd-Chiari syndrome, erythromelalgia), high cell turnover (gout), and progression to myelofibrosis or acute myeloid leukemia. NPLEX + USMLE | Pathology | DOK 3

HE-9-A.9 Predict hemoglobin, reticulocyte, and iron changes after acute blood loss (normal hemoglobin at first, fall after fluid shift, reticulocytosis in days) and after chronic blood loss (iron deficiency). USMLE extension | Physiology | DOK 3

HE-9-A.10 Classify neutropenia by mechanism (marrow failure, drug-induced agranulocytosis such as with propylthiouracil, immune destruction, cyclic neutropenia from ELANE mutation) and relate an absolute neutrophil count below 500/µL to severe bacterial and fungal infection risk. USMLE extension | Pathology | DOK 3

HE-9-A.11 Distinguish a leukemoid reaction (neutrophilia with left shift, toxic granulation, high leukocyte alkaline phosphatase) from chronic myeloid leukemia (basophilia, low leukocyte alkaline phosphatase, BCR-ABL1). USMLE extension · Pathology · DOK 3

HE-9-B Diseases of blood cell lysis

1 Hemolytic anemia (sickle cell, thalassemia, G6PD deficiency, spherocytosis)

2 Hemolytic disease of the newborn (erythroblastosis fetalis)

HE-9-B.1 Predict complications of sickle cell disease from vaso-occlusion and hemolysis: dactylitis, acute chest syndrome, stroke, splenic autoinfarction, Salmonella osteomyelitis, renal papillary necrosis, priapism, pigment gallstones, and aplastic crisis. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.2 Explain how beta thalassemia major causes ineffective erythropoiesis, extramedullary hematopoiesis (crew-cut skull, maxillary overgrowth), hepatosplenomegaly, and iron overload, and interpret electrophoresis showing high HbA2 in beta thalassemia minor. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.3 Explain how G6PD deficiency causes episodic intravascular hemolysis after oxidant stress (infection, fava beans, sulfonamides, primaquine, dapsone), and identify Heinz bodies and bite cells on smear. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.4 Explain how ankyrin, spectrin, or band 3 defects in hereditary spherocytosis cause membrane loss and splenic trapping, and predict spherocytes, high MCHC, splenomegaly, pigment gallstones, abnormal osmotic fragility or EMA binding, and a negative Coombs test. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.5 Explain how maternal anti-D IgG formed after prior Rh-positive fetal exposure crosses the placenta to cause fetal hemolysis, anemia, and hydrops fetalis, and contrast milder ABO hemolytic disease in infants of group O mothers. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.6 Interpret a positive direct Coombs test in the newborn and indirect Coombs test in the mother, and predict rising unconjugated bilirubin and kernicterus risk after birth when placental clearance ends. NPLEX + USMLE · Pathology · DOK 3

HE-9-B.7 Explain how an acquired PIGA mutation in paroxysmal nocturnal hemoglobinuria removes GPI-anchored CD55 and CD59, causing complement-mediated intravascular hemolysis, venous thrombosis, and cytopenias detected by flow cytometry. USMLE extension · Pathology · DOK 3

HE-9-B.8 Describe pyruvate kinase deficiency (autosomal recessive, ATP depletion, rigid red cells, extravascular hemolysis) and hereditary elliptocytosis (spectrin defects, elliptocytes, usually mild hemolysis). USMLE extension · Pathology · DOK 2

HE-9-B.9 Distinguish thrombotic thrombocytopenic purpura (ADAMTS13 deficiency, ultralarge von Willebrand multimers) from hemolytic uremic syndrome (Shiga toxin of E. coli O157:H7 injuring renal endothelium), noting schistocytes, thrombocytopenia, and normal PT and PTT in both. USMLE extension · Pathology · DOK 3

HE-9-B.10 Explain how prosthetic heart valves and severe aortic stenosis cause mechanical intravascular hemolysis with schistocytes, and predict the iron deficiency that can follow chronic hemoglobinuria. USMLE extension · Pathology · DOK 3

HE-9-B.11 Identify causes of splenic rupture (blunt trauma, mononucleosis splenomegaly), infarct (sickle cell disease, emboli), and abscess (septic emboli from endocarditis), and predict post-splenectomy smear findings (Howell-Jolly bodies, target cells, thrombocytosis). USMLE extension · Pathology · DOK 2

HE-9-B.12 Explain how hypersplenism from portal hypertension or infiltrative disease causes cytopenias by splenic sequestration, and distinguish it from marrow failure by marrow cellularity and splenomegaly. USMLE extension · Pathology · DOK 3

HE-9-C Clotting abnormalities

1 Disseminated intravascular coagulation (DIC) **2** Hemophilia **3** Immune thrombocytopenic purpura (ITP)

4 Von Willebrand disease **5** Vitamin K deficiency

HE-9-C.1 Identify triggers of DIC (sepsis, trauma, placental abruption, amniotic fluid embolism, acute promyelocytic leukemia, mucin-secreting adenocarcinoma) and predict thrombocytopenia, prolonged PT and PTT, low fibrinogen, high D-dimer, and schistocytes. NPLEX + USMLE · Pathology · DOK 3

HE-9-C.2 Relate microthrombi in DIC to ischemic injury of kidney, lung, brain, and adrenal glands, and consumption of platelets and clotting factors to simultaneous bleeding from venipuncture sites and mucosa. NPLEX + USMLE · Pathology · DOK 3

HE-9-C.3 Distinguish hemophilia A (factor VIII) from hemophilia B (factor IX) as X-linked deficiencies with hemarthroses and deep muscle bleeds, isolated prolonged PTT, and correction on mixing, unlike an acquired factor VIII inhibitor. NPLEX + USMLE Pathology DOK 3

HE-9-C.4 Explain how IgG autoantibodies against platelet GpIIb/IIIa cause splenic platelet destruction in ITP, and contrast acute childhood ITP after viral infection with chronic adult ITP, both showing isolated thrombocytopenia and increased marrow megakaryocytes. NPLEX + USMLE Pathology DOK 3

HE-9-C.5 Explain how von Willebrand factor deficiency impairs platelet adhesion and factor VIII stability, and predict mucosal bleeding, prolonged bleeding time, normal platelet count, normal or prolonged PTT, and decreased ristocetin cofactor activity. NPLEX + USMLE Pathology DOK 3

HE-9-C.6 Identify causes of vitamin K deficiency (newborns not given prophylaxis, fat malabsorption, prolonged broad-spectrum antibiotics, poor intake) and explain why the PT prolongs first, since factor VII has the shortest half-life. NPLEX + USMLE Pathology DOK 3

HE-9-C.7 Contrast the coagulation profile of liver failure (prolonged PT and PTT, low fibrinogen, thrombocytopenia, normal or high factor VIII) with that of vitamin K deficiency (prolonged PT, normal fibrinogen and platelets). NPLEX + USMLE Pathology DOK 3

HE-9-C.8 Distinguish Bernard-Soulier syndrome (GpIb deficiency, large platelets, defective adhesion), Glanzmann thrombasthenia (GpIIb/IIIa deficiency, defective aggregation), and uremic platelet dysfunction by mechanism and platelet findings. USMLE extension Pathology DOK 3

HE-9-C.9 Explain how inherited antithrombin III, protein C, or protein S deficiency and hyperhomocysteinemia increase venous thrombosis risk, and identify features that suggest an inherited thrombophilia (young age, recurrent or unusual-site clots, family history). USMLE extension Pathology DOK 2

HE-9-C.10 Identify hematologic paraneoplastic syndromes: migratory thrombophlebitis (Trousseau syndrome) with pancreatic and other adenocarcinomas, erythrocytosis from ectopic erythropoietin (renal cell and hepatocellular carcinoma), and pure red cell aplasia with thymoma. USMLE extension Pathology DOK 2

HE-9-D Blood and lymph neoplasms

1 Leukemias **2** Lymphomas (Hodgkin, non-Hodgkin) **3** Multiple myeloma

HE-9-D.1 Describe acute lymphoblastic leukemia as a TdT-positive lymphoblast neoplasm, mainly of children, presenting with marrow failure and bone pain, with CNS and testicular spread and increased risk in Down syndrome. NPLEX + USMLE Pathology DOK 2

HE-9-D.2 Describe acute myeloid leukemia (usually at least 20% marrow blasts, myeloperoxidase-positive Auer rods, mainly adults), its risk factors (prior chemotherapy, radiation, benzene, myelodysplasia, Down syndrome), and the link between acute promyelocytic leukemia and DIC. NPLEX + USMLE Pathology DOK 3

HE-9-D.3 Describe chronic lymphocytic leukemia as a CD5–positive, CD23–positive mature B cell neoplasm of older adults with smudge cells, hypogammaglobulinemia, warm autoimmune hemolytic anemia, and Richter transformation to diffuse large B cell lymphoma. NPLEX + USMLE Pathology DOK 2

HE-9-D.4 Describe chronic myeloid leukemia (neutrophilia across all maturation stages, basophilia, splenomegaly), its confirmation by detecting t(9;22) BCR-ABL1, and progression from chronic phase to accelerated phase and blast crisis. NPLEX + USMLE Pathology DOK 2

HE-9-D.5 Identify hairy cell leukemia by mature B cells with hairlike projections, TRAP and CD11c positivity, BRAF V600E mutation, a dry tap from marrow fibrosis, and massive splenomegaly without lymphadenopathy. NPLEX + USMLE Pathology DOK 2

HE-9-D.6 Distinguish Hodgkin lymphoma (CD15 and CD30 positive Reed-Sternberg cells, contiguous nodal spread, bimodal age, EBV association, nodular sclerosis most common) from non-Hodgkin lymphoma (noncontiguous spread, frequent extranodal involvement). NPLEX + USMLE Pathology DOK 3

HE-9-D.7 Relate major non-Hodgkin lymphomas to their features: diffuse large B cell (most common, aggressive), follicular (t(14;18), indolent), Burkitt (t(8;14), starry sky, EBV), mantle cell (t(11;14), CD5 positive), and MALT lymphoma (Helicobacter pylori). NPLEX + USMLE Pathology DOK 3

HE-9-D.8 Describe T cell lymphomas: adult T cell leukemia/lymphoma (HTLV-1, hypercalcemia, lytic bone lesions) and mycosis fungoides (skin-homing CD4 T cells, Pautrier microabscesses, Sezary cells in blood). NPLEX + USMLE Pathology DOK 2

HE-9-D.9 Explain how clonal plasma cells in multiple myeloma cause hypercalcemia and punched-out lytic lesions (osteoclast activation through RANKL), renal injury from light chain casts, anemia, rouleaux, infections, and an M spike on electrophoresis. NPLEX + USMLE Pathology DOK 3

HE-9-D.10 Distinguish multiple myeloma from MGUS (M protein below 3 g/dL, under 10% marrow plasma cells, no end-organ damage) and from Waldenström macroglobulinemia (IgM, hyperviscosity, no lytic lesions), and relate light chains to AL amyloidosis. NPLEX + USMLE | Pathology | DOK 3

HE-9-D.11 Describe myelodysplastic syndromes as clonal, ineffective hematopoiesis in older adults or after alkylating agents or radiation, with cytopenias, dysplastic cells (pseudo-Pelger-Huet neutrophils, ringed sideroblasts), and risk of progression to AML. USMLE extension | Pathology | DOK 2

HE-9-D.12 Describe primary myelofibrosis (JAK2, CALR, or MPL mutation, megakaryocyte-driven marrow fibrosis, dry tap, teardrop cells, leukoerythroblastic smear) and massive splenomegaly from extramedullary hematopoiesis. USMLE extension | Pathology | DOK 2

HE-9-D.13 Describe Langerhans cell histiocytosis as a clonal proliferation of CD1a-positive, S100–positive Langerhans cells with Birbeck granules, presenting in children with lytic bone lesions (eosinophilic granuloma) or multisystem disease with diabetes insipidus. USMLE extension | Pathology | DOK 2

HE-9-D.14 Explain how rapid lysis of high-turnover tumors (Burkitt lymphoma, acute leukemias) during treatment causes tumor lysis syndrome, and predict hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia, and acute kidney injury. USMLE extension | Pathology | DOK 3

HE-9-E Infectious diseases of the blood

1 Babesiosis **2** Malaria **3** Schistosomiasis

HE-9-E.1 Identify risk factors for severe babesiosis (asplenia, older age, immunosuppression) and predict fever, hemolytic anemia, and thrombocytopenia after a tick bite in the northeastern or upper midwestern United States. NPLEX + USMLE | Microbiology & Immunology | DOK 3

HE-9-E.2 Relate the cyclic fevers of malaria to synchronized rupture of infected red cells (about 48 hours for P. vivax and P. ovale, 72 hours for P. malariae) and predict hemolytic anemia, splenomegaly, and thrombocytopenia. NPLEX + USMLE | Microbiology & Immunology | DOK 3

HE-9-E.3 Predict complications of Plasmodium falciparum malaria (cerebral malaria, severe anemia, hypoglycemia, acute kidney injury, hemoglobinuria, acute respiratory distress) and identify multiple ring forms and banana-shaped gametocytes on a thin smear. NPLEX + USMLE | Microbiology & Immunology | DOK 3

HE-9-E.4 Explain how granulomatous reactions to trapped Schistosoma eggs cause presinusoidal portal hypertension with periportal pipestem fibrosis (S. mansoni) and hematuria, bladder fibrosis, and squamous cell carcinoma (S. haematobium), with eosinophilia. NPLEX | Pathology | DOK 3

HE-9-E.5 Compare dengue, chikungunya, and Zika as Aedes-transmitted viruses: severe dengue with thrombocytopenia and plasma leakage after reinfection with another serotype, persistent arthralgia in chikungunya, and congenital microcephaly with Zika. USMLE extension | Microbiology & Immunology | DOK 2

HE-9-E.6 Describe Ebola and Marburg filoviruses: spread by contact with infected body fluids, fruit bats as the likely reservoir, and fever with vomiting, diarrhea, hemorrhage, DIC, and multiorgan failure. USMLE extension | Microbiology & Immunology | DOK 2

HE-9-E.7 Describe Yersinia pestis (rodent flea vector, bipolar staining, painful suppurative bubo, septicemic and pneumonic spread) and Bartonella henselae (cat scratch, regional lymphadenopathy with stellate necrotizing granulomas) as causes of lymphadenitis. USMLE extension | Microbiology & Immunology | DOK 2

HE-9-E.8 Describe acute lymphangitis from Streptococcus pyogenes (red streaks tracking toward regional nodes) and distinguish reactive lymphadenitis patterns (follicular hyperplasia, paracortical hyperplasia, sinus histiocytosis) from lymphoma. USMLE extension | Pathology | DOK 2

HOW THE BODY WORKS

IN-1 Skin layers and the dermal-epidermal junction

Describe microscopic anatomy, derivations, and differentiating features of skin layers and dermal-epidermal junction.

- **IN-1.1** Identify the epidermal layers from stratum basale to stratum corneum and describe how keratinocytes change in each (keratins 5 and 14 to 1 and 10, keratohyalin granules, lamellar bodies, loss of nuclei), noting that stratum lucidum appears only in thick skin. NPLEX · Anatomy · DOK 2
- **IN-1.2** Identify melanocytes (neural crest origin, melanin transfer to keratinocytes), Langerhans cells (bone marrow origin, Birbeck granules, CD1a, antigen presentation), and Merkel cells (basal layer touch receptors) by epidermal location and function. NPLEX · Anatomy · DOK 1
- **IN-1.3** Compare desmosomes (desmogleins), hemidesmosomes (BP180 or collagen XVII, BP230, integrin alpha6 beta4), and adherens junctions (E-cadherin), and predict the blistering disease that results when desmosomal or hemidesmosomal proteins are targeted by autoantibodies or mutation. NPLEX · Anatomy · DOK 3
- **IN-1.4** Describe the dermal-epidermal junction (lamina lucida with laminin 332 and collagen XVII anchoring filaments, lamina densa with type IV collagen, type VII collagen anchoring fibrils) and predict the level of the split when each component fails. NPLEX · Anatomy · DOK 3
- **IN-1.5** Compare the papillary dermis, reticular dermis, and hypodermis by connective tissue density, vessels, and sensory receptors (Meissner corpuscles in dermal papillae, Pacinian corpuscles in deep dermis and subcutis, free nerve endings). NPLEX · Anatomy · DOK 2
- **IN-1.6** Distinguish eccrine, apocrine, and sebaceous glands by location, secretion, and control (eccrine glands by sympathetic cholinergic fibers, apocrine glands opening into axillary and anogenital hair follicles, sebaceous glands holocrine and androgen driven), and relate hyperhidrosis to eccrine glands and acne to sebaceous glands. NPLEX + USMLE · Anatomy · DOK 2
- **IN-1.7** Describe the hair follicle (bulb, dermal papilla, bulge stem cells, arrector pili) and nail unit (matrix, plate, bed, folds), and relate anagen, catagen, and telogen to telogen effluvium and the lateral nail fold to an ingrown nail. NPLEX + USMLE · Anatomy · DOK 2
- **IN-1.8** Define acanthosis, hyperkeratosis, parakeratosis, spongiosis, acantholysis, and dyskeratosis, and match each to an example disease such as psoriasis, eczema, or pemphigus. NPLEX · Pathology · DOK 2

IN-2 Ectoderm embryology

Describe embryological development of the ectoderm.

- **IN-2.1** Distinguish derivatives of surface ectoderm (epidermis, hair, nails, cutaneous and mammary glands, lens, anterior pituitary, tooth enamel) from those of neuroectoderm (central nervous system, retina, posterior pituitary) and neural crest. NPLEX · Anatomy · DOK 1
- **IN-2.2** Identify neural crest derivatives relevant to skin and nerves (melanocytes, Schwann cells, dorsal root and autonomic ganglia, adrenal medulla, odontoblasts), and explain why facial dermis arises from neural crest while dermis of the back arises from somite dermatomes. NPLEX · Anatomy · DOK 2
- **IN-2.3** Describe fetal skin development, including the periderm, epidermal stratification, melanoblast migration from the neural crest, hair bud and dermal papilla formation, lanugo, and vernix caseosa as a protective coat of sebum and shed cells. NPLEX · Anatomy · DOK 2
- **IN-2.4** Explain why melanocytes arrested in the dermis during migration produce dermal melanocytosis (Mongolian spots), and distinguish it from congenital melanocytic nevi, infantile hemangiomas, and port-wine stains in newborns. USMLE extension · Anatomy · DOK 2
- **IN-2.5** Explain why failed melanoblast migration in piebaldism produces congenital white patches lacking melanocytes, whereas albinism shows normal melanocyte numbers with reduced melanin synthesis. NPLEX · Anatomy · DOK 3

IN-3 Membrane transport and membrane potential

Explain function of membrane constituents and mechanisms governing transport across cell membranes, osmosis, membrane potential, and ionic equilibrium.

IN-3.1 Describe the plasma membrane components (phospholipid bilayer, cholesterol, integral and peripheral proteins, glycocalyx) and explain how cholesterol buffers membrane fluidity and how unsaturated fatty acids increase it. NPLEX Physiology DOK 2

IN-3.2 Compare simple diffusion, facilitated diffusion, primary active transport, secondary active transport, and endocytosis by energy source, saturability, and direction, using the Na+/K+-ATPase, GLUT transporters, and SGLT1 as examples. NPLEX Physiology DOK 2

IN-3.3 Predict how inhibiting the Na+/K+-ATPase (3 Na+ out, 2 K+ in) changes intracellular sodium, cell volume, and membrane potential, and relate this to cellular swelling in hypoxic cell injury. NPLEX Physiology DOK 3

IN-3.4 Explain osmosis, distinguish osmolarity from tonicity, and predict cell volume in isotonic, hypotonic, and hypertonic solutions, including why urea is an ineffective osmole. NPLEX Physiology DOK 3

IN-3.5 Calculate an ion's equilibrium potential with the Nernst equation, and explain how potassium leak channels, the Goldman equation, and the Gibbs-Donnan effect of impermeant intracellular anions establish the resting membrane potential. NPLEX Physiology DOK 3

IN-3.6 Predict how hyperkalemia depolarizes and hypokalemia hyperpolarizes the resting membrane potential, and explain why sustained depolarization inactivates voltage-gated sodium channels and lowers nerve and muscle excitability. NPLEX + USMLE Physiology DOK 3

IN-3.7 Explain how CFTR mutation blocks chloride reabsorption in the eccrine sweat duct, raising sweat chloride to 60 mmol/L or higher, while it blocks chloride secretion in airway epithelium. NPLEX + USMLE Physiology DOK 3

IN-4 Thermoregulation

Explain principles of thermal physiology, including regulation of body temperature.

IN-4.1 Explain how the hypothalamus integrates core and skin thermoreceptor input around a set point, with the anterior (preoptic) hypothalamus driving heat loss and the posterior hypothalamus driving heat conservation and production. NPLEX Physiology DOK 2

IN-4.2 Compare heat loss by radiation, conduction, convection, and evaporation, and predict why sweat evaporation becomes the only route when ambient temperature exceeds skin temperature and why high humidity impairs cooling. NPLEX Physiology DOK 3

IN-4.3 Explain heat conservation and production by cutaneous vasoconstriction, piloerection, shivering, thyroid hormone, and nonshivering thermogenesis through UCP1 (thermogenin) in brown fat, and why neonates depend on brown fat. NPLEX Physiology DOK 2

IN-4.4 Explain how IL-1, IL-6, and TNF-alpha induce hypothalamic PGE2 to raise the set point in fever, and distinguish fever from hyperthermia, in which the set point is unchanged. NPLEX + USMLE Physiology DOK 2

IN-4.5 Distinguish heat exhaustion from heat stroke (core temperature above 40 °C (104 °F) with central nervous system dysfunction), compare classic and exertional heat stroke, and predict rhabdomyolysis, acute kidney injury, and DIC. NPLEX + USMLE Pathology DOK 3

IN-4.6 Explain how an RYR1 mutation causes malignant hyperthermia after volatile anesthetics or succinylcholine through uncontrolled sarcoplasmic reticulum calcium release, producing muscle rigidity, hypercarbia, and rising core temperature. USMLE extension Biochemistry & Genetics DOK 2

IN-4.7 Describe hypothermia (core below 35 °C (95 °F)), including loss of shivering when severe, bradycardia, Osborn (J) waves, and arrhythmias, and distinguish frostbite (ice crystals, endothelial injury, thrombosis) from perniosis (nonfreezing damp cold). NPLEX + USMLE Pathology DOK 3

IN-5 Skin biochemistry and nutrition

Explain biochemistry of proteins, carbohydrates, lipids, vitamins, minerals, and co-factors as they relate to the integumentary system.

IN-5.1 Trace collagen synthesis (prolyl and lysyl hydroxylation needing vitamin C and Fe2+, glycosylation, triple helix formation, procollagen peptidase cleavage, lysyl oxidase cross-linking needing copper) and predict the connective tissue defect when each step fails. NPLEX + USMLE Biochemistry & Genetics DOK 3

IN-5.2 Relate vitamin C deficiency to scurvy findings (perifollicular hemorrhages, corkscrew hairs, bleeding gums, poor wound healing) and defective copper transport in Menkes disease to brittle kinky hair and connective tissue weakness. NPLEX + USMLE Biochemistry & Genetics DOK 2

IN-5.3 Explain the photosensitive dermatitis of niacin deficiency (pellagra) and its causes, including corn-based diets, Hartnup disease, carcinoid syndrome, and isoniazid-induced vitamin B6 deficiency. NPLEX + USMLE Biochemistry & Genetics DOK 2

IN-5.4 Distinguish skin findings of deficiency of vitamin A (follicular hyperkeratosis, xerosis), zinc (acrodermatitis enteropathica with periorificial and acral dermatitis, alopecia, poor healing), biotin (dermatitis, alopecia), riboflavin (angular cheilitis), and essential fatty acids (scaly dermatitis). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-5.5 Trace vitamin D synthesis from 7-dehydrocholesterol in the epidermis by UVB to cholecalciferol, then 25-hydroxylation in the liver and 1-alpha-hydroxylation in the kidney, and predict how dark skin, high latitude, and sunscreen reduce it. NPLEX + USMLE | Biochemistry & Genetics | DOK 2

IN-5.6 Describe the stratum corneum lipid barrier (ceramides, cholesterol, free fatty acids from lamellar bodies), sebum composition (triglycerides, wax esters, squalene), and dermal glycosaminoglycans such as hyaluronic acid and dermatan sulfate. NPLEX | Biochemistry & Genetics | DOK 2

IN-5.7 Relate xanthoma type to lipid disorder: eruptive xanthomas with severe hypertriglyceridemia (lipoprotein lipase or apoC-II deficiency), tendon xanthomas and xanthelasma with familial hypercholesterolemia (LDL receptor), and palmar xanthomas with dysbetalipoproteinemia (apoE2). NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-5.8 Explain how uroporphyrinogen decarboxylase deficiency in porphyria cutanea tarda causes blistering, fragility, and hypertrichosis on sun-exposed skin, and identify triggers such as alcohol, hepatitis C, iron overload, and estrogen. USMLE extension | Biochemistry & Genetics | DOK 3

IN-6 Antioxidants and free radical scavengers

Explain biochemistry of non-vitamin antioxidants and free radical scavengers as they relate to integumentary function and pathology.

IN-6.1 Describe how superoxide, hydrogen peroxide, and hydroxyl radical form (mitochondrial electron leak, NADPH oxidase, UV and ionizing radiation, Fenton reaction with Fe2+) and how they damage membrane lipids, proteins, and DNA. NPLEX | Biochemistry & Genetics | DOK 2

IN-6.2 Explain how superoxide dismutase (Cu/Zn cytosolic, Mn mitochondrial), catalase, and selenium-dependent glutathione peroxidase detoxify reactive oxygen species, and why glutathione regeneration needs NADPH from G6PD in the pentose phosphate pathway. NPLEX | Biochemistry & Genetics | DOK 2

IN-6.3 Explain how non-vitamin antioxidants protect skin, including reduced glutathione, ubiquinol (coenzyme Q10), dietary carotenoids such as lycopene, and melanin, and why eumelanin is more photoprotective than pheomelanin. NPLEX | Biochemistry & Genetics | DOK 2

IN-6.4 Compare UVB injury (direct DNA damage forming cyclobutane pyrimidine dimers, sunburn) with UVA injury (deeper penetration, oxidative DNA damage), and explain photoaging through matrix metalloproteinase breakdown of collagen and solar elastosis. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-6.5 Explain how the NADPH oxidase respiratory burst and myeloperoxidase kill microbes, and predict recurrent skin abscesses and granulomas with catalase-positive organisms such as Staphylococcus aureus in chronic granulomatous disease. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-7 Skin genetics

Describe features and explain principles of gene expression, control, cell cycle regulation, and consequences of genetic abnormalities underlying integumentary disease.

IN-7.1 Explain how cyclins, cyclin-dependent kinases, Rb, and p53 control the G1 to S checkpoint, and how UV-signature C to T mutations in TP53 drive actinic keratosis and squamous cell carcinoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-7.2 Explain how defective nucleotide excision repair in xeroderma pigmentosum (autosomal recessive) causes extreme photosensitivity, early freckling, and basal cell carcinoma, squamous cell carcinoma, and melanoma in childhood. USMLE extension | Biochemistry & Genetics | DOK 3

IN-7.3 Relate driver mutations to skin tumors: PTCH1 loss activating hedgehog signaling in basal cell carcinoma and Gorlin syndrome, BRAF V600E or NRAS in nevi and melanoma, and germline CDKN2A (p16) in familial melanoma. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-7.4 Interpret pedigrees of skin disorders with autosomal dominant (NF1, tuberous sclerosis, epidermolysis bullosa simplex), autosomal recessive (oculocutaneous albinism, xeroderma pigmentosum), and X-linked recessive (X-linked ichthyosis) inheritance, and calculate recurrence risk for offspring. NPLEX | Biochemistry & Genetics | DOK 3

IN-7.5 Explain the gene products in NF1 (neurofibromin, a Ras GTPase-activating protein on chromosome 17) and tuberous sclerosis (hamartin and tuberin, which inhibit mTOR), and identify café-au-lait macules, neurofibromas, ash-leaf macules, and facial angiofibromas. USMLE extension · Biochemistry & Genetics · DOK 3

IN-7.6 Explain how loss-of-function filaggrin (FLG) mutations impair the epidermal barrier in ichthyosis vulgaris and atopic dermatitis, and how steroid sulfatase deficiency causes X-linked ichthyosis in males. USMLE extension · Biochemistry & Genetics · DOK 3

IN-7.7 Explain how MSH binding the MC1R receptor raises cAMP and MITF-driven tyrosinase expression to favor eumelanin, and why MC1R loss-of-function variants cause red hair, fair skin, and higher melanoma risk. NPLEX · Biochemistry & Genetics · DOK 2

IN-8 Cell adaptation, injury, pigmentation, and neoplasia

Explain processes relating to adaptive changes, cellular injury, pigmentation, infiltration, and neoplasia.

IN-8.1 Distinguish hypertrophy, hyperplasia, atrophy, and metaplasia by mechanism and example (epidermal hyperplasia in a callus, squamous metaplasia of bronchi in smokers, columnar metaplasia in Barrett esophagus), and explain how metaplasia and pathologic hyperplasia can progress to dysplasia. NPLEX · Pathology · DOK 2

IN-8.2 Explain how ATP depletion, mitochondrial damage, calcium influx, oxidative stress, and membrane damage injure cells, and distinguish reversible changes (swelling, fatty change, blebbing, ribosome detachment) from irreversible ones (membrane rupture, mitochondrial densities, pyknosis, karyorrhexis, karyolysis). NPLEX · Pathology · DOK 3

IN-8.3 Classify necrosis as coagulative, liquefactive, caseous, fat, fibrinoid, or gangrenous, and predict the type from its setting (myocardial infarct, brain infarct or abscess, tuberculosis, acute pancreatitis, vasculitis, limb ischemia). NPLEX · Pathology · DOK 3

IN-8.4 Compare intrinsic (BCL-2 family, cytochrome c, caspase-9) and extrinsic (Fas ligand, TNF, caspase-8) apoptosis and perforin-granzyme killing by cytotoxic T cells, and distinguish apoptosis (shrinkage, apoptotic bodies, no inflammation) from necrosis. NPLEX · Pathology · DOK 3

IN-8.5 Distinguish endogenous pigments (lipofuscin from lipid peroxidation, hemosiderin staining with Prussian blue, melanin, bilirubin) from exogenous pigments (inhaled carbon in anthracosis, tattoo ink held in dermal macrophages), and relate each to its source. NPLEX + USMLE · Pathology · DOK 2

IN-8.6 Distinguish dystrophic calcification (necrotic tissue, normal serum calcium) from metastatic calcification (hypercalcemia, normal tissues), and identify infiltrates of fat (steatosis), cholesterol (foam cells, xanthomas), and amyloid (Congo red with apple-green birefringence). NPLEX · Pathology · DOK 3

IN-8.7 Distinguish benign from malignant neoplasms by differentiation, anaplasia (pleomorphism, high nuclear to cytoplasmic ratio, atypical mitoses), invasion, and metastasis, apply tumor naming rules, and distinguish dysplasia, carcinoma in situ, grade, and stage. NPLEX + USMLE · Pathology · DOK 3

IN-8.8 Explain the hallmarks of cancer, including oncogene growth signaling (BRAF in melanoma), two-hit loss of tumor suppressors (RB, TP53), evasion of apoptosis, telomerase activity, VEGF-driven angiogenesis, invasion, Warburg metabolism, and immune evasion. NPLEX + USMLE · Pathology · DOK 3

IN-9 Infection principles in skin disease

Explain principles of infectious disease in dermatological pathologies, including normal flora, stages of infection, and characteristics of pathogenesis.

IN-9.1 Identify normal skin flora (Staphylococcus epidermidis, Cutibacterium acnes, Corynebacterium, Malassezia, nasal Staphylococcus aureus carriage) and explain how flora, acidic pH, sebum fatty acids, desquamation, and antimicrobial peptides resist pathogen colonization. NPLEX · Microbiology & Immunology · DOK 2

IN-9.2 Trace the stages of a skin infection (entry through a breach, adherence, colonization, invasion, spread, tissue damage) and relate the depth reached to impetigo, folliculitis, erysipelas, cellulitis, abscess, and necrotizing fasciitis. NPLEX + USMLE · Microbiology & Immunology · DOK 2

IN-9.3 Distinguish skin disease from direct invasion versus toxins, explaining how exfoliative toxins cleave desmoglein 1 in staphylococcal scalded skin syndrome and how superantigens (TSST-1, streptococcal pyrogenic exotoxins) cause diffuse rash and shock. NPLEX + USMLE · Microbiology & Immunology · DOK 3

IN-9.4 Explain how diabetes, venous insufficiency, lymphedema, tinea pedis fissures, injection drug use, neutropenia, and T cell deficiency predispose to skin infection, and predict the likely organism from the host context. NPLEX · Microbiology & Immunology · DOK 3

IN-10 Skin trauma and wound healing

Explain complications and clinical characteristics of skin trauma and healing mechanisms.

IN-10.1 Sequence wound healing: hemostasis, inflammation (neutrophils, then macrophages by day 3), proliferation (granulation tissue, VEGF-driven angiogenesis, type III collagen), and remodeling (type I collagen, zinc-dependent metalloproteinases), noting scars regain only about 70 to 80 percent of normal strength. NPLEX + USMLE | Pathology | DOK 2

IN-10.2 Compare healing by primary and secondary intention (more granulation tissue, myofibroblast contraction), and explain how infection, diabetes, ischemia, glucocorticoids, malnutrition, vitamin C or zinc deficiency, and foreign bodies delay healing. NPLEX | Pathology | DOK 3

IN-10.3 Distinguish hypertrophic scars (parallel type III collagen, confined to the wound, may regress) from keloids (disorganized type I and III collagen beyond wound borders, earlobes, darker skin), and identify exuberant granulation tissue, dehiscence, and contracture. NPLEX + USMLE | Pathology | DOK 3

IN-10.4 Classify burns as superficial, superficial partial-thickness, deep partial-thickness, or full-thickness by blisters, blanching, and sensation, estimate burned area by the rule of nines, and predict hypovolemia, Pseudomonas sepsis, hypermetabolism, inhalation injury, and contracture. NPLEX + USMLE | Pathology | DOK 3

IN-10.5 Explain why electrical injury causes small entry and exit skin wounds with extensive deep muscle and nerve necrosis, rhabdomyolysis, myoglobinuric kidney injury, and arrhythmias, and identify Lichtenberg figures and cardiac arrest after lightning strike. NPLEX + USMLE | Pathology | DOK 3

IN-10.6 Explain how ionizing radiation damages DNA directly and through free radicals, and distinguish acute radiation dermatitis (erythema, desquamation) from chronic changes (atrophy, telangiectasia, fibrosis, obliterative endarteritis) and later skin cancers or angiosarcoma. NPLEX + USMLE | Pathology | DOK 3

IN-10.7 Explain how sustained pressure and shear over bony prominences cause ischemic decubitus ulcers, stage them from nonblanching erythema to exposed bone, and explain how an auricular hematoma separating perichondrium from cartilage leads to cauliflower ear. NPLEX + USMLE | Pathology | DOK 3

IN-10.8 Explain how brown recluse spider venom (sphingomyelinase D) causes dermonecrosis with a central necrotic ulcer and, rarely, systemic hemolysis. USMLE extension | Pathology | DOK 2

IN-11 Immune responses in skin disease

Describe activation of innate and adaptive immune mechanisms by microbial pathogens, cancer/tumors, antigens, and vaccines; describe inflammatory, autoimmune, and hypersensitivity immune responses.

IN-11.1 Explain how keratinocytes (Toll-like receptors, IL-1, TNF, defensins, cathelicidin), mast cells, and Langerhans cells provide innate defense, and trace Langerhans cell antigen capture and migration to regional lymph nodes to prime naive T cells. NPLEX | Microbiology & Immunology | DOK 2

IN-11.2 Trace acute inflammation from vasodilation and venular leak (histamine, bradykinin, prostaglandins) through rolling (selectins), firm adhesion (integrins binding ICAM-1), transmigration (PECAM-1), and chemotaxis (C5a, LTB4, IL-8), and predict findings in leukocyte adhesion deficiency type 1. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-11.3 Explain how acute inflammation resolves, forms an abscess, or becomes chronic, and how Th1 cells and IFN-gamma activate macrophages to form granulomas, distinguishing caseating (tuberculosis) from noncaseating (sarcoidosis) granulomas. NPLEX + USMLE | Microbiology & Immunology | DOK 2

IN-11.4 Classify skin reactions by hypersensitivity type: type I (urticaria, angioedema), type II (pemphigus, bullous pemphigoid), type III (Arthus reaction, leukocytoclastic vasculitis), and type IV (allergic contact dermatitis, tuberculin test), and predict the timing and mediators of each. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-11.5 Relate autoimmune skin findings to disease: malar and discoid lesions with interface dermatitis and junctional immune deposits in lupus, Gottron papules and heliotrope rash in dermatomyositis, and sclerodactyly with anticentromere or anti-Scl-70 antibodies in systemic sclerosis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-11.6 Explain how CD8 T cells and NK cells eliminate tumor cells, how tumors escape by losing MHC class I or expressing PD-L1, and why organ transplant recipients develop many cutaneous squamous cell carcinomas. NPLEX | Microbiology & Immunology | DOK 3

IN-11.7 Explain how the recombinant HPV L1 virus-like particle vaccine induces neutralizing antibody, how live attenuated varicella vaccine induces both antibody and cell-mediated immunity, and why an Arthus (type III) local reaction can follow a toxoid booster. NPLEX + USMLE | Microbiology & Immunology | DOK 2

IN-12 Skin pathogens

Explain morphology, replication/life cycle, transmission (including vectors), mechanisms of infection, virulence factors, and genetic characteristics of common microbial pathogens causing listed conditions.

IN-12.1 Identify Staphylococcus aureus (gram-positive cocci in clusters, catalase and coagulase positive) and explain its virulence factors (protein A, Panton-Valentine leukocidin, exfoliative toxins, TSST-1) and how mecA-encoded PBP2a makes MRSA resistant to beta-lactams. NPLEX + USMLE Microbiology & Immunology DOK 2

IN-12.2 Identify Streptococcus pyogenes (beta-hemolytic, catalase negative, bacitracin sensitive, PYR positive), explain M protein, hyaluronic acid capsule, streptolysin O, streptokinase, and pyrogenic exotoxins, and explain why skin infection can cause glomerulonephritis but not rheumatic fever. NPLEX + USMLE Microbiology & Immunology DOK 3

IN-12.3 Explain how dermatophytes (Trichophyton, Microsporum, Epidermophyton) use keratinases to live in stratum corneum, hair, and nails, identify septate hyphae on KOH prep, and contrast Malassezia (spaghetti and meatballs, hypopigmented or hyperpigmented tinea versicolor). NPLEX + USMLE Microbiology & Immunology DOK 2

IN-12.4 Identify Candida albicans as a dimorphic commensal forming budding yeast and pseudohyphae (germ tube positive), and explain how moisture, diabetes, antibiotics, glucocorticoids, and T cell or neutrophil defects allow mucocutaneous or invasive disease. NPLEX + USMLE Microbiology & Immunology DOK 2

IN-12.5 Compare human papillomavirus (nonenveloped circular dsDNA, infects basal keratinocytes, koilocytes, E6 degrading p53 and E7 inactivating Rb in high-risk types) with molluscum contagiosum poxvirus (enveloped dsDNA replicating in cytoplasm, molluscum bodies) in structure, replication, and transmission. NPLEX + USMLE Microbiology & Immunology DOK 2

IN-12.6 Explain how HSV-1, HSV-2, and varicella zoster virus (enveloped dsDNA) infect epithelium, establish latency in sensory ganglia, and reactivate as herpes labialis, genital herpes, or dermatomal zoster, and identify multinucleated giant cells on Tzanck smear. USMLE extension Microbiology & Immunology DOK 2

IN-12.7 Explain how Cutibacterium acnes (gram-positive anaerobic rod) lipases release free fatty acids and activate TLR2 in acne, and how Pseudomonas aeruginosa (gram-negative, oxidase positive) causes hot tub folliculitis. NPLEX + USMLE Microbiology & Immunology DOK 2

IN-12.8 Identify vectors of skin-presenting infections: Ixodes ticks (Borrelia burgdorferi, erythema migrans), Dermacentor ticks (Rickettsia rickettsii, rash starting at wrists and ankles and spreading to the trunk, palms, and soles), sandflies (Leishmania), and dog and cat hookworm larvae in sand or soil (cutaneous larva migrans). NPLEX + USMLE Microbiology & Immunology DOK 2

DISEASES AND CONDITIONS

IN-13-A Pigmentation changes of the skin

1 Nevocellular nevus **2** Vitiligo

IN-13-A.1 Explain how activating BRAF or NRAS mutations followed by oncogene-induced senescence produce nevocellular nevi, and classify junctional, compound, and intradermal nevi by nest location, relating this to patient age and to maturation of cells with depth. NPLEX + USMLE Pathology DOK 3

IN-13-A.2 Distinguish a benign nevus (small, symmetric, uniform color, stable) from melanoma using ABCDE criteria, and identify large congenital melanocytic nevi as carrying increased melanoma risk. NPLEX + USMLE Pathology DOK 3

IN-13-A.3 Explain vitiligo as CD8 T cell destruction of melanocytes, identify sharply demarcated depigmented macules accentuated by Wood lamp, Koebner phenomenon, and absent melanocytes on biopsy, and relate it to autoimmune thyroid disease and type 1 diabetes. NPLEX + USMLE Pathology DOK 3

IN-13-A.4 Distinguish vitiligo (melanocytes absent) from albinism (melanocytes present, melanin reduced), tinea versicolor, and postinflammatory hypopigmentation using distribution, Wood lamp, and KOH findings. NPLEX + USMLE Pathology DOK 3

IN-13-A.5 Distinguish ephelides (normal melanocyte number with more melanin, darken in sun) from lentigo simplex and solar lentigines (increased basal melanocytes) and from melasma linked to pregnancy and estrogen. USMLE extension Pathology DOK 2

IN-13-A.6 Explain how high ACTH in primary adrenal insufficiency causes diffuse hyperpigmentation through POMC-derived MSH acting on melanocortin receptors, and why secondary adrenal insufficiency does not. USMLE extension Pathology DOK 3

IN-13-B Acute inflammatory skin conditions

1 Atopic dermatitis (eczema) **2** Contact dermatitis **3** Erythema multiforme **4** Urticaria

IN-13-B.1 Explain atopic dermatitis as filaggrin-related barrier failure plus Th2 inflammation (IL-4, IL-13, IL-31 itch), and identify infant facial and extensor versus older flexural distribution, lichenification, the atopic triad, and Staphylococcus aureus and eczema herpeticum complications. NPLEX + USMLE Pathology DOK 3

IN-13-B.2 Distinguish allergic contact dermatitis (type IV, hapten sensitization via Langerhans cells, nickel or urushiol from poison ivy and oak, eruption 48 to 72 hours after re-exposure, positive patch test) from irritant contact dermatitis (direct damage, no sensitization). NPLEX + USMLE Pathology DOK 3

IN-13-B.3 Predict the distribution of contact dermatitis from the exposure (earlobes or wrist from nickel, linear vesicles from plants, hands from wet work) and identify spongiosis as its histologic hallmark. NPLEX + USMLE Pathology DOK 3

IN-13-B.4 Explain erythema multiforme as a cytotoxic T cell reaction most often triggered by HSV (also Mycoplasma pneumoniae), and identify acral target lesions with dusky centers, limited mucosal involvement, and interface dermatitis with keratinocyte necrosis. NPLEX + USMLE Pathology DOK 3

IN-13-B.5 Explain urticaria as mast cell degranulation (IgE-mediated, physical, or direct) causing superficial dermal edema with wheals lasting under 24 hours, and distinguish angioedema, chronic urticaria (over 6 weeks), and bradykinin-mediated C1 inhibitor deficiency without hives. NPLEX + USMLE Pathology DOK 3

IN-13-B.6 Distinguish Stevens-Johnson syndrome (under 10 percent body surface) from toxic epidermal necrolysis (over 30 percent) by extent of full-thickness epidermal necrosis, and identify mucosal involvement, positive Nikolsky sign, and drug triggers (sulfonamides, anticonvulsants, allopurinol). USMLE extension Pathology DOK 3

IN-13-B.7 Identify pityriasis rosea by a herald patch followed by oval scaly patches along skin cleavage lines in a Christmas tree pattern on the trunk, resolving on its own within weeks. USMLE extension Pathology DOK 2

IN-13-B.8 Explain seborrheic dermatitis as an inflammatory response to Malassezia in sebaceous areas (scalp, eyebrows, nasolabial folds), identify greasy yellow scale and infant cradle cap, and relate severe disease to HIV and Parkinson disease. USMLE extension Pathology DOK 2

IN-13-B.9 Identify recurrent aphthous ulcers as painful shallow ulcers with a yellow-gray base and red halo on nonkeratinized oral mucosa, and relate them to Behçet disease, inflammatory bowel disease, and celiac disease. USMLE extension Pathology DOK 2

IN-13-C Chronic inflammatory skin conditions

1 Acne rosacea **2** Lichen planus **3** Psoriasis

IN-13-C.1 Identify rosacea by central facial flushing, persistent erythema, telangiectasias, and papules and pustules without comedones, identify triggers (heat, sunlight, alcohol, spicy food), and predict rhinophyma and ocular rosacea as complications. NPLEX + USMLE Pathology DOK 3

IN-13-C.2 Distinguish rosacea from acne vulgaris (comedones, adolescents), seborrheic dermatitis (greasy scale), and the malar rash of lupus, which spares the nasolabial folds. NPLEX + USMLE Pathology DOK 3

IN-13-C.3 Identify lichen planus by pruritic, purple, polygonal, planar papules with Wickham striae and reticular oral lesions, explain its band-like T cell infiltrate with sawtooth rete ridges and Civatte bodies, and relate it to hepatitis C. NPLEX + USMLE Pathology DOK 3

IN-13-C.4 Explain psoriasis as IL-23 and Th17 (IL-17) driven keratinocyte hyperproliferation with TNF and HLA-Cw6 risk, and identify silvery-scaled plaques on extensor surfaces and scalp, Koebner phenomenon, nail pitting, and psoriatic arthritis. NPLEX + USMLE Pathology DOK 3

IN-13-C.5 Interpret psoriasis histology (acanthosis with regularly elongated rete ridges, parakeratosis, reduced stratum granulosum, thinned suprapapillary plates over dilated capillaries, Munro microabscesses) and relate the thin suprapapillary plates to the Auspitz sign. NPLEX + USMLE Pathology DOK 3

IN-13-C.6 Identify pyoderma gangrenosum as a neutrophilic dermatosis forming painful ulcers with violaceous undermined borders that worsen after trauma (pathergy), and relate it to inflammatory bowel disease, rheumatoid arthritis, and hematologic malignancy. USMLE extension Pathology DOK 2

IN-13-C.7 Explain hidradenitis suppurativa as chronic follicular occlusion in axillary, inguinal, and perineal skin that produces painful nodules, abscesses, sinus tracts, and scarring, and identify obesity and smoking as risk factors. USMLE extension Pathology DOK 2

IN-13-C.8 Distinguish nonscarring alopecias (alopecia areata from autoimmune attack on anagen follicles with exclamation point hairs, androgenetic alopecia from DHT-driven follicle miniaturization, telogen effluvium) from scarring alopecias that destroy follicles. USMLE extension Pathology DOK 3

IN-13-D Blistering diseases

1 Epidermolysis bullosa **2** Pemphigoid **3** Pemphigus

IN-13-D.1 Compare the main types of epidermolysis bullosa by defective protein and split level: simplex (keratins 5 and 14, basal keratinocytes), junctional (laminin 332 or collagen XVII, lamina lucida), and dystrophic (type VII collagen, below the lamina densa). NPLEX + USMLE Pathology DOK 3

IN-13-D.2 Predict the clinical course of epidermolysis bullosa, including blisters from minor friction beginning at birth, mucosal and nail involvement, and, in recessive dystrophic disease, scarring, fused digits, and aggressive squamous cell carcinoma. NPLEX + USMLE Pathology DOK 3

IN-13-D.3 Explain bullous pemphigoid as IgG against hemidesmosomal BP180 and BP230 with complement and eosinophils, and identify tense subepidermal bullae in older adults, infrequent oral lesions, negative Nikolsky sign, and linear IgG and C3 along the basement membrane. NPLEX + USMLE Pathology DOK 3

IN-13-D.4 Explain pemphigus vulgaris as IgG against desmoglein 3 (often also 1) causing suprabasal acantholysis with a tombstone basal layer, and identify flaccid bullae, early painful oral erosions, positive Nikolsky sign, and net-like intercellular IgG on immunofluorescence. NPLEX + USMLE Pathology DOK 3

IN-13-D.5 Distinguish pemphigus vulgaris, pemphigus foliaceus (desmoglein 1, subcorneal split, no mucosal lesions), and bullous pemphigoid by target antigen, split level, Nikolsky sign, mucosal involvement, and immunofluorescence pattern. NPLEX + USMLE Pathology DOK 3

IN-13-D.6 Explain dermatitis herpetiformis as IgA against epidermal transglutaminase linked to celiac disease, and identify grouped pruritic vesicles on extensor surfaces, neutrophilic microabscesses at dermal papillary tips, and granular IgA deposits. USMLE extension Pathology DOK 3

IN-13-E Genetic skin conditions

1 Albinism **2** Ehlers-Danlos syndrome

IN-13-E.1 Explain oculocutaneous albinism as autosomal recessive loss of tyrosinase or other melanosome proteins with normal melanocyte numbers, and predict photosensitivity, skin cancer risk, and visual problems such as nystagmus and reduced acuity. NPLEX + USMLE Biochemistry & Genetics DOK 3

IN-13-E.2 Distinguish oculocutaneous albinism from X-linked ocular albinism and from the partial albinism of Chediak-Higashi syndrome (LYST defect, giant granules, recurrent pyogenic infections, neuropathy). NPLEX + USMLE Biochemistry & Genetics DOK 3

IN-13-E.3 Explain how defects in collagen synthesis or processing cause Ehlers-Danlos syndrome, linking the classic type to type V collagen (COL5A1), the vascular type to type III collagen (COL3A1), and the kyphoscoliotic type to lysyl hydroxylase. NPLEX + USMLE Biochemistry & Genetics DOK 3

IN-13-E.4 Identify skin features of Ehlers-Danlos syndrome (hyperextensible, soft, fragile skin, easy bruising, wide atrophic cigarette-paper scars) with joint hypermobility, and predict arterial, bowel, and uterine rupture and berry aneurysms in the vascular type. NPLEX + USMLE | Pathology | DOK 3

IN-13-E.5 Distinguish Ehlers-Danlos syndrome from Marfan syndrome (fibrillin-1, tall stature, lens subluxation, aortic root dilation) and osteogenesis imperfecta (type I collagen, fractures, blue sclerae) by defective protein and findings. NPLEX + USMLE | Biochemistry & Genetics | DOK 3

IN-13-F Benign and premalignant skin lesions

1 Actinic keratosis **2** Dysplastic nevi **3** Seborrheic keratosis

IN-13-F.1 Explain actinic keratosis as UV-induced keratinocyte dysplasia, identify rough sandpaper-like scaly papules on sun-exposed skin of older fair-skinned adults, lower epidermal atypia with parakeratosis and solar elastosis, and predict progression of some lesions to squamous cell carcinoma. NPLEX + USMLE | Pathology | DOK 3

IN-13-F.2 Identify dysplastic nevi as larger than common nevi with irregular borders and variable pigment, with cytologic and architectural atypia on biopsy, and explain why multiple dysplastic nevi and familial CDKN2A mutation mark high melanoma risk. NPLEX + USMLE | Pathology | DOK 3

IN-13-F.3 Identify seborrheic keratosis as a benign epidermal proliferation of older adults with a stuck-on waxy appearance, basaloid cells, horn cysts, and FGFR3 activating mutations, and interpret sudden eruption of many lesions as the Leser-Trélat sign. NPLEX + USMLE | Pathology | DOK 3

IN-13-F.4 Relate mucocutaneous paraneoplastic signs to cancers: acanthosis nigricans and Leser-Trélat sign with gastric adenocarcinoma, dermatomyositis with ovarian, lung, and GI cancers, and paraneoplastic pemphigus with lymphoma, noting acanthosis nigricans more often reflects insulin resistance. USMLE extension | Pathology | DOK 3

IN-13-F.5 Distinguish an epidermal inclusion cyst (stratified squamous lining with granular layer, laminated keratin, foreign body reaction on rupture) from a trichilemmal (pilar) cyst of the scalp, and identify lipoma as a soft, mobile, benign adipocyte tumor. USMLE extension | Pathology | DOK 2

IN-13-F.6 Distinguish infantile hemangioma (appears in the first weeks, proliferates, then involutes), cherry hemangioma of older adults, pyogenic granuloma (rapid, friable, bleeds, often in pregnancy), and port-wine stain (persistent capillary malformation, Sturge-Weber association). USMLE extension | Pathology | DOK 3

IN-13-F.7 Distinguish oral leukoplakia (white plaque that cannot be scraped off, linked to tobacco and alcohol, may show dysplasia and progress to squamous cell carcinoma) from oral hairy leukoplakia (EBV, lateral tongue, HIV) and thrush (scrapes off). USMLE extension | Pathology | DOK 3

IN-13-G Malignant skin neoplasms

1 Basal cell carcinoma **2** Squamous cell carcinoma **3** Melanoma

IN-13-G.1 Explain basal cell carcinoma as the most common skin cancer, driven by UV and hedgehog activation, identify pearly papules with telangiectasias, rolled borders, or ulceration and basaloid nests with peripheral palisading, and predict local invasion with rare metastasis. NPLEX + USMLE | Pathology | DOK 3

IN-13-G.2 Identify risk factors for cutaneous squamous cell carcinoma (cumulative UV, actinic keratosis, transplant immunosuppression, arsenic, HPV, xeroderma pigmentosum, chronic scars or ulcers giving Marjolin ulcer) and explain why it favors the lower lip and sun-exposed skin. NPLEX + USMLE | Pathology | DOK 3

IN-13-G.3 Identify squamous cell carcinoma as a scaly, crusted, or ulcerated nodule with keratin pearls and intercellular bridges, distinguish it from squamous cell carcinoma in situ (Bowen disease) and keratoacanthoma, and predict higher metastatic risk than basal cell carcinoma. NPLEX + USMLE | Pathology | DOK 3

IN-13-G.4 Explain melanoma risk factors (intense intermittent UV, fair skin, many or dysplastic nevi, family history, CDKN2A, xeroderma pigmentosum) and distinguish superficial spreading, nodular, lentigo maligna, acral lentiginous, and mucosal (including genital) subtypes by site and behavior. NPLEX + USMLE | Pathology | DOK 3

IN-13-G.5 Explain why entry into the vertical growth phase allows melanoma to metastasize, identify Breslow thickness as the key prognostic factor for localized melanoma, and identify S-100, HMB-45, and Melan-A as melanocytic immunostains. NPLEX + USMLE | Pathology | DOK 3

IN-13-G.6 Describe mycosis fungoides as a cutaneous CD4 T cell lymphoma progressing through patch, plaque, and tumor stages with Pautrier microabscesses of cerebriform cells, and identify Sézary syndrome by erythroderma with circulating Sézary cells. USMLE extension | Pathology | DOK 3

IN-13-H Infectious skin diseases

1 Acne vulgaris **2** Candidiasis **3** Cellulitis **4** Erysipelas **5** Erythema nodosum **6** Folliculitis **7** Impetigo **8** Methicillin-resistant Staphylococcus aureus (MRSA) **9** Molluscum contagiosum **10** Tinea **11** Verrucae **12** Wound infection/needle stick

IN-13-H.1 Explain the four steps of acne vulgaris (follicular hyperkeratinization, androgen-driven sebum, Cutibacterium acnes growth, inflammation), distinguish comedones from inflammatory papules, pustules, and nodules, and identify hyperandrogenism (polycystic ovary syndrome) and glucocorticoids as aggravating causes. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.2 Identify cutaneous candidiasis by beefy red intertriginous plaques with satellite pustules, distinguish candidal diaper dermatitis (involves folds) from irritant diaper dermatitis (spares folds), and relate thrush and chronic mucocutaneous candidiasis to host risk factors. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-13-H.3 Distinguish erysipelas (upper dermis and superficial lymphatics, raised sharply demarcated erythema, usually Streptococcus pyogenes) from cellulitis (deep dermis and subcutaneous fat, ill-defined borders, Streptococcus pyogenes or Staphylococcus aureus) and relate each to its portal of entry. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.4 Predict complications of cellulitis, including abscess, lymphangitis, bacteremia, and necrotizing fasciitis, suspected when pain is out of proportion to findings, with crepitus and rapid spread along fascial planes. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.5 Explain erythema nodosum as a septal panniculitis without vasculitis, identify tender red nodules on the anterior shins that heal without scarring, and relate it to streptococcal infection, sarcoidosis, tuberculosis, coccidioidomycosis, histoplasmosis, inflammatory bowel disease, and pregnancy. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.6 Distinguish folliculitis, furuncles, and carbuncles by depth and extent of follicular infection, mostly by Staphylococcus aureus, and identify hot tub folliculitis from Pseudomonas aeruginosa and cutaneous or pilonidal abscess as related suppurative lesions. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.7 Distinguish nonbullous impetigo (honey-colored crusts, Staphylococcus aureus or Streptococcus pyogenes) from bullous impetigo (staphylococcal exfoliative toxin splitting the granular layer) in children, and predict poststreptococcal glomerulonephritis as a complication. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.8 Explain how MRSA carries mecA on SCCmec, distinguish community-acquired MRSA (Panton-Valentine leukocidin, recurrent abscesses and furuncles in athletes, crowded settings, and injection drug use) from hospital-acquired MRSA, and identify nasal colonization as a reservoir. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-13-H.9 Identify molluscum contagiosum by firm, flesh-colored, dome-shaped papules with central umbilication, spread by skin contact, autoinoculation, and sexual contact, with a self-limited course in children and extensive lesions in HIV. NPLEX + USMLE | Pathology | DOK 2

IN-13-H.10 Classify tinea by site (capitis with kerion and alopecia, corporis with annular scaly border and central clearing, cruris, pedis, barbae, unguium or onychomycosis), and distinguish tinea versicolor as a Malassezia infection, not a dermatophytosis. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.11 Distinguish common, plantar, flat, and anogenital warts (condyloma acuminatum from HPV 6 and 11), identify papillomatosis, hyperkeratosis, and koilocytes on histology, and predict extensive warts with T cell immunodeficiency. NPLEX + USMLE | Pathology | DOK 3

IN-13-H.12 Rank bloodborne pathogen transmission risk after a needle stick (hepatitis B highest, then hepatitis C, then HIV), and explain how a hollow-bore needle, deep injury, and high source viral load raise risk. NPLEX | Microbiology & Immunology | DOK 2

IN-13-H.13 Predict the wound infection organism from context: Pasteurella multocida (cat or dog bite), Capnocytophaga canimorsus (dog bite with asplenia), Eikenella corrodens (human bite), Vibrio vulnificus and Mycobacterium marinum (water exposure), and Sporothrix schenckii (rose thorn). NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-13-H.14 Explain how Clostridium perfringens alpha toxin (lecithinase) causes gas gangrene with crepitus and myonecrosis, and how Clostridium tetani in a puncture wound releases tetanospasmin that blocks GABA and glycine release, causing spastic paralysis. NPLEX + USMLE | Microbiology & Immunology | DOK 3

IN-13-H.15 Identify herpes simplex skin infections (herpes labialis, genital herpes, herpetic whitlow, eczema herpeticum) and herpes zoster (painful dermatomal vesicles, Ramsay Hunt syndrome with facial palsy and ear vesicles), and predict postherpetic neuralgia. USMLE extension | Pathology | DOK 3

IN-13-H.16 Distinguish viral exanthems: measles (cough, coryza, conjunctivitis, Koplik spots, rash spreading down from the head), rubella (postauricular nodes, milder rash), roseola (HHV-6, high fever then rash), and erythema infectiosum (parvovirus B19, slapped cheek, lacy rash). USMLE extension Pathology DOK 3

IN-13-H.17 Distinguish varicella (crops of vesicles in different stages starting on the trunk), hand-foot-and-mouth disease and herpangina (coxsackievirus A), and scarlet fever (sandpaper rash, strawberry tongue, circumoral pallor, desquamation from streptococcal pyrogenic exotoxin). USMLE extension Pathology DOK 3

IN-13-H.18 Distinguish staphylococcal scalded skin syndrome (young children, toxin cleaves desmoglein 1 in the granular layer, mucosa spared) from toxic epidermal necrolysis (drug-triggered full-thickness epidermal necrosis with mucosal involvement) by split level and cause. USMLE extension Pathology DOK 3

IN-13-H.19 Compare tuberculoid leprosy (Th1, few bacilli, few hypopigmented anesthetic plaques) with lepromatous leprosy (Th2, many bacilli, diffuse nodules, leonine facies), and explain why Mycobacterium leprae favors cool skin and peripheral nerves and cannot be cultured in vitro. USMLE extension Microbiology & Immunology DOK 3

IN-13-H.20 Explain how Bacillus anthracis edema factor (adenylate cyclase), lethal factor, protective antigen, and polypeptide capsule produce cutaneous anthrax, a painless papule that becomes a black eschar surrounded by edema. USMLE extension Microbiology & Immunology DOK 3

IN-13-H.21 Identify scabies (Sarcoptes scabiei burrows in web spaces, wrists, and genitals, nocturnal itch, crusted scabies with immunosuppression), head and pubic lice, body lice as vectors of Rickettsia prowazekii and Bartonella quintana, and bedbug bites in linear clusters. USMLE extension Microbiology & Immunology DOK 2

APPENDIX

USMLE Step 1 crosswalk

Every topic the USMLE Step 1 outline names for each system, and where it lives in this course. Topics outside NPLEX Part I are listed as excluded with the reason.

Cardio 130 mapped · 10 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Cardiovascular: Infectious, immunologic, and inflammatory disorders: bacterial endocarditis	CV-18-K.1	
Cardiovascular: Infectious, immunologic, and inflammatory disorders: myocarditis	CV-18-E.5	
Cardiovascular: Infectious, immunologic, and inflammatory disorders: atherosclerosis, including aortic	CV-18-I.2	
Cardiovascular: Neoplasms: myxoma	CV-18-J.5	Added as USMLE extension
Cardiovascular: Neoplasms: metastases	CV-18-F.6	
Cardiovascular: Dysrhythmias: premature beats	CV-7.6	
Cardiovascular: Dysrhythmias: atrial flutter and fibrillation	CV-7.6	
Cardiovascular: Dysrhythmias: tachycardias, including torsades de pointes and wide-complex	CV-7.6	
Cardiovascular: Dysrhythmias: bradycardias	CV-7.3	
Cardiovascular: Dysrhythmias: atrioventricular blocks	CV-7.5	
Cardiovascular: Dysrhythmias: conduction disorders	CV-7.5	
Cardiovascular: Dysrhythmias: cardiac arrest	CV-7.6	
Cardiovascular: Dysrhythmias: sick sinus syndrome	CV-7.7	Added as USMLE extension
Cardiovascular: Dysrhythmias: prolonged QT interval	CV-14.4	
Cardiovascular: Dysrhythmias: Wolff-Parkinson-White syndrome	CV-7.7	Added as USMLE extension
Cardiovascular: Dysrhythmias: carotid sinus hypersensitivity	CV-6.5	
Cardiovascular: Dysrhythmias: pacemaker dysfunction	Excluded	See excluded
Cardiovascular: Heart failure: chordae tendineae rupture	CV-18-B.7	Added as USMLE extension
Cardiovascular: Heart failure: congestive heart failure	CV-18-B.1	
Cardiovascular: Heart failure: high-output heart failure	CV-18-B.6	Added as USMLE extension
Cardiovascular: Heart failure: cor pulmonale	CV-15.4	
Cardiovascular: Heart failure: systolic and diastolic dysfunction	CV-18-B.3	
Cardiovascular: Heart failure: mitral valve dysfunction	CV-18-D.4	
Cardiovascular: Heart failure: cardiogenic pulmonary edema	CV-18-B.7	Added as USMLE extension
Cardiovascular: Ischemic heart disease: acute coronary syndrome and myocardial infarction	CV-18-C.3	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Cardiovascular: Ischemic heart disease: stable angina	CV-18-C.2	
Cardiovascular: Ischemic heart disease: unstable angina	CV-18-C.2	
Cardiovascular: Ischemic heart disease: coronary artery disease	CV-18-C.1	
Cardiovascular: Ischemic heart disease: coronary artery spasm	CV-18-C.2	
Cardiovascular: Diseases of the myocardium: dilated cardiomyopathy	CV-18-E.1	
Cardiovascular: Diseases of the myocardium: hypertrophic cardiomyopathy	CV-18-E.2	
Cardiovascular: Diseases of the myocardium: restrictive cardiomyopathy	CV-18-E.3	
Cardiovascular: Diseases of the myocardium: hypertensive heart disease	CV-18-A.3	
Cardiovascular: Diseases of the myocardium: complications of myocardial infarction (papillary muscle and free wall rupture)	CV-18-C.6	
Cardiovascular: Diseases of the myocardium: myocarditis	CV-18-E.5	
Cardiovascular: Diseases of the pericardium: acute pericarditis	CV-18-F.1	
Cardiovascular: Diseases of the pericardium: constrictive pericarditis	CV-18-F.3	
Cardiovascular: Diseases of the pericardium: pericardial effusion	CV-18-F.4	
Cardiovascular: Diseases of the pericardium: cardiac tamponade	CV-18-F.5	
Cardiovascular: Valvular heart disease: mitral valve disorders	CV-18-D.3	
Cardiovascular: Valvular heart disease: aortic valve disorders	CV-18-D.1	
Cardiovascular: Valvular heart disease: tricuspid valve disorders	CV-18-D.9	
Cardiovascular: Valvular heart disease: pulmonic valve disorders	CV-18-D.9	
Cardiovascular: Valvular heart disease: functional murmurs	CV-4.6	
Cardiovascular: Valvular heart disease: rheumatic heart disease	CV-18-D.8	
Cardiovascular: Valvular heart disease: prosthetic valve complications	CV-18-D.10	Added as USMLE extension
Cardiovascular: Hypotension: orthostatic hypotension	CV-6.7	Added as USMLE extension
Cardiovascular: Hypertension: essential hypertension	CV-18-A.1	
Cardiovascular: Hypertension: secondary hypertension	CV-18-A.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Cardiovascular: Hypertension: hypertensive emergency	CV-18-A.4	
Cardiovascular: Hypertension: elevated blood pressure without diagnosis of hypertension	CV-18-A.1	
Cardiovascular: Dyslipidemia: hypercholesterolemia	CV-18-I.3	
Cardiovascular: Dyslipidemia: hyperlipidemia	CV-13.2	
Cardiovascular: Dyslipidemia: hypertriglyceridemia	CV-13.2	
Cardiovascular: Dyslipidemia: lipoprotein lipase deficiency	CV-13.2	
Cardiovascular: Vascular disorders: aortic and iliac aneurysm	CV-18-I.4	
Cardiovascular: Vascular disorders: aortic dissection	CV-18-I.5	
Cardiovascular: Vascular disorders: aortic and iliac aneurysm rupture	CV-18-I.4	
Cardiovascular: Vascular disorders: aortoiliac disease	CV-18-I.6	
Cardiovascular: Vascular disorders: arterial embolism and thrombosis	CV-18-H.3	
Cardiovascular: Vascular disorders: arteriovenous fistula	CV-18-B.6	Added as USMLE extension
Cardiovascular: Vascular disorders: atheroembolic disease	CV-18-I.6	
Cardiovascular: Vascular disorders: claudication	CV-18-I.6	
Cardiovascular: Vascular disorders: thromboangiitis obliterans	CV-18-I.11	
Cardiovascular: Vascular disorders: deep venous thrombosis and venous thromboembolism	CV-18-I.8	
Cardiovascular: Vascular disorders: thrombophlebitis	CV-18-I.8	
Cardiovascular: Vascular disorders: varicose veins	CV-18-I.9	
Cardiovascular: Vascular disorders: venous insufficiency	CV-18-I.9	
Cardiovascular: Vascular disorders: stasis ulcers	CV-18-I.9	
Cardiovascular: Traumatic and mechanical disorders: ventricular puncture	CV-3.6	Added as USMLE extension
Cardiovascular: Traumatic and mechanical disorders: myocardial contusion and rupture	CV-3.6	Added as USMLE extension
Cardiovascular: Traumatic and mechanical disorders: traumatic aortic dissection	CV-3.6	Added as USMLE extension
Cardiovascular: Traumatic and mechanical disorders: traumatic cardiac tamponade	CV-18-F.5	
Cardiovascular: Congenital disorders, including disease in adults: anomalous left coronary artery	CV-8.6	Added as USMLE extension
Cardiovascular: Congenital disorders, including disease in adults: atrial septal defect	CV-18-G.3	
Cardiovascular: Congenital disorders, including disease in adults: ventricular septal defect	CV-18-G.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Cardiovascular: Congenital disorders, including disease in adults: patent ductus arteriosus	CV-18-G.4	
Cardiovascular: Congenital disorders, including disease in adults: patent foramen ovale	CV-18-G.3	
Cardiovascular: Congenital disorders, including disease in adults: coarctation of the aorta	CV-18-G.7	Added as USMLE extension
Cardiovascular: Congenital disorders, including disease in adults: endocardial cushion defect	CV-1.3	
Cardiovascular: Congenital disorders, including disease in adults: tetralogy of Fallot	CV-18-G.5	
Cardiovascular: Congenital disorders, including disease in adults: transposition of the great vessels	CV-18-G.8	Added as USMLE extension
Cardiovascular: Adverse effects of drugs: doxorubicin	Excluded	See excluded
Cardiovascular: Adverse effects of drugs: sympathomimetic stimulants (cocaine, amphetamine, phencyclidine)	Excluded	See excluded
Cardiovascular: Adverse effects of drugs: ACE inhibitors	Excluded	See excluded
Cardiovascular: Adverse effects of drugs: calcium channel blockers	Excluded	See excluded
Cardiovascular: Adverse effects of drugs: alpha blockers	Excluded	See excluded
Cardiovascular: Adverse effects of drugs: minoxidil	Excluded	See excluded
Multisystem Processes & Disorders: Infectious: Chagas disease (acute and chronic)	CV-18-K.3	
Multisystem Processes & Disorders: Immunologic: acute rheumatic fever	CV-16.2	
Multisystem Processes & Disorders: Immunologic: autoimmune arteritis and vasculitis	CV-16.3	
Multisystem Processes & Disorders: Immunologic: Behcet syndrome	CV-18-I.17	
Multisystem Processes & Disorders: Immunologic: Churg-Strauss syndrome (eosinophilic granulomatosis with polyangiitis)	CV-18-I.16	
Multisystem Processes & Disorders: Immunologic: Kawasaki disease	CV-18-I.15	
Multisystem Processes & Disorders: Immunologic: polyarteritis nodosa	CV-18-I.14	
Multisystem Processes & Disorders: Immunologic: polymyalgia rheumatica and temporal arteritis	CV-18-I.12	
Multisystem Processes & Disorders: Immunologic: Takayasu arteritis	CV-18-I.13	
Multisystem Processes & Disorders: Immunologic: granulomatosis with polyangiitis	CV-18-I.16	
Multisystem Processes & Disorders: Immunologic: IgA vasculitis	CV-18-I.17	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem Processes & Disorders: Immunologic: Raynaud disease and syndrome	CV-18-I.10	
Multisystem Processes & Disorders: Immunologic: sarcoidosis (cardiac)	CV-18-E.3	
Multisystem Processes & Disorders: Neoplasms: nonbacterial thrombotic endocarditis	CV-18-D.6	
Multisystem Processes & Disorders: Shock: cardiogenic	CV-18-H.7	
Multisystem Processes & Disorders: Shock: hypovolemic	CV-18-H.7	
Multisystem Processes & Disorders: Shock: neurogenic	CV-18-H.7	
Multisystem Processes & Disorders: Shock: septic	CV-18-H.7	
Multisystem Processes & Disorders: Shock: refractory shock	CV-18-H.8	
Multisystem Processes & Disorders: Shock: systemic inflammatory response syndrome	CV-18-H.8	
Multisystem Processes & Disorders: Shock: multiorgan dysfunction	CV-18-H.8	
Multisystem Processes & Disorders: Nutrition: thiamine deficiency, including beriberi	CV-13.5	
Multisystem Processes & Disorders: Genetic: Marfan syndrome	CV-14.3	
Multisystem Processes & Disorders: Genetic: Ehlers-Danlos syndrome	CV-14.3	
Multisystem Processes & Disorders: Genetic: Down syndrome	CV-14.5	
Multisystem Processes & Disorders: Genetic: amyloidosis (cardiac)	CV-18-E.3	
Multisystem Processes & Disorders: Genetic: VATER/VACTERL association	CV-14.5	
Cardiovascular: Dysrhythmias: multifocal atrial tachycardia	CV-7.6	
Cardiovascular: Dysrhythmias: paroxysmal tachycardias	CV-7.6	
Cardiovascular: Heart failure: heart failure secondary to myocardial infarction	CV-18-C.7	
Cardiovascular: Heart failure: tachycardia-induced	CV-18-E.6	Added as USMLE extension
Cardiovascular: Diseases of the myocardium: takotsubo cardiomyopathy	CV-18-E.6	Added as USMLE extension
Cardiovascular: Diseases of the myocardium: familial dilated cardiomyopathy	CV-18-E.1	
Cardiovascular: Diseases of the myocardium: nontraumatic tamponade post-myocardial infarction	CV-18-F.5	
Cardiovascular: Diseases of the pericardium: pericarditis following myocardial infarction, surgery, trauma	CV-18-F.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Cardiovascular: Vascular disorders: cholesterol emboli	CV-18-I.6	
Cardiovascular: Vascular disorders: hypertensive vascular disease	CV-18-A.3	
Cardiovascular: Vascular disorders: peripheral arterial disease	CV-18-I.6	
Cardiovascular: Vascular disorders: stasis dermatitis	CV-18-I.9	
Immune System: HIV/AIDS: Kaposi sarcoma	CV-18-J.3	Outside the Cardiovascular section; on both outlines
Skin & Subcutaneous Tissue: Neoplasms: hemangioma	CV-18-J.1	Outside the Cardiovascular section; on both outlines
Multisystem Processes & Disorders: Infectious: Lyme disease	CV-18-K.4	
Multisystem Processes & Disorders: Infectious: Rocky Mountain spotted fever	CV-18-K.5	
Multisystem Processes & Disorders: Infectious: yellow fever	CV-18-K.6	
Blood & Lymphoreticular System: Infectious: dengue fever	CV-18-K.6	Outside the Cardiovascular section; on both outlines
Blood & Lymphoreticular System: Infectious: viral hemorrhagic fever, Ebola virus	CV-18-K.7	Outside the Cardiovascular section; on both outlines
Respiratory System: Pulmonary vascular disorders: pulmonary hypertension	CV-18-A.5	Outside the Cardiovascular section; on both outlines
Blood & Lymphoreticular System: Coagulation disorders: hypercoagulable states	CV-18-H.2	Outside the Cardiovascular section; on both outlines
Multisystem Processes & Disorders: Signs and symptoms: lymphedema	CV-10.5	
Cardiovascular: Adverse effects of drugs on the cardiovascular system (doxorubicin, sympathomimetic stimulants, ACE inhibitors, calcium channel blockers, alpha blockers, minoxidil)	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I. Doxorubicin and cocaine are kept only as named causes of dilated cardiomyopathy, aortic dissection, and coronary vasospasm.
Cardiovascular: Dysrhythmias: pacemaker dysfunction	Excluded	Device management is clinical practice, not biomedical science on NPLEX Part I.
Multisystem Processes & Disorders: Signs and symptoms (chest pain, palpitations, syncope, dizziness)	Excluded	Symptom work-up is clinical diagnosis; these findings are taught inside the condition objectives (angina, arrhythmias, aortic stenosis, orthostatic hypotension) rather than as separate objectives.

GI 161 mapped · 16 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: pseudomembranous colitis (Clostridioides difficile)	GI-13-L.1	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Staphylococcus aureus enteric infection	GI-12.1	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Escherichia coli enteric infection	GI-12.3	EHEC; ETEC in GI-12.1
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Listeria monocytogenes	GI-12.3	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Yersinia enterocolitica	GI-12.3	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Campylobacter species	GI-12.2	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Vibrio cholerae	GI-12.1	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Salmonella species	GI-12.2	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Shigella species	GI-12.2	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: traveler's or infectious diarrhea	GI-13-L.4	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: hepatic, subhepatic, and subphrenic abscess	GI-13-L.10	Added as USMLE extension; spaces in GI-4.6
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: peritonitis, primary and secondary	GI-13-I.2	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: infectious esophagitis (CMV, herpes)	GI-13-L.3	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: hepatitis A, B, C, D, E	GI-12.5	Outcomes in GI-13-L.9, serology in GI-13-L.8
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: coxsackievirus enteritis or colitis	GI-12.4	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: echovirus enteritis or colitis	GI-12.4	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: rotavirus enteritis	GI-12.4	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: mumps	GI-13-A.1	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: herpetic gingivostomatitis	GI-13-L.7	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: thrush	GI-13-L.6	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Cryptosporidium, Cyclospora, Isospora belli	GI-12.7	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Entamoeba histolytica, Giardia	GI-12.7	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: Strongyloides stercoralis	GI-12.7	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: autoimmune hepatitis	GI-13-C.9	Autoantibodies in GI-11.4
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: celiac disease	GI-13-E.2	Immunopathogenesis in GI-11.2
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: eosinophilic esophagitis	GI-13-G.7	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: granulomatous enteritis, regional enteritis (Crohn disease)	GI-13-G.10	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: microscopic colitis (collagenous, lymphocytic)	GI-13-G.4	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: ulcerative colitis	GI-13-G.10	
Gastrointestinal: Infectious, immunologic, and inflammatory disorders: toxic megacolon	GI-13-F.5	
Gastrointestinal: Neoplasms: benign polyps and cysts of stomach, intestine, colon, rectum, anus	GI-13-K.11	
Gastrointestinal: Neoplasms: oral cancer (lips, mouth, tongue)	GI-13-K.8	
Gastrointestinal: Neoplasms: salivary gland neoplasms	GI-13-K.12	Added as USMLE extension
Gastrointestinal: Neoplasms: esophageal squamous cell carcinoma and adenocarcinoma	GI-13-K.1	
Gastrointestinal: Neoplasms: Barrett esophagus	GI-13-G.2	
Gastrointestinal: Neoplasms: gastrinoma, Zollinger-Ellison syndrome	GI-13-K.4	
Gastrointestinal: Neoplasms: GI carcinoid tumors	GI-13-K.5	
Gastrointestinal: Neoplasms: GI stromal tumors	GI-13-K.3	
Gastrointestinal: Neoplasms: small intestine neoplasms	GI-13-K.5	
Gastrointestinal: Neoplasms: stomach adenocarcinoma	GI-13-K.2	
Gastrointestinal: Neoplasms: stomach lymphoma, MALT lymphoma	GI-13-K.3	Immune mechanism in GI-11.5
Gastrointestinal: Neoplasms: colon and rectal cancer	GI-13-K.10	
Gastrointestinal: Neoplasms: anal cancer	GI-13-K.13	Added as USMLE extension
Gastrointestinal: Neoplasms: familial adenomatous polyposis, Gardner, Turcot	GI-10.5	
Gastrointestinal: Neoplasms: Peutz-Jeghers syndrome	GI-10.5	
Gastrointestinal: Neoplasms: MUTYH-associated polyposis	GI-10.5	
Gastrointestinal: Neoplasms: gallbladder cancer	GI-13-K.13	Porcelain gallbladder risk in GI-13-D.2
Gastrointestinal: Neoplasms: cholangiocarcinoma, adenocarcinoma of the ampulla of Vater	GI-13-K.13	Added as USMLE extension
Gastrointestinal: Neoplasms: liver, hepatoma	GI-13-K.6	
Gastrointestinal: Neoplasms: peritoneal cancer	GI-13-K.7	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Neoplasms: pancreatic cancer	GI-13-K.9	
Gastrointestinal: Neoplasms: metastatic neoplasms	GI-13-K.7	
Gastrointestinal: Signs, symptoms, and ill-defined disorders: abdominal pain	GI-1.1	Referred visceral pain
Gastrointestinal: Signs, symptoms, and ill-defined disorders: constipation	GI-13-J.2	As a hemorrhoid risk; motility in GI-6.2
Gastrointestinal: Signs, symptoms, and ill-defined disorders: diarrhea	GI-7.6	
Gastrointestinal: Signs, symptoms, and ill-defined disorders: nausea and vomiting	GI-6.7	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: abscessed tooth	GI-13-L.5	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: dental caries	GI-13-L.5	Salivary protection in GI-9.6
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: malocclusion	Excluded	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: salivary gland stones, sialadenitis	GI-13-A.2	Added as USMLE extension
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: parotitis	GI-13-A.1	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: achalasia, cardiospasm	GI-13-F.1	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: dysphagia	GI-13-K.1	Also GI-13-F.1, GI-13-H.2
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: diverticulum (Zenker)	GI-13-F.6	Added as USMLE extension
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: periapical abscess without sinus	GI-13-L.5	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: esophagitis, GERD	GI-13-G.9	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: pill esophagitis	GI-13-G.7	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: Mallory-Weiss syndrome	GI-13-G.12	Added as USMLE extension
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: paraesophageal (hiatal) hernia	GI-13-F.3	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: stricture and stenosis of esophagus	GI-13-H.2	Peptic stricture in GI-13-G.9
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: dyspepsia, hyperacidity	GI-13-G.5	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: gastric ulcer	GI-13-G.5	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: gastritis	GI-13-G.8	Acute gastritis in GI-13-G.6

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: peptic ulcer	GI-13-G.5	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: peptic ulcer perforation	GI-13-G.5	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: gastroparesis	GI-13-F.7	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: appendicitis	GI-13-G.1	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: angiodysplasia	GI-13-J.4	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: diverticula, diverticulitis, diverticulosis	GI-13-G.3	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: duodenitis, duodenal ulcer	GI-13-G.5	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: noninfectious gastroenteritis and colitis	GI-13-G.4	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: granulomatous enterocolitis	GI-13-G.10	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: Hirschsprung disease	GI-13-F.5	Embryology in GI-1.5
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: impaction of intestine	GI-13-F.2	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: intestinal obstruction, stricture	GI-13-F.2	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: intussusception	GI-13-F.4	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: irritable bowel syndrome	GI-13-F.7	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: mesenteric ischemia, ischemic bowel, ischemic colitis	GI-13-J.3	Watershed zones in GI-5.2
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: necrotizing enterocolitis	GI-13-J.5	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: paralytic ileus	GI-13-F.2	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: volvulus	GI-13-F.4	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: malabsorption, lactose intolerance	GI-13-E.4	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: short bowel syndrome	GI-13-E.5	Added as USMLE extension

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: abscess of anal and rectal regions	GI-13-G.13	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: anal fissure	GI-13-G.13	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: anal fistula	GI-13-G.13	Added as USMLE extension
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: anal ulcer	GI-13-G.13	Grouped with fissure
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: fecal incontinence	GI-6.6	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: rectal and anal hemorrhage	GI-13-J.2	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: proctitis	GI-13-G.13	
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: hemorrhoids	GI-13-J.2	Anatomy in GI-5.5
Gastrointestinal: Disorders of the stomach, small intestine, colon, rectum, anus: rectal prolapse	GI-13-G.13	
Gastrointestinal: Disorders of the liver and biliary system: cirrhosis	GI-13-C.3	
Gastrointestinal: Disorders of the liver and biliary system: Dubin-Johnson syndrome, Rotor syndrome	GI-13-C.11	Added as USMLE extension
Gastrointestinal: Disorders of the liver and biliary system: end-stage liver disease	GI-13-C.4	Synthetic failure in GI-9.5
Gastrointestinal: Disorders of the liver and biliary system: Gilbert syndrome, Crigler-Najjar syndrome	GI-13-C.6	
Gastrointestinal: Disorders of the liver and biliary system: hepatic coma, hepatic encephalopathy	GI-13-C.4	
Gastrointestinal: Disorders of the liver and biliary system: noninfectious hepatitis	GI-13-C.9	
Gastrointestinal: Disorders of the liver and biliary system: alcoholic fatty liver	GI-13-C.7	
Gastrointestinal: Disorders of the liver and biliary system: hepatorenal syndrome	GI-13-C.5	
Gastrointestinal: Disorders of the liver and biliary system: hepatopulmonary syndrome	GI-13-C.5	
Gastrointestinal: Disorders of the liver and biliary system: jaundice	GI-9.2	
Gastrointestinal: Disorders of the liver and biliary system: nonalcoholic fatty liver disease	GI-13-C.8	
Gastrointestinal: Disorders of the liver and biliary system: portal hypertension, esophageal varices	GI-13-C.10	Varices in GI-13-J.1
Gastrointestinal: Disorders of the liver and biliary system: bile duct obstruction, cholestasis	GI-13-C.1	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Disorders of the liver and biliary system: cholangitis, including ascending	GI-13-D.4	
Gastrointestinal: Disorders of the liver and biliary system: choledocholithiasis	GI-13-D.4	
Gastrointestinal: Disorders of the liver and biliary system: cholelithiasis, cholecystitis	GI-13-D.3	Cholecystitis in GI-13-D.1
Gastrointestinal: Disorders of the liver and biliary system: cholestasis due to parenteral nutrition	GI-13-C.1	
Gastrointestinal: Disorders of the liver and biliary system: gallstone ileus	GI-13-D.4	
Gastrointestinal: Disorders of the liver and biliary system: Mirizzi syndrome	GI-13-D.5	Added as USMLE extension
Gastrointestinal: Disorders of the liver and biliary system: primary biliary cirrhosis (cholangitis)	GI-13-C.2	
Gastrointestinal: Disorders of the liver and biliary system: primary sclerosing cholangitis	GI-13-C.2	
Gastrointestinal: Disorders of the pancreas: acute pancreatitis	GI-13-B.1	
Gastrointestinal: Disorders of the pancreas: chronic pancreatitis	GI-13-B.4	
Gastrointestinal: Disorders of the pancreas: hereditary pancreatitis	GI-13-B.4	
Gastrointestinal: Disorders of the pancreas: pancreatic cyst, pseudocyst	GI-13-B.3	
Gastrointestinal: Disorders of the pancreas: pancreatic duct obstruction	GI-13-K.9	Also as a cause in GI-13-B.4
Gastrointestinal: Disorders of the pancreas: pancreatic insufficiency	GI-13-E.3	Also GI-13-B.4
Gastrointestinal: Disorders of the peritoneal cavity: ascites	GI-13-I.1	
Gastrointestinal: Traumatic and mechanical disorders: abdominal wall defects	GI-13-H.6	Added as USMLE extension
Gastrointestinal: Traumatic and mechanical disorders: postsurgical adhesions	GI-13-I.3	
Gastrointestinal: Traumatic and mechanical disorders: digestive system complications of surgery	Excluded	
Gastrointestinal: Traumatic and mechanical disorders: post-gastric surgery syndromes (blind loop syndrome)	GI-13-E.6	Added as USMLE extension
Gastrointestinal: Traumatic and mechanical disorders: duodenal tear	Excluded	
Gastrointestinal: Traumatic and mechanical disorders: foreign body in digestive system	Excluded	
Gastrointestinal: Traumatic and mechanical disorders: inguinal, femoral, and abdominal wall hernias	GI-4.5	Strangulation in GI-13-F.3
Gastrointestinal: Traumatic and mechanical disorders: open abdominal wound	Excluded	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Gastrointestinal: Traumatic and mechanical disorders: perforation of hollow viscus and blunt trauma	GI-13-G.5	Perforation only; blunt trauma excluded
Gastrointestinal: Traumatic and mechanical disorders: Boerhaave syndrome	GI-13-G.12	Added as USMLE extension
Gastrointestinal: Traumatic and mechanical disorders: umbilical hernia	GI-4.5	
Gastrointestinal: Congenital disorders: annular pancreas	GI-1.4	
Gastrointestinal: Congenital disorders: biliary atresia	GI-13-H.5	Added as USMLE extension
Gastrointestinal: Congenital disorders: cleft lip and palate	GI-13-H.7	Added as USMLE extension
Gastrointestinal: Congenital disorders: esophageal atresia	GI-13-H.1	
Gastrointestinal: Congenital disorders: malrotation without volvulus	GI-13-H.6	Embryology in GI-1.2
Gastrointestinal: Congenital disorders: Meckel diverticulum	GI-13-H.3	
Gastrointestinal: Congenital disorders: pyloric stenosis	GI-13-H.4	
Gastrointestinal: Congenital disorders: tracheoesophageal fistula	GI-13-H.1	
Gastrointestinal: Adverse effects of drugs: motility changes (chronic laxatives, opioids)	Excluded	
Gastrointestinal: Adverse effects of drugs: gastritis, duodenitis, peptic ulcer (NSAIDs)	GI-13-G.5	NSAIDs only as a cause of ulcer
Gastrointestinal: Adverse effects of drugs: hepatitis (acetaminophen, isoniazid)	GI-13-C.9	Only as causes of hepatitis; mechanism in GI-9.3
Gastrointestinal: Adverse effects of drugs: pancreatitis (thiazide diuretics)	GI-13-B.1	Only as a cause of pancreatitis
Multisystem: Nutrition: protein-calorie malnutrition (kwashiorkor, marasmus)	GI-8.1	
Multisystem: Nutrition: vitamin A deficiency and toxicity	GI-8.2	Toxicity in GI-8.7
Multisystem: Nutrition: vitamin B1, B3, B6 deficiency	GI-8.4	
Multisystem: Nutrition: vitamin B9 and B12 deficiency	GI-8.3	
Multisystem: Nutrition: vitamin C deficiency	GI-8.4	
Multisystem: Nutrition: vitamin D, E, K deficiency	GI-8.2	Vitamin D toxicity in GI-8.7
Multisystem: Nutrition: mineral deficiencies and toxicities	GI-8.6	
Multisystem: Nutrition: obesity	GI-13-C.8	Only as a risk for fatty liver disease
Multisystem: Nutrition: enteral and parenteral nutrition	Excluded	
Multisystem: Genetic, metabolic, and developmental disorders: cystic fibrosis	GI-10.6	
Multisystem: Genetic, metabolic, and developmental disorders: hemochromatosis	GI-10.1	
Multisystem: Genetic, metabolic, and developmental disorders: Wilson disease	GI-10.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem: Genetic, metabolic, and developmental disorders: alpha-1 antitrypsin deficiency	GI-10.3	
Multisystem: Toxins and environmental injury: acetaminophen	GI-9.3	
Multisystem: Toxins and environmental injury: carbon tetrachloride	GI-9.4	
Gastrointestinal: Disorders of the oral cavity, salivary glands, and esophagus: malocclusion	Excluded	Dental and orthodontic topic with no biomedical mechanism tested on NPLEX Part I
Gastrointestinal: Traumatic and mechanical disorders: digestive system complications of surgery	Excluded	Surgical management, not biomedical science; blind loop syndrome and adhesions are covered elsewhere
Gastrointestinal: Traumatic and mechanical disorders: duodenal tear	Excluded	Trauma management, not biomedical science
Gastrointestinal: Traumatic and mechanical disorders: foreign body in digestive system	Excluded	Clinical and emergency management, not biomedical science
Gastrointestinal: Traumatic and mechanical disorders: open abdominal wound	Excluded	Trauma management, not biomedical science
Gastrointestinal: Traumatic and mechanical disorders: blunt abdominal trauma	Excluded	Trauma management, not biomedical science; perforation is covered in GI-13-G.5
Gastrointestinal: Adverse effects of drugs: motility changes (chronic laxatives, opioids)	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I; opioids appear only as a cause of ileus in GI-13-F.2
Gastrointestinal: Adverse effects of drugs (NSAIDs, acetaminophen, isoniazid, thiazides) as drug topics	Excluded	Pharmacology is not on NPLEX Part I; each drug is kept only as a cause of a listed condition
Multisystem: Nutrition: enteral and parenteral nutrition	Excluded	Therapeutic nutrition support is not on NPLEX Part I; parenteral nutrition cholestasis is covered in GI-13-C.1

MSK 91 mapped · 10 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Musculoskeletal: Congenital disorders: disorders of limb development (HOX gene mutation, phocomelia)	MS-1.3	
Musculoskeletal: Traumatic and mechanical disorders: kyphoscoliosis, scoliosis	MS-3.2	
Musculoskeletal: Congenital disorders: foot deformities (flat foot, valgus and varus deformities)	MS-3.5	Flat foot (pes planus) named in the objective
Musculoskeletal: Inflammatory disorders: temporomandibular joint disorders	MS-4.5	
Musculoskeletal: Traumatic and mechanical disorders: rotator cuff syndrome	MS-5.1	
Multisystem: Nutrition: vitamin deficiencies: vitamin D (rickets)	MS-8.1	Physiology here; deficiency in MS-13-A

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Musculoskeletal: Congenital disorders: McArdle disease	MS-8.4	
Musculoskeletal: Congenital disorders: achondroplasia, dwarfism	MS-9.2	Mechanism here; clinical in MS-13-D
Musculoskeletal: Congenital disorders: mitochondrial myopathies	MS-9.6	
Musculoskeletal: Degenerative and metabolic disorders: muscles, ligaments, fascia: muscle wasting and diffuse atrophy	MS-10.4	
Musculoskeletal: Congenital disorders: dislocation of hip in infantile spinal muscular atrophy	MS-10.4	Spinal muscular atrophy mechanism covered here
Musculoskeletal: Immunologic disorders: rheumatoid arthritis	MS-11.1	
Musculoskeletal: Immunologic disorders: Felty syndrome	MS-11.2	
Multisystem: Immunologic and inflammatory disorders: Sjogren syndrome	MS-11.3	Autoantibodies only; NPLEX covers it in IM-15-D3
Multisystem: Immunologic and inflammatory disorders: systemic lupus erythematosus	MS-11.4	
Multisystem: Immunologic and inflammatory disorders: scleroderma (systemic sclerosis)	MS-11.4	
Multisystem: Immunologic and inflammatory disorders: mixed connective tissue disease	MS-11.4	
Multisystem: Infectious disorders: bacterial: Lyme disease	MS-12.6	Arthritis and organism here; NPLEX lists Lyme disease in CV-18-K3
Musculoskeletal: Infectious disorders: spondylitis, tuberculous	MS-12.7	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteomalacia	MS-13-A.1	
Musculoskeletal: Congenital disorders: genu valgum or varum	MS-13-A.4	Bowing from rickets covered here
Multisystem: Nutrition: vitamin deficiencies: vitamin C (scurvy)	MS-13-A.5	
Musculoskeletal: Immunologic disorders: ankylosing spondylitis	MS-13-B.1	
Musculoskeletal: Inflammatory disorders: ankylosis, spondylopathy (inflammatory)	MS-13-B.1	
Musculoskeletal: Inflammatory disorders: bursitis	MS-13-B.3	
Musculoskeletal: Inflammatory disorders: fibrositis, myofascial pain syndrome	MS-13-B.4	
Musculoskeletal: Degenerative and metabolic disorders: joints: gout, gouty arthritis	MS-13-B.5	
Musculoskeletal: Immunologic disorders: dermatomyositis, polymyositis	MS-13-B.7	
Multisystem: Paraneoplastic syndromes: musculoskeletal: polymyositis	MS-13-B.7	Malignancy association covered here

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Musculoskeletal: Inflammatory disorders: myositis	MS-13-B.7	
Multisystem: Immunologic and inflammatory disorders: polymyalgia rheumatica, temporal arteritis	MS-13-B.9	
Multisystem: Immunologic and inflammatory disorders: reactive arthritis (Reiter arthritis)	MS-13-B.10	
Musculoskeletal: Inflammatory disorders: tendinitis, supraspinatus syndrome, enthesopathy of spine, elbow, ankle	MS-13-B.11	
Musculoskeletal: Inflammatory disorders: tenosynovitis	MS-13-B.12	
Musculoskeletal: Inflammatory disorders: synovitis	MS-13-B.12	
Musculoskeletal: Inflammatory disorders: fasciitis	MS-13-B.12	
Musculoskeletal: Immunologic disorders: psoriatic arthropathy	MS-13-B.13	
Musculoskeletal: Immunologic disorders: juvenile idiopathic arthritis	MS-13-B.14	
Musculoskeletal: Degenerative and metabolic disorders: joints: pseudogout	MS-13-B.15	
Musculoskeletal: Inflammatory disorders: adhesive capsulitis of shoulder	MS-13-B.16	
Musculoskeletal: Inflammatory disorders: ganglion cyst	MS-13-B.16	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteopenia	MS-13-C.3	Lab comparison here; DEXA in MS-13-E
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteodystrophy	MS-13-C.4	
Multisystem: Genetic disorders: structural protein disorders: Marfan syndrome	MS-13-D.1	
Musculoskeletal: Congenital disorders: osteogenesis imperfecta	MS-13-D.5	
Multisystem: Genetic disorders: structural protein disorders: Ehlers-Danlos syndrome	MS-13-D.7	Extension; NPLEX lists it under IN-13-E2
Musculoskeletal: Degenerative and metabolic disorders: joints: osteoarthritis	MS-13-E.1	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteoporosis	MS-13-E.3	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteopenia	MS-13-E.4	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: pathologic fracture	MS-13-E.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteitis deformans (Paget disease of bone)	MS-13-E.5	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: osteonecrosis (avascular), bone infarct	MS-13-E.7	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: Legg-Calve-Perthes disease	MS-13-E.8	
Musculoskeletal: Congenital disorders: developmental dysplasia of the hip	MS-13-E.8	
Musculoskeletal: Traumatic and mechanical disorders: slipped capital femoral epiphysis	MS-13-E.8	
Musculoskeletal: Traumatic and mechanical disorders: dislocation of hip	MS-13-E.8	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: disc degeneration, herniated disc	MS-13-E.9	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: radiculopathies	MS-13-E.9	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: spondylolisthesis, spondylosis (degenerative)	MS-13-E.9	
Musculoskeletal: Traumatic and mechanical disorders: backache, including low back pain	MS-13-E.9	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: Osgood-Schlatter disease	MS-13-E.10	
Musculoskeletal: Degenerative and metabolic disorders: bone, tendon, cartilage: chondromalacia	MS-13-E.10	
Musculoskeletal: Inflammatory disorders: osteochondritis, osteochondritis dissecans	MS-13-E.10	
Musculoskeletal: Degenerative and metabolic disorders: muscles, ligaments, fascia: Dupuytren contracture	MS-13-E.11	
Musculoskeletal: Neoplasms: sarcoma	MS-13-F.1	
Musculoskeletal: Neoplasms: osteoid osteoma	MS-13-F.3	
Musculoskeletal: Neoplasms: osteosarcoma	MS-13-F.4	
Musculoskeletal: Neoplasms: rhabdomyosarcoma	MS-13-F.6	
Musculoskeletal: Neoplasms: enchondroma	MS-13-F.7	
Musculoskeletal: Neoplasms: tenosynovial giant cell tumor	MS-13-F.8	
Musculoskeletal: Neoplasms: metastases to bone, secondary malignant neoplasm of bone	MS-13-F.9	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Musculoskeletal: Neoplasms: leiomyosarcoma	MS-13-F.10	
Multisystem: Paraneoplastic syndromes: musculoskeletal: pulmonary osteoarthropathy	MS-13-F.11	
Musculoskeletal: Infectious disorders: septic arthritis	MS-13-G.1	
Musculoskeletal: Degenerative and metabolic disorders: joints: joint effusion	MS-13-G.2	
Multisystem: Signs and symptoms: arthralgias, joint pain	MS-13-G.2	
Musculoskeletal: Infectious disorders: necrotizing fasciitis	MS-13-G.3	
Musculoskeletal: Infectious disorders: osteomyelitis	MS-13-G.4	
Musculoskeletal: Infectious disorders: discitis	MS-13-G.4	
Musculoskeletal: Infectious disorders: gangrene, dry and wet	MS-13-G.6	
Musculoskeletal: Infectious disorders: clostridial myonecrosis	MS-13-G.7	
Musculoskeletal: Infectious disorders: myositis, infective	MS-13-G.8	
Musculoskeletal: Traumatic and mechanical disorders: fractures	MS-13-H.1	
Musculoskeletal: Traumatic and mechanical disorders: sprains, strains	MS-13-H.3	
Musculoskeletal: Traumatic and mechanical disorders: contusions	MS-13-H.3	
Musculoskeletal: Traumatic and mechanical disorders: dislocations	MS-13-H.3	
Musculoskeletal: Traumatic and mechanical disorders: compartment syndrome	MS-13-H.4	
Musculoskeletal: Degenerative and metabolic disorders: muscles, ligaments, fascia: rhabdomyolysis	MS-13-H.5	
Musculoskeletal: Degenerative and metabolic disorders: muscles, ligaments, fascia: muscle calcification and ossification	MS-13-H.6	
Musculoskeletal: Adverse effects of drugs: drug-induced myopathy (steroids, statins, cocaine, AZT)	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I; statin and steroid causes appear only where they cause a listed condition (rhabdomyolysis, type II atrophy)
Musculoskeletal: Adverse effects of drugs: malignant hyperthermia	Excluded	Drug adverse effect (anesthetic trigger); pharmacology is not on NPLEX Part I
Musculoskeletal: Traumatic and mechanical disorders: amputation and care of amputees	Excluded	Clinical management and rehabilitation, not biomedical science
Musculoskeletal: Traumatic and mechanical disorders: blast injuries	Excluded	Emergency trauma care, not biomedical science
Musculoskeletal: Traumatic and mechanical disorders: contractures, hospital-acquired	Excluded	Nursing care and prevention; contracture mechanisms covered in MS-13-H.4 and MS-13-E.11

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Neuromuscular disorders: muscular dystrophy	MS-13-D.3	Listed in the USMLE Nervous System section; NPLEX lists it in MS-13-D2
Nervous System: Immunologic and inflammatory disorders: myasthenia gravis	MS-10.2	Listed in the USMLE Nervous System section
Musculoskeletal: Adverse effects of drugs: drug-induced myopathy (steroids, statins, cocaine, AZT)	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I; statin and steroid causes appear only where they cause a listed condition (rhabdomyolysis, type II atrophy)
Musculoskeletal: Adverse effects of drugs: malignant hyperthermia	Excluded	Drug adverse effect (anesthetic trigger); pharmacology is not on NPLEX Part I
Musculoskeletal: Traumatic and mechanical disorders: amputation and care of amputees	Excluded	Clinical management and rehabilitation, not biomedical science
Musculoskeletal: Traumatic and mechanical disorders: blast injuries	Excluded	Emergency trauma care, not biomedical science
Musculoskeletal: Traumatic and mechanical disorders: contractures, hospital-acquired	Excluded	Nursing care and prevention; contracture mechanisms covered in MS-13-H.4 and MS-13-E.11

Neuro 174 mapped · 31 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Infectious disorders: meningitis, bacterial (H. influenzae, Listeria, N. meningitidis, S. agalactiae, S. pneumoniae)	NE-20-G.9	
Nervous System: Infectious disorders: meningitis, bacterial (M. tuberculosis)	NE-20-G.10	Tuberculous basilar meningitis
Nervous System: Infectious disorders: meningitis, bacterial (S. aureus, S. epidermidis, Actinomyces israelii)	NE-20-G.3	Staphylococci covered with brain abscess organisms; Actinomyces not written as a separate fact
Nervous System: Infectious disorders: meningitis, viral (enteroviruses, polioviruses, arboviruses, HSV, VZV, HIV, LCMV, measles, mumps, adenovirus)	NE-19.5	Organism biology; clinical CSF pattern in NE-4.5
Nervous System: Infectious disorders: meningitis, fungal (Cryptococcus, Blastomyces)	NE-20-G.6	Blastomyces not written as a separate fact
Nervous System: Infectious disorders: meningitis, spirochetal: Treponema pallidum (neurosyphilis)	NE-20-G.11	
Nervous System: Infectious disorders: meningitis, spirochetal: Borrelia burgdorferi, Leptospira	NE-20-G.18	USMLE extension
Nervous System: Infectious disorders: meningitis, protozoal and helminthic (Naegleria fowleri, Acanthamoeba, Strongyloides, Angiostrongylus, Baylisascaris)	NE-20-G.5	Naegleria written; Acanthamoeba in NE-20-J.5; helminthic eosinophilic meningitis not written
Nervous System: Infectious disorders: encephalitis (HSV-1, VZV, EBV, CMV)	NE-20-G.7	
Nervous System: Infectious disorders: encephalitis (West Nile, St. Louis, eastern and western equine)	NE-20-G.1	
Nervous System: Infectious disorders: encephalitis (rabies virus)	NE-20-G.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Infectious disorders: encephalitis (Taenia, Toxoplasma gondii)	NE-20-G.5	
Nervous System: Infectious disorders: prion disease (Creutzfeldt-Jakob disease)	NE-20-G.13	
Nervous System: Infectious disorders: botulism	NE-20-G.2	
Nervous System: Infectious disorders: tetanus	NE-20-G.16	
Nervous System: Infectious disorders: CNS disorders associated with AIDS: progressive multifocal leukoencephalopathy	NE-20-G.14	
Nervous System: Immunologic and inflammatory disorders: myasthenia gravis, including thymoma	NE-18.4	NMJ physiology in NE-13.5
Nervous System: Immunologic and inflammatory disorders: multiple sclerosis	NE-20-B.8	Immunology in NE-18.2
Nervous System: Immunologic and inflammatory disorders: transverse myelitis	NE-20-B.12	USMLE extension
Nervous System: Neoplasms: meningioma	NE-20-F.2	
Nervous System: Neoplasms: neurofibromatosis	NE-20-F.8	
Nervous System: Neoplasms: glioblastoma	NE-20-F.4	
Nervous System: Neoplasms: astrocytoma	NE-20-F.5	
Nervous System: Neoplasms: medulloblastoma	NE-20-F.1	
Nervous System: Neoplasms: primary CNS lymphoma	NE-20-F.9	USMLE extension
Nervous System: Neoplasms: metastatic (breast, lung, pancreatic, testicular, melanoma)	NE-20-F.9	USMLE extension
Nervous System: Cerebrovascular disease: arteriovenous malformations	NE-20-A.9	USMLE extension
Nervous System: Cerebrovascular disease: ectatic cerebral vessels	NE-20-A.9	Covered only as other vascular malformations
Nervous System: Cerebrovascular disease: transient ischemic attack	NE-20-A.3	
Nervous System: Cerebrovascular disease: stroke, thrombotic (cerebral artery occlusion, cerebral infarction)	NE-20-A.1	
Nervous System: Cerebrovascular disease: stroke, embolic	NE-20-A.1	
Nervous System: Cerebrovascular disease: intracerebral hemorrhage	NE-20-A.7	
Nervous System: Cerebrovascular disease: subarachnoid hemorrhage	NE-20-A.6	
Nervous System: Cerebrovascular disease: traumatic intracranial hemorrhage	NE-20-H.3	
Nervous System: Cerebrovascular disease: cerebral artery aneurysm	NE-20-A.6	
Nervous System: Cerebrovascular disease: carotid artery stenosis, atherosclerosis, occlusion, dissection	NE-20-A.10	Atherosclerotic carotid disease also in NE-20-A.1
Nervous System: Cerebrovascular disease: vertebral artery deficiency, dissection	NE-20-A.10	Vertebrobasilar syndromes in NE-10.3

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Cerebrovascular disease: subclavian steal syndrome	NE-20-A.10	
Nervous System: Cerebrovascular disease: vascular dementia	NE-20-B.11	
Nervous System: Cerebrovascular disease: hypertensive encephalopathy	NE-20-A.11	
Nervous System: Cerebrovascular disease: posterior reversible encephalopathy syndrome	NE-20-A.11	
Nervous System: Cerebrovascular disease: venous sinus thrombosis	NE-20-A.11	
Nervous System: Spine, spinal cord, and roots: cauda equina syndrome	NE-20-H.5	
Nervous System: Spine, spinal cord, and roots: spinal artery thrombosis, embolus, infarct	NE-20-A.8	
Nervous System: Spine, spinal cord, and roots: spinal cord compression	NE-20-H.5	
Nervous System: Spine, spinal cord, and roots: spinal cord transection	NE-20-H.4	
Nervous System: Spine, spinal cord, and roots: paraplegia, quadriplegia	NE-20-H.4	
Nervous System: Spine, spinal cord, and roots: autonomic dysreflexia	NE-9.5	
Nervous System: Spine, spinal cord, and roots: spinal stenosis (cervical, lumbar)	NE-20-I.9	USMLE extension
Nervous System: Spine, spinal cord, and roots: syringomyelia	NE-5.3	
Nervous System: Cranial nerve injury/disorders: cranial nerve injury	NE-8.1	
Nervous System: Cranial nerve injury/disorders: Bell palsy	NE-20-I.1	
Nervous System: Cranial nerve injury/disorders: anisocoria, miosis, mydriasis	NE-6.2	
Nervous System: Cranial nerve injury/disorders: internuclear ophthalmoplegia	NE-8.6	
Nervous System: Cranial nerve injury/disorders: nystagmus	NE-6.6	
Nervous System: Cranial nerve injury/disorders: vestibular neuritis, labyrinthitis	NE-20-J.11	
Nervous System: Cranial nerve injury/disorders: ptosis of the eyelid, Horner syndrome	NE-9.3	
Nervous System: Peripheral nerve/plexus disorders: peripheral nerve injury, including brachial plexus	NE-8.3	
Nervous System: Peripheral nerve/plexus disorders: carpal tunnel syndrome	NE-20-I.2	
Nervous System: Peripheral nerve/plexus disorders: cubital, tarsal, and peroneal tunnel syndromes	NE-20-I.8	USMLE extension
Nervous System: Peripheral nerve/plexus disorders: mononeuritis, neuropathy	NE-20-D.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Peripheral nerve/plexus disorders: Guillain-Barré syndrome	NE-20-B.7	
Nervous System: Peripheral nerve/plexus disorders: Miller Fisher syndrome	NE-18.3	
Nervous System: Peripheral nerve/plexus disorders: Charcot-Marie-Tooth disease	NE-20-D.6	USMLE extension
Nervous System: Peripheral nerve/plexus disorders: herpes zoster	NE-20-G.17	
Nervous System: Neurologic pain syndromes: complex regional pain syndrome	NE-20-I.10	USMLE extension
Nervous System: Neurologic pain syndromes: postherpetic neuralgia	NE-20-G.17	
Nervous System: Neurologic pain syndromes: phantom limb pain	NE-20-I.10	USMLE extension
Nervous System: Neurologic pain syndromes: thalamic pain syndrome	NE-3.4	
Nervous System: Neurologic pain syndromes: trigeminal neuralgia	NE-20-I.7	
Nervous System: Neurologic pain syndromes: fibromyalgia	Excluded	See excluded
Nervous System: Degenerative disorders/amnestic syndromes: Alzheimer disease	NE-20-B.1	
Nervous System: Degenerative disorders/amnestic syndromes: frontotemporal dementia	NE-20-B.10	Frontal function in NE-11.1
Nervous System: Degenerative disorders/amnestic syndromes: progressive supranuclear palsy	NE-20-B.10	
Nervous System: Degenerative disorders/amnestic syndromes: Lewy body disease	NE-20-B.10	
Nervous System: Degenerative disorders/amnestic syndromes: mild cognitive impairment	NE-20-B.11	
Nervous System: Degenerative disorders/amnestic syndromes: mild neurocognitive disorder	Excluded	See excluded
Nervous System: Global cerebral dysfunction: altered states of consciousness, coma, brain death	NE-3.6	
Nervous System: Global cerebral dysfunction: delirium	NE-3.6	Defined against coma; metabolic causes in NE-20-D.1
Nervous System: Neuromuscular disorders: amyotrophic lateral sclerosis	NE-20-B.3	
Nervous System: Neuromuscular disorders: spinal muscular atrophy	NE-16.5	USMLE extension
Nervous System: Neuromuscular disorders: muscular dystrophy (Duchenne, myotonic)	NE-16.1	Myotonic dystrophy repeat genetics only; Duchenne excluded here
Nervous System: Neuromuscular disorders: muscle channelopathies (hypokalemic periodic paralysis)	NE-13.7	USMLE extension
Nervous System: Movement disorders: essential tremor	NE-20-B.13	USMLE extension
Nervous System: Movement disorders: Huntington disease	NE-20-B.5	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Movement disorders: Parkinson disease, including Parkinson dementia	NE-20-B.4	
Nervous System: Movement disorders: acute dystonia	Excluded	See excluded
Nervous System: Movement disorders: adult tic disease	Excluded	See excluded
Nervous System: Metabolic disorders: adrenoleukodystrophy	NE-20-E.2	
Nervous System: Metabolic disorders: metabolic encephalopathy	NE-20-D.1	Osmotic demyelination added in NE-20-D.7
Nervous System: Paroxysmal disorders: headache, including migraine, tension, cluster	NE-5.7	USMLE extension
Nervous System: Paroxysmal disorders: mixed headache, ice-pick headache	Excluded	See excluded
Nervous System: Paroxysmal disorders: medication withdrawal and caffeine withdrawal headache	Excluded	See excluded
Nervous System: Paroxysmal disorders: seizure disorders (generalized tonic-clonic, partial, absence, febrile)	NE-13.6	USMLE extension; EEG patterns in NE-11.5
Nervous System: Sleep disorders: cataplexy and narcolepsy	NE-12.4	
Nervous System: Sleep disorders: circadian rhythm sleep-wake disorder	NE-12.4	
Nervous System: Sleep disorders: insomnia, primary	Excluded	See excluded
Nervous System: Sleep disorders: sleep terror disorder and sleepwalking	NE-12.5	USMLE extension
Nervous System: Sleep disorders: REM sleep behavior disorder	NE-12.5	USMLE extension
Nervous System: Sleep disorders: restless legs syndrome	NE-12.5	USMLE extension
Nervous System: Traumatic and mechanical disorders: anoxic brain damage, cerebral hypoxia	NE-20-A.4	
Nervous System: Traumatic and mechanical disorders: epidural hematoma (cerebral and spinal)	NE-20-A.5	Spinal epidural compression in NE-20-H.5
Nervous System: Traumatic and mechanical disorders: subdural hematoma (cerebral and spinal)	NE-20-A.5	
Nervous System: Traumatic and mechanical disorders: intraparenchymal hemorrhage	NE-20-A.7	
Nervous System: Traumatic and mechanical disorders: traumatic subarachnoid hemorrhage	NE-20-H.2	
Nervous System: Traumatic and mechanical disorders: cerebral edema	NE-20-C.1	
Nervous System: Traumatic and mechanical disorders: idiopathic intracranial hypertension	NE-20-C.4	USMLE extension
Nervous System: Traumatic and mechanical disorders: torticollis/cervical dystonia	NE-20-B.13	USMLE extension
Nervous System: Traumatic and mechanical disorders: hydrocephalus, including normal-pressure	NE-20-C.3	
Nervous System: Traumatic and mechanical disorders: traumatic brain injury (concussion)	NE-20-H.1	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Traumatic and mechanical disorders: postconcussion syndrome	NE-20-H.1	
Nervous System: Traumatic and mechanical disorders: dementia pugilistica	NE-20-H.1	
Nervous System: Traumatic and mechanical disorders: traumatic brain syndrome	NE-20-H.2	
Nervous System: Congenital disorders: Friedreich ataxia	NE-20-B.6	
Nervous System: Congenital disorders: neural tube defects (spina bifida, anencephaly)	NE-1.4	
Nervous System: Congenital disorders: neural tube defects (holoprosencephaly)	NE-1.5	
Nervous System: Congenital disorders: microcephaly	NE-20-E.10	USMLE extension
Nervous System: Congenital disorders: Sturge-Weber syndrome	NE-16.4	USMLE extension
Nervous System: Congenital disorders: tuberous sclerosis	NE-20-F.11	Genetics in NE-16.3
Nervous System: Congenital disorders: von Hippel-Lindau disease	NE-20-F.11	Genetics in NE-16.3
Nervous System: Congenital disorders: hydrocephalus, obstructive (Arnold-Chiari)	NE-1.6	
Nervous System: Adverse effects of drugs on the nervous system: acute dystonic reaction, extrapyramidal effects, tardive dyskinesia	Excluded	See excluded
Nervous System: Adverse effects of drugs on the nervous system: drug-induced meningitis	Excluded	See excluded
Nervous System: Adverse effects of drugs on the nervous system: drug-induced neuropathy	NE-20-D.2	Only as a cause of peripheral neuropathy (neurotoxic chemotherapy)
Nervous System: Adverse effects of drugs on the nervous system: neuroleptic malignant syndrome, serotonin syndrome, psychotropic poisoning	Excluded	See excluded
Nervous System: Disorders of the eye: blepharitis/eyelid inflammation	NE-20-J.1	
Nervous System: Disorders of the eye: chalazion, hordeolum	NE-20-J.1	
Nervous System: Disorders of the eye: chorioretinitis	NE-20-J.7	
Nervous System: Disorders of the eye: conjunctivitis (adenovirus), keratoconjunctivitis	NE-20-J.3	
Nervous System: Disorders of the eye: dacryocystitis	NE-20-J.17	USMLE extension
Nervous System: Disorders of the eye: endophthalmitis	NE-20-J.17	USMLE extension
Nervous System: Disorders of the eye: iridocyclitis	NE-20-J.5	
Nervous System: Disorders of the eye: optic neuritis	NE-20-B.8	As a presentation of multiple sclerosis
Nervous System: Disorders of the eye: periorbital cellulitis	NE-20-J.17	USMLE extension
Nervous System: Disorders of the eye: uveitis	NE-20-J.7	
Nervous System: Disorders of the eye: neoplasms: melanoma, retinoblastoma	NE-20-F.10	USMLE extension; RB1 genetics in NE-16.6
Nervous System: Disorders of the eye: cataract	NE-20-J.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Disorders of the eye: glaucoma	NE-20-J.4	
Nervous System: Disorders of the eye: lacrimal system disorders	NE-20-J.17	Dacryocystitis; normal lacrimal function in NE-6.4
Nervous System: Disorders of the eye: pterygium	NE-20-J.18	USMLE extension
Nervous System: Disorders of the eye: refractive disorders (presbyopia, myopia, hyperopia, astigmatism)	NE-20-J.15	USMLE extension
Nervous System: Disorders of the eye: amblyopia, strabismus	NE-20-J.15	USMLE extension
Nervous System: Disorders of the retina: hypertensive retinopathy	NE-20-J.13	USMLE extension
Nervous System: Disorders of the retina: macular degeneration	NE-20-J.6	
Nervous System: Disorders of the retina: papilledema	NE-20-J.13	Also NE-20-C.1
Nervous System: Disorders of the retina: retinal detachment	NE-20-J.14	USMLE extension
Nervous System: Disorders of the retina: retinitis pigmentosa	NE-20-J.14	USMLE extension
Nervous System: Disorders of the retina: central retinal artery embolus, embolus, carotid artery stenosis	NE-20-J.12	USMLE extension
Nervous System: Disorders of the retina: retinal hemorrhage	NE-20-J.13	
Nervous System: Disorders of the retina: amaurosis fugax	NE-20-J.12	
Nervous System: Disorders of the retina: central retinal vein occlusion	NE-20-J.12	
Nervous System: Disorders of the retina: visual impairment/blindness	NE-6.1	Localization of visual loss
Nervous System: Disorders of the retina: night blindness	NE-6.3	
Nervous System: Traumatic disorders of the eye: corneal abrasion, foreign body, hyphema, orbital fracture, subconjunctival hemorrhage, burn	NE-20-J.19	USMLE extension
Nervous System: Traumatic disorders of the eye: corneal ulcer	NE-20-J.5	
Nervous System: Traumatic disorders of the eye: dislocated lens	NE-20-J.18	
Nervous System: Traumatic disorders of the eye: injury to optic nerve and pathways	NE-6.1	
Nervous System: Traumatic disorders of the eye: black eye, laceration of eye and eyelid, ocular open wounds	Excluded	See excluded
Nervous System: Adverse effects of drugs on the eyes: ethambutol, hydroxychloroquine, prednisone	Excluded	See excluded
Nervous System: Disorders of the ear: chondritis	NE-20-J.10	
Nervous System: Disorders of the ear: mastoiditis	NE-20-J.9	
Nervous System: Disorders of the ear: otitis externa, malignant otitis externa	NE-20-J.10	
Nervous System: Disorders of the ear: otitis media, serous otitis, suppurative otitis	NE-20-J.9	
Nervous System: Disorders of the ear: otitis interna	NE-20-J.11	As labyrinthitis

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Disorders of the ear: acoustic neuroma, neurofibromatosis type 2	NE-20-F.7	
Nervous System: Disorders of the ear: cholesteatoma	NE-20-J.16	USMLE extension
Nervous System: Disorders of the ear: hearing loss, including noise-induced	NE-20-J.16	USMLE extension; Weber and Rinne in NE-6.5
Nervous System: Disorders of the ear: otosclerosis	NE-20-J.16	
Nervous System: Disorders of the ear: tinnitus	NE-20-J.16	
Nervous System: Disorders of the ear: Ménière disease	NE-20-J.8	
Nervous System: Disorders of the ear: motion sickness	NE-6.6	
Nervous System: Disorders of the ear: vertigo, including benign positional vertigo	NE-20-J.11	
Nervous System: Disorders of the ear: barotrauma, perforation of tympanic membrane, eustachian tube disorders, impacted cerumen	NE-20-J.16	
Nervous System: Disorders of the ear: foreign body, laceration, avulsion	Excluded	See excluded
Nervous System: Adverse effects of drugs on the ear: cisplatin, aminoglycosides, furosemide, salicylates	Excluded	See excluded
Multisystem: Storage diseases: Tay-Sachs disease	NE-20-E.4	
Multisystem: Storage diseases: Fabry disease, mucopolysaccharidoses	NE-20-E.5	
Multisystem: Storage diseases: glycogen storage disease	Excluded	See excluded
Multisystem: Inborn errors of metabolism: maple syrup urine disease, urea cycle diseases	NE-20-E.8	USMLE extension; ammonia neurotoxicity in NE-14.5
Multisystem: Enzymatic/metabolic disorders: porphyria	NE-20-E.9	USMLE extension, acute intermittent porphyria only
Multisystem: Vitamin deficiencies: vitamin B1, thiamine (Wernicke-Korsakoff syndrome, beriberi)	NE-20-D.5	Biochemistry in NE-14.2
Multisystem: Vitamin deficiencies: vitamin B12, cobalamins (pernicious anemia)	NE-20-D.4	Biochemistry in NE-14.3
Multisystem: Vitamin deficiencies: vitamin B3, B6, E (neurologic effects)	NE-14.4	
Multisystem: Vitamin deficiencies: vitamin A (night blindness)	NE-6.3	
Multisystem: Genetic disorders: Down syndrome	NE-20-E.1	Chromosomal mechanism in NE-16.2
Multisystem: Genetic disorders: fragile X syndrome	NE-20-E.7	USMLE extension; repeat genetics in NE-16.1
Multisystem: Transport disorders: Wilson disease (neurologic features)	NE-20-E.6	
Multisystem: Paraneoplastic syndromes: myasthenic syndrome (Lambert-Eaton)	NE-18.4	
Multisystem: Paraneoplastic syndromes: paraneoplastic cerebellar degeneration, limbic encephalitis, pure sensory neuropathy, anti-NMDA receptor encephalitis	NE-18.5	
Nervous System: Adverse effects of drugs on the nervous system: acute dystonic reaction, extrapyramidal effects (akathisia, drug-induced parkinsonism), tardive dyskinesia	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I; drug-induced parkinsonism is named only as a cause in NE-20-B.4

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Nervous System: Adverse effects of drugs on the nervous system: drug-induced meningitis (NSAIDs, sulfa drugs)	Excluded	Drug adverse effect; pharmacology is not on NPLEX Part I
Nervous System: Adverse effects of drugs on the nervous system: neuroleptic malignant syndrome, serotonin syndrome, poisoning by psychotropic agents	Excluded	Pharmacology and psychiatric drug toxicity are not on NPLEX Part I
Nervous System: Disorders of the eye: adverse effects of drugs (ethambutol, hydroxychloroquine, prednisone)	Excluded	Drug adverse effect; pharmacology is not on NPLEX Part I (steroid cataract named only as a risk factor in NE-20-J.2)
Nervous System: Disorders of the ear: adverse effects of drugs (cisplatin, aminoglycosides, furosemide, salicylates)	Excluded	Drug ototoxicity is pharmacology, not on NPLEX Part I
Nervous System: Movement disorders: acute dystonia	Excluded	Drug-induced reaction; pharmacology is not on NPLEX Part I
Nervous System: Movement disorders: adult tic disease	Excluded	Tic disorders are classified with behavioral health and neurodevelopmental disorders, outside NPLEX Part I
Nervous System: Degenerative disorders/amnestic syndromes: mild neurocognitive disorder	Excluded	DSM psychiatric diagnosis (behavioral health); the biomedical equivalent, mild cognitive impairment, is in NE-20-B.11
Nervous System: Sleep disorders: insomnia, primary	Excluded	Classified as a behavioral health and sleep-wake disorder treated behaviorally; no biomedical mechanism to test at Part I depth
Nervous System: Paroxysmal disorders: medication withdrawal headache, caffeine withdrawal headache	Excluded	Drug and substance withdrawal effects; pharmacology is not on NPLEX Part I
Nervous System: Paroxysmal disorders: mixed headache, ice-pick headache	Excluded	Clinical headache descriptors with no established mechanism to test at Step 1 depth
Nervous System: Neurologic pain syndromes: fibromyalgia	Excluded	Covered with the musculoskeletal system objectives
Nervous System: Neuromuscular disorders: muscular dystrophy (Duchenne)	Excluded	Covered with the musculoskeletal system objectives; myotonic dystrophy repeat genetics kept in NE-16.1
Nervous System: Disorders of the eye: traumatic (black eye, laceration of eye and eyelid, ocular open wounds)	Excluded	Emergency injury care with no distinct biomedical mechanism beyond the eye injuries in NE-20-J.19
Nervous System: Disorders of the ear: traumatic (foreign body, laceration, avulsion)	Excluded	Emergency injury care, not biomedical science
Multisystem: Storage diseases: glycogen storage disease	Excluded	Hepatic and muscle effects rather than neurologic; belongs with metabolic and musculoskeletal systems

Repro 124 mapped · 69 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Breast: breast abscess	RE-13-E.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Breast: inflammatory disease of breast	RE-13-E.14	Inflammatory carcinoma vs mastitis; periductal mastitis also covered
Female and Transgender Reproductive System & Breast: Breast: fat necrosis	RE-13-E.6	
Female and Transgender Reproductive System & Breast: Breast: mastitis	RE-13-E.4	
Female and Transgender Reproductive System & Breast: Breast: nipple discharge	RE-13-E.9	Also duct ectasia
Female and Transgender Reproductive System & Breast: Breast: breast cyst, solitary	RE-13-E.1	
Female and Transgender Reproductive System & Breast: Breast: fibrocystic changes	RE-13-E.1	
Female and Transgender Reproductive System & Breast: Breast: fibroadenoma	RE-13-E.7	
Female and Transgender Reproductive System & Breast: Breast: hypertrophy of breast	Excluded	Low-yield; no clear biomedical mechanism objective
Female and Transgender Reproductive System & Breast: Breast: intraductal papilloma	RE-13-E.9	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Breast: breast cancer	RE-13-E.10	
Female and Transgender Reproductive System & Breast: Breast: intraductal carcinoma	RE-13-E.11	
Female and Transgender Reproductive System & Breast: Breast: Paget disease of breast	RE-13-E.13	
Female and Transgender Reproductive System & Breast: Breast: phyllodes tumors	RE-13-E.8	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: bacterial vaginosis	RE-13-K.2	
Female and Transgender Reproductive System & Breast: Female: Bartholin gland abscess	RE-13-I.1	
Female and Transgender Reproductive System & Breast: Female: pelvic cellulitis	RE-13-B.7	Covered within ascending pelvic infection
Female and Transgender Reproductive System & Breast: Female: candidiasis of vulva or vagina	RE-13-B.9	
Female and Transgender Reproductive System & Breast: Female: lichen sclerosus	RE-13-B.10	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: sexually transmitted infections and exposure	RE-13-K.4	
Female and Transgender Reproductive System & Breast: Female: cervicitis; endocervicitis	RE-13-B.2	
Female and Transgender Reproductive System & Breast: Female: chancroid (Haemophilus ducreyi)	RE-13-K.1	
Female and Transgender Reproductive System & Breast: Female: genital herpes	RE-13-K.7	
Female and Transgender Reproductive System & Breast: Female: gonorrhea (Neisseria gonorrhoeae)	RE-13-K.6	
Female and Transgender Reproductive System & Breast: Female: human papillomavirus infection; genital warts; condylomata acuminata	RE-13-K.8	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Female: lymphogranuloma venereum (Chlamydia trachomatis)	RE-13-K.5	
Female and Transgender Reproductive System & Breast: Female: non-lymphogranuloma venereum chlamydial infection	RE-13-K.4	
Female and Transgender Reproductive System & Breast: Female: pelvic inflammatory disease	RE-13-B.7	
Female and Transgender Reproductive System & Breast: Female: Fitz-Hugh-Curtis syndrome	RE-13-B.8	
Female and Transgender Reproductive System & Breast: Female: salpingitis; oophoritis	RE-13-B.8	Oophoritis also in tubo-ovarian objective
Female and Transgender Reproductive System & Breast: Female: syphilis (Treponema pallidum)	RE-13-K.10	
Female and Transgender Reproductive System & Breast: Female: trichomoniasis (Trichomonas vaginalis)	RE-13-K.14	
Female and Transgender Reproductive System & Breast: Female: urethritis	RE-13-K.9	
Female and Transgender Reproductive System & Breast: Female: vaginitis; vulvovaginitis	RE-13-K.3	
Female and Transgender Reproductive System & Breast: Female: abnormal cervical cytology	RE-13-J.1	
Female and Transgender Reproductive System & Breast: Female: benign neoplasm of ovary	RE-13-J.11	
Female and Transgender Reproductive System & Breast: Female: endocervical polyps; endometrial polyps	RE-13-J.6	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: leiomyomata uteri	RE-13-J.7	
Female and Transgender Reproductive System & Breast: Female: ovarian cyst	RE-13-G.1	
Female and Transgender Reproductive System & Breast: Female: cervical cancer; HPV-caused cancer	RE-13-J.3	
Female and Transgender Reproductive System & Breast: Female: cervical dysplasia; HPV-caused dysplasia	RE-13-J.1	
Female and Transgender Reproductive System & Breast: Female: endometrial hyperplasia	RE-13-J.4	
Female and Transgender Reproductive System & Breast: Female: endometrial/uterine cancer	RE-13-J.5	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: gestational trophoblastic disease (hydatidiform mole)	RE-13-H.1	
Female and Transgender Reproductive System & Breast: Female: ovarian cancer	RE-13-J.12	
Female and Transgender Reproductive System & Breast: Female: vulvar dysplasia and cancer	RE-13-J.21	
Female and Transgender Reproductive System & Breast: Female: assisted reproductive techniques	Excluded	Clinical management, not biomedical science
Female and Transgender Reproductive System & Breast: Female: contraception (oral contraceptives, IUD, barrier methods, implant, emergency contraception)	Excluded	Contraception counseling and pharmacology are not on NPLEX Part I

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Female: female infertility; infertility	RE-13-A.9	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: Turner syndrome	RE-13-C.11	
Female and Transgender Reproductive System & Breast: Female: sterilization	Excluded	Procedure and contraception counseling, not biomedical science
Female and Transgender Reproductive System & Breast: Female: tubal factors	RE-11.7	
Female and Transgender Reproductive System & Breast: Female: ovarian failure; premature menopause	RE-13-A.6	
Female and Transgender Reproductive System & Breast: Female: perimenopause; vasomotor symptoms; postmenopausal atrophic vaginitis	RE-13-A.5	
Female and Transgender Reproductive System & Breast: Female: premenopausal menorrhagia; postmenopausal bleeding	RE-13-A.4	Postmenopausal bleeding also in endometrial carcinoma objective
Female and Transgender Reproductive System & Breast: Female: abnormal uterine bleeding	RE-13-A.4	
Female and Transgender Reproductive System & Breast: Female: primary amenorrhea; secondary amenorrhea	RE-13-A.1	
Female and Transgender Reproductive System & Breast: Female: undiagnosed pregnancy	RE-13-A.1	
Female and Transgender Reproductive System & Breast: Female: anovulation	RE-13-A.3	
Female and Transgender Reproductive System & Breast: Female: dysmenorrhea	RE-13-A.8	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: endometriosis	RE-13-B.3	
Female and Transgender Reproductive System & Breast: Female: hirsutism; virilization	RE-13-A.7	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: mittelschmerz	RE-13-A.8	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: pelvic pain	Excluded	Symptom-based clinical approach; covered through specific disorders
Female and Transgender Reproductive System & Breast: Female: polycystic ovary syndrome	RE-13-G.4	
Female and Transgender Reproductive System & Breast: Female: postcoital bleeding	RE-13-J.3	
Female and Transgender Reproductive System & Breast: Female: premenstrual syndrome	Excluded	Largely behavioral and clinical; mechanism not established at board level
Female and Transgender Reproductive System & Breast: Female: dyspareunia; orgasmic dysfunction; sexual desire/arousal syndrome; vaginismus	Excluded	Behavioral health and sexual counseling, not on NPLEX Part I
Female and Transgender Reproductive System & Breast: Female: Asherman syndrome	RE-13-A.2	
Female and Transgender Reproductive System & Breast: Female: chronic inversion of uterus	Excluded	Obstetric management topic

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Female: chronic pelvic pain syndrome	Excluded	Symptom-based clinical approach
Female and Transgender Reproductive System & Breast: Female: cystocele; rectocele; urethrocele	RE-13-I.3	
Female and Transgender Reproductive System & Breast: Female: imperforate hymen	RE-13-C.4	
Female and Transgender Reproductive System & Breast: Female: injuries, wounds, and burns (including blast injuries)	Excluded	Trauma management, not biomedical science
Female and Transgender Reproductive System & Breast: Female: ovarian torsion	RE-13-G.2	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: pelvic relaxation	RE-4.3	
Female and Transgender Reproductive System & Breast: Female: mullerian agenesis	RE-13-C.9	Added as USMLE extension
Female and Transgender Reproductive System & Breast: Female: uterus didelphys; bicornuate uterus	RE-1.3	
Female and Transgender Reproductive System & Breast: Female: short cervix	Excluded	Obstetric management (cervical insufficiency)
Female and Transgender Reproductive System & Breast: Female: adverse drug effects (antihistamines, H2 blockers, benzodiazepines, beta blockers, hormone replacement, opioids, spironolactone, SSRIs, TCAs)	Excluded	Pharmacology is not on NPLEX Part I
Male and Transgender Reproductive System: balanitis	RE-13-B.1	
Male and Transgender Reproductive System: epididymitis	RE-13-B.6	
Male and Transgender Reproductive System: orchitis	RE-13-B.6	
Male and Transgender Reproductive System: prostatitis	RE-13-F.3	
Male and Transgender Reproductive System: sexually transmitted infections (chancroid, genital herpes, gonorrhea, HPV, LGV, syphilis, trichomoniasis)	RE-13-K.4	Each organism has its own RE-13-K objective
Male and Transgender Reproductive System: chlamydial and nonchlamydial nongonococcal urethritis	RE-13-K.9	
Male and Transgender Reproductive System: Immunologic: autoimmune hypogonadism	RE-11.1	
Male and Transgender Reproductive System: Neoplasms: penile cancer	RE-13-J.16	
Male and Transgender Reproductive System: Neoplasms: prostate cancer	RE-13-J.9	
Male and Transgender Reproductive System: Neoplasms: testicular cancer	RE-13-J.17	
Male and Transgender Reproductive System: Neoplasms: breast cancer	RE-13-E.15	Added as USMLE extension
Male and Transgender Reproductive System: erectile dysfunction; impotence	RE-13-D.1	
Male and Transgender Reproductive System: male factor infertility	RE-13-D.9	Added as USMLE extension
Male and Transgender Reproductive System: male sexual dysfunction; premature ejaculation	Excluded	Behavioral health and sexual counseling, not on NPLEX Part I
Male and Transgender Reproductive System: priapism	RE-13-D.7	Added as USMLE extension
Male and Transgender Reproductive System: benign prostatic hyperplasia	RE-13-F.1	
Male and Transgender Reproductive System: circumcision, including complications	Excluded	Procedural management

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Male and Transgender Reproductive System: epididymal cyst	RE-13-D.4	
Male and Transgender Reproductive System: hydrocele (including neonatal)	RE-13-D.3	
Male and Transgender Reproductive System: injuries, wounds, and burns to male genitalia; penile, scrotal, and urethral laceration; testicular rupture	Excluded	Trauma management, not biomedical science
Male and Transgender Reproductive System: penile fracture	RE-4.6	
Male and Transgender Reproductive System: Peyronie disease	RE-13-D.8	Added as USMLE extension
Male and Transgender Reproductive System: phimosis	RE-13-C.6	
Male and Transgender Reproductive System: spermatocele	RE-13-D.4	
Male and Transgender Reproductive System: torsion of testis	RE-13-D.6	Added as USMLE extension
Male and Transgender Reproductive System: varicocele	RE-13-D.5	
Male and Transgender Reproductive System: Congenital: hypospadias	RE-13-C.2	
Male and Transgender Reproductive System: Congenital: Klinefelter syndrome	RE-13-C.5	
Male and Transgender Reproductive System: Congenital: undescended testicle	RE-13-C.1	
Male and Transgender Reproductive System: adverse drug effects (alcohol, androgens, antipsychotics, antidepressants, beta blockers, thiazides, finasteride, dutasteride, PDE-5 inhibitors, marijuana)	Excluded	Pharmacology is not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Prenatal care: preconception counseling and care; immunizations; nutritional assessment; Rh screening; genetic screening; surveillance and ultrasonography; biophysical profile; adolescent pregnancy	Excluded	Prenatal care and screening (care of the well patient)
Pregnancy, Childbirth, & the Puerperium: Prenatal care: folate deficiency prevention; neural tube defects	RE-9.1	
Pregnancy, Childbirth, & the Puerperium: Prenatal care: iron deficiency prevention	RE-9.2	
Pregnancy, Childbirth, & the Puerperium: Prenatal care: alpha-fetoprotein	Excluded	Prenatal screening schedule
Pregnancy, Childbirth, & the Puerperium: Prenatal care: diabetes mellitus	RE-9.4	
Pregnancy, Childbirth, & the Puerperium: Prenatal care: Rh isoimmunization	RE-11.4	
Pregnancy, Childbirth, & the Puerperium: Prenatal care: infections (CMV, HSV, HIV, rubella, parvovirus B19, VZV, Toxoplasma, T. pallidum, S. agalactiae, C. trachomatis, hepatitis B)	RE-12.7	
Pregnancy, Childbirth, & the Puerperium: Prenatal care: amnionitis; asymptomatic urinary tract infection	Excluded	Obstetric management; urinary infection covered in the renal system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: abortion (induced, septic, missed, spontaneous, threatened)	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: acute fatty liver of pregnancy; intrahepatic cholestasis of pregnancy	Excluded	Maternal hepatic disorders; obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: anemia of pregnancy	RE-6.4	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: sickle cell disease and thalassemia in pregnancy	Excluded	Hematologic disorders covered in the blood system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: antepartum hemorrhage, third-trimester bleeding	RE-13-H.7	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: cardiomyopathy of pregnancy	Excluded	Covered in the cardiovascular system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: cervical insufficiency	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: maternal congenital abnormalities (bicornuate uterus)	RE-1.3	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: ectopic pregnancy	RE-3.5	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: fetal abnormality affecting management; fetal growth restriction; prolonged pregnancy; maternal mortality; trauma in pregnancy	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: gestational diabetes	RE-6.5	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: multiple gestation	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: placental abnormalities (abruptio placentae, placenta previa)	RE-13-H.7	Added as USMLE extension
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: polyhydramnios, oligohydramnios	RE-13-H.9	Added as USMLE extension
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: preeclampsia, eclampsia, HELLP syndrome, gestational hypertension	RE-13-H.5	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: Rh isoimmunization affecting management of mother	RE-11.4	
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: hyperemesis gravidarum, vomiting in pregnancy	RE-13-H.2	Only as a feature of molar pregnancy; routine management excluded
Pregnancy, Childbirth, & the Puerperium: Labor and delivery: uncomplicated labor	RE-6.7	
Pregnancy, Childbirth, & the Puerperium: Labor and delivery: complicated labor, shoulder dystocia, cesarean delivery, cord compression and prolapse, malpresentation, intrapartum fetal evaluation and prophylaxis, PROM, preterm and postdates labor	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Puerperium: lactation problems; breast-feeding problems	RE-6.6	
Pregnancy, Childbirth, & the Puerperium: Puerperium: lochia; postpartum hemorrhage; uterine atony; retained products	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Puerperium: retained placenta (placenta accreta)	RE-13-H.8	Added as USMLE extension
Pregnancy, Childbirth, & the Puerperium: Puerperium: postpartum sepsis	RE-13-B.5	
Pregnancy, Childbirth, & the Puerperium: Puerperium: postpartum cardiomyopathy	Excluded	Covered in the cardiovascular system
Pregnancy, Childbirth, & the Puerperium: Puerperium: postpartum blues	Excluded	Behavioral health is not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Newborn: normal newborn examination and screening	Excluded	Care of the well patient and screening
Pregnancy, Childbirth, & the Puerperium: Newborn: ABO and Rh hemolytic disease of the newborn	RE-11.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Pregnancy, Childbirth, & the Puerperium: Newborn: congenital or peripartum infections (TORCH, GBS, HSV, HIV, hepatitis B, C. trachomatis)	RE-12.7	
Pregnancy, Childbirth, & the Puerperium: Newborn: neonatal conjunctivitis; ophthalmic gonorrhea	RE-13-K.6	
Pregnancy, Childbirth, & the Puerperium: Newborn: neonatal hypoglycemia; macrosomia	RE-9.4	
Pregnancy, Childbirth, & the Puerperium: Newborn: other neonatal disorders (birth asphyxia, birth trauma, drug withdrawal, feeding problems, GI obstruction, hypocalcemia, jaundice, laryngomalacia, meconium aspiration, neonatal acne, thrush, PKU, prematurity, RDS, BPD, TE fistula, retinopathy of prematurity, seizures, SIDS, tetanus neonatorum)	Excluded	Neonatal disorders belong to other systems or are clinical management
Pregnancy, Childbirth, & the Puerperium: Adverse effects of drugs: alcohol, tobacco, and other drugs; prenatal radiation exposure	Excluded	Pharmacology and toxicology are not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Adverse effects of drugs: teratology (timing of sensitivity)	RE-3.6	Timing only; individual drug teratogens excluded as pharmacology
Pregnancy, Childbirth, & the Puerperium: Systemic disorders affecting pregnancy (appendicitis, asthma, carpal tunnel, cirrhosis, DVT, diabetes, heart disease, hypertension, myasthenia gravis, obesity, pancreatitis, psychiatric, renal, seizure, thyroid disorders)	Excluded	Covered in their own organ systems; obstetric management
Multisystem Processes & Disorders: hereditary nonpolyposis colorectal cancer (female reproductive)	RE-10.5	
Multisystem Processes & Disorders: Genetic, metabolic, and developmental disorders: triplet repeat/RNA disorders: fragile X syndrome	RE-13-C.3	Reproductive features (macroorchidism) covered here
Multisystem Processes & Disorders: Infectious disorders: bacterial: toxic shock syndrome	RE-13-K.13	
Endocrine System: Congenital disorders: disorders of sexual differentiation	RE-13-C.7	Also androgen insensitivity in RE-13-C.8
Musculoskeletal System: Neoplasms, malignant: leiomyosarcoma	RE-13-J.8	Uterine leiomyosarcoma
Female and Transgender Reproductive System & Breast: Neoplasms of the vulva, vagina, cervix, uterus, or ovaries (vaginal)	RE-13-J.20	
Female and Transgender Reproductive System & Breast: Breast: hypertrophy of breast	Excluded	Low-yield; no clear biomedical mechanism objective
Female and Transgender Reproductive System & Breast: Female: assisted reproductive techniques	Excluded	Clinical management, not biomedical science
Female and Transgender Reproductive System & Breast: Female: contraception (oral contraceptives, IUD, barrier methods, implant, emergency contraception)	Excluded	Contraception counseling and pharmacology are not on NPLEX Part I
Female and Transgender Reproductive System & Breast: Female: sterilization	Excluded	Procedure and contraception counseling, not biomedical science
Female and Transgender Reproductive System & Breast: Female: pelvic pain	Excluded	Symptom-based clinical approach; covered through specific disorders

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Female and Transgender Reproductive System & Breast: Female: premenstrual syndrome	Excluded	Largely behavioral and clinical; mechanism not established at board level
Female and Transgender Reproductive System & Breast: Female: dyspareunia; orgasmic dysfunction; sexual desire/arousal syndrome; vaginismus	Excluded	Behavioral health and sexual counseling, not on NPLEX Part I
Female and Transgender Reproductive System & Breast: Female: chronic inversion of uterus	Excluded	Obstetric management topic
Female and Transgender Reproductive System & Breast: Female: chronic pelvic pain syndrome	Excluded	Symptom-based clinical approach
Female and Transgender Reproductive System & Breast: Female: injuries, wounds, and burns (including blast injuries)	Excluded	Trauma management, not biomedical science
Female and Transgender Reproductive System & Breast: Female: short cervix	Excluded	Obstetric management (cervical insufficiency)
Female and Transgender Reproductive System & Breast: Female: adverse drug effects (antihistamines, H2 blockers, benzodiazepines, beta blockers, hormone replacement, opioids, spironolactone, SSRIs, TCAs)	Excluded	Pharmacology is not on NPLEX Part I
Male and Transgender Reproductive System: male sexual dysfunction; premature ejaculation	Excluded	Behavioral health and sexual counseling, not on NPLEX Part I
Male and Transgender Reproductive System: circumcision, including complications	Excluded	Procedural management
Male and Transgender Reproductive System: injuries, wounds, and burns to male genitalia; penile, scrotal, and urethral laceration; testicular rupture	Excluded	Trauma management, not biomedical science
Male and Transgender Reproductive System: adverse drug effects (alcohol, androgens, antipsychotics, antidepressants, beta blockers, thiazides, finasteride, dutasteride, PDE-5 inhibitors, marijuana)	Excluded	Pharmacology is not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Prenatal care: preconception counseling and care; immunizations; nutritional assessment; Rh screening; genetic screening; surveillance and ultrasonography; biophysical profile; adolescent pregnancy	Excluded	Prenatal care and screening (care of the well patient)
Pregnancy, Childbirth, & the Puerperium: Prenatal care: alpha-fetoprotein	Excluded	Prenatal screening schedule
Pregnancy, Childbirth, & the Puerperium: Prenatal care: amnionitis; asymptomatic urinary tract infection	Excluded	Obstetric management; urinary infection covered in the renal system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: abortion (induced, septic, missed, spontaneous, threatened)	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: acute fatty liver of pregnancy; intrahepatic cholestasis of pregnancy	Excluded	Maternal hepatic disorders; obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: sickle cell disease and thalassemia in pregnancy	Excluded	Hematologic disorders covered in the blood system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: cardiomyopathy of pregnancy	Excluded	Covered in the cardiovascular system
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: cervical insufficiency	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: fetal abnormality affecting management; fetal growth restriction; prolonged pregnancy; maternal mortality; trauma in pregnancy	Excluded	Obstetric management

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Pregnancy, Childbirth, & the Puerperium: Obstetric complications: multiple gestation	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Labor and delivery: complicated labor, shoulder dystocia, cesarean delivery, cord compression and prolapse, malpresentation, intrapartum fetal evaluation and prophylaxis, PROM, preterm and postdates labor	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Puerperium: lochia; postpartum hemorrhage; uterine atony; retained products	Excluded	Obstetric management
Pregnancy, Childbirth, & the Puerperium: Puerperium: postpartum cardiomyopathy	Excluded	Covered in the cardiovascular system
Pregnancy, Childbirth, & the Puerperium: Puerperium: postpartum blues	Excluded	Behavioral health is not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Newborn: normal newborn examination and screening	Excluded	Care of the well patient and screening
Pregnancy, Childbirth, & the Puerperium: Newborn: other neonatal disorders (birth asphyxia, birth trauma, drug withdrawal, feeding problems, GI obstruction, hypocalcemia, jaundice, laryngomalacia, meconium aspiration, neonatal acne, thrush, PKU, prematurity, RDS, BPD, TE fistula, retinopathy of prematurity, seizures, SIDS, tetanus neonatorum)	Excluded	Neonatal disorders belong to other systems or are clinical management
Pregnancy, Childbirth, & the Puerperium: Adverse effects of drugs: alcohol, tobacco, and other drugs; prenatal radiation exposure	Excluded	Pharmacology and toxicology are not on NPLEX Part I
Pregnancy, Childbirth, & the Puerperium: Systemic disorders affecting pregnancy (appendicitis, asthma, carpal tunnel, cirrhosis, DVT, diabetes, heart disease, hypertension, myasthenia gravis, obesity, pancreatitis, psychiatric, renal, seizure, thyroid disorders)	Excluded	Covered in their own organ systems; obstetric management
Female and Transgender Reproductive System & Breast and Male and Transgender Reproductive System: gender-affirming hormone and surgical therapy	Excluded	Gender-affirming therapy is clinical management and pharmacology, not on NPLEX Part I

Urology 83 mapped · 8 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Renal & Urinary System: Infectious disorders: upper urinary tract: pyelonephritis	UR-12-H.1	
Renal & Urinary System: Infectious disorders: upper urinary tract: perinephric and renal abscess	UR-12-H.1	
Renal & Urinary System: Infectious disorders: upper urinary tract: pyonephrosis	UR-12-H.1	
Renal & Urinary System: Infectious disorders: upper urinary tract: granulomatous pyelonephritis	UR-12-D.4	Xanthogranulomatous pyelonephritis
Renal & Urinary System: Infectious disorders: upper urinary tract: renal tuberculosis	UR-11.7	
Renal & Urinary System: Infectious disorders: lower urinary tract: cystitis	UR-12-H.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Renal & Urinary System: Infectious disorders: lower urinary tract: chlamydial and nonchlamydial infection	UR-12-H.3	Organisms in UR-11.5
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: Alport syndrome	UR-12-E.1	Genetics in UR-9.2
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: hepatitis B/C-related glomerular disease	UR-12-A.7	Also UR-11.6
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: glomerulonephritis	UR-12-A.5	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: poststreptococcal glomerulonephritis	UR-12-A.3	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: IgA nephropathy	UR-12-A.4	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: lupus nephritis	UR-12-A.6	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: minimal change disease	UR-12-A.11	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: nephrotic syndrome	UR-12-A.10	
Renal & Urinary System: Immunologic and inflammatory disorders: glomerular: thin basement membrane disease	UR-9.2	
Renal & Urinary System: Immunologic and inflammatory disorders: tubulointerstitial: acute tubular necrosis	UR-12-B.1	
Renal & Urinary System: Immunologic and inflammatory disorders: tubulointerstitial: acute interstitial nephritis	UR-12-D.1	
Renal & Urinary System: Immunologic and inflammatory disorders: tubulointerstitial: papillary necrosis	UR-12-D.2	Also UR-5.5, UR-12-F.5
Renal & Urinary System: Immunologic and inflammatory disorders: tubulointerstitial: HIV nephropathy	UR-12-A.9	
Renal & Urinary System: Immunologic and inflammatory disorders: lower urinary tract: interstitial cystitis	UR-12-D.5	Added as USMLE extension
Renal & Urinary System: Neoplasms: benign neoplasms and cysts: polycystic kidney disease	UR-12-E.2	Genetics in UR-9.1
Renal & Urinary System: Neoplasms: malignant: Wilms tumor	UR-12-G.4	
Renal & Urinary System: Neoplasms: malignant: renal cell carcinoma	UR-12-G.1	
Renal & Urinary System: Neoplasms: malignant: tumors linked to hereditary conditions	UR-9.3	Also UR-9.4, UR-9.5
Renal & Urinary System: Neoplasms: malignant: urinary bladder and collecting system	UR-12-G.5	Added as USMLE extension; also UR-12-G.6
Renal & Urinary System: Abnormalities of urine/urination: dysuria	UR-12-H.2	
Renal & Urinary System: Abnormalities of urine/urination: hematuria	UR-12-A.2	
Renal & Urinary System: Abnormalities of urine/urination: oliguria	UR-12-A.1	Also UR-12-B.2

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Renal & Urinary System: Abnormalities of urine/urination: anuria	UR-12-C.2	Bilateral obstruction; also UR-12-A.14
Renal & Urinary System: Abnormalities of urine/urination: proteinuria	UR-8.2	
Renal & Urinary System: Metabolic and regulatory disorders: acute kidney injury	UR-12-A.14	
Renal & Urinary System: Metabolic and regulatory disorders: renal insufficiency	UR-12-A.15	
Renal & Urinary System: Metabolic and regulatory disorders: azotemia/uremic syndrome	UR-12-A.15	
Renal & Urinary System: Metabolic and regulatory disorders: chronic kidney disease, end-stage renal disease	UR-12-A.15	Also UR-8.3, UR-8.5
Renal & Urinary System: Metabolic and regulatory disorders: cystinuria	UR-9.6	Added as USMLE extension
Renal & Urinary System: Metabolic and regulatory disorders: Fanconi syndrome	UR-12-B.4	Added as USMLE extension
Renal & Urinary System: Metabolic and regulatory disorders: hypertensive renal disease	UR-12-F.2	Also UR-7.7
Renal & Urinary System: Metabolic and regulatory disorders: renal and ureteral calculi	UR-12-C.3	
Renal & Urinary System: Metabolic and regulatory disorders: renal tubular acidosis	UR-7.4	
Renal & Urinary System: Vascular disorders: renal artery stenosis, atherosclerotic	UR-12-F.3	
Renal & Urinary System: Vascular disorders: renal artery stenosis, fibromuscular dysplasia	UR-12-F.3	
Renal & Urinary System: Vascular disorders: nephrosclerosis	UR-12-F.2	
Renal & Urinary System: Vascular disorders: renal vein thrombosis	UR-12-F.6	Added as USMLE extension
Renal & Urinary System: Vascular disorders: renal infarction	UR-12-F.4	
Renal & Urinary System: Traumatic and mechanical disorders: bladder rupture	UR-4.1	Added as USMLE extension
Renal & Urinary System: Traumatic and mechanical disorders: neurogenic bladder	UR-6.7	
Renal & Urinary System: Traumatic and mechanical disorders: obstructive uropathy	UR-12-C.1	
Renal & Urinary System: Traumatic and mechanical disorders: posterior urethral valves	UR-1.6	Added as USMLE extension
Renal & Urinary System: Traumatic and mechanical disorders: renal laceration and vascular injury	Excluded	
Renal & Urinary System: Traumatic and mechanical disorders: ureteral injury	UR-4.5	Added as USMLE extension
Renal & Urinary System: Traumatic and mechanical disorders: urethral laceration/disruption	UR-3.6	Added as USMLE extension
Renal & Urinary System: Traumatic and mechanical disorders: urethral diverticulum	Excluded	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Renal & Urinary System: Traumatic and mechanical disorders: urethral, ureteral, vaginal, uterine obstruction or prolapse	UR-4.6	Cystocele named; other prolapse types not written
Renal & Urinary System: Traumatic and mechanical disorders: urinary incontinence	UR-4.6	Added as USMLE extension
Renal & Urinary System: Traumatic and mechanical disorders: secondary enuresis	Excluded	
Renal & Urinary System: Traumatic and mechanical disorders: vesicoureteral reflux	UR-12-E.5	
Renal & Urinary System: Congenital disorders: duplex ureters	UR-1.4	Added as USMLE extension
Renal & Urinary System: Congenital disorders: horseshoe kidney	UR-1.3	Added as USMLE extension
Renal & Urinary System: Congenital disorders: hydronephrosis/reflux	UR-12-C.1	Also UR-12-E.5
Renal & Urinary System: Congenital disorders: renal agenesis	UR-12-E.4	
Renal & Urinary System: Congenital disorders: renal hypoplasia/dysplasia	UR-12-E.3	Multicystic dysplastic kidney; hypoplasia not written separately
Renal & Urinary System: Congenital disorders: single kidney	UR-12-E.4	
Renal & Urinary System: Adverse effects of drugs: aminoglycosides, cisplatin, iodinated contrast	UR-12-B.3	Only as causes of tubular necrosis
Renal & Urinary System: Adverse effects of drugs: NSAIDs, penicillins, sulfa drugs	UR-12-D.1	Only as causes of drug-induced nephritis; NSAIDs also UR-12-D.2
Renal & Urinary System: Adverse effects of drugs: heroin	UR-12-A.9	Only as a cause of FSGS
Renal & Urinary System: Adverse effects of drugs: ACE inhibitors, amphotericin B, furosemide, gadolinium, lithium, tenofovir, drug-induced urinary retention	Excluded	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: dehydration	UR-6.6	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hypovolemia	UR-6.6	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: volume overload	UR-6.6	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hyponatremia, hypernatremia	UR-6.5	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hypokalemia, hyperkalemia	UR-8.4	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hypocalcemia, hypercalcemia	UR-8.3	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hypophosphatemia, hyperphosphatemia	UR-8.3	
Multisystem Processes & Disorders: Fluid volume and electrolyte disorders: hypomagnesemia	UR-8.3	
Multisystem Processes & Disorders: Acid-base disorders: metabolic acidosis	UR-7.3	
Multisystem Processes & Disorders: Acid-base disorders: metabolic alkalosis	UR-7.5	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem Processes & Disorders: Acid-base disorders: respiratory acidosis, respiratory alkalosis	UR-7.2	
Multisystem Processes & Disorders: Acid-base disorders: mixed disorders	UR-7.2	
Multisystem Processes & Disorders: Immunologic and inflammatory disorders: Goodpasture syndrome	UR-10.1	
Multisystem Processes & Disorders: Immunologic and inflammatory disorders: IgA vasculitis (renal involvement)	UR-10.4	
Multisystem Processes & Disorders: Immunologic and inflammatory disorders: polyangiitis (renal involvement)	UR-10.3	
Multisystem Processes & Disorders: Immunologic and inflammatory disorders: polyarteritis nodosa (renal involvement)	UR-10.3	
Multisystem Processes & Disorders: Paraneoplastic syndromes: membranous glomerulonephritis	UR-12-A.12	
Multisystem Processes & Disorders: Other chemical agents: ethylene glycol	UR-8.6	Also UR-12-B.3
Multisystem Processes & Disorders: Enzymatic/metabolic: Fabry disease (renal involvement)	UR-9.7	Added as USMLE extension
Multisystem Processes & Disorders: Structural protein disorders: amyloidosis (renal involvement)	UR-12-A.13	
Renal & Urinary System: Adverse effects of drugs: ACE inhibitors, amphotericin B, furosemide, gadolinium (nephrogenic systemic fibrosis), lithium, tenofovir, drug-induced urinary retention	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I; these are not causes of a listed NPLEX urinary condition
Renal & Urinary System: Traumatic and mechanical disorders: renal laceration and vascular injury	Excluded	Clinical trauma grading and management; the anatomy that predicts injury sites is covered in UR-3 and UR-4
Renal & Urinary System: Traumatic and mechanical disorders: urethral diverticulum	Excluded	Low-yield clinical diagnosis not named on the NPLEX blueprint; left out rather than written without a firm standard-text basis
Renal & Urinary System: Traumatic and mechanical disorders: secondary enuresis	Excluded	Pediatric behavioral and clinical management topic outside Part I biomedical science

Endo 83 mapped · 7 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Endocrine: Diabetes mellitus: Type 1 diabetes mellitus	EN-12-D.1	
Endocrine: Diabetes mellitus: Type 1.5 diabetes mellitus	EN-10.3	LADA covered with type 1 autoimmunity
Endocrine: Diabetes mellitus: Type 2 diabetes mellitus	EN-12-D.1	
Endocrine: Diabetes mellitus: Acute complications: hyperosmolar coma	EN-12-D.3	
Endocrine: Diabetes mellitus: Acute complications: hypoglycemic shock	EN-12-G.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Endocrine: Diabetes mellitus: Acute complications: ketoacidosis	EN-12-D.3	Biochemistry of ketogenesis in EN-7.2
Endocrine: Diabetes mellitus: Acute complications: cerebral edema	EN-12-D.3	Treatment-related cerebral edema not written as a separate objective
Endocrine: Diabetes mellitus: Acute complications: electrolyte abnormalities	EN-12-D.3	
Endocrine: Diabetes mellitus: Chronic complications: gastroparesis	EN-12-D.5	
Endocrine: Diabetes mellitus: Chronic complications: neuropathy	EN-12-D.4	
Endocrine: Diabetes mellitus: Chronic complications: retinopathy	EN-12-D.5	
Endocrine: Diabetes mellitus: Chronic complications: peripheral vascular disease	EN-12-D.5	
Endocrine: Diabetes mellitus: Chronic complications: nephropathy	EN-12-D.5	
Endocrine: Diabetes mellitus: Metabolic syndrome	EN-12-D.6	Added as USMLE extension
Endocrine: Hypoglycemia and islet cell disorders: Hypoglycemia (insulinoma, surreptitious insulin use, sepsis, liver failure)	EN-12-G.4	
Endocrine: Hypoglycemia and islet cell disorders: Hyperglycemia secondary to glucagonoma	EN-12-G.5	Added as USMLE extension
Endocrine: Hypoglycemia and islet cell disorders: Hyperinsulinism	EN-12-G.3	
Endocrine: Hypoglycemia and islet cell disorders: Islet cell tumors, insulinoma	EN-12-G.3	
Endocrine: Hypoglycemia and islet cell disorders: somatostatinoma	EN-12-G.5	Added as USMLE extension
Endocrine: Hypoglycemia and islet cell disorders: Pancreatic neuroendocrine tumors	EN-12-G.5	Added as USMLE extension
Endocrine: Thyroid disorders: Cyst, nodule	EN-12-G.13	Thyroglossal duct cyst in EN-12-E.1
Endocrine: Thyroid disorders: Nonthyroidal illness syndrome (euthyroid sick syndrome)	EN-12-B.13	Added as USMLE extension
Endocrine: Thyroid disorders: Goiter	EN-12-B.10	Euthyroid goiter in EN-12-G.13; toxic multinodular goiter in EN-12-A.9
Endocrine: Thyroid disorders: Hypothyroidism	EN-12-B.10	
Endocrine: Thyroid disorders: Hyperthyroidism, thyrotoxicosis	EN-12-A.9	
Endocrine: Thyroid disorders: Thyroid storm	EN-12-A.10	
Endocrine: Thyroid disorders: Thyroiditis, including Hashimoto	EN-12-C.1	Painless and Riedel thyroiditis added in EN-12-C.5
Endocrine: Thyroid disorders: Graves disease	EN-12-A.9	Immunology in EN-10.1
Endocrine: Thyroid disorders: Neoplasms: benign cysts and nodules	EN-12-G.13	
Endocrine: Thyroid disorders: Neoplasms: papillary thyroid cancer	EN-12-G.11	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Endocrine: Thyroid disorders: Neoplasms: follicular thyroid cancer	EN-12-G.10	
Endocrine: Thyroid disorders: Neoplasms: medullary thyroid cancer	EN-12-G.12	
Endocrine: Thyroid disorders: Neoplasms: anaplastic thyroid cancer	EN-12-G.14	Added as USMLE extension
Endocrine: Thyroid disorders: Thyroid deficiency from pituitary disorder	EN-12-B.11	
Endocrine: Thyroid disorders: Infertility due to thyroid disease	EN-12-B.11	
Endocrine: Thyroid disorders: Secondary hypothyroidism and hyperthyroidism	EN-12-B.11	
Endocrine: Parathyroid disorders: Hyperparathyroidism	EN-12-A.5	Secondary in EN-12-A.6; FHH added in EN-12-A.12
Endocrine: Parathyroid disorders: Hypoparathyroidism	EN-12-B.6	
Endocrine: Parathyroid disorders: Metabolic bone disease	EN-12-A.6	Vitamin D deficiency bone disease in EN-7.4
Endocrine: Adrenal disorders: Corticoadrenal insufficiency (Addison disease)	EN-12-B.3	
Endocrine: Adrenal disorders: Adrenal insufficiency, secondary	EN-12-B.5	
Endocrine: Adrenal disorders: Hypocortisolism	EN-12-B.4	
Endocrine: Adrenal disorders: Cushing syndrome	EN-12-A.1	
Endocrine: Adrenal disorders: Hyperaldosteronism	EN-12-A.3	
Endocrine: Adrenal disorders: Neoplasms: adrenal neuroblastoma	EN-12-G.2	
Endocrine: Adrenal disorders: Neoplasms: pheochromocytoma	EN-12-G.16	
Endocrine: Adrenal disorders: Neoplasms: adrenal carcinoma, adrenal adenoma, aldosteronoma, adrenal incidentaloma	EN-12-G.1	
Endocrine: Adrenal disorders: Delayed and precocious puberty	EN-12-A.13	Precocious puberty added as extension; delayed puberty causes covered by hypogonadism objective EN-9.6
Endocrine: Adrenal disorders: Hypertensive endocrine disease	EN-12-A.3	Also EN-8.6 (apparent mineralocorticoid excess), EN-12-G.16 (pheochromocytoma)
Endocrine: Pituitary disorders: Acromegaly/gigantism	EN-12-A.7	
Endocrine: Pituitary disorders: Diabetes insipidus	EN-12-B.1	
Endocrine: Pituitary disorders: Galactorrhea not associated with childbirth	EN-12-G.7	
Endocrine: Pituitary disorders: Panhypopituitarism	EN-12-B.8	
Endocrine: Pituitary disorders: Pituitary apoplexy (Sheehan syndrome)	EN-12-F.3	Sheehan in EN-12-F.1
Endocrine: Pituitary disorders: Growth hormone deficiency	EN-12-B.14	Added as USMLE extension

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Endocrine: Pituitary disorders: Short stature	EN-12-B.14	Added as USMLE extension
Endocrine: Pituitary disorders: SIADH	EN-12-A.11	Added as USMLE extension
Endocrine: Pituitary disorders: Neoplasms: pituitary adenomas	EN-12-G.8	
Endocrine: Pituitary disorders: Neoplasms: craniopharyngioma	EN-12-G.9	
Endocrine: Pituitary disorders: Neoplasms: metastatic disease	Excluded	
Endocrine: Pituitary disorders: Prolactinoma and hyperprolactinemia	EN-12-G.7	Physiology in EN-6.7
Endocrine: Pituitary disorders: Hypogonadism, primary and secondary	EN-9.6	Added as USMLE extension
Endocrine: Hypothalamic endocrine disorders	EN-12-B.9	
Endocrine: Multiple endocrine neoplasia (MEN1, MEN2)	EN-12-G.15	Genetics in EN-9.1
Endocrine: Congenital disorders: Disorders of sexual differentiation	EN-9.5	Added as USMLE extension
Endocrine: Congenital disorders: Congenital adrenal hyperplasia	EN-12-A.4	Enzymology EN-8.2, genetics EN-9.2
Endocrine: Congenital disorders: Androgen insensitivity/resistance syndrome	EN-9.5	Added as USMLE extension
Endocrine: Congenital disorders: Congenital hypothyroidism	EN-12-B.12	Added as USMLE extension
Endocrine: Adverse effects of drugs: Drug, medicinal, and biologic substance effects	Excluded	
Endocrine: Adverse effects of drugs: Exogenous steroid suppression of adrenal glands	EN-12-B.5	Kept only as a cause of secondary adrenal insufficiency, a listed NPLEX condition
Endocrine: Adverse effects of drugs: Anabolic steroids	Excluded	
Multisystem: Immunologic and inflammatory disorders: polyglandular autoimmune syndrome, type 1	EN-10.4	
Multisystem: Neoplasms: Paraneoplastic syndromes: Endocrine (SIADH, Cushing syndrome, hypercalcemia of malignancy, PTH-related protein)	EN-12-G.17	Added as USMLE extension
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hyponatremia	EN-12-A.11	Endocrine causes only; also Addison EN-12-B.3
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypernatremia	EN-12-B.1	Endocrine causes only
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypokalemia	EN-12-A.3	Endocrine causes only
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hyperkalemia	EN-12-B.3	Endocrine causes only
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypocalcemia	EN-12-B.6	PTH physiology EN-5.3

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypercalcemia	EN-12-A.5	PTH physiology EN-5.3
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypophosphatemia	EN-12-A.5	As it relates to PTH
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hyperphosphatemia	EN-12-B.6	As it relates to PTH; also CKD in EN-12-A.6
Multisystem: Fluid, electrolyte, and acid-base balance disorders: hypomagnesemia	EN-7.5	
Multisystem: Nutrition: vitamin D (rickets)	EN-7.4	
Multisystem: Nutrition: Obesity	EN-7.6	
Multisystem: Shock: Meningococcemia	EN-12-H.2	Organism in EN-11.1; vaccine immunology in EN-10.6
Multisystem: Adverse effects of drugs: drugs inducing SIADH or diabetes insipidus	EN-12-B.1	Lithium named only as a cause of nephrogenic diabetes insipidus
Endocrine: Adverse effects of drugs on the endocrine system: Drug, medicinal, and biologic substance effects	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I
Endocrine: Adverse effects of drugs on the endocrine system: Anabolic steroids	Excluded	Drug adverse effect; pharmacology is not on NPLEX Part I
Endocrine: Pituitary disorders: Neoplasms: metastatic disease	Excluded	Low yield; no standard Step 1 review fact specific enough to write an accurate objective beyond the mass effects already covered in EN-12-G.8
Multisystem: Adverse effects of drugs: drug-induced electrolyte and acid-base disorders (albuterol, prednisone, diuretics, alcohol)	Excluded	Pharmacology is not on NPLEX Part I

Immuno 52 mapped · 26 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: common variable immunodeficiency	IM-15-A.1	
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: hyper-IgM syndrome	IM-12.4	Also IM-5.3
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: X-linked (Bruton) agammaglobulinemia	IM-15-A.5	
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: selective IgA deficiency	IM-15-A.3	
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: selective IgM and IgE deficiencies	Excluded	Rare; see excluded
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: adenosine deaminase deficiency	IM-10.3	Also IM-12.2
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: DiGeorge syndrome	IM-15-A.2	Embryology in IM-1.3
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: severe combined immunodeficiency	IM-15-A.4	Genetics in IM-12.1 and IM-12.2

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: Wiskott-Aldrich syndrome	IM-12.3	Added in immunogenetics unit
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: granulomatous disease	IM-10.2	Chronic granulomatous disease; also IM-10.1
Immune System: Disorders associated with immunodeficiency: primarily cell-mediated deficiency: allergic reactions and skin manifestations	IM-12.3	Eczema of Wiskott-Aldrich
Immune System: Disorders associated with immunodeficiency: complement deficiencies: alternative pathway components	IM-7.4	
Immune System: Disorders associated with immunodeficiency: complement deficiencies: classical pathway components (C1q, C1r, C1 to C5)	IM-7.4	
Immune System: Disorders associated with immunodeficiency: complement deficiencies: terminal components (C5b to C9, membrane attack complex)	IM-7.4	
Immune System: Disorders associated with immunodeficiency: complement deficiencies: C1 esterase inhibitor deficiency (hereditary angioedema)	IM-7.3	
Immune System: Disorders associated with immunodeficiency: complement deficiencies: mannose-binding lectin deficiency	IM-7.4	
Immune System: Disorders associated with immunodeficiency: phagocyte and NK cell deficiencies: Chediak-Higashi disease	IM-6.4	
Immune System: Disorders associated with immunodeficiency: phagocyte and NK cell deficiencies: chronic granulomatous disease and other phagocytic disorders	IM-10.2	
Immune System: Disorders associated with immunodeficiency: phagocyte and NK cell deficiencies: leukocyte adhesion deficiency	IM-6.2	
Immune System: HIV/AIDS: HIV-1 and HIV-2	IM-15-B.2	
Immune System: HIV/AIDS: AIDS complications (neuropathy, dementia, renal insufficiency)	IM-15-B.6	
Immune System: HIV/AIDS: immunology of AIDS	IM-15-B.3	
Immune System: HIV/AIDS: immune reconstitution syndrome	IM-15-B.6	
Immune System: HIV/AIDS: secondary infections	IM-15-B.5	
Immune System: HIV/AIDS: noninfectious complications (Kaposi sarcoma)	IM-15-B.6	
Immune System: Immunologically mediated disorders: hypersensitivity: types 1, 2, and 3 including anaphylaxis	IM-13.6	Disease detail in IM-15-C.1 to IM-15-C.10
Immune System: Immunologically mediated disorders: hypersensitivity: type 4	IM-15-C.11	
Immune System: Immunologically mediated disorders: hypersensitivity: drug reactions	IM-15-C.15	Added as USMLE extension
Immune System: Immunologically mediated disorders: hypersensitivity: serum sickness	IM-15-C.14	Added as USMLE extension
Immune System: Immunologically mediated disorders: transplantation: rejection	IM-15-C.13	HLA basis in IM-3.3 and IM-3.6
Immune System: Immunologically mediated disorders: transplantation: graft-versus-host disease	IM-15-C.13	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Immune System: Adverse effects of drugs: Jarisch-Herxheimer reaction	Excluded	Pharmacology
Immune System: Adverse effects of drugs on the immune system: immunosuppressive and immune-modulating drugs	IM-15-B.1	Only as a cause of immunodeficiency; drug pharmacology excluded
Immune System: Adverse effects of drugs: monoclonal antibodies	IM-15-B.1	Rituximab and TNF inhibitors as causes of immunodeficiency only
Immune System: Adverse effects of drugs: vaccine adverse effects	Excluded	Pharmacology and preventive care
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: acute rheumatic fever	IM-15-D.3	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: Goodpasture syndrome	IM-15-C.4	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: polyarteritis nodosa	IM-15-C.8	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: systemic lupus erythematosus	IM-15-C.6	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: scleroderma (systemic sclerosis)	IM-15-D.1	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: Sjogren syndrome	IM-15-D.5	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: sarcoidosis, Lofgren syndrome	IM-15-C.12	Also IM-10.5; NPLEX also lists sarcoidosis under PU-14-A4
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: polyglandular autoimmune syndrome, type 1	IM-12.5	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: mixed connective tissue disease	IM-15-D.7	Added as USMLE extension
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: familial Mediterranean fever	IM-15-E.3	As a cause of AA amyloidosis
Multisystem Processes & Disorders: Genetic, metabolic, and developmental disorders: amyloidosis	IM-15-E.1	Types in IM-15-E.2 to IM-15-E.4
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: autoimmune arteritis and vasculitis (granulomatosis with polyangiitis, Churg-Strauss, polyangiitis, IgA vasculitis, Kawasaki, Takayasu, temporal arteritis, Behcet)	Excluded	NPLEX places vasculitis in CV-18-I
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: Raynaud disease and syndrome	Excluded	NPLEX places in CV-18-I8

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: polymyalgia rheumatica	Excluded	NPLEX places in MS-13-B6
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: reactive arthritis	Excluded	NPLEX places in MS-13-B7
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: eosinophilic granuloma, Langerhans cell histiocytosis	Excluded	Hematopoietic neoplasm; not on the IM blueprint
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: viral infections: infectious mononucleosis (Epstein-Barr virus)	IM-15-F.5	
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: viral infections: cytomegalovirus infection	IM-15-F.13	Added as USMLE extension
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: viral infections: human herpesvirus 8	IM-15-B.6	Kaposi sarcoma
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: viral infections: yellow fever	Excluded	NPLEX places viral hemorrhagic fevers in CV-18-K5
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial infections: toxic shock syndrome	IM-6.7	Superantigen immunology; NPLEX lists the condition in RE-13-K9
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial infections: Lyme disease, Rocky Mountain spotted fever	Excluded	NPLEX places in CV-18-K3 and CV-18-K4
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial infections: miliary tuberculosis	Excluded	NPLEX places tuberculosis in PU-14-H9; granuloma immunology in IM-15-C.11
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial infections: brucellosis, leptospirosis, melioidosis, tularemia, Q fever, anaplasmosis and ehrlichiosis	Excluded	Not on the NPLEX blueprint for this system
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: fungal infections (blastomycosis, candidiasis, coccidioidomycosis, histoplasmosis)	Excluded	Not in IM blueprint; candidiasis in IN-13-H2
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: parasitic infections (schistosomiasis, visceral leishmaniasis, Chagas disease)	Excluded	Schistosomiasis in HE-9-E3, Chagas in CV-18-K2; leishmaniasis not on NPLEX blueprint
Multisystem Processes & Disorders: Shock: septic shock	IM-6.6	
Multisystem Processes & Disorders: Shock: sepsis and bacteremia	IM-6.6	
Multisystem Processes & Disorders: Shock: systemic inflammatory response syndrome	IM-6.6	
Multisystem Processes & Disorders: Shock: multiorgan dysfunction syndrome	IM-6.6	Immunologic mechanism only
Multisystem Processes & Disorders: Shock: meningococcemia	IM-7.4	Terminal complement deficiency link
Multisystem Processes & Disorders: Shock: cardiogenic, hypovolemic, neurogenic, and refractory shock	Excluded	Hemodynamics; NPLEX places shock in CV-18-H5
Immune System: Adverse effects of drugs on the immune system: Jarisch-Herxheimer reaction	Excluded	Pharmacology (reaction to antimicrobial treatment) is not on NPLEX Part I

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Immune System: Adverse effects of drugs on the immune system: immunosuppressive and immune-modulating drugs (mechanisms, dosing, adverse effects)	Excluded	Pharmacology is not on NPLEX Part I; drugs appear only as causes of immunodeficiency in IM-15-B.1
Immune System: Adverse effects of drugs on the immune system: vaccine adverse effects	Excluded	Pharmacology and preventive care are not on NPLEX Part I
Immune System: Disorders associated with immunodeficiency: primarily humoral deficiency: selective IgM and IgE deficiencies	Excluded	Rare and not standard review-level content; accuracy could not be held to the same level as selective IgA deficiency
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: vasculitides other than polyarteritis nodosa, Raynaud, polymyalgia rheumatica, reactive arthritis	Excluded	NPLEX blueprint places these in Cardiovascular (CV-18-I) and Musculoskeletal (MS-13-B)
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: immunologic and inflammatory disorders: eosinophilic granuloma, Langerhans cell histiocytosis	Excluded	Hematopoietic neoplasm outside the IM blueprint
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial zoonoses (brucellosis, leptospirosis, melioidosis, tularemia, Q fever, anaplasmosis, ehrlichiosis)	Excluded	Not named on the NPLEX blueprint for this system
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: bacterial infections: Lyme disease, Rocky Mountain spotted fever, miliary tuberculosis	Excluded	NPLEX places these in CV-18-K and PU-14-H9
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: viral infections: yellow fever	Excluded	NPLEX places viral hemorrhagic fevers in CV-18-K5
Multisystem Processes & Disorders: Infectious, immunologic, and inflammatory disorders: fungal and parasitic infections	Excluded	Not in the IM blueprint; candidiasis, schistosomiasis, and Chagas disease sit in IN, HE, and CV units
Multisystem Processes & Disorders: Shock: cardiogenic, hypovolemic, neurogenic, and refractory shock	Excluded	Hemodynamic shock belongs to CV-18-H5; only sepsis and SIRS immunology is covered here

Pulm 99 mapped · 15 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Respiratory: Upper airways: acute upper respiratory infection; viral infections (adenovirus, coxsackievirus, influenza, parainfluenza, rhinoviruses)	PU-14-C.4	Rhinovirus in rhinitis; others in PU-13.5
Respiratory: Upper airways: coronaviruses	PU-14-H.11	
Respiratory: Upper airways: sinusitis	PU-14-C.5	
Respiratory: Upper airways: nasopharyngitis	PU-14-C.4	Common cold
Respiratory: Upper airways: epiglottitis	PU-14-C.2	
Respiratory: Upper airways: Bordetella pertussis pneumonia	PU-14-H.7	
Respiratory: Upper airways: croup, acute laryngitis, acute laryngotracheitis, tracheitis	PU-14-C.3	Croup and tracheitis added as extensions

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Respiratory: Upper airways: pharyngitis, streptococcal throat infections	PU-14-C.6	
Respiratory: Upper airways: tonsillitis, peritonsillar abscess	PU-14-C.7	
Respiratory: Upper airways: rhinitis (allergic, chronic)	PU-14-C.4	
Respiratory: Upper airways: ulcers of nasal cavity and sinuses	PU-12.5	Granulomatosis with polyangiitis as main biomedical cause
Respiratory: Lower airways: hospital-acquired and ventilator-associated pneumonia	PU-14-H.12	Added as USMLE extension
Respiratory: Lower airways: community-acquired pneumonia	PU-14-H.1	
Respiratory: Lower airways: acute bronchiolitis	PU-14-H.8	
Respiratory: Lower airways: bronchiolitis obliterans with organizing pneumonia	PU-14-A.7	Added as USMLE extension
Respiratory: Lower airways: anthrax (pulmonary)	PU-14-H.14	Added as USMLE extension
Respiratory: Lower airways: aspiration pneumonia, pneumonitis	PU-14-H.12	Aspiration abscess in PU-14-H.6; aspiration anatomy in PU-4.2
Respiratory: Lower airways: bronchitis (acute)	PU-14-C.1	
Respiratory: Lower airways: bronchopneumonia	PU-14-H.1	
Respiratory: Lower airways: pneumonia (Streptococcus pneumoniae)	PU-13.1	
Respiratory: Lower airways: pneumonia (Haemophilus influenzae, Klebsiella, Pseudomonas, Legionella, other gram-negative bacteria)	PU-13.2	
Respiratory: Lower airways: pneumonia (Mycoplasma pneumoniae, Chlamydophila pneumoniae)	PU-14-H.2	
Respiratory: Lower airways: pneumonia (Burkholderia pseudomallei, Coxiella burnetii, Francisella tularensis)	PU-14-H.14	Added as USMLE extension
Respiratory: Lower airways: pneumonia (Moraxella catarrhalis)	PU-14-C.5	Covered with sinusitis organisms
Respiratory: Lower airways: pneumonia (MSSA, MRSA)	PU-14-H.12	Also post-influenza in PU-14-H.13
Respiratory: Lower airways: viral infection (influenza A and B, H1N1, avian influenza)	PU-14-H.13	Virology in PU-13.5
Respiratory: Lower airways: viral infection (adenovirus, parainfluenza virus)	PU-14-C.3	Parainfluenza in croup
Respiratory: Lower airways: respiratory syncytial virus	PU-14-H.8	
Respiratory: Lower airways: aspergillosis, allergic bronchopulmonary aspergillosis, aspergilloma	PU-14-H.5	
Respiratory: Lower airways: histoplasmosis, coccidioidomycosis	PU-13.6	Clinical disease in PU-14-H.4
Respiratory: Lower airways: Pneumocystis jirovecii	PU-14-H.5	
Respiratory: Lower airways: pulmonary tuberculosis	PU-14-H.9	
Respiratory: Lower airways: lung abscess	PU-14-H.6	
Respiratory: Neoplasms: vocal cord polyps, nasal polyps	PU-14-F.7	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Respiratory: Neoplasms: juvenile papillomatosis	PU-14-F.4	
Respiratory: Neoplasms: solitary pulmonary nodule	PU-14-F.12	Added as USMLE extension
Respiratory: Neoplasms: lip, oral cavity, and pharynx; head and neck cancer	PU-14-F.13	Added as USMLE extension
Respiratory: Neoplasms: larynx	PU-14-F.4	
Respiratory: Neoplasms: trachea	Excluded	Rare primary tumor; no standard Step 1 depth content beyond squamous and adenoid cystic types
Respiratory: Neoplasms: squamous cell carcinoma of lung	PU-14-F.9	
Respiratory: Neoplasms: adenocarcinoma of lung	PU-14-F.1	
Respiratory: Neoplasms: large cell carcinoma	PU-14-F.9	
Respiratory: Neoplasms: small cell carcinoma	PU-14-F.8	
Respiratory: Neoplasms: bronchial carcinoid tumors	PU-14-F.2	
Respiratory: Neoplasms: mesothelioma	PU-14-F.5	
Respiratory: Neoplasms: secondary malignant neoplasms of lung	PU-14-F.12	
Respiratory: Neoplasms: secondary malignant neoplasms of pleura; metastatic neoplasms including pleural	PU-14-F.5	
Respiratory: Obstructive airway disease: asthma, reactive airway disease	PU-14-B.1	
Respiratory: Obstructive airway disease: bronchiectasis	PU-14-B.3	
Respiratory: Obstructive airway disease: chronic airway obstruction, COPD	PU-14-B.6	
Respiratory: Obstructive airway disease: chronic bronchitis	PU-14-B.4	
Respiratory: Obstructive airway disease: emphysema	PU-14-B.5	
Respiratory: Pneumoconiosis and ILD: pneumoconiosis, silicosis	PU-14-A.4	
Respiratory: Pneumoconiosis and ILD: asbestosis	PU-14-D.4	Also PU-14-A.4
Respiratory: Pneumoconiosis and ILD: berylliosis	PU-14-A.4	
Respiratory: Pneumoconiosis and ILD: silo-filler's lung, byssinosis, bagassosis	PU-14-A.8	Added as USMLE extension
Respiratory: Pneumoconiosis and ILD: hypersensitivity pneumonitis	PU-12.4	Added as USMLE extension
Respiratory: Pneumoconiosis and ILD: hypereosinophilic syndromes, Loeffler syndrome	Excluded	See excluded list
Respiratory: Pneumoconiosis and ILD: usual interstitial pneumonia	PU-14-A.3	
Respiratory: Pneumoconiosis and ILD: desquamative and nonspecific interstitial pneumonia	PU-14-A.7	Added as USMLE extension
Respiratory: Respiratory failure: acute respiratory distress syndrome	PU-14-A.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Respiratory: Respiratory failure: pulmonary hypertension	PU-14-E.5	Added as USMLE extension; physiology in PU-5.2
Respiratory: Respiratory failure: arteriovenous fistula	PU-14-E.6	Added as USMLE extension
Respiratory: Respiratory failure: pulmonary edema	PU-14-E.1	
Respiratory: Respiratory failure: pulmonary embolism	PU-14-E.2	
Respiratory: Respiratory failure: air and fat embolism	PU-14-E.3	Added as USMLE extension
Respiratory: Respiratory failure: respiratory failure due to enteral feeding	PU-14-H.12	Aspiration pneumonitis
Respiratory: Metabolic and structural: disorders of gas exchange	PU-7.1	
Respiratory: Metabolic and structural: hypoventilation	PU-6.2	
Respiratory: Metabolic and structural: hypoxia	PU-7.5	
Respiratory: Metabolic and structural: ventilation-perfusion imbalance	PU-7.3	
Respiratory: Pleura: chylothorax	PU-14-D.1	
Respiratory: Pleura: hemothorax	PU-14-D.2	
Respiratory: Pleura: pneumothorax	PU-14-D.5	
Respiratory: Pleura: pleural effusion	PU-14-D.6	Added as USMLE extension
Respiratory: Pleura: pleuritis	PU-4.5	
Respiratory: Pleura: empyema, mediastinitis, pneumomediastinum, costochondritis	PU-14-D.7	Added as USMLE extension
Respiratory: Traumatic: epistaxis	PU-3.3	
Respiratory: Traumatic: barotrauma (sinus)	Excluded	See excluded list
Respiratory: Traumatic: laryngeal and pharyngeal obstruction; foreign body	PU-4.2	Aspiration site; obstructive atelectasis in PU-14-D.3
Respiratory: Traumatic: tracheoesophageal fistula	PU-14-G.3	
Respiratory: Traumatic: tracheal stenosis, tracheomalacia, tracheal injury, septal perforation	Excluded	See excluded list
Respiratory: Traumatic: atelectasis	PU-14-D.3	
Respiratory: Traumatic: diaphragm and chest wall injury, penetrating chest wounds, pulmonary contusion	PU-14-D.8	Added as USMLE extension
Respiratory: Traumatic: drowning and near drowning	Excluded	See excluded list
Respiratory: Traumatic: sleep apnea (obstructive, central), hypoventilation syndrome, obesity-hypoventilation syndrome	PU-6.7	Added as USMLE extension
Respiratory: Congenital: bronchogenic cysts, congenital cysts, pulmonary sequestration	PU-14-G.5	Added as USMLE extension
Respiratory: Congenital: congenital diaphragmatic hernia	PU-14-G.4	Added as USMLE extension; hypoplasia in PU-1.5
Respiratory: Congenital: immotile cilia syndrome	PU-10.3	
Respiratory: Adverse effects of drugs: bleomycin, amiodarone	PU-14-A.9	Drug as cause of pulmonary fibrosis only
Respiratory: Adverse effects of drugs: adverse effects of 100% oxygen	PU-9.5	Oxidant injury mechanism

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Respiratory: Adverse effects of drugs: tobacco and nicotine, inhalants, cocaine	Excluded	See excluded list
Multisystem: Genetic: alpha-1 antitrypsin deficiency	PU-10.2	
Multisystem: Genetic: immotile cilia syndrome (Kartagener syndrome; primary ciliary dyskinesia)	PU-10.3	
Multisystem: Genetic: cystic fibrosis	PU-10.1	Clinical features in PU-14-G.1
Multisystem: Genetic: VACTERL association	PU-14-G.3	
Multisystem: Immunologic: sarcoidosis, Lofgren syndrome	PU-14-A.5	
Multisystem: Immunologic: Goodpasture syndrome, granulomatosis with polyangiitis	PU-12.5	Added as USMLE extension
Multisystem: Infectious: miliary (disseminated) tuberculosis	PU-14-H.10	
Multisystem: Toxins: decreased atmospheric pressure, high-altitude sickness	PU-7.6	Added as USMLE extension
Multisystem: Toxins: gases, vapors, smoke inhalation	PU-11.2	Carbon monoxide also in PU-8.5
Multisystem: Acid-base: respiratory acidosis, respiratory alkalosis, mixed disturbances	PU-8.6	
Multisystem: Paraneoplastic syndromes: SIADH, Cushing syndrome, hypercalcemia of malignancy, myasthenic syndrome, pulmonary osteoarthropathy	PU-14-F.10	
Multisystem: Paraneoplastic syndromes: paraneoplastic cerebellar degeneration, limbic encephalitis, sensory neuropathy (anti-Hu)	PU-12.7	
Multisystem: Signs and symptoms: cough, dyspnea, hemoptysis	PU-14-E.4	Hemoptysis and dyspnea are tied to specific disorders throughout PU-14
Respiratory System: Adverse effects of drugs (acute effects of tobacco and nicotine, inhalants, cocaine)	Excluded	Drug and substance adverse effects are pharmacology, not on NPLEX Part I; tobacco as a cause of listed diseases is covered within those objectives
Respiratory System: Adverse effects of drugs (bleomycin, amiodarone) as pharmacology	Excluded	Drug pharmacology excluded; kept only as a cause of pulmonary fibrosis in PU-14-A.9
Respiratory System: Pneumoconiosis and ILD: hypereosinophilic syndromes, Loeffler syndrome	Excluded	Not named on the NPLEX blueprint and lower yield than the included interstitial diseases; left out rather than risk thin coverage
Respiratory System: Traumatic: barotrauma (sinus)	Excluded	Not on the NPLEX blueprint and low yield for biomedical science items
Respiratory System: Traumatic: tracheal stenosis, tracheomalacia, tracheal injury, septal perforation	Excluded	Clinical and surgical topics outside NPLEX Part I biomedical scope
Respiratory System: Traumatic: drowning and near drowning	Excluded	Emergency care topic, not on the NPLEX blueprint
Respiratory System: Neoplasms: malignant tumors of the trachea	Excluded	Rare; not on the NPLEX blueprint
Respiratory System: management of respiratory failure, oxygen therapy, and ventilator settings	Excluded	Therapeutics are not on NPLEX Part I
Respiratory System: screening for lung cancer and vaccination schedules	Excluded	Care of the well patient and screening schedules are not on NPLEX Part I

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USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Blood & Lymphoreticular System: Infectious and immunologic: bacterial infections	HE-9-E.7	Covered by plague and cat-scratch lymphadenitis; bacterial sepsis sits in other systems
Blood & Lymphoreticular System: Infectious and immunologic: viral hemorrhagic fever (Ebola, Marburg)	HE-9-E.6	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: chikungunya	HE-9-E.5	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: dengue fever	HE-9-E.5	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: Zika virus disease	HE-9-E.5	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: malaria	HE-9-E.2	Also HE-8.1 to HE-8.3, HE-9-E.3
Blood & Lymphoreticular System: Infectious and immunologic: babesiosis	HE-9-E.1	Also HE-8.4
Blood & Lymphoreticular System: Infectious and immunologic: lymphadenitis (viral, bacterial, fungal, parasitic)	HE-9-E.8	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: lymphangitis	HE-9-E.8	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: bubonic plague	HE-9-E.7	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: cat-scratch disease	HE-9-E.7	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: cryoglobulinemia, essential mixed cryoglobulinemia	HE-7.2	
Blood & Lymphoreticular System: Infectious and immunologic: autoimmune hemolytic anemia	HE-7.1	
Blood & Lymphoreticular System: Infectious and immunologic: paroxysmal nocturnal hemoglobinuria	HE-9-B.7	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: thrombotic thrombocytopenic purpura	HE-9-B.9	Added as USMLE extension
Blood & Lymphoreticular System: Infectious and immunologic: hemolytic uremic syndrome	HE-9-B.9	Added as USMLE extension
Blood & Lymphoreticular System: Neoplasms: acute lymphoblastic leukemia	HE-9-D.1	
Blood & Lymphoreticular System: Neoplasms: acute myeloid leukemia	HE-9-D.2	
Blood & Lymphoreticular System: Neoplasms: chronic lymphocytic leukemia	HE-9-D.3	
Blood & Lymphoreticular System: Neoplasms: chronic myeloid leukemia	HE-9-D.4	Also HE-6.4, HE-9-A.11
Blood & Lymphoreticular System: Neoplasms: Hodgkin lymphoma	HE-9-D.6	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Blood & Lymphoreticular System: Neoplasms: non-Hodgkin lymphoma	HE-9-D.7	
Blood & Lymphoreticular System: Neoplasms: Burkitt lymphoma	HE-9-D.7	Also HE-6.5
Blood & Lymphoreticular System: Neoplasms: T-cell lymphoma	HE-9-D.8	
Blood & Lymphoreticular System: Neoplasms: multiple myeloma	HE-9-D.9	
Blood & Lymphoreticular System: Neoplasms: dysproteinemias	HE-9-D.10	
Blood & Lymphoreticular System: Neoplasms: monoclonal gammopathy of unknown significance	HE-9-D.10	
Blood & Lymphoreticular System: Neoplasms: Waldenström macroglobulinemia	HE-9-D.10	
Blood & Lymphoreticular System: Neoplasms: myelofibrosis	HE-9-D.12	Added as USMLE extension
Blood & Lymphoreticular System: Neoplasms: myelodysplastic syndrome	HE-9-D.11	Added as USMLE extension
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: iron deficiency anemia (decreased production)	HE-9-A.1	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: anemia of chronic disease	HE-9-A.2	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: G6PD deficiency	HE-9-B.3	Also HE-5.5
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: pyruvate kinase deficiency	HE-9-B.8	Added as USMLE extension
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: hereditary spherocytosis	HE-9-B.4	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: hereditary elliptocytosis	HE-9-B.8	Added as USMLE extension
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: congenital methemoglobinemia	HE-5.4	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: sickle cell disease	HE-9-B.1	Also HE-6.1
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: sideroblastic anemia	HE-9-A.3	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: thalassemias	HE-9-B.2	Also HE-6.2
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: blood loss anemia (acute and chronic)	HE-9-A.9	Added as USMLE extension
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: aplastic anemia	HE-9-A.6	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: leukopenia, neutropenia, cyclic neutropenia, agranulocytosis	HE-9-A.10	Added as USMLE extension
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: pancytopenia	HE-3.5	Also HE-9-A.6, HE-9-B.12
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: quantitative thrombocytopenia	HE-9-C.4	Also HE-3.5
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: immune thrombocytopenic purpura	HE-9-C.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: leukocytosis	HE-9-A.11	Also HE-2.6
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: polycythemia vera	HE-9-A.7	
Blood & Lymphoreticular System: Anemia, cytopenias, and polycythemia: secondary polycythemia	HE-9-A.7	
Blood & Lymphoreticular System: Coagulation disorders: disseminated intravascular coagulation	HE-9-C.1	
Blood & Lymphoreticular System: Coagulation disorders: hemophilia A, hemophilia B, congenital hemophilia	HE-9-C.3	
Blood & Lymphoreticular System: Coagulation disorders: hypofibrinogenemia	HE-9-C.7	Acquired hypofibrinogenemia of liver failure and DIC
Blood & Lymphoreticular System: Coagulation disorders: von Willebrand disease	HE-9-C.5	
Blood & Lymphoreticular System: Coagulation disorders: qualitative platelet dysfunction	HE-9-C.8	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: heparin-induced thrombocytopenia	HE-7.4	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: homocysteinemia	HE-9-C.9	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: hypoplasminogenemia	Excluded	See excluded
Blood & Lymphoreticular System: Coagulation disorders: antithrombin III deficiency	HE-9-C.9	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: protein C deficiency	HE-9-C.9	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: protein S deficiency	HE-9-C.9	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: factor V Leiden	HE-6.7	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: anticardiolipin antibodies	HE-7.5	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: lupus anticoagulant	HE-7.5	Added as USMLE extension
Blood & Lymphoreticular System: Coagulation disorders: prothrombin G20210A mutation	HE-6.7	Added as USMLE extension
Blood & Lymphoreticular System: Reactions to blood components: ABO incompatibility	HE-7.3	Also HE-2.7, HE-9-B.5
Blood & Lymphoreticular System: Reactions to blood components: Rh incompatibility	HE-9-B.5	
Blood & Lymphoreticular System: Reactions to blood components: delayed hemolytic transfusion reaction	HE-7.3	
Blood & Lymphoreticular System: Reactions to blood components: transfusion reaction	HE-7.3	
Blood & Lymphoreticular System: Reactions to blood components: bacterially contaminated transfusion	HE-7.3	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Blood & Lymphoreticular System: Reactions to blood components: transfusion-related acute lung injury	HE-7.3	
Blood & Lymphoreticular System: Reactions to blood components: anaphylactoid reaction (IgA deficiency)	HE-7.3	
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: mechanical erythrocyte injury (cardiac valve hemolysis)	HE-9-B.10	Added as USMLE extension
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: splenic rupture/laceration	HE-9-B.11	Added as USMLE extension
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: splenic infarct	HE-9-B.11	Added as USMLE extension
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: splenic abscess	HE-9-B.11	Added as USMLE extension
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: splenectomy effects/complications	HE-9-B.11	Also HE-7.7
Blood & Lymphoreticular System: Traumatic, mechanical, and vascular disorders: hypersplenism	HE-9-B.12	Added as USMLE extension
Blood & Lymphoreticular System: Adverse effects of drugs: antiplatelet drugs	Excluded	Pharmacology
Blood & Lymphoreticular System: Adverse effects of drugs: antithrombin drugs (dabigatran)	Excluded	Pharmacology
Blood & Lymphoreticular System: Adverse effects of drugs: chemotherapeutic agents	Excluded	Pharmacology
Blood & Lymphoreticular System: Adverse effects of drugs: coagulation factor inhibitors	Excluded	Pharmacology
Blood & Lymphoreticular System: Adverse effects of drugs: acquired methemoglobinemia	HE-5.4	Kept as a drug cause of a listed condition
Blood & Lymphoreticular System: Adverse effects of drugs: propylthiouracil	HE-9-A.10	Kept only as a cause of agranulocytosis
Blood & Lymphoreticular System: Adverse effects of drugs: tumor lysis syndrome	HE-9-D.14	Added as USMLE extension
Blood & Lymphoreticular System: Adverse effects of drugs: warfarin	Excluded	Pharmacology
Multisystem Processes & Disorders: Genetic, metabolic, and developmental disorders: porphyria	HE-5.2	Also HE-5.1
Multisystem Processes & Disorders: Toxins and environmental extremes: metals (lead)	HE-9-A.3	Heme synthesis effects only; neurologic and renal effects belong to other systems
Multisystem Processes & Disorders: Neoplasms: inherited cancer syndromes: DNA repair abnormalities (Fanconi anemia)	HE-6.6	
Multisystem Processes & Disorders: Neoplasms: paraneoplastic syndromes: hematologic	HE-9-C.10	Added as USMLE extension
Multisystem Processes & Disorders: Nutrition: vitamin deficiencies and/or toxicities: vitamin B12, cobalamins (pernicious anemia)	HE-9-A.4	Also HE-5.6
Multisystem Processes & Disorders: Nutrition: vitamin deficiencies and/or toxicities: vitamin B9, folic acid	HE-9-A.5	Also HE-5.6

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Multisystem Processes & Disorders: Nutrition: mineral deficiencies and/or toxicities: iron	HE-9-A.1	Also HE-3.4
Multisystem Processes & Disorders: Nutrition: vitamin deficiencies and/or toxicities: vitamin K	HE-9-C.6	Also HE-5.7
Multisystem Processes & Disorders: Immunologic and inflammatory disorders: Langerhans cell histiocytosis, eosinophilic granuloma	HE-9-D.13	Added as USMLE extension
Blood & Lymphoreticular System: Adverse effects of drugs: antiplatelet drugs, antithrombin drugs (dabigatran), chemotherapeutic agents, coagulation factor inhibitors, warfarin	Excluded	Pharmacology and drug adverse effects are not on NPLEX Part I
Blood & Lymphoreticular System: Coagulation disorders: hypoplasminogenemia	Excluded	Left out because its link to thrombosis is not consistent standard review-level teaching; accuracy rule
Blood & Lymphoreticular System: transfusion and anticoagulant therapy selection, chemotherapy regimens	Excluded	Therapeutics are not on NPLEX Part I

Derm 95 mapped · 6 excluded

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: cellulitis	IN-13-H.3	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: erysipelas	IN-13-H.3	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: impetigo	IN-13-H.7	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: staphylococcal scalded skin syndrome	IN-13-H.18	Added as USMLE extension; toxin mechanism also in IN-9.3
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: abscess, cutaneous, including septic abscess	IN-13-H.6	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: anthrax	IN-13-H.20	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: carbuncle	IN-13-H.6	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: folliculitis	IN-13-H.6	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: pilonidal cyst, infected	IN-13-H.6	
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: MSSA and MRSA skin infections	IN-13-H.8	Organism detail in IN-12.1
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: mycobacterial infections (leprosy, draining sinus)	IN-13-H.19	Added as USMLE extension; Mycobacterium marinum in IN-13-H.13
Skin & Subcutaneous Tissue: Infectious disorders: bacterial: scarlet fever	IN-13-H.17	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: herpes simplex type 1 and type 2	IN-13-H.15	Added as USMLE extension; virology in IN-12.6

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Skin & Subcutaneous Tissue: Infectious disorders: viral: herpes zoster	IN-13-H.15	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: Ramsay-Hunt syndrome	IN-13-H.15	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: molluscum contagiosum	IN-13-H.9	
Skin & Subcutaneous Tissue: Infectious disorders: viral: hand-foot-and-mouth disease	IN-13-H.17	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: herpangina	IN-13-H.17	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: parvovirus; erythema infectiosum	IN-13-H.16	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: chickenpox	IN-13-H.17	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: rubella	IN-13-H.16	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: measles	IN-13-H.16	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: roseola (exanthema subitum)	IN-13-H.16	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: viral: verrucae vulgaris	IN-13-H.11	
Skin & Subcutaneous Tissue: Infectious disorders: fungal: candidiasis, skin	IN-13-H.2	
Skin & Subcutaneous Tissue: Infectious disorders: fungal: dermatophytosis; tinea corporis; dermatomycoses	IN-13-H.10	
Skin & Subcutaneous Tissue: Infectious disorders: fungal: diaper rash	IN-13-H.2	
Skin & Subcutaneous Tissue: Infectious disorders: fungal: onychomycosis	IN-13-H.10	
Skin & Subcutaneous Tissue: Infectious disorders: parasitic: cutaneous larva migrans	IN-12.8	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: parasitic: cutaneous leishmaniasis	IN-12.8	Added as USMLE extension
Skin & Subcutaneous Tissue: Infectious disorders: infestations and nonvenomous bites: scabies, lice, insect bites including bed bugs	IN-13-H.21	Added as USMLE extension
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: papulosquamous and eczematous: psoriasis	IN-13-C.4	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: papulosquamous and eczematous: lichen planus and lichenoid dermatoses	IN-13-C.3	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: papulosquamous and eczematous: allergic and irritant contact dermatitis (nickel)	IN-13-B.2	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: papulosquamous and eczematous: dermatoses caused by plants (poison ivy, poison oak)	IN-13-B.2	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: vesiculobullous: epidermolysis bullosa	IN-13-D.1	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: vesiculobullous: dermatitis herpetiformis	IN-13-D.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: vesiculobullous: pemphigus	IN-13-D.4	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: vesiculobullous: pemphigoid	IN-13-D.3	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: vesiculobullous: pyoderma gangrenosum	IN-13-C.6	Added as USMLE extension under chronic inflammatory conditions
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: erythema nodosum	IN-13-H.5	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: atopic dermatitis	IN-13-B.1	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: pityriasis rosea	IN-13-B.7	Added as USMLE extension
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: urticaria	IN-13-B.5	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: Stevens-Johnson syndrome	IN-13-B.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: erythema multiforme	IN-13-B.4	
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: urticaria, erythema, exanthema, purpura: toxic epidermal necrolysis	IN-13-B.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Immunologic and inflammatory disorders: autoimmune: vitiligo	IN-13-A.3	
Skin & Subcutaneous Tissue: Neoplasms: benign: actinic keratoses	IN-13-F.1	
Skin & Subcutaneous Tissue: Neoplasms: benign: cysts, including epidermal	IN-13-F.5	Added as USMLE extension
Skin & Subcutaneous Tissue: Neoplasms: benign: hemangiomas	IN-13-F.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Neoplasms: benign: lipoma	IN-13-F.5	Added as USMLE extension
Skin & Subcutaneous Tissue: Neoplasms: benign: pigmented nevi	IN-13-A.1	
Skin & Subcutaneous Tissue: Neoplasms: benign: seborrheic keratosis	IN-13-F.3	
Skin & Subcutaneous Tissue: Neoplasms: benign: xanthomas	IN-5.7	
Skin & Subcutaneous Tissue: Neoplasms: malignant: basal cell carcinoma	IN-13-G.1	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Skin & Subcutaneous Tissue: Neoplasms: malignant: squamous cell carcinoma	IN-13-G.2	
Skin & Subcutaneous Tissue: Neoplasms: malignant: melanoma, including genital	IN-13-G.4	
Skin & Subcutaneous Tissue: Neoplasms: malignant: cutaneous T-cell lymphoma, mycosis fungoides	IN-13-G.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: hair: alopecia	IN-13-C.8	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: hair: seborrhea capitis, seborrheic dermatitis	IN-13-B.8	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: hair: tinea barbae and capitis	IN-13-H.10	
Skin & Subcutaneous Tissue: Integumentary disorders: nails: ingrowing nail	IN-1.7	Covered as nail fold anatomy
Skin & Subcutaneous Tissue: Integumentary disorders: sweat and sebaceous glands: acne vulgaris	IN-13-H.1	
Skin & Subcutaneous Tissue: Integumentary disorders: sweat and sebaceous glands: hidradenitis suppurativa	IN-13-C.7	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: sweat and sebaceous glands: hyperhidrosis	IN-1.6	Covered with eccrine gland control
Skin & Subcutaneous Tissue: Integumentary disorders: sweat and sebaceous glands: ichthyosis	IN-7.6	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: sweat and sebaceous glands: rosacea	IN-13-C.1	
Skin & Subcutaneous Tissue: Integumentary disorders: oral disease: aphthous ulcers	IN-13-B.9	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: oral disease: leukoplakia	IN-13-F.7	Added as USMLE extension
Skin & Subcutaneous Tissue: Integumentary disorders: disorders of pigmentation: albinism	IN-13-E.1	
Skin & Subcutaneous Tissue: Integumentary disorders: disorders of pigmentation: lentigo	IN-13-A.5	Added as USMLE extension
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: animal bites	IN-13-H.13	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: burns or wounds, including sunburn and blast injuries	IN-10.4	Blast injury mechanics not given a separate objective; sunburn in IN-6.4
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: cauliflower ear	IN-10.7	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: effects of ultraviolet light	IN-6.4	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: keloids	IN-10.3	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: tattoo	IN-8.5	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: thermal injury	IN-10.4	

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: perniosis	IN-4.7	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: frostbite	IN-4.7	
Skin & Subcutaneous Tissue: Traumatic and mechanical disorders: ulcers, decubitus	IN-10.7	
Skin & Subcutaneous Tissue: Congenital disorders: xeroderma pigmentosum	IN-7.2	Added as USMLE extension
Skin & Subcutaneous Tissue: Congenital disorders: benign lesions in neonates, infants, and children (congenital nevi)	IN-2.4	Congenital nevus risk also in IN-13-A.2
Skin & Subcutaneous Tissue: Adverse effects of drugs: drug reactions and eruptions	Excluded	
Skin & Subcutaneous Tissue: Adverse effects of drugs: local reaction to vaccine	IN-11.7	Covered only as Arthus (type III) immunology
Multisystem Processes & Disorders: Toxins and environmental extremes: temperature (hypothermia, hyperthermia, heat stroke)	IN-4.5	
Multisystem Processes & Disorders: Toxins and environmental extremes: thermal injury, burns	IN-10.4	
Multisystem Processes & Disorders: Toxins and environmental extremes: electrocution, lightning	IN-10.5	
Multisystem Processes & Disorders: Toxins and environmental extremes: radiation	IN-10.6	Skin effects only; radon and uranium mining excluded
Multisystem Processes & Disorders: Neoplasms and related disorders: paraneoplastic syndromes: mucocutaneous	IN-13-F.4	Added as USMLE extension
Multisystem Processes & Disorders: Neoplasms and related disorders: inherited cancer syndromes (familial melanoma, Gorlin syndrome)	IN-7.3	
Multisystem Processes & Disorders: Genetic, metabolic, and developmental disorders: structural protein disorders: Ehlers-Danlos syndrome	IN-13-E.3	
Multisystem Processes & Disorders: Genetic, metabolic, and developmental disorders: enzymatic/metabolic: porphyria (cutaneous)	IN-5.8	Added as USMLE extension
Multisystem Processes & Disorders: Nutrition: vitamins A, B3, C, D and mineral deficiencies (skin findings)	IN-5.4	
Multisystem Processes & Disorders: Venomous bites and stings: spider bites	IN-10.8	Brown recluse dermonecrosis only
Multisystem Processes & Disorders: Venomous bites and stings: hymenoptera, scorpion, snake, jellyfish	Excluded	
Skin & Subcutaneous Tissue: Adverse effects of drugs: drug reactions and eruptions (fixed drug eruption, morbilliform eruptions, photosensitivity reactions)	Excluded	Drug adverse effects are pharmacology, not on NPLEX Part I; SJS and TEN kept as conditions with drug causes (IN-13-B.6)
Multisystem Processes & Disorders: Venomous bites and stings: hymenoptera stings, scorpion stings, snake bites, jellyfish stings, black widow spider	Excluded	Systemic toxicology and anaphylaxis without primary skin pathology; outside this system's skin scope
Multisystem Processes & Disorders: Toxins and environmental extremes: radiation (radon, uranium mining, imaging studies)	Excluded	Lung cancer and imaging exposure belong to the respiratory system and radiology, not skin

USMLE STEP 1 TOPIC	WHERE IN THIS COURSE	NOTE
Human Development: care of the well patient (skin cancer screening, sun protection counseling)	Excluded	Care of the well patient and screening schedules are not on NPLEX Part I

About this list

NPLEX source. NABNE, NPLEX Part I Biomedical Sciences Study Guide, revised March 2026, for the August 2026 administrations. Open the study guide.

USMLE source. USMLE Step 1 Content Outline and Specifications. Open the specifications. USMLE removed its General Principles section, so foundational science sits inside each organ system here too.

Not included. Pharmacology, behavioral and social sciences, biostatistics and epidemiology, and care of the well patient. NPLEX Part I does not test them. A drug appears only when it causes a listed condition.

Codes. Unit codes come from the blueprint (CV-7, CV-18–C). Competency codes add a number (CV-7.3). Condition codes add a digit to the category (CV-18–C3 is myocardial infarction).

Added by MedMasters, not NABNE. Unit short names, exam-area tags, depth (DOK) levels, the USMLE extensions, and the crosswalk. Each question still carries its own exam area.

MedMasters Collaborative · Dr. Sharilyn Rennie